THE **COMPLETE VISUAL GUIDE** TO

MACHINE LEARNING



★★★★★ *With Best-Selling Instructors* ***Josh MacCarty*** *&* ***Chris Dutton***

MAVEN®
ANALYTICS

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SETTING EXPECTATIONS

- This is **NOT** a coding/programming course; it's about introducing and demystifying essential machine learning topics
  - *Our goal is to break down complex techniques using simple and intuitive explanations and demos*
- We'll focus on tools commonly applied to **business intelligence** use cases
  - *We'll focus on techniques like data profiling, linear/logistic regression, forecasting, and unsupervised learning, but will not cover some more advanced or specialized techniques (deep learning, NLP, etc.)*
- We'll use **Microsoft Excel** as a tool to help explain key concepts
  - *Excel's intuitive, visual interface allows us to expose the nuts and bolts of each technique to understand HOW and WHY these algorithms work (rather than simply running lines of code)*
- You do **NOT** need a math or stats background to take this course
  - *We'll cover the basics as needed, but won't dive deep into statistics or econometric theory*

SETTING EXPECTATIONS

Who this is for:

- Analysts or BI professionals looking to transition into a ML/data science role
- Students looking to develop a deep conceptual understanding of core machine learning topics
- Anyone who wants to understand WHEN, WHY, and HOW to deploy machine learning tools & techniques

Who this is NOT for:

- Senior data professionals looking to master advanced topics in ML/AI
- Students looking for a hands-on coding course or bootcamp (i.e. python/R)
- Anyone who would rather copy and paste code than become fluent in the underlying algorithms

PART 1:

DATA PROFILING



ABOUT THIS SERIES

This is **Part 1** of a **4-Part series** designed to help you build a deep, foundational understanding of machine learning, including **data QA & profiling**, **classification**, **forecasting** and **unsupervised learning**



PART 1

QA & Data Profiling

PART 2

Classification

PART 3

Regression & Forecasting

PART 4

Unsupervised Learning

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COURSE OUTLINE

1 ML Intro & Landscape	*Machine Learning introduction, definition, process & landscape*
2 Preliminary Data QA	*Tools to explore **data quality** (variable types, empty values, range & count calculations, table structure, left/right censored data, etc.)*
3 Univariate Profiling	*Tools to understand **individual variables** (distribution, histograms & kernel densities, data profiling metrics, etc.)*
4 Multivariate Profiling	*Tools to understand **multiple variables** (kernel densities, violin & box plots, correlation, variance, etc.)*

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ML INTRO & LANDSCAPE

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INTRO TO MACHINE LEARNING (ML)



MACHINE LEARNING [m*uh*-**sheen**-lur-ning]

noun

~~1. The capacity of a computer to process and evaluate data beyond programmed algorithms, through contextualized inference*~~

Using statistical models to find patterns and make predictions

*Dictionary.com

COMMON ML QUESTIONS



Which customers are most likely to churn next month?



What will sales look like for the next 12 months?



What patterns do we see in terms of product cross-selling?



How can we use online customer reviews to monitor changes in sentiment?

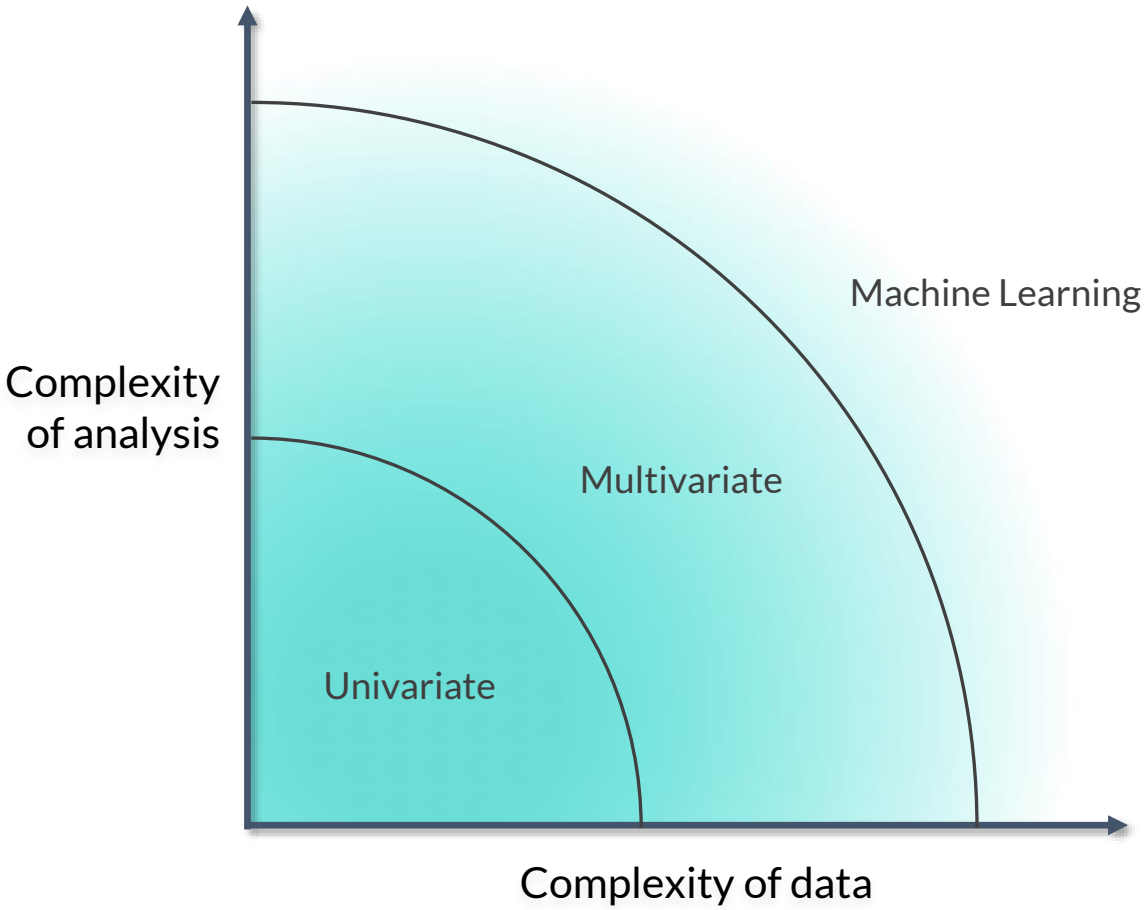


When we adjusted tactics last month, did we drive any incremental revenue?



Which product is customer X most likely to purchase next?

WHEN IS ML THE RIGHT FIT?



- Machine Learning is a natural extension of data profiling and basic visual analysis
- ML is required when the underlying data or analysis is **too complex for basic data profiling** *(i.e. visualizing relationships between 3+ variables)*
- Machine Learning is ideal for finding optimal solutions that would be **impossible or impractical to derive through human trial-and-error**

THE MACHINE LEARNING PROCESS

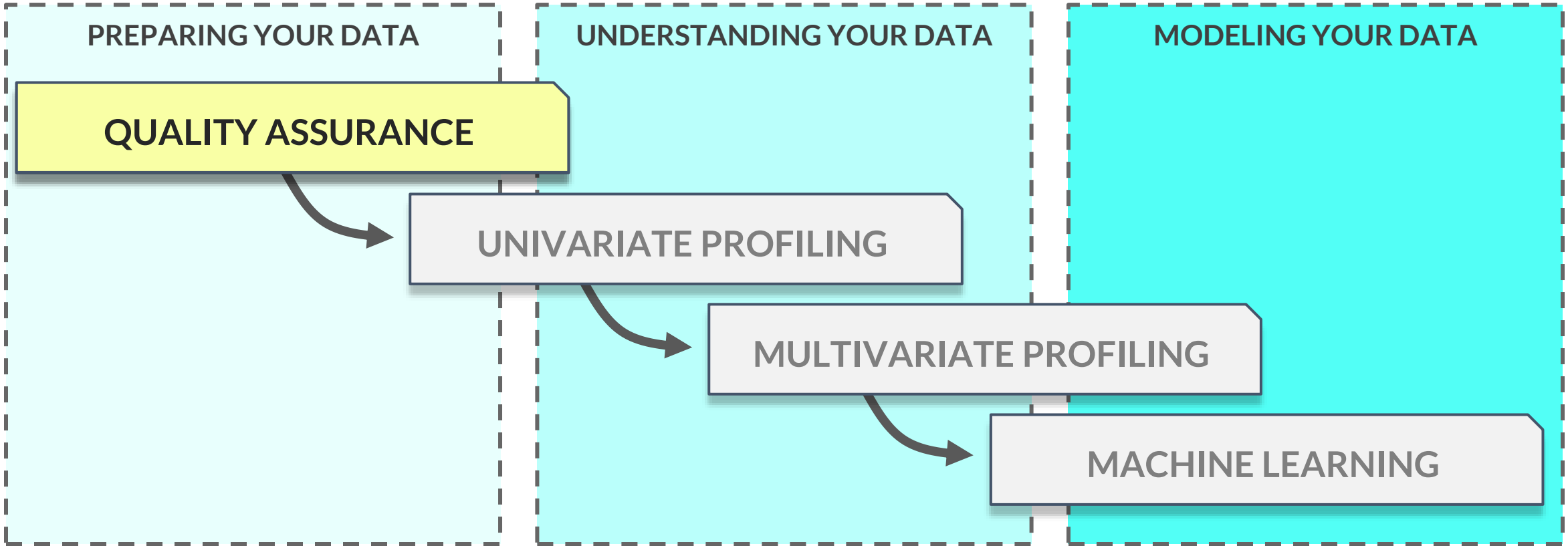
Building models
(the "fun" stuff)

Data Prep, QA & Profiling
(the boring but really, really, REALLY important stuff)



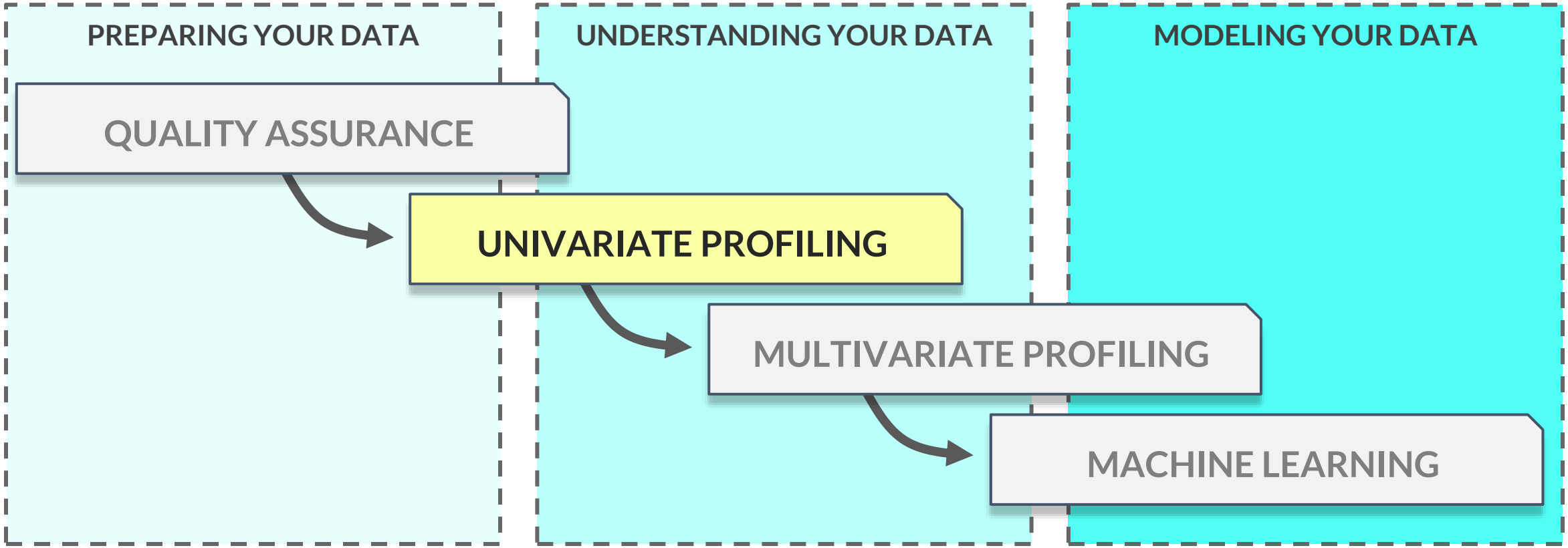
- ML models are only as good as the data they are built on *("garbage in, garbage out")*
- Data prep & QA happens behind the scenes, but typically accounts for the majority (**80%+**) of the machine learning workflow
- While it's tempting to start with the "fun" stuff, data prep and QA skills are ***absolutely critical!***

MACHINE LEARNING PROCESS



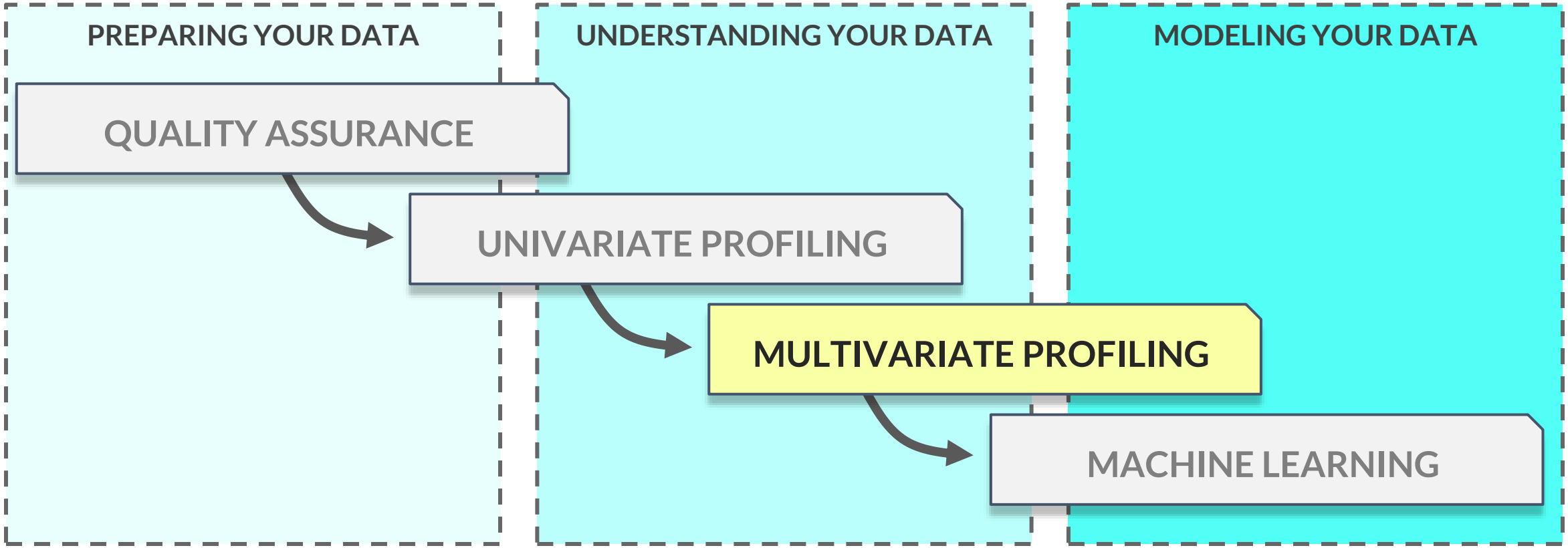
Quality Assurance (QA) is about **preparing & cleaning data prior to analysis**. We'll cover common QA topics including variable types, empty/missing values, range & count calculations, censored data, etc.

MACHINE LEARNING PROCESS



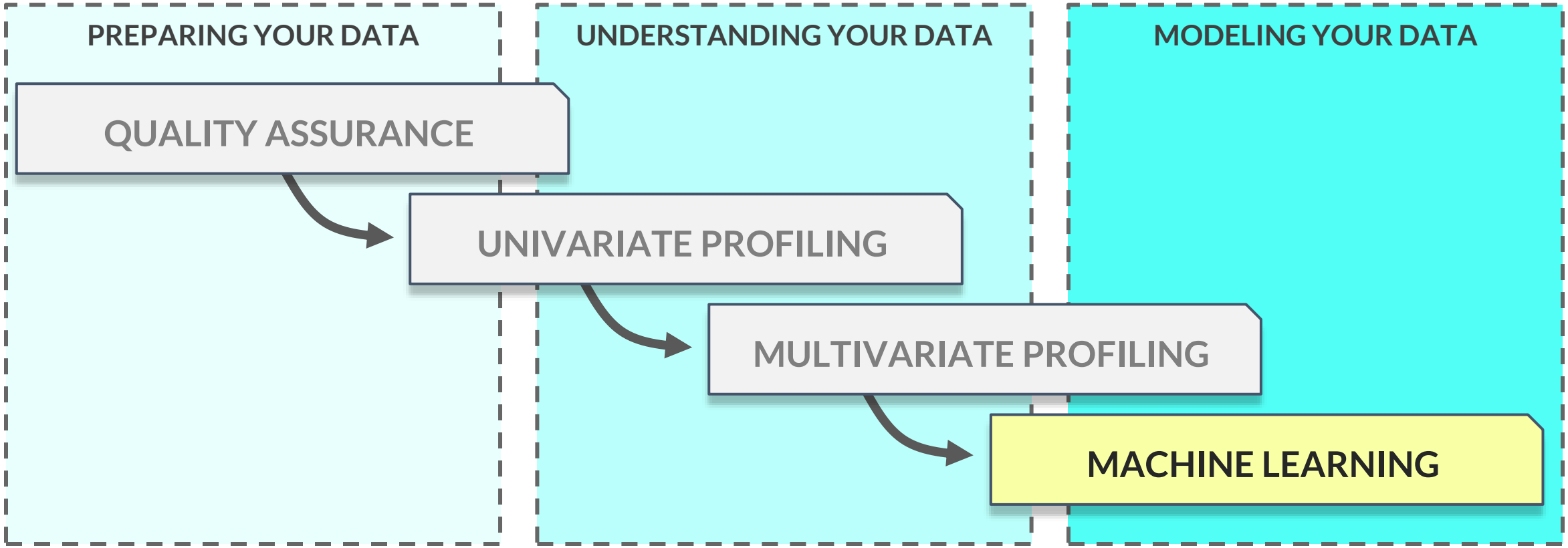
Univariate profiling is about **exploring individual variables** to build an understanding of your data. We'll cover common topics like normal distributions, frequency tables, histograms, etc.

MACHINE LEARNING PROCESS



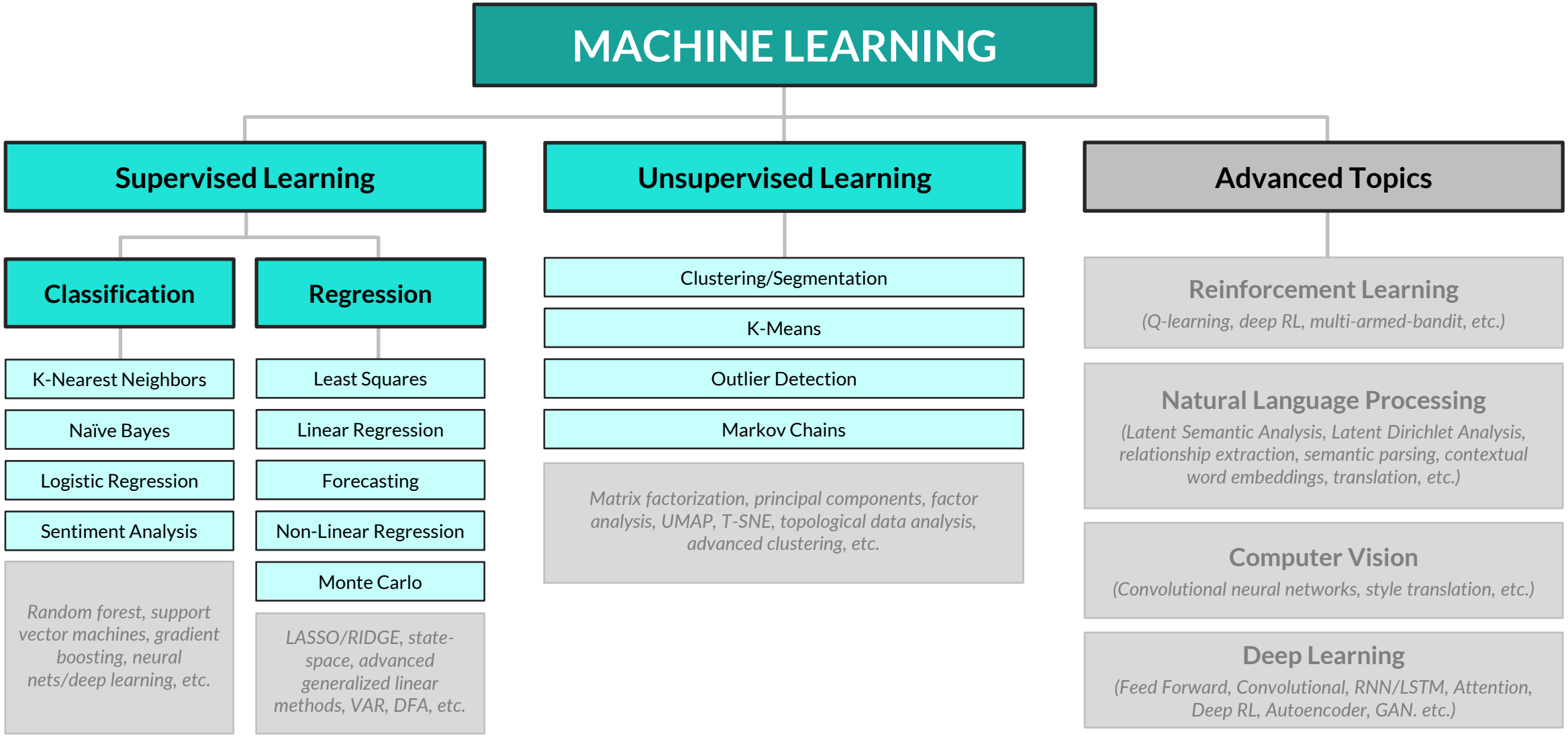
Multivariate profiling is about **understanding relationships between *multiple* variables**. We'll cover common tools for exploring categorical & numerical data, including kernel densities, violin & box plots, scatterplots, etc.

MACHINE LEARNING PROCESS



Machine learning is a natural extension of multivariate profiling, and uses **statistical models and methods** to answer questions which are too complex to solve using simple visual analysis or trial-and-error

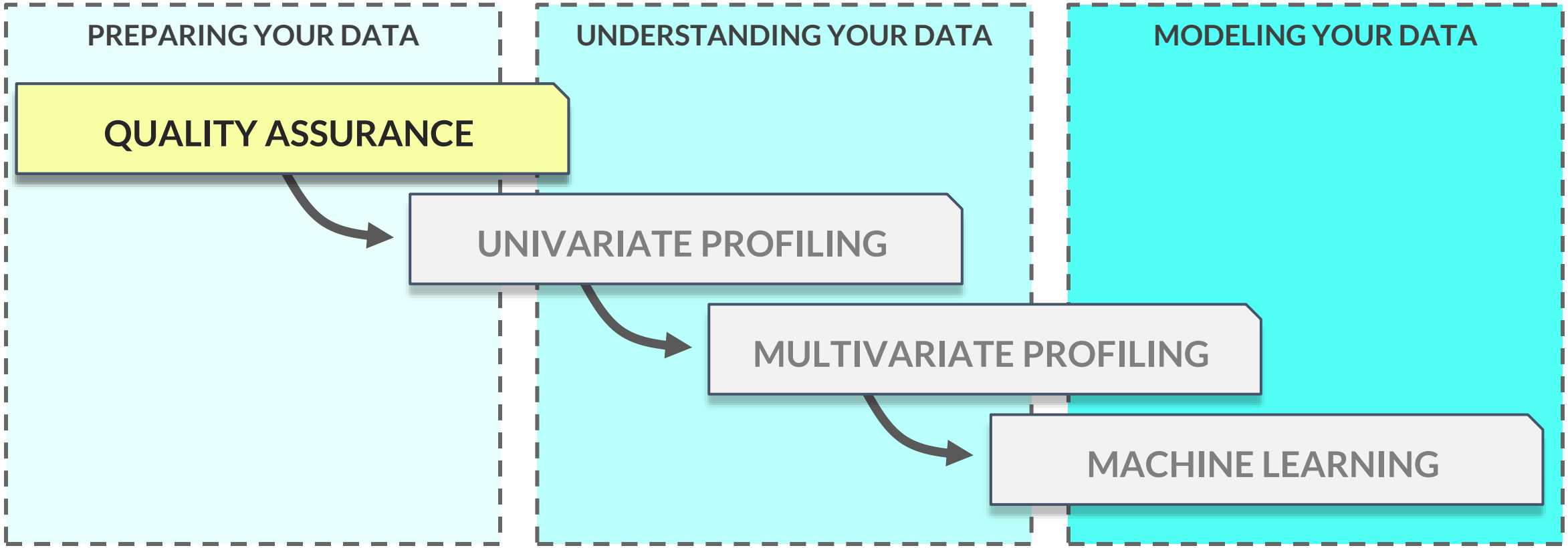
MACHINE LEARNING LANDSCAPE



PRELIMINARY DATA QA

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MACHINE LEARNING PROCESS



Quality Assurance (QA) is about **preparing & cleaning data prior to analysis**. We'll cover common QA topics including variable types, empty/missing values, range & count calculations, censored data, etc.

PRELIMINARY DATA QA



Data QA (otherwise known as **Quality Assurance** or **Quality Control**) is the first step in the analytics and machine learning process; QA allows you to identify and correct underlying data issues *(blanks, errors, incorrect formats, etc.)* prior to analysis

TOPICS WE'LL COVER:

- Variable Types
- Empty Values
- Range Calculations
- Count Calculations
- Table Structures
- Left/Right Censors

COMMON USE CASES:

- Minimizing the risk of drawing false conclusions or misunderstanding your data
- Confirming that dates are properly formatted for analysis *(as date values, not text strings)*
- Replacing blanks or errors with appropriate values to prevent summarization errors
- Calculating ranges *(max & min)* to spot-check for outliers or unexpected values

PRELIMINARY DATA QA



POP QUIZ: *When should you QA your data?*

EVERY. SINGLE. TIME.

(no exceptions!)

WHY IS QA IMPORTANT?

As an analyst, **Preliminary Data QA** will help you answer questions like:



Are there any missing or empty values in the data shared by the HR team?



Was the data our client captured from the online survey encoded properly?



Is there any risk that the data capture process was biased in some way?



Are there any outliers that might skew the results of our analysis?

VARIABLE TYPES

- Variable Types
- Empty Values
- Range Calculations
- Count Calculations
- Left/Right Censored
- Table Structure

Variable types give us information about our variables

- January 1, 2000 as a **number** simply displays a date
- January 1, 2000 as a **date** implies a set of information we can use *(first day of the year*, *winter*, *Saturday*, *weekend*, *etc.)* all of which can be used to build strong machine learning models

Common **variable types** include:

Numeric
- Customer count

Ordinal
- Small, Medium, Large

Interval
- Temperature

Ratio
- Weight

Discrete
- Count

Categorical
- Gender

Nominal
- Nationality

Binary
- Yes/No

Date
- 1/1/2020

Complex
- 4 + 2i

Logical
- True/False

Monetary
- $4.50

VARIABLE TYPES

- Variable Types
- Empty Values
- Range Calculations
- Count Calculations
- Left/Right Censored
- Table Structure

Understanding **variable types** is fundamental to the machine learning process, and can help avoid common QA issues, including:

- Numeric variables formatted as characters or strings, preventing proper aggregation or analysis
- String/character variables formatted as numeric, preventing proper text-based operations
- Values that Less obvious cases like variables that are re-coded from raw values to buckets (surveys)

ZIP CODES	
Numeric	**String**
90210	90210
32992	32992
1730	01730
2215	02215
92120	92120

*Here we're formatting **zip codes** (which will never be analyzed as values), as a **numeric** rather than **string***



When we talk about **variables**, we might also refer to them as metrics, KPIs, dimensions, or columns. Similarly, we may use rows, observations, and records interchangeably

EMPTY VALUES

- Variable Types
- **Empty Values**
- Range Calculations
- Count Calculations
- Left/Right Censored
- Table Structure

Investigating **empty values**, and how they are recorded in your data, is a prerequisite for *every single* analysis

Empty values can be recorded in many ways (*NA*, *N/A*, *#N/A*, *NaN*, *Null*, *"-"*, *"Invalid"*, *blank*, *etc.)*, but the most common mistake is **turning empty numerical values into zeros (0)**

Sales Leaders		
Name	Age	Sales
John	33	278
Sally	30	173
Mark	38	210
Gina	0	122
Phil	42	246

Average Age = **28.6** ✗

Sales Leaders		
Name	Age	Sales
John	33	278
Sally	30	173
Mark	38	210
Gina		122
Phil	42	246

Average Age = **35.8** ✓

EMPTY VALUES

- Variable Types
- Empty Values
- Range Calculations
- Count Calculations
- Left/Right Censored
- Table Structure

Empty values can be handled in 3 ways: **keep**, **remove**, or **impute**

- **Keep** empty or zero values if you are certain that they are accurate and meaningful *(i.e. no sales of a product on a specific date)*
- **Remove** empty values if you have a large volume of data and can confirm that there is no pattern or bias to the missing values *(i.e. all sales from a specific product category are missing)*
- **Impute** (substitute) empty values if you can accurately populate the data or if you are working with limited data and can use statistical methods *(mean, conditional mean, linear interpolation, etc.)* without introducing bias

PRODUCT SALES				
Date	**Product Name**	**Product ID**	**Retail Price**	**Units Sold**
5-Jan-20	Club Chocolate Bar	10660	$1.35	4
5-Jan-20	Big Time Waffles	62736	$3.69	
6-Jan-20	Excellent Apple Drink	30389	$2.98	7
7-Jan-20	High Top Oranges	98772		11
7-Jan-20	Jeffers Oatmeal	60396	$1.54	3

*For a missing Units Sold value, you would likely **remove** the row unless you are certain that an empty value represents 0 sales*

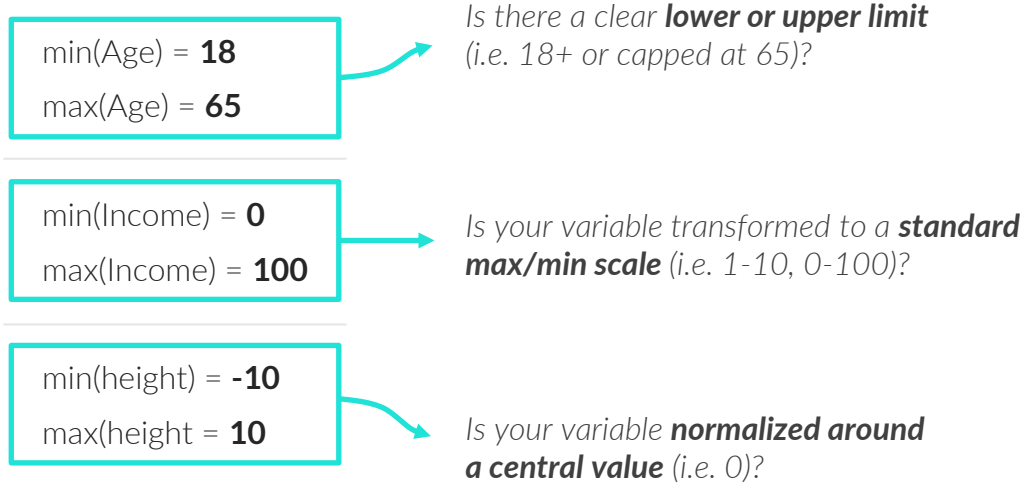
*For a missing Retail Price, you would likely be able to **impute** the value since you know the product name/ID*

RANGE CALCULATIONS

- Variable Types
- Empty Values
- **Range Calculations**
- Count Calculations
- Left/Right Censored
- Table Structure

One of the simplest QA tools for numerical variables is to calculate the **range of values** in a column *(minimum and maximum values)*

Range calculation is a helpful tool to understand your variables, confirm that ranges are realistic, and identify potential outliers



COUNT CALCULATIONS

- Variable Types
- Empty Values
- Range Calculations
- **Count Calculations**
- Left/Right Censored
- Table Structure

Count calculations help you understand the number of records or observations that fall within specific categories, and can be used to:

- Identify categories you were not expecting to see
- Begin understanding how your data is distributed
- Gather knowledge for building accurate ML models *(more on this later)*

Distinct counts can be particularly useful for QA, and help to:

- Understand the granularity or "grain" of your data
- Identify how many unique values a field contains
- Ensure consistency by identifying misspellings or categorization errors which might otherwise be difficult to catch *(i.e. leading or trailing spaces)*



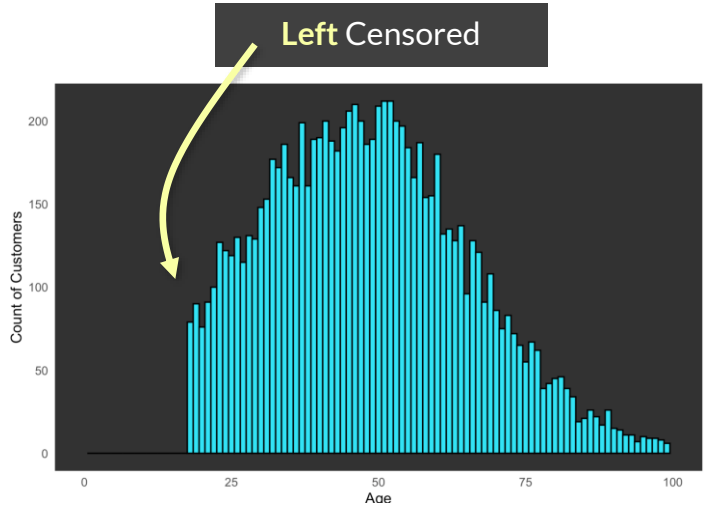
PRO TIP: For numerical variables with many unique values *(i.e. long decimals)*, use a histogram to plot frequency based on custom ranges or "bins" *(more on that soon!)*

LEFT/RIGHT CENSORED

- Variable Types
- Empty Values
- Range Calculations
- Count Calculations
- **Left/Right Censored**
- Table Structure

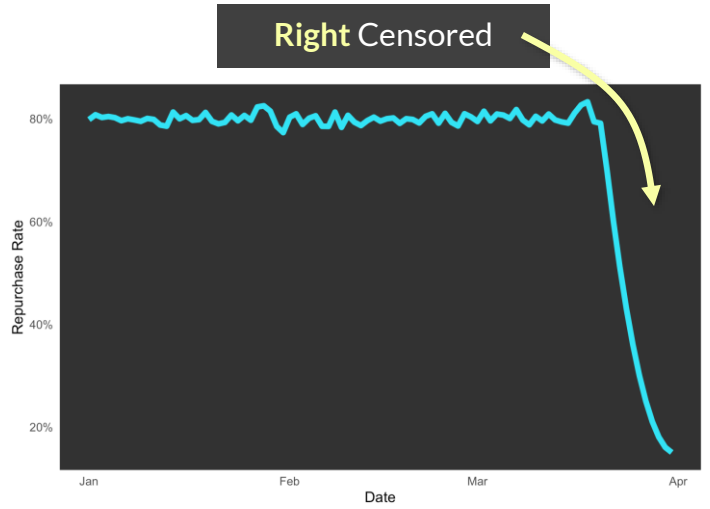
When data is **left or right censored**, it means that due to some circumstance the min or max value observed is **not the *natural* minimum or maximum** of that metric

- This can be difficult to spot unless you are aware of how the data is being recorded *(which means it's a particularly dangerous issue to watch out for!)*



Mall Shopper Survey Results

Only tracks shoppers over the age of 18 due to legal reasons, so anyone under 18 is excluded (even though there are plenty of mall shoppers under 18)



Ecommerce Repeat Purchase Rate

Sharp drop as you approach the current date has nothing to do with customer behavior, but the fact that recent customers haven't have the opportunity or need to repurchase yet

TABLE STRUCTURE

- Variable Types
- Empty Values
- Range Calculations
- Count Calculations
- Left/Right Censored
- Table Structure

Table structures generally come in two flavors: **long** or **wide**

Long Table

Product Details				
ID	Size	Section	Price	Quantity
1	L	Camping	5	2000
2	L	Camping	10	1600
3	L	Camping	30	1100
4	L	Camping	100	500
5	L	Camping	200	20
6	L	Camping	15	1700
7	L	Camping	20	1600
8	L	Camping	25	1500
9	L	Biking	600	100
10	L	Biking	50	200
11	S	Camping	4	1000
12	S	Camping	8	1000
13	S	Camping	26	600
14	S	Camping	80	900
15	S	Camping	190	10
16	S	Camping	12	500
17	S	Biking	600	150
18	S	Biking	50	1000
19	S	Biking	20	300
20	S	Biking	30	700

PIVOT

UNPIVOT

Wide Table

Product Details				
Section	Large Price	Small Price	Large Quantity	Small Quantity
Camping	400	300	10	20
Biking	200	150	100	110



Pivoting is the process of adjusting a table from **long** to **wide** by transforming **rows** into **columns**, and **Unpivoting** is the opposite *(wide to long)*

TABLE STRUCTURE

- Variable Types
- Empty Values
- Range Calculations
- Count Calculations
- Left/Right Censored
- Table Structure

Long tables typically contain a ***single, distinct column*** for each field *(Date, Product, Category, Quantity, Profit, etc.)*

- Easy to see all available fields and variable types
- Great for exploratory data analysis and aggregation *(i.e. PivotTables)*

Wide tables typically split the same metric into ***multiple columns*** or categories *(i.e. 2018 Sales, 2019 Sales, 2020 Sales, etc.)*

- Typically not ideal for human readability, since wide tables may contain thousands of columns *(vs. only a handful if pivoted to a long format)*
- Often *(but not always)* the best format for machine learning model input
- Great format for visualizing categorical data *(i.e. sales by product category)*



There's no ***right*** or ***wrong*** table structure; each type has strengths & weaknesses!

CASE STUDY: PRELIMINARY QA



THE SITUATION

You've just been hired as a Data Analyst for **Maven Market**, a local grocery store looking for help with basic data management and analysis.



THE ASSIGNMENT

The store manager would like you to conduct some analyses on product inventory and sales, but the data is a mess.

You'll need to **explore the data**, **conduct a preliminary QA**, and **help clean it up** to prepare the data for further analysis.



THE OBJECTIVES

1. Look at common data profiling metrics to identify potential issues
2. Take note of any issues you find in the product inventory sample
3. Correct the issues to prepare the data for further analysis

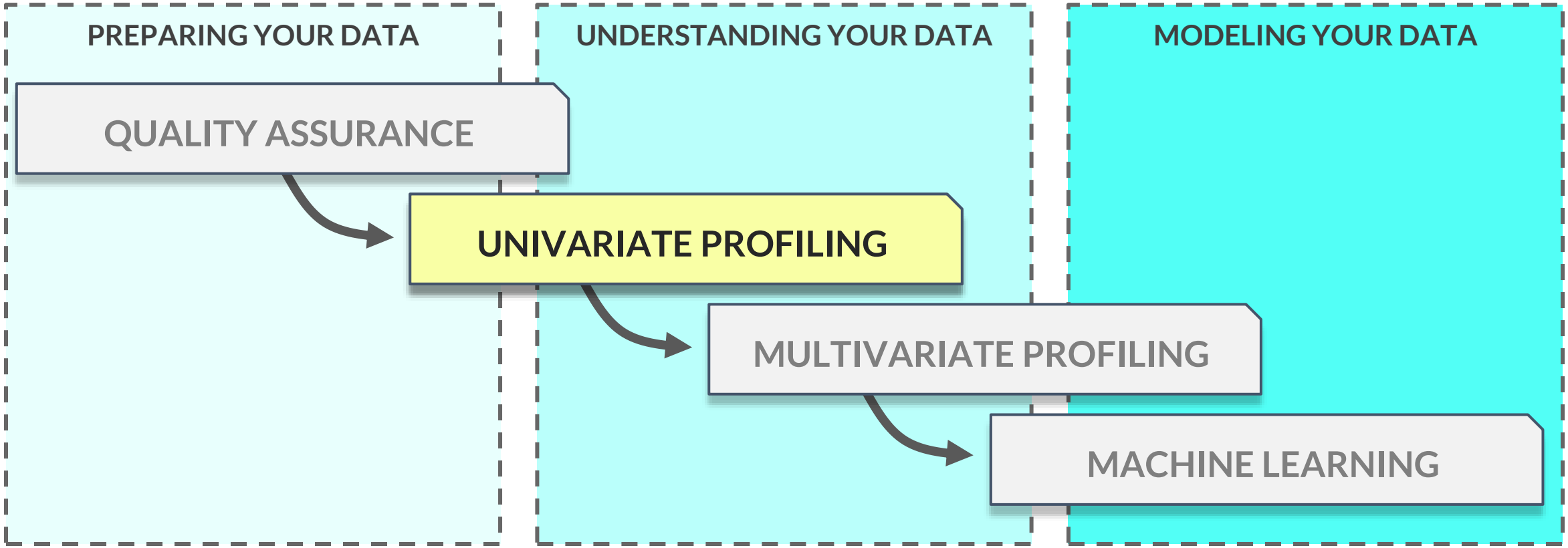
BEST PRACTICES: PRELIMINARY QA

- Review all fields to ensure that **variable types** are configured for proper analysis *(i.e. no dates formatted as strings, text formatted as values, etc.)*
- Remember that **NA** and **0** do *not* mean the same thing! Think carefully about how to handle missing data and the impact it may have on your analysis
- Run basic diagnostics like **Range**, **Count**, and **Left/Right Censored** checks against all columns in your data set…*every time*
- Understand your **table structure** *before* conducting any analysis to reduce the risk of double counting, inaccurate calculations, omitted data, etc.

UNIVARIATE PROFILING

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MACHINE LEARNING PROCESS



Univariate profiling is about **exploring individual variables** to build an understanding of your data. We'll cover common topics like normal distributions, frequency tables, histograms, etc.

UNIVARIATE PROFILING



Univariate profiling is the next step after preliminary QA; think of univariate profiling as conducting a descriptive analysis of each variable *by itself*

TOPICS WE'LL COVER:

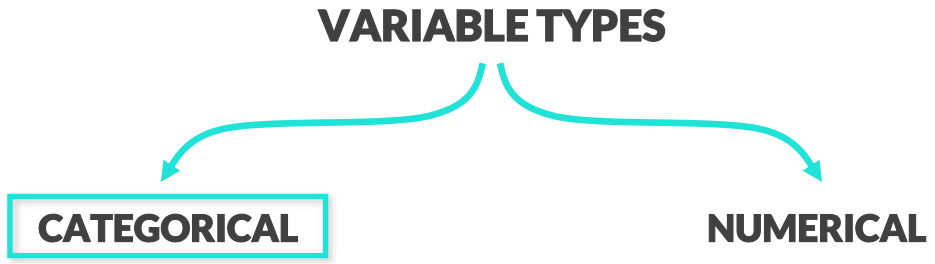
- Categorical Variables
- Categorical Distributions
- Numerical Variables
- Histograms & Kernel Densities
- Normal Distribution
- Data Profiling

COMMON USE CASES:

- Developing a deeper understanding of the fields you're working with *(outliers, distribution, etc.)*
- Preparing to build ML models *(necessary for selecting the correct model type)*
- Exploring individual variables before conducting a deeper multivariate analysis

CATEGORICAL VARIABLES

- Categorical Variables
- Categorical Distributions
- Numerical Variables
- Histograms & Kernel Densities
- Normal Distribution
- Data Profiling



Categorical variables contain *categories* as values, instead of numbers *(i.e. Product Type, Customer Name, Country, Month, etc.)*

- Categorical fields are exceptionally important for data analysis, and are typically used as dimensions by which we filter or "cut" numerical values *(i.e. sales by store)*
- Many types of predictive models are used to predict categorical dependent variables, or "classes" *(more on that later!)*



Terms like **discrete**, **categorical**, **multinomial**, and **classes** may all be used interchangeably. **Binary** is a special type of categorical variable which takes only 1 of 2 cases: *true* for *false (or 1 or 0)* and is also known as a **logical** variable or a **binary flag**

DISCRETIZATION

- Categorical Variables
- Categorical Distributions
- Numerical Variables
- Histograms & Kernel Densities
- Normal Distribution
- Data Profiling

Discretization is the process of creating a new ***categorical*** variable from an existing ***numerical*** variable, based on the values of the numerical variable

Product Details					
ID	Size	Section	Price	Quantity	Discretized Price
1	L	Camping	5	2000	Low
2	L	Camping	10	1600	Low
3	L	Camping	30	1100	Low
4	L	Camping	100	500	Med
5	L	Camping	200	20	Med
6	L	Camping	15	1700	Low
7	L	Camping	20	1600	Low
8	L	Camping	25	1500	Low
9	L	Biking	600	100	High
10	S	Biking	50	200	Low
11	S	Camping	4	1000	Low
12	S	Camping	8	1000	Low
13	S	Camping	26	600	Low
14	S	Camping	80	900	Low
15	S	Camping	190	10	Med
16	S	Camping	12	500	Low
17	S	Biking	600	150	High
18	S	Biking	50	1000	Low
19	S	Biking	20	300	Low
20	S	Biking	30	700	Low

Discretization Rules:

If Price <100 then Price Level = Low

If Price >=100 & Price <500 then Price Level = Med

If Price >=500 then Price Level = High

NOMINAL VS. ORDINAL VARIABLES

Categorical Variables

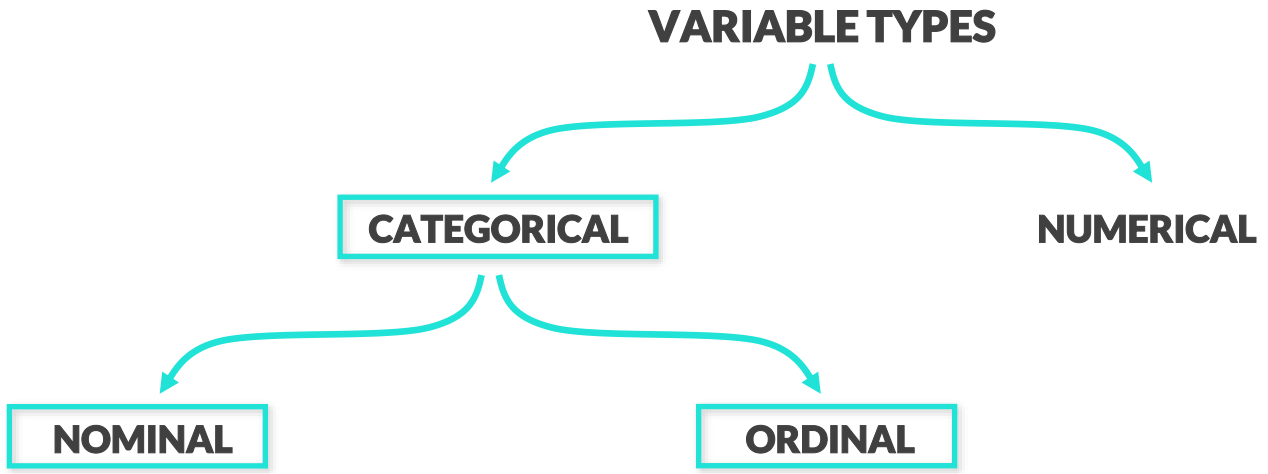
Categorical Distributions

Numerical Variables

Histograms & Kernel Densities

Normal Distribution

Data Profiling



There are two types of categorical variables: **nominal** and **ordinal**

- **Nominal variables** contain categories with no inherent logical rank, which can be re-ordered with no consequence *(i.e. **Product Type** = Camping, Biking or Hiking)*
- **Ordinal variables** contain categories with a logical order *(i.e. **Size** = Small, Medium, Large)*, but the interval between those categories has no logical interpretation

CATEGORICAL DISTRIBUTIONS

- Categorical Variables
- **Categorical Distributions**
- Numerical Variables
- Histograms & Kernel Densities
- Normal Distribution
- Data Profiling

Categorical distributions are visual and/or numeric representations of the unique values a variable contains, and how often each occurs

Common categorical distributions include:

- **Frequency tables:** Show the count *(or frequency)* of each distinct value
- **Proportions tables:** Show the count of each value as a % of the total
- **Heat maps:** Formatted to visualize patterns *(typically used for multiple variables)*

Understanding categorical distributions will help us gather knowledge for building accurate machine learning models *(more on this later!)*

CATEGORICAL DISTRIBUTIONS

- Categorical Variables
- **Categorical Distributions**
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Product Details				
ID	Size	Section	Price	Quantity
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13	S	Camping	26	600
14	S	Camping	80	900
15	S	Camping	190	10
16	S	Camping	12	500
17	S	Biking	600	150
18	S	Biking	50	1000
19	S	Biking	20	300
20	S	Biking	30	700

Section Distribution:

Camping	Biking
14	6

Frequency table

Camping	Biking
70%	30%

Proportions table

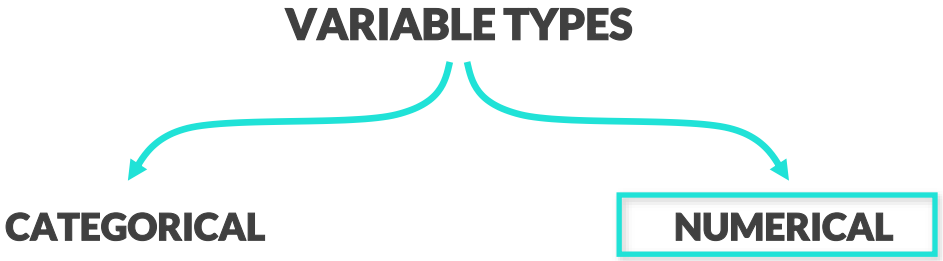
Size & **Section** Distribution:

	Camping	Biking
S	6	4
L	8	2

Heat Map

NUMERICAL VARIABLES

- Categorical Variables
- Categorical Distributions
- Numerical Variables
- Histograms & Kernel Densities
- Normal Distribution
- Data Profiling



Numerical variables contain *numbers* as values, instead of categories *(i.e. Gross Revenue, Pageviews, Quantity, Retail Price, etc.)*

- Numerical fields are typically **aggregated** *(as a sum, count, average, max, min, etc.)* and broken down by different dimensions or categories
- Numerical data is also known as **quantitative** data, while categorical data is often referred to as "qualitative"

You may hear numeric variables described further as interval and ratio, but the distinction is trivial and rarely makes a difference in common use cases

HISTOGRAMS

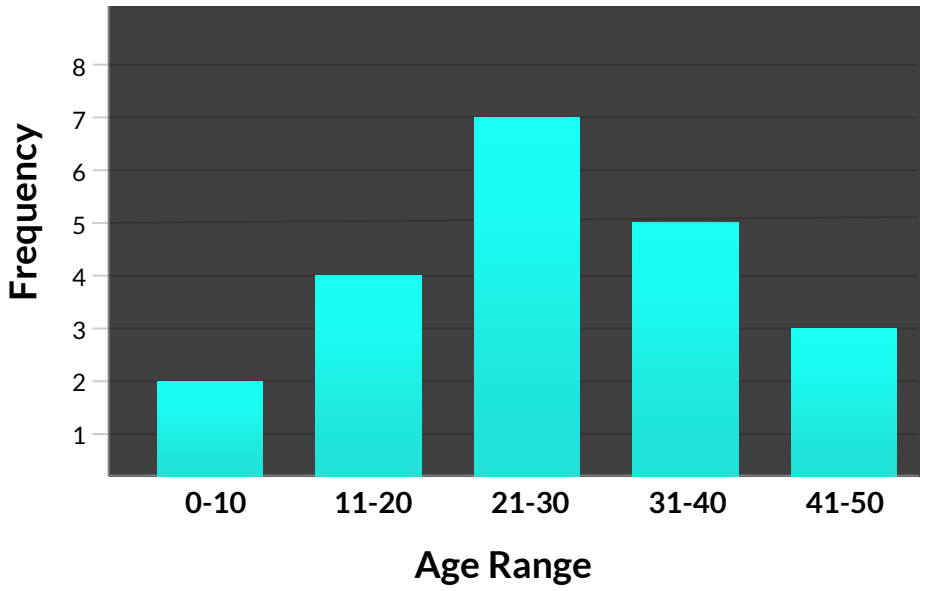
- Categorical Variables
- Categorical Distributions
- Numerical Variables
- Histograms & Kernel Densities
- Normal Distribution
- Data Profiling

Histograms are used to plot a single, discretized numerical variable

Imagine taking a numerical variable *(like **age**)*, defining ranges or "bins" *(1-5, 6-10, etc.)*, and counting the number of observations which fall into each bin; ***this is exactly what histograms are designed to do!***

Age Values:

8	29	45
25	33	37
19	43	21
28	32	40
24	17	28
5	22	39
15	47	12



KERNEL DENSITIES

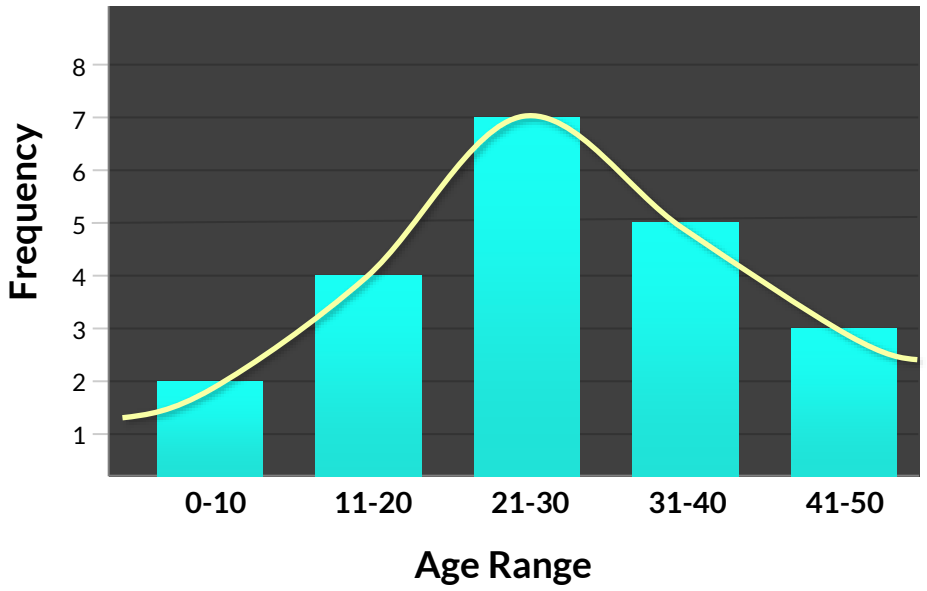
- Categorical Variables
- Categorical Distributions
- Numerical Variables
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- Normal Distribution
- Data Profiling

Kernel densities are "smooth" versions of histograms, which can help to prevent users from over-interpreting breaks between bins

- ***Technical** definition:* *"Non-parametric density estimation via smoothing"*
- ***Intuitive** definition:* *"Wet noodle laying on a histogram"*

Age Values:

8	29	45
25	33	37
19	43	21
28	32	40
24	17	28
5	22	39
15	47	12



HISTOGRAMS & KERNEL DENSITIES

Categorical Variables

Categorical Distributions

Numerical Variables

Histograms & Kernel Densities

Normal Distribution

Data Profiling

When to use histograms and kernel density charts:

- Visualizing how a given variable is distributed
- Providing a visual glimpse of profiling metrics like **mean**, **mode**, and **skewness**

Things to watch out for:

- **Bin sensitivity**: Bin size can significantly change the shape and ”smoothness” of a histogram, so select a bin width that accurately shows the data distribution
- **Outliers**: Histograms can be used to identify outliers in your data set, but you may need to remove them to avoid skewing the distribution
- **Sample size**: Histograms are best suited for variables with many observations, to reflect the true population distribution



PRO TIP: If your data is relatively symmetrical *(not skewed)*, you can use **Sturge’s Rule** as a quick “rule of thumb” to determine an appropriate number of bins: **K = 1 + 3.322 log (N)** (where **K** = number of bins, **N** = number of observations)

CASE STUDY: HISTOGRAMS



THE SITUATION

You've just been promoted as the new Pit Boss at **The Lucky Roll Casino**. Your mission? Use data to help expose cheats on the casino floor.



THE ASSIGNMENT

Profits at the craps tables have been unusually low, and you've been asked to investigate the possible use of loaded die *(weighted towards specific numbers)*.

Your plan is to **track the outcome of each roll**, then **compare your results against the expected probability distribution** to see how closely they match.



THE OBJECTIVES

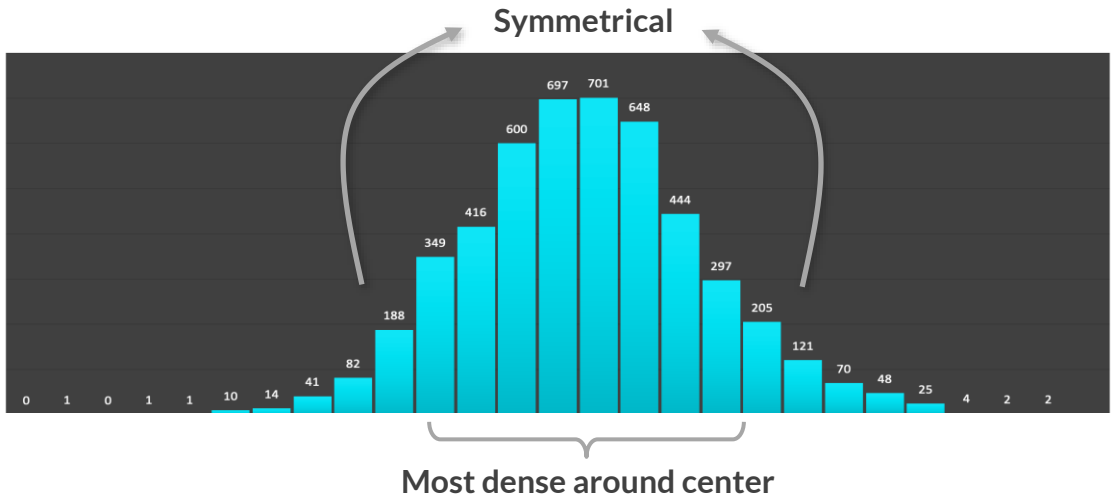
1. Record the outcomes for a series of individual dice rolls
2. Plot the frequency of each result *(1-12)* using a histogram
3. Compare your plot against the expected frequency distribution

NORMAL DISTRIBUTION

- Categorical Variables
- Categorical Distributions
- Numerical Variables
- Histograms & Kernel Densities
- Normal Distribution
- Data Profiling

Many numerical variables naturally follow a **normal distribution**, also known as a Gaussian distribution or "bell curve"

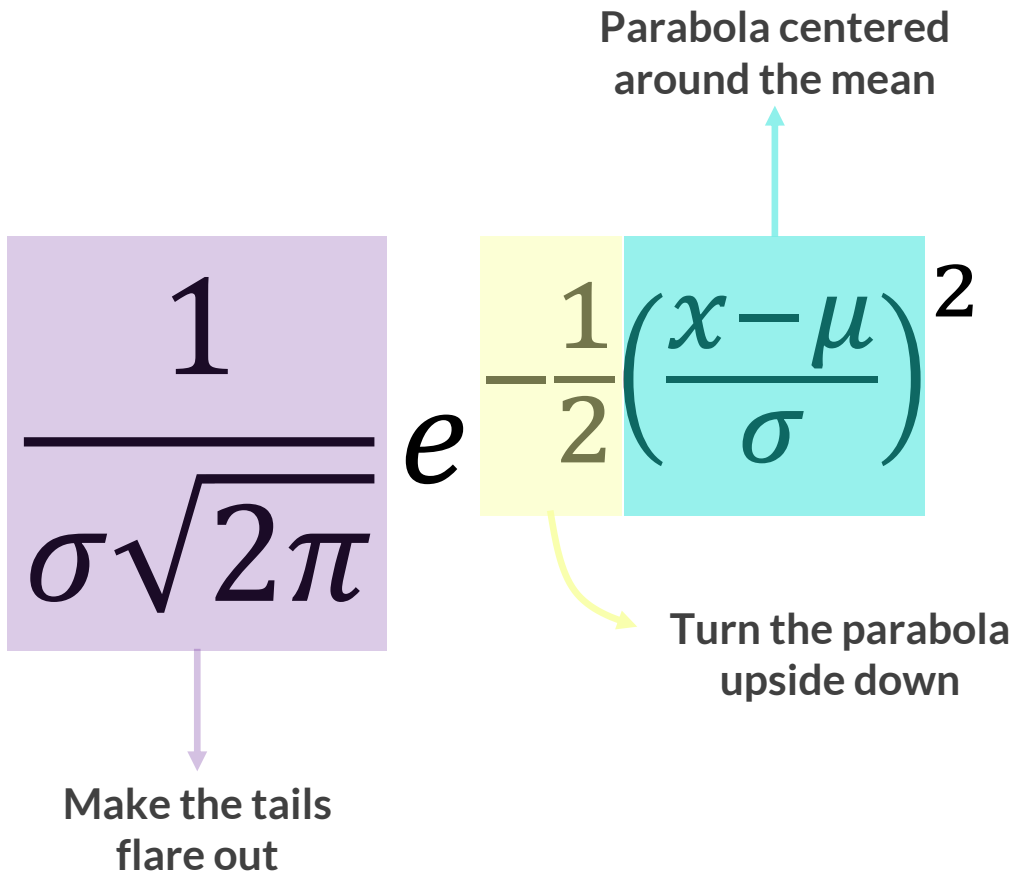
- Normal distributions are symmetrical and dense around the center, with flared out "tails" on both ends *(like a bell!)*
- Normal distributions are essential to many underlying assumptions in ML, and a helpful tool for comparing distributions or testing differences between them



You can find normal distributions in many real-world examples: heights, weights, test scores, etc.

NORMAL DISTRIBUTION

- Categorical Variables
- Categorical Distributions
- Numerical Variables
- Histograms & Kernel Densities
- Normal Distribution
- Data Profiling



CASE STUDY: NORMAL DISTRIBUTION



THE SITUATION

It's August 2016, and you've been invited to Rio de Janeiro as a Data Analyst for the **Global Olympic Committee**.



THE ASSIGNMENT

Your job is to collect demographic data for all female athletes competing in the Summer Games and determine how the **distribution of Olympic athlete heights compares against the general public**.



THE OBJECTIVES

1. Gather heights for all female athletes competing in the 2016 Games
2. Plot height frequencies using a Histogram, and test various bin widths
3. Determine if athlete heights follow a normal distribution, or "bell curve"
4. Compare the distributions for athletes vs. the general public

DATA PROFILING

- Categorical Variables
- Categorical Distributions
- Numerical Variables
- Histograms & Kernel Densities
- Normal Distribution
- Data Profiling

Data profiling describes the process of using statistics to describe or summarize information about a particular variable

- Data profiling is critical for understanding variable characteristics which cannot be *seen* or easily visualized using tools like histograms
- Data profiling communicates **meaning**, using simple, concise, and universally understood metrics

Common data profiling metrics include:

- **Mode**
- **Mean**
- **Median**
- **Percentile**
- **Variance**
- **Standard Deviation**
- **Skewness**



PRO TIP: Combining visual distributions with data profiling metrics is a powerful way to communicate meaning in advanced analytics

MODE

Categorical Variables

Categorical Distributions

Numerical Variables

Histograms & Kernel Densities

Normal Distribution

Data Profiling

The **mode** is the most frequently observed value

- With a ***numerical*** variable *(most common)*, it's the general range around the highest "peak" in the distribution
- With a ***categorical*** variable, it's simply the value that appears most often

Customer ID	First Name	Last Name	Gender	City	Sessions
1	Jennifer	Mcgrath	F	Seattle	24
2	Chad	Lewis	M	Seattle	12
3	Susan	Rodriguez	F	Houston	9
4	Wayne	Nielson	M	Miami	31
5	John	Depaul	M	Houston	24
6	Joseph	Martinez	M	Houston	11
7	Diane	Henry	F	Miami	3
8	Veronica	Comerford	F	Miami	24
9	Beverly	Nixon	F	Houston	15
10	Ivan	Layton	M	Houston	27

Mode of **City** = "**Houston**"

Mode of **Sessions** = **24**

Mode of **Gender** = **F, M**
*(this is a **bimodal** field!)*

Common uses:

- Understanding the most common values within a dataset
- Diagnosing if one variable is influenced by another

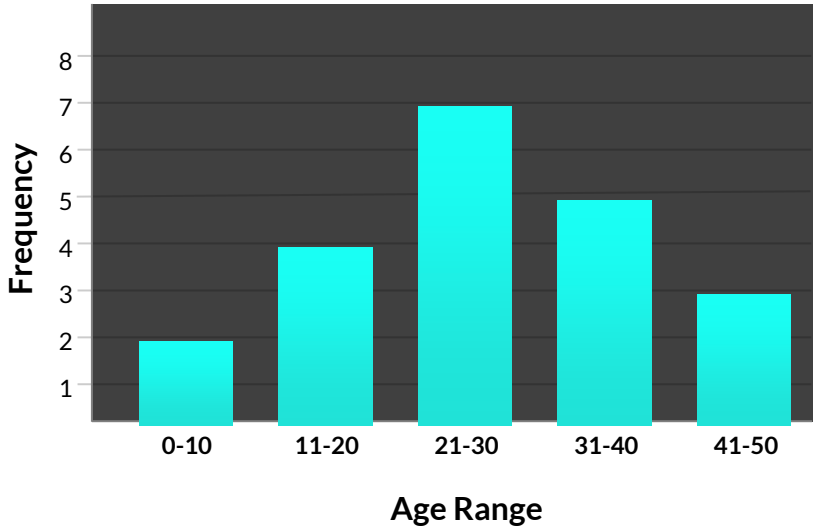
MODE

- Categorical Variables
- Categorical Distributions
- Numerical Variables
- Histograms & Kernel Densities
- Normal Distribution
- Data Profiling

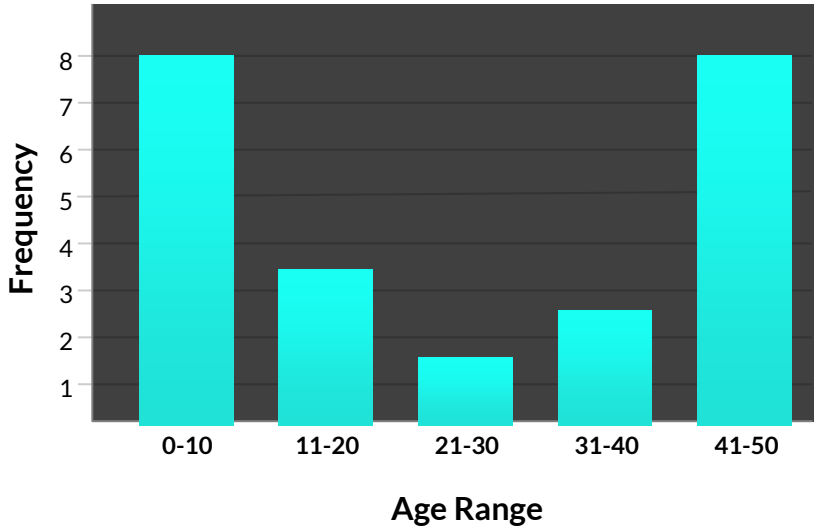
While **modes** typically aren't very useful on their own, they can provide helpful hints for deeper data exploration

- For example, the right histogram below shows a *multi-modal* distribution, which indicates that there may be another variable impacting the age distribution

Single Mode (***21-30***)



Two Modes (***0-10***, ***41-50***)



MEAN

Categorical Variables

Categorical Distributions

Numerical Variables

Histograms & Kernel Densities

Normal Distribution

Data Profiling

The **mean** is the calculated "central" value in a discrete set on numbers

- Mean is what most people think of when they hear the word "average", and is calculated by dividing the sum of all values by the count of all observations
- Means can only be applied to *numerical* variables *(not categorical)*

Product Details				
ID	Size	Section	Price	Quantity
1	L	Camping	5	2000
2	L	Camping	10	1600
3	L	Camping	30	1100
4	L	Camping	100	500
5	L	Camping	200	20

$$mean = \frac{sum\ of\ all\ values}{count\ of\ observations}$$

$$= \frac{5{,}220}{5} = \mathbf{1,044}$$

Common uses:

- Making a "best-guess" estimate of a value
- Calculating a central value when outliers are not present

MEDIAN

- Categorical Variables
- Categorical Distributions
- Numerical Variables
- Histograms & Kernel Densities
- Normal Distribution
- Data Profiling

The **median** is the *middle value* in a list of values sorted from highest to lowest *(or vice versa)*

- When there are two middle-ranked values, the median is the ***average*** of the two
- Medians can only be applied to ***numerical*** variables *(not categorical)*

Customer ID	First Name	Last Name	Sessions
4	Wayne	Nielson	31
10	Ivan	Layton	27
1	Jennifer	Mcgrath	24
5	John	Depaul	24
8	Veronica	Comerford	24
9	Beverly	Nixon	15
2	Chad	Lewis	12
6	Joseph	Martinez	11
3	Susan	Rodriguez	9
7	Diane	Henry	3

Median = 19.5

(average of 15 and 24)

Common uses:

- Identifying the "center" of a distribution
- Calculating a central value when outliers may be present

PERCENTILE

Categorical Variables

Categorical Distributions

Numerical Variables

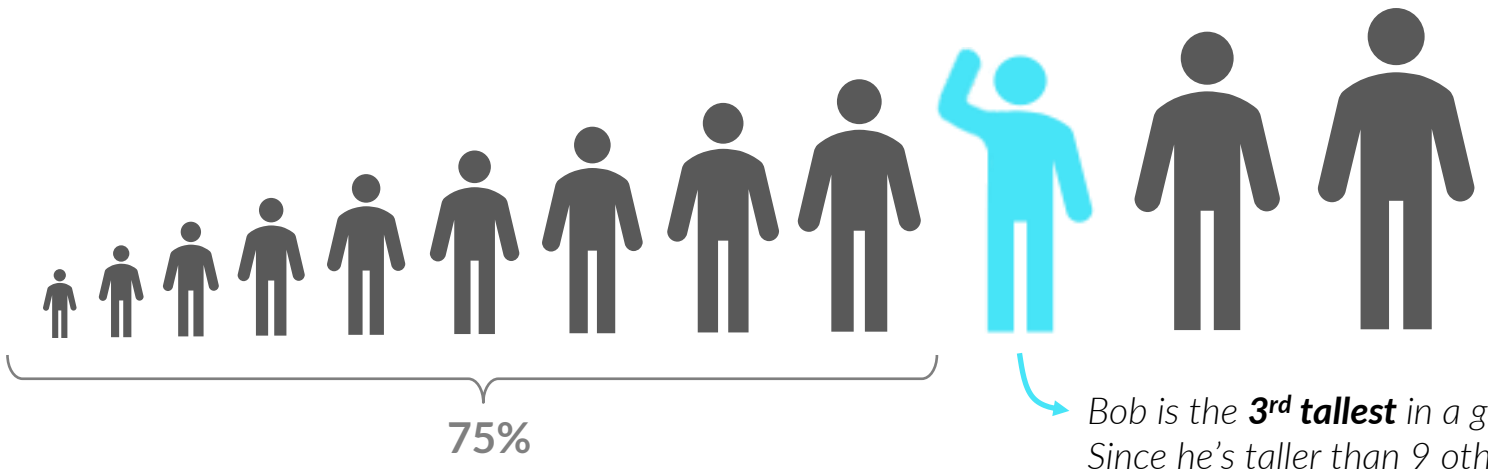
Histograms & Kernel Densities

Normal Distribution

Data Profiling

Percentiles are used to describe the *percent of values* in a column which fall below a particular number

- If you are in the **90th** percentile for height, you're taller than **90%** of the population
- If the **50th** percentile for test scores is **86**, half of the students scored lower *(which means 86 is also the **median**!)*



Common uses:

- Providing context and intuitive benchmarks for how values "rank" within a sample *(test scores, height/weight, blood pressure, etc.)*

VARIANCE

Categorical Variables

Categorical Distributions

Numerical Variables

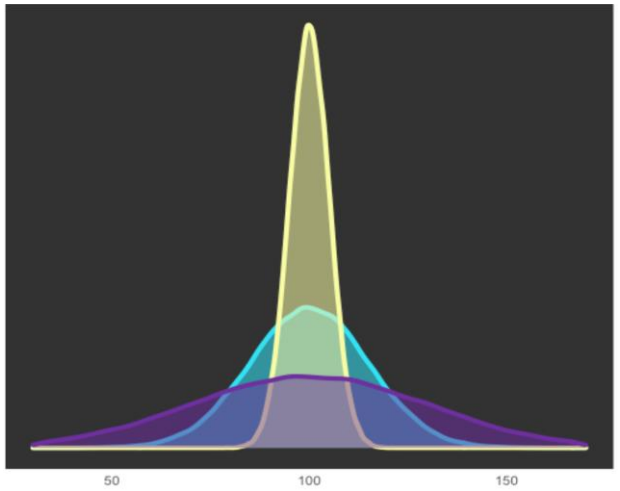
Histograms & Kernel Densities

Normal Distribution

Data Profiling

Variance, in simple terms, describes how ***thin*** or ***wide*** a distribution is

- Variance measures how far the observations are from the mean, on average, and help us concisely describe a variable's distribution
- The ***wider*** a distribution, the ***higher*** the variance *(and vice versa)*



Variance = **5**

Variance = **15**

Variance = **30**

Common uses:

- Comparing the numerical distributions of two different groups *(i.e. prices of products ordered online vs. in store)*

VARIANCE

- Categorical Variables
- Categorical Distributions
- Numerical Variables
- Histograms & Kernel Densities
- Normal Distribution
- Data Profiling

$$\frac{\sum_{i=1}^{n}(x_i - \mu)^2}{n-1}$$

Calculation Steps:

1) Calculate the average of the variable
2) Subtract that average from the first row
3) Square that difference
4) Do steps 1-3 for every row and sum it up
5) Divide by the number of observations (-1)

Average squared distance from the mean

STANDARD DEVIATION

Categorical Variables

Categorical Distributions

Numerical Variables

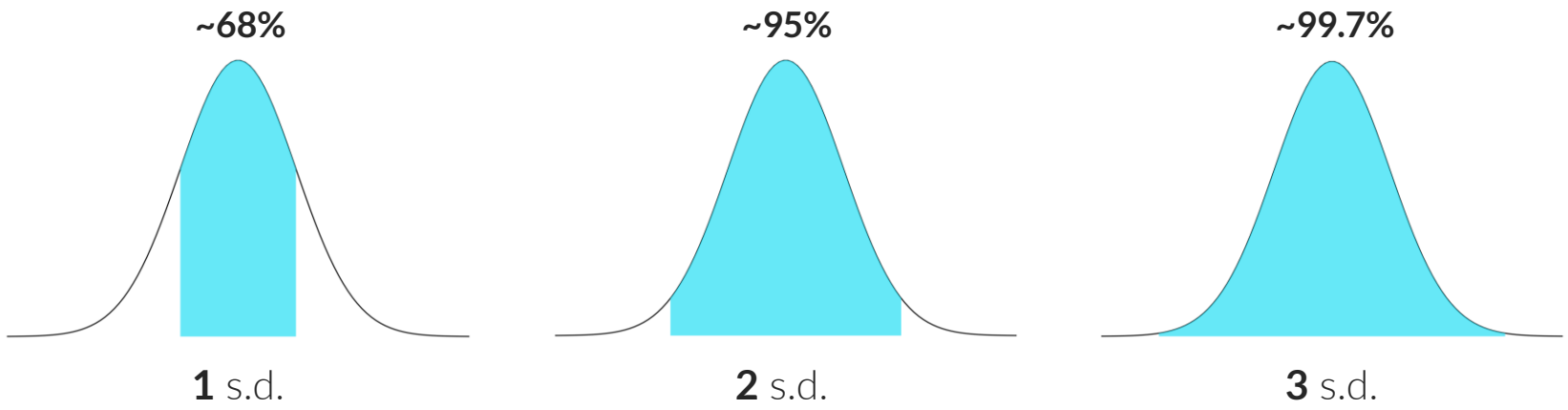
Histograms & Kernel Densities

Normal Distribution

Data Profiling

Standard Deviation is the square root of the variance

- This converts the variance back to the scale of the variable itself
- In a normal distribution, **~68%** of values fall within 1 standard deviation of the mean, **~95%** fall within 2, and **~99.7%** fall within 3 *(known as the "**empirical rule**")*



Common uses:

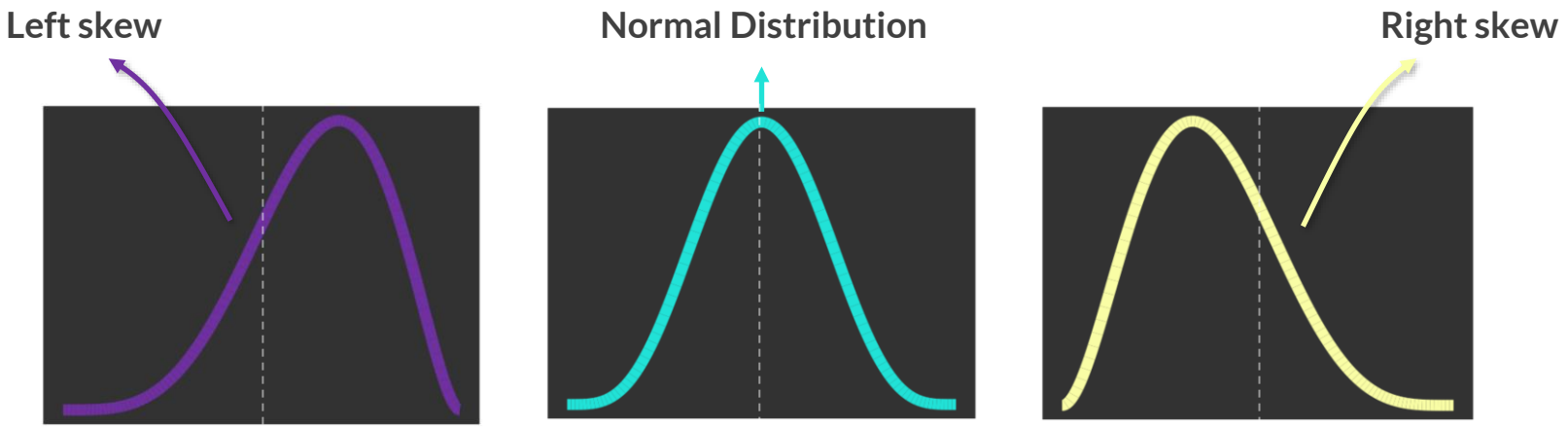
- Comparing segments for a given metric *(i.e. time on site for mobile users vs. desktop)*
- Understanding how likely certain values are bound to occur

SKEWNESS

- Categorical Variables
- Categorical Distributions
- Numerical Variables
- Histograms & Kernel Densities
- Normal Distribution
- Data Profiling

Skewness tells us how a distribution varies from a normal distribution

- This is commonly used to mathematically describe skew to the left or right



Common uses:

- Identifying non-normal distributions, and describing them mathematically

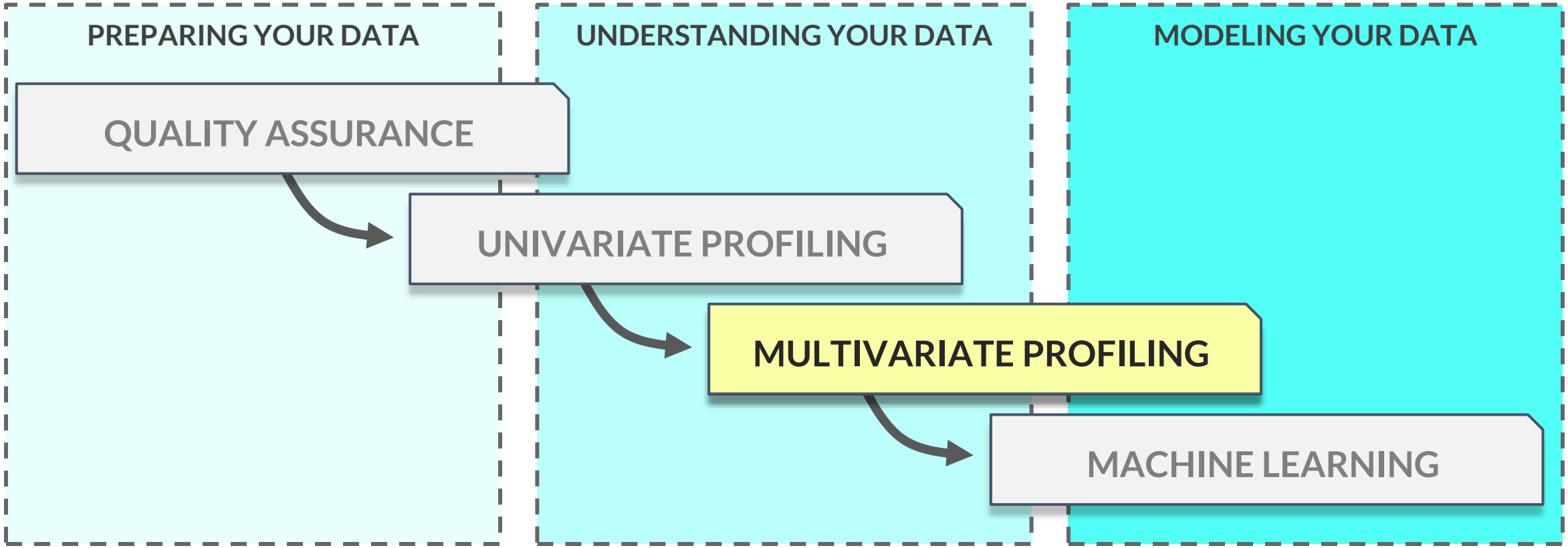
BEST PRACTICES: UNIVARIATE PROFILING

- Make sure you are using the appropriate tools for profiling **categorical** variables vs. **numerical** variables
- **Distributions** are a great way to quickly and visually explore variables
- **QA still comes first!** Profiling metrics are important, but can lead to misleading results without proper QA *(i.e. handling outliers or missing values)*
- One single distribution, visualization, or metric isn't enough to fully understand a variable; **always explore your data from multiple angles**

MULTIVARIATE DISTRIBUTIONS

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MACHINE LEARNING PROCESS



Multivariate profiling is about **understanding relationships between *multiple* variables**. We'll cover common tools for exploring categorical & numerical data, including kernel densities, violin & box plots, scatterplots, etc.

MULTIVARIATE PROFILING



Multivariate profiling is the next step after univariate profiling, since single-metric distributions are rarely enough to draw meaningful insights or conclusions

TOPICS WE'LL COVER:

- Categorical-Categorical Distributions
- Categorical-Numerical Distributions
- Multivariate Kernel Densities
- Violin & Box Plots
- Numerical-Numerical Distributions
- Scatter Plots & Correlation

COMMON USE CASES:

- Exploring relationships between multiple categorical and/or numerical variables
- Informing which specific machine learning models or techniques to use



Multivariate distributions are generally called "joint" distributions, and it's clear why—it's joining two variables into the same distribution!

CATEGORICAL-CATEGORICAL DISTRIBUTIONS

- Categorical-Categorical Distributions
- Categorical-Numerical Distributions
- Multivariate Kernel Densities
- Violin & Box Plots
- Numerical-Numerical Distributions
- Scatter Plots & Correlation

Categorical/categorical distributions represent the frequency of *unique combinations* between two or more categorical variables

This is one of the simplest forms of multivariate profiling, and leverages the same tools we used to analyze univariate distributions:

- **Frequency tables:** Show the count *(or frequency)* of each distinct combination
- **Proportions tables:** Show the count of each combination as a % of the total
- **Heat maps:** Frequency or proportions table formatted to visualize patterns

Common uses:

- Understanding product mix based on multiple characteristics *(i.e. size & category)*
- Exploring customer demographics based on attributes like location, gender, etc.

CATEGORICAL-CATEGORICAL DISTRIBUTIONS

- Categorical-Categorical Distributions
- Categorical-Numerical Distributions
- Multivariate Kernel Densities
- Violin & Box Plots
- Numerical-Numerical Distributions
- Scatter Plots & Correlation

In this example we're looking at product inventory using a distribution of two categorical fields: ***Design*** *and* ***Size***

We can show this joint distribution as a simple count (*Frequency Table*), as a percentage of the total (*Proportions Table*), or as a conditionally formatted table (*Heat Map*)

Product Inventory

	Design 1	Design 2	Design 3	Design 4	Design 5
X-Small	2	31	44	48	3
Small	5	33	39	33	19
Medium	5	48	28	36	37
Large	16	27	6	38	42
X-Large	35	28	1	31	3

Frequency Table

Product Inventory

	Design 1	Design 2	Design 3	Design 4	Design 5
X-Small	0.3%	4.9%	6.9%	7.5%	0.5%
Small	0.8%	5.2%	6.1%	5.2%	3.0%
Medium	0.8%	7.5%	4.4%	5.6%	5.8%
Large	2.5%	4.2%	0.9%	6.0%	6.6%
X-Large	5.5%	4.4%	0.2%	4.9%	0.5%

Proportions Table

Product Inventory

	Design 1	Design 2	Design 3	Design 4	Design 5
X-Small	2	31	44	48	3
Small	5	33	39	33	19
Medium	5	48	28	36	37
Large	16	27	6	38	42
X-Large	35	28	1	31	3

Heat Map

CASE STUDY: HEAT MAPS



THE SITUATION

You've just been hired by the **New York Department of Transportation** (DOT) to help analyze traffic accidents in New York City from 2019-2020



THE ASSIGNMENT

The DOT commissioner would like to understand accident frequency by **time of day** and **day of week**, in order to support a public service campaign promoting safe driving habits.

Your role is to provide the data that she needs to understand **when traffic accidents are most likely to occur**.



THE OBJECTIVES

1. Create a table to plot accident frequency by time of day and day of week
2. Apply conditional formatting to the table to create a heatmap showing the days and times with the fewest *(green)* and most *(red)* accidents in the sample

CATEGORICAL-NUMERICAL DISTRIBUTIONS

- Categorical-Categorical Distributions
- **Categorical-Numerical Distributions**
- Multivariate Kernel Densities
- Violin & Box Plots
- Numerical-Numerical Distributions
- Scatter Plots & Correlation

Categorical-numerical distributions are used for comparing numerical distributions across classes in a category *(i.e. age distribution by gender)*

These are typically visualized using variations of familiar univariate numerical distributions, including:

- **Histograms & Kernel Densities:** Show the count *(or frequency)* of values
- **Violin Plots:** Kernel density "glued" to its mirror image, and tilted on its side
- **Box Plots:** Like a kernel density, but formatted to visualize key statistical values *(min/max, median, quartiles)* and outliers

Common uses:

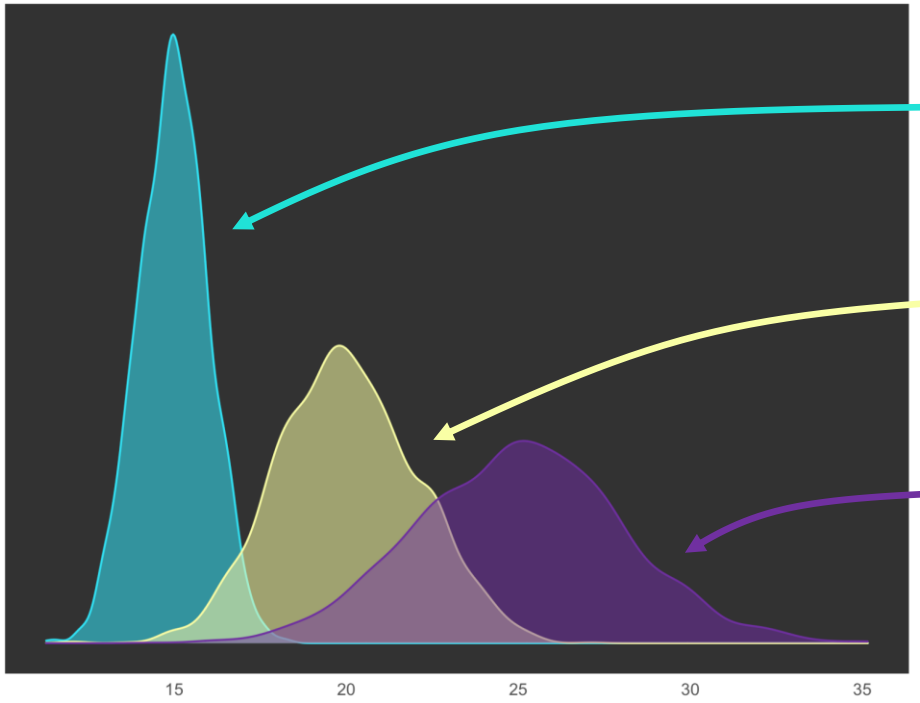
- Comparing key business metrics *(i.e. customer lifetime value, average order size, purchase frequency, etc.)* by customer class *(gender, loyalty status, location, etc.)*
- Comparing sales performance by day of week or hour of day

KERNEL DENSITIES

- Categorical-Categorical Distributions
- Categorical-Numerical Distributions
- **Multivariate Kernel Densities**
- Violin & Box Plots
- Numerical-Numerical Distributions
- Scatter Plots & Correlation

Remember: **kernel densities** are just smooth versions of histograms

- To visualize a categorical-numerical distribution, kernel densities can be repeated to represent each class within a particular category



Teal class has a mean of **~15** and relatively **low variance** *(highly concentrated around the mean)*

Yellow class has a mean of **~20** and **moderate variance** relative to other categories

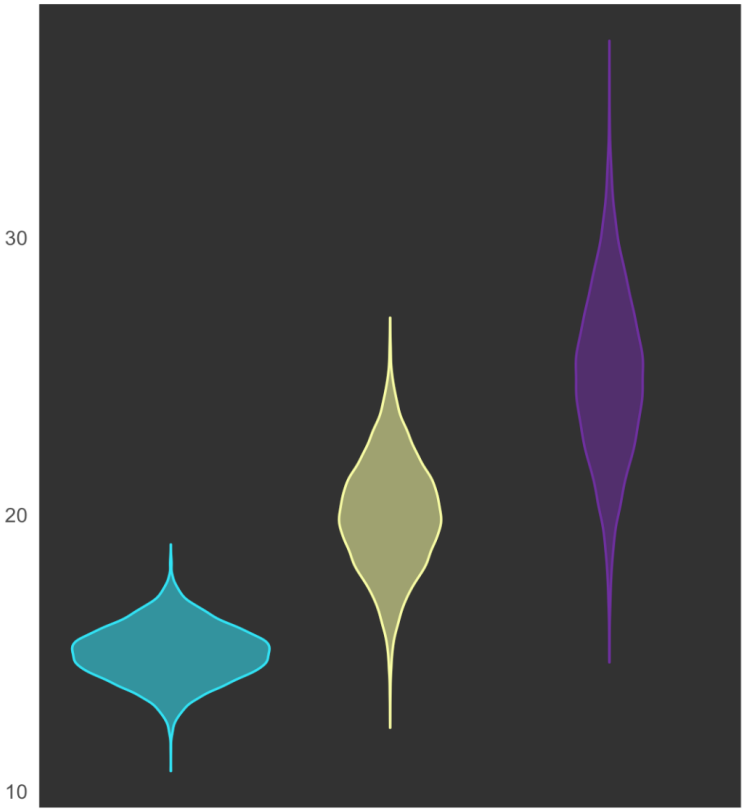
Purple class has a mean of **~25**, overlaps with yellow, and has relatively **high variance**

VIOLIN PLOTS

- Categorical-Categorical Distributions
- Categorical-Numerical Distributions
- Multivariate Kernel Densities
- **Violin & Box Plots**
- Numerical-Numerical Distributions
- Scatter Plots & Correlation

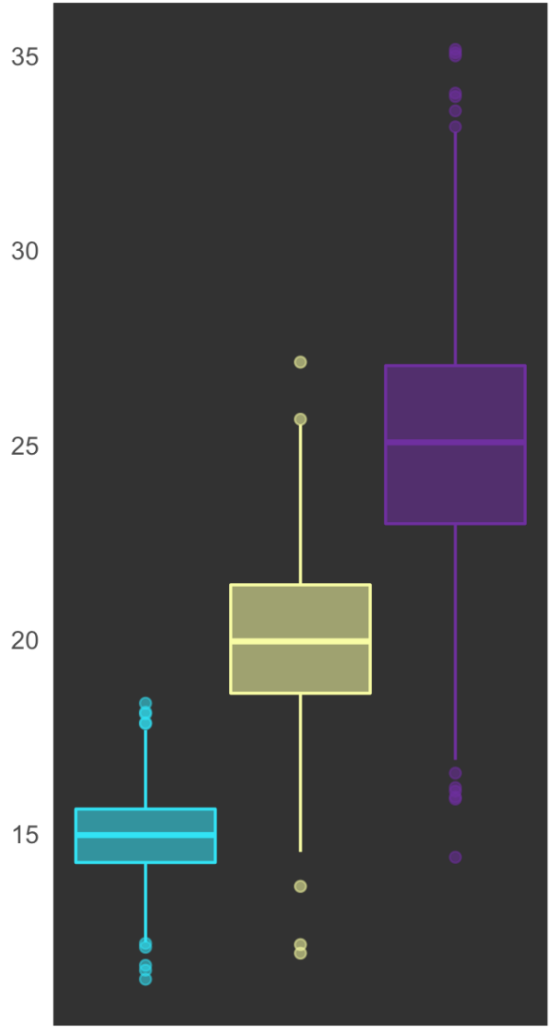
A **violin plot** is essentially a kernel density flipped vertically and combined with its mirror image

- Violin plots use the exact same data as kernel densities, just visualized slightly differently
- These can help you visualize the shape of each distribution, and compare them across classes more clearly



BOX PLOTS

- Categorical-Categorical Distributions
- Categorical-Numerical Distributions
- Multivariate Kernel Densities
- **Violin & Box Plots**
- Numerical-Numerical Distributions
- Scatter Plots & Correlation



Box plots are like violin plots, but designed to show **key statistical attributes** rather than smooth distributions, including:

- **Median**
- **Min** & **Max** *(excluding outliers)*
- **25th** & **75th Percentiles**
- **Outliers**

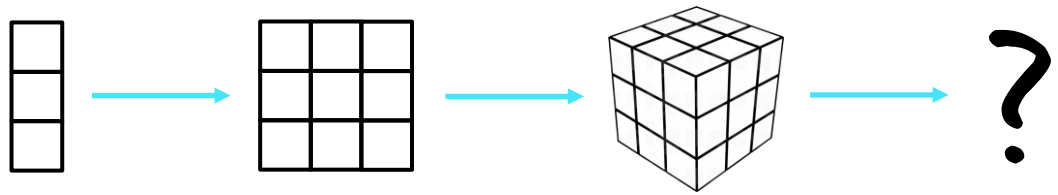
Box plots provide a *ton* of information in a single visual, and can be used to quickly compare statistical characteristics between classes

LIMITATIONS OF CATEGORICAL DISTRIBUTIONS

- Categorical-Categorical Distributions
- Categorical-Numerical Distributions
- Multivariate Kernel Densities
- **Violin & Box Plots**
- Numerical-Numerical Distributions
- Scatter Plots & Correlation

Categorical profiling works for simple cases, but breaks down quickly

- Humans are pretty good at visualizing 1, 2, or maybe even 3 variables, but how would you visualize a joint distribution for **10** variables? **100**?



Categorical profiling can't answer ***prescriptive*** or ***predictive*** questions

- Suppose you randomized several elements on your sales page (*font*, *image*, *layout*, *button*, *copy*, *etc.*) to understand which ones drive conversions
- You could count conversions for individual elements, or some combinations of elements, but categorical distribution alone can't measure *causation*

This is when you need machine learning!

NUMERICAL-NUMERICAL DISTRIBUTIONS

- Categorical-Categorical Distributions
- Categorical-Numerical Distributions
- Multivariate Kernel Densities
- Violin & Box Plots
- **Numerical-Numerical Distributions**
- Scatter Plots & Correlation

Numerical-Numerical distributions are common and intuitive, but the most complex mathematically

They are typically visualized using **scatter plots**, which plot points along the X and Y axis to show the relationship between two variables

- Scatter plots allow for simple, visual intuition: ***when one variable increases or decreases, how does the other variable change?***
- There are many possibilities: no relationship, positive, negative, linear, non-linear, cubic, exponential, etc.

Common uses:

- Quickly visualizing how two numerical variables relate
- Predicting how a change in one variable will impact another *(i.e. square footage and house price, marketing spend and sales, etc.)*

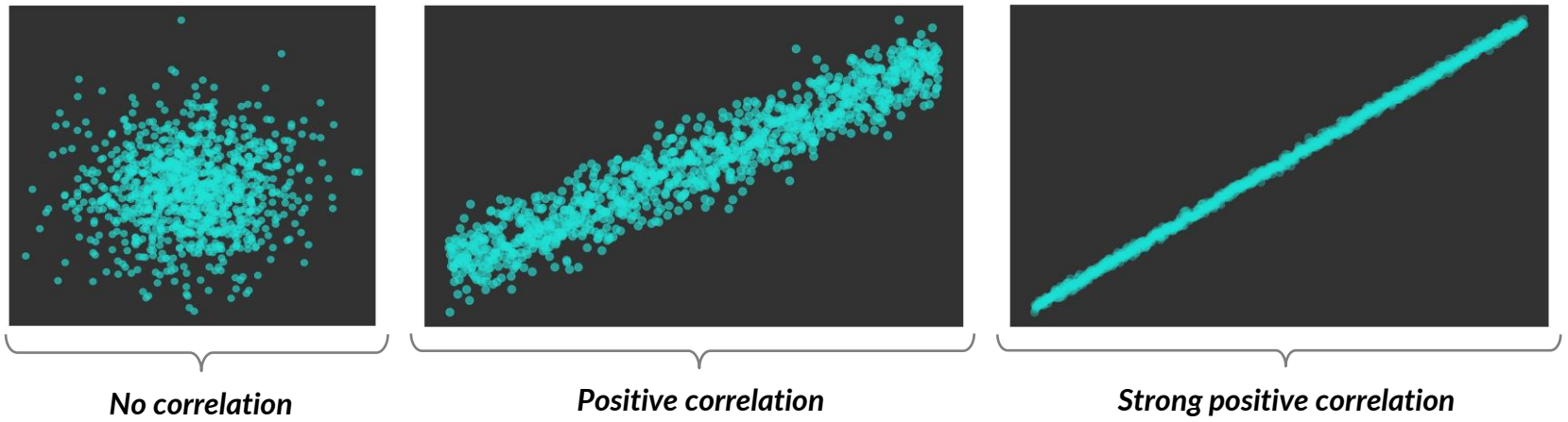
CORRELATION

- Categorical-Categorical Distributions
- Categorical-Numerical Distributions
- Multivariate Kernel Densities
- Violin & Box Plots
- Numerical-Numerical Distributions
- Scatter Plots & Correlation

Univariate profiling metrics aren't much help in the multivariate world; we now need a way to describe ***relationships*** between variables

Correlation is the most common multivariate profiling metric, and is used to describe how a pair of variables are linearly related

- In other words: *for a given row, when one variable's observation goes above its mean, does the other variable's observation also go above its mean (and vice versa)?*



No correlation ***Positive correlation*** ***Strong positive correlation***

CORRELATION

- Categorical-Categorical Distributions
- Categorical-Numerical Distributions
- Multivariate Kernel Densities
- Violin & Box Plots
- Numerical-Numerical Distributions
- Scatter Plots & Correlation

Correlation is an **extension of variance**

- Think of correlation as a way to measure the **variance** of both variables at one time *(called "**co-variance**")*, while controlling for the scales of each variable

***Variance** formula*

$$\frac{\sum_{i=1}^{n}(x_i - \mu)^2}{n-1}$$

***Correlation** formula*

$$\frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{(n-1)S_xS_y}$$

- Here we multiply **variable X's difference from its mean** with **variable Y's difference from its mean**, instead of squaring a single variable *(like we do with variance)*
- S_x and S_y are the **standard deviations** of X and Y, which puts them on the same scale

CORRELATION VS. CAUSATION

- Categorical-Categorical Distributions
- Categorical-Numerical Distributions
- Multivariate Kernel Densities
- Violin & Box Plots
- Numerical-Numerical Distributions
- Scatter Plots & Correlation

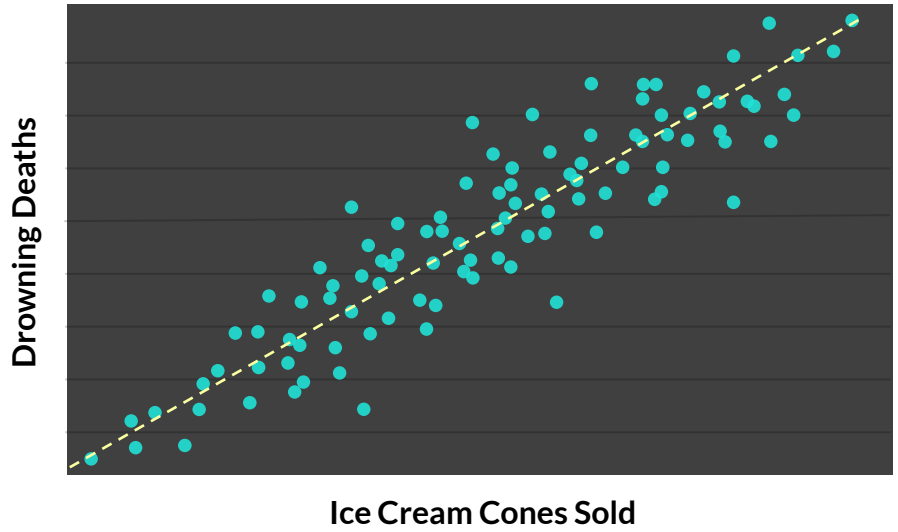
CORRELATION

⚠ DOES **NOT** IMPLY ⚠

CAUSATION

CORRELATION VS. CAUSATION

- Categorical-Categorical Distributions
- Categorical-Numerical Distributions
- Multivariate Kernel Densities
- Violin & Box Plots
- Numerical-Numerical Distributions
- Scatter Plots & Correlation



Consider the scatter plot above, showing daily **ice cream sales** and **drowning deaths** in a popular New England vacation town

- These two variables are clearly ***correlated***, but do ice cream cones CAUSE people to drown? Do drowning deaths CAUSE a surge in ice cream sales?

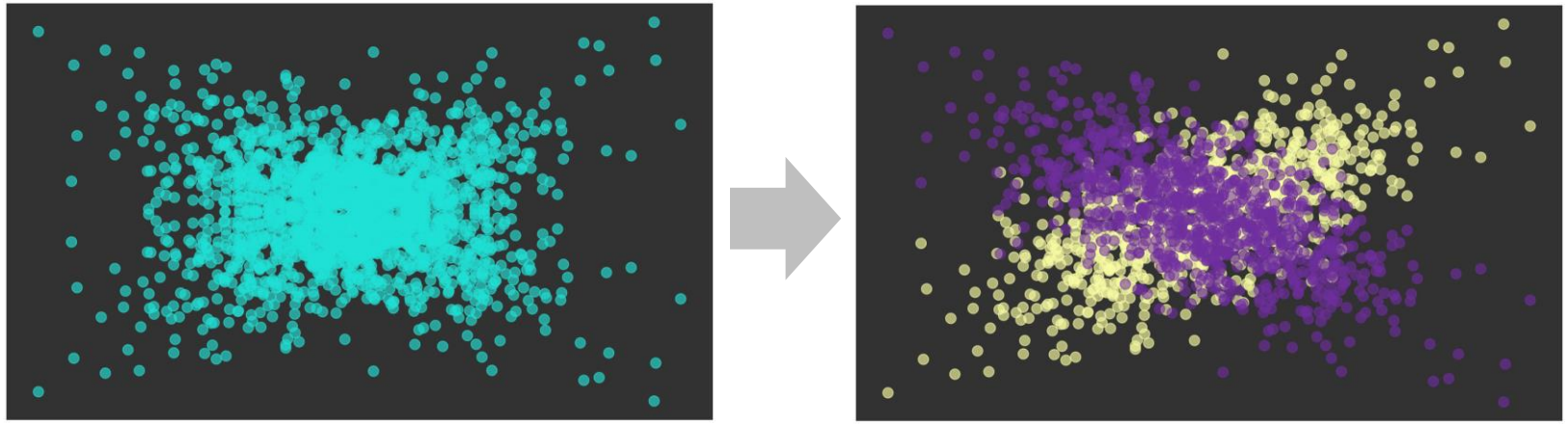
Of course not, because correlation does NOT imply causation!

*So what do you think is **really** going on here?*

PRO TIP: VISUALIZING A THIRD DIMENSION

- Categorical-Categorical Distributions
- Categorical-Numerical Distributions
- Multivariate Kernel Densities
- Violin & Box Plots
- Numerical-Numerical Distributions
- Scatter Plots & Correlation

Scatter plots show two dimensions by default *(X and Y)*, but using **symbols** or **color** allows you to visualize additional variables and expose otherwise hidden patterns or trends



Visualizing more than 3-4 dimensions is beyond human capability.
This is where you need machine learning!

CASE STUDY: CORRELATION



THE SITUATION

You've just landed your dream job as a Marketing Analyst at **Loud & Clear**, the hottest ad agency in San Diego.



THE ASSIGNMENT

Your client would like to understand the **impact of their digital media spend**, and how it relates to website traffic, offline spend, site load time, and sales.

Your role is to **collect and visualize these metrics** at the weekly-level in order to begin exploring the relationships between them.



THE OBJECTIVES

1. Gather weekly spend, traffic, website, and sales data
2. Create a scatterplot to visualize any two given variables
3. Compare the relationship between Digital Media Spend and each variable
4. What patterns do you see, and how would you interpret them?

BEST PRACTICES: MULTIVARIATE PROFILING

- Univariate profiling is a great start, but multivariate profiling is necessary when working with **more than one variable** *(pretty much always)*
- Understand the **types of variables** you're working with *(categorical vs. numerical)* to determine which profiling and visualization techniques to use
- Use categorical variables to **filter or "cut"** your data and quickly compare profiling metrics or distributions across classes
- Remember that **correlation does not imply causation**, and that variables can be related without one *causing* a change in the other

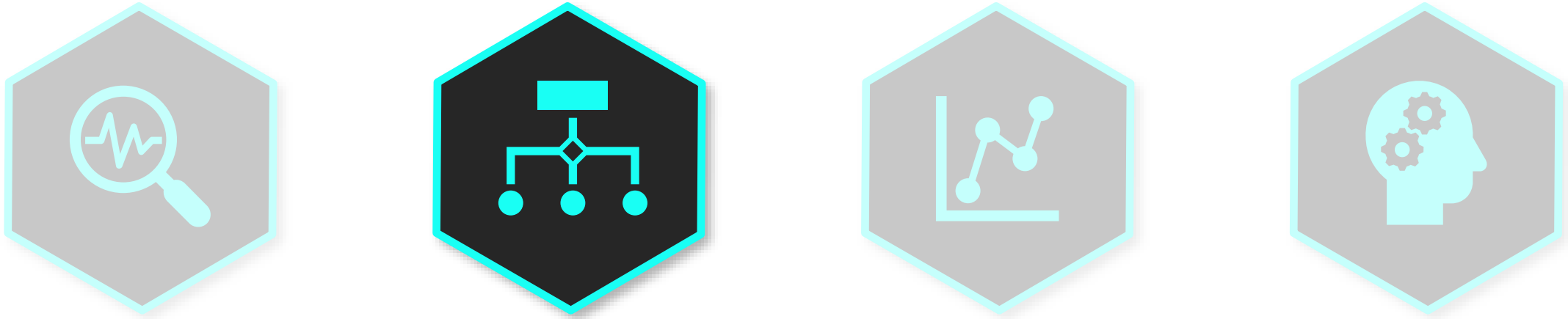
PART 2:

CLASSIFICATION



ABOUT THIS SERIES

This is **Part 2** of a **4-Part series** designed to help you build a deep, foundational understanding of machine learning, including data QA & profiling, classification, forecasting and unsupervised learning



PART 1
QA & Data Profiling

PART 2
Classification

PART 3
Regression & Forecasting

PART 4
Unsupervised Learning

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COURSE OUTLINE

1 Intro to Classification

Machine Learning landscape & key concepts, classification process, feature engineering, data splitting, overfitting, etc.

2 Classification Models



K-Nearest Neighbors



Naïve Bayes



Decision Trees



Random Forests



Logistic Regression



Sentiment Analysis

3 Model Selection & Tuning

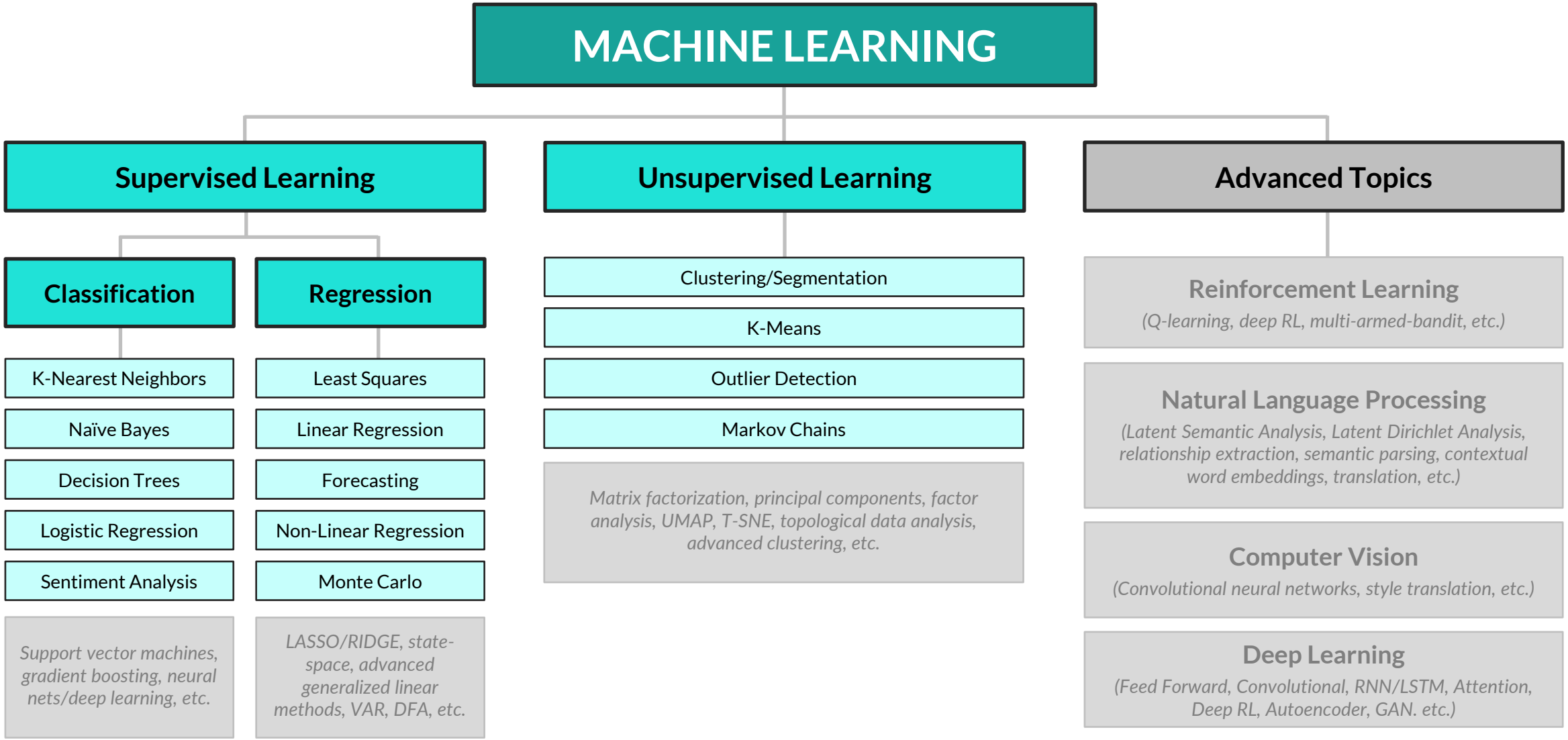
Model performance, hyperparameter tuning, confusion matrices, imbalanced classes, model drift, etc.

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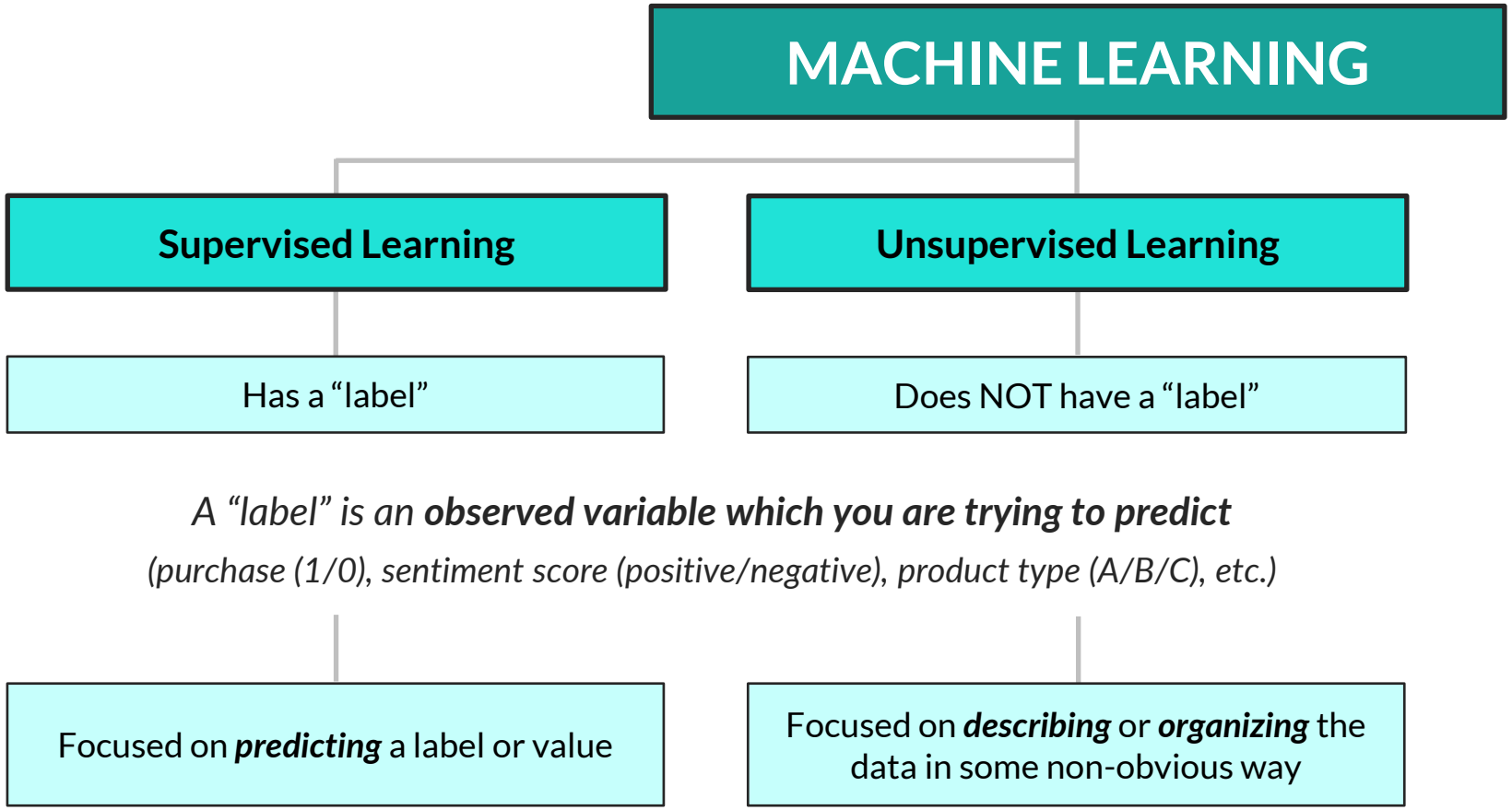
INTRO TO CLASSIFICATION

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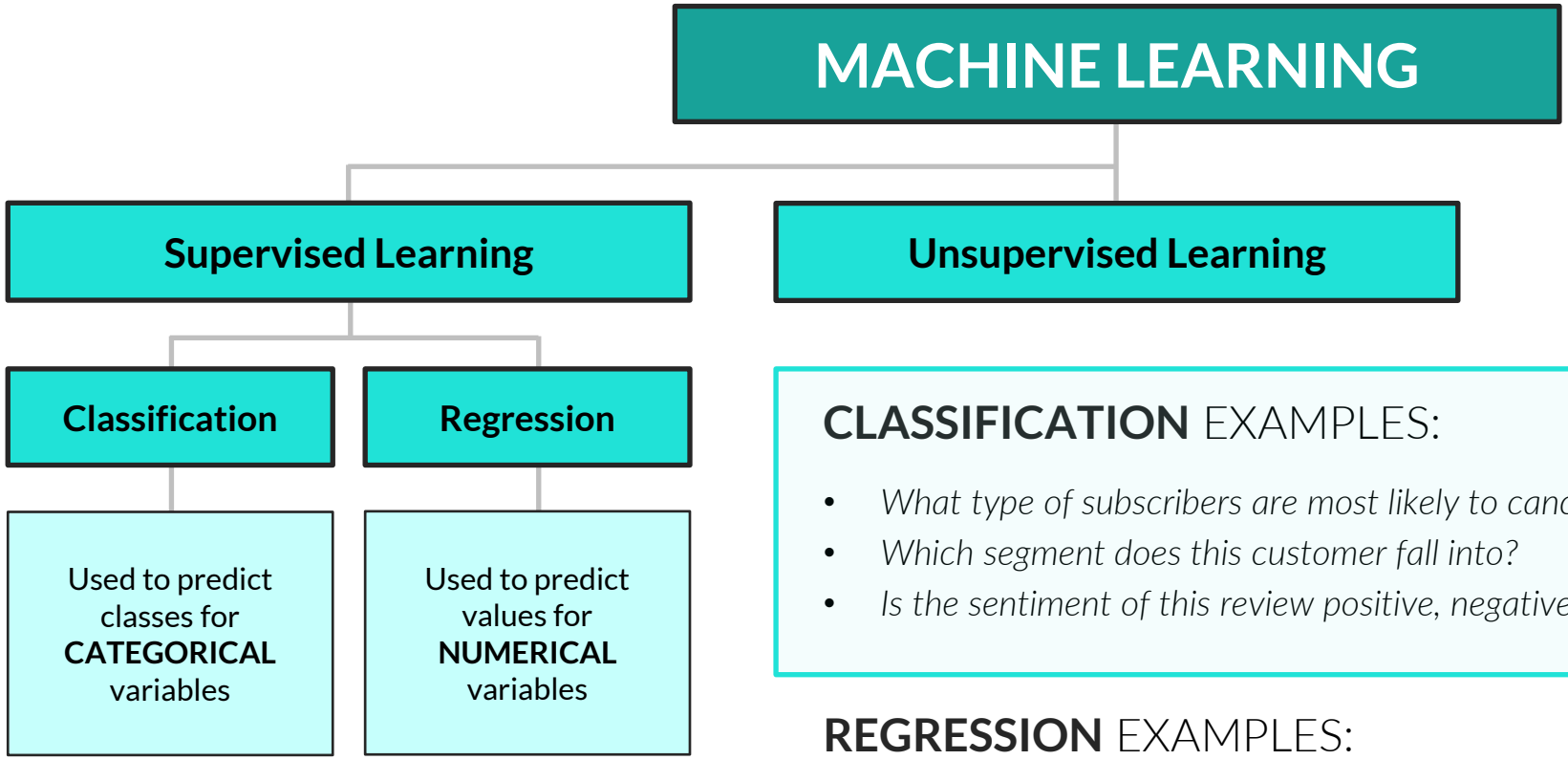
MACHINE LEARNING LANDSCAPE



MACHINE LEARNING LANDSCAPE



MACHINE LEARNING LANDSCAPE



CLASSIFICATION EXAMPLES:

- *What type of subscribers are most likely to cancel or churn?*
- *Which segment does this customer fall into?*
- *Is the sentiment of this review positive, negative or neutral?*

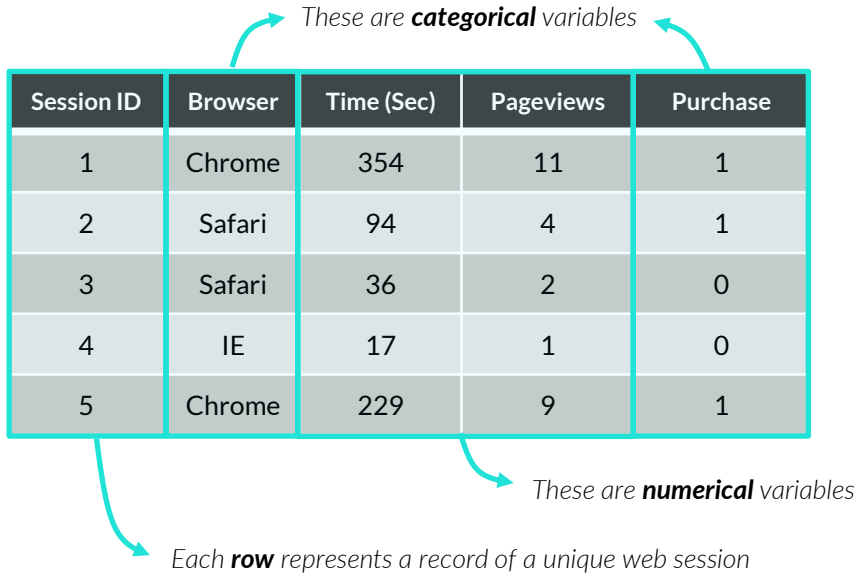
REGRESSION EXAMPLES:

- *How much revenue are we forecasting for the quarter?*
- *How many units do we expect to sell next season?*
- *How will seasonality impact our customer base next year?*

RECAP: KEY CONCEPTS

All tables contain rows and columns

- Each row represents an individual **record**/**observation**
- Each columns represent an individual **variable**



Variables can be categorical or numerical

- Categorical variables contain **classes** or **categories** *(used for filtering)*
- Numerical variables contain **numbers** *(used for aggregation)*

QA and data profiling comes first, every time

- Without quality data, you can't build quality models *("garbage in, garbage out")*
- Profiling is the first step towards **conditioning** *(or filtering)* on key variables to understand their impact

Machine Learning is needed when visual analysis falls short

- Humans aren't capable of thinking beyond 3+ dimensions, but machines are built for it
- ML is a natural extension of visual analysis and univariate/multivariate profiling

CLASSIFICATION 101

The goal of any classification model is to predict a ***dependent* variable** using ***independent* variables**

y **Dependent variable** (DV)

- This is the variable **you're trying to predict**
- The dependent variable is commonly referred to as *"Y"*, *"predicted"*, *"output"*, or *"target"* variable
- Classification is about understanding how the DV is impacted by, or *dependent* on, other variables in the model

x **Independent variables** (IVs)

- These are the variables **which help you predict the dependent variable**
- Independent variables are commonly referred to as *"X's"*, *"predictors"*, *"features"*, *"dimensions"*, *"explanatory variables"*, *or "covariates"*
- Classification is about understanding how the IVs impact, or *predict*, the DV

CLASSIFICATION 101

EXAMPLE: Using data from a CRM database *(sample below)* to predict if a customer will churn next month

Cust. ID	Gender	Status	HH Income	Age	Sign-Up Date	Newsletter	Churn
1	M	Bronze	$30,000	29	Jan 17, 2019	1	0
2	F	Gold	$60,000	37	Mar 18, 2017	1	1
3	F	Bronze	$15,000	24	Oct 1, 2020	0	0
4	F	Silver	$75,000	41	Apr 12, 2019	1	0
5	M	Bronze	$40,000	36	Jul 23, 2020	1	1
6	M	Gold	$35,000	31	Oct 22, 2017	0	1
7	F	Gold	$80,000	46	May 2, 2019	0	0
8	F	Bronze	$20,000	33	Feb 20, 2020	0	0
9	M	Silver	$100,000	51	Aug 5, 2020	1	1
10	M	Silver	$45,000	29	Sep 15, 2019	1	0

Churn *is our* ***dependent variable****, since it's what we want to predict*

Gender, ***Status***, ***HH Income***, ***Age***, ***Sign-Up Date*** *and* ***Newsletter*** *are all* ***independent variables****, since they can help us explain, or predict, Churn*

CLASSIFICATION 101

EXAMPLE: Using data from a CRM database *(sample below)* to predict if a customer will churn next month

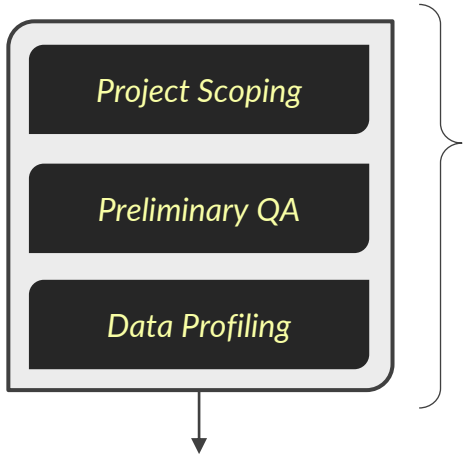
Cust. ID	Gender	Status	HH Income	Age	Sign-Up Date	Newsletter	Churn
1	M	Bronze	$30,000	29	Jan 17, 2019	1	0
2	F	Gold	$60,000	37	Mar 18, 2017	1	1
3	F	Bronze	$15,000	24	Oct 1, 2020	0	0
4	F	Silver	$75,000	41	Apr 12, 2019	1	0
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8	F	Bronze	$20,000	33	Feb 20, 2020	0	0
9	M	Silver	$100,000	51	Aug 5, 2020	1	1
10	M	Silver	$45,000	29	Sep 15, 2019	1	0
11	F	Bronze	$40,000	40	Dec 12, 2020	0	???

We use records with **observed values** for both independent and dependent variables to *"train"* our classification model…

…then apply that model to new, **unobserved values** containing IVs but no DV

This is what our model will predict!

CLASSIFICATION WORKFLOW



Remember, these steps ALWAYS come first

*Before building a model, you should have a deep understanding of both the **project scope** (stakeholders, framework, desired outcome, etc.) and the **data at hand** (variable types, table structure, data quality, profiling metrics, etc.)*

Feature Engineering → **Data Splitting** → **Model Training** → **Selection & Tuning**

Adding new, calculated variables (or "features") to a data set based on existing fields

Splitting records into "Training" and "Test" data sets, to validate accuracy and avoid overfitting

Building classification models from Training data and applying to Test data to maximize prediction accuracy

Choosing the best performing model for a given prediction, and tuning it to prevent drift over time

FEATURE ENGINEERING



Feature engineering is the process of enriching a data set by creating additional independent variables based on existing fields

- New features can help improve the accuracy and predictive power of your ML models
- Feature engineering is often used to convert fields into *"model-friendly"* formats; for example, **one-hot encoding** transforms categorical variables into binary (1/0) fields

***Original** features*

***Engineered** features*

Cust. ID	Status	HH Income	Age	Sign-Up Date	Newsletter	Gold	Silver	Bronze	Scaled Income	Log Income	Age Group	Sign-Up Year	Priority
1	Bronze	$30,000	29	Jan 17, 2019	1	0	0	1	.25	10.3	20-30	2019	1
2	Gold	$60,000	37	Mar 18, 2017	1	1	0	0	.75	11.0	30-40	2017	1
3	Bronze	$15,000	24	Oct 1, 2020	0	0	0	1	0	9.6	20-30	2020	0
4	Silver	$75,000	41	Apr 12, 2019	1	0	1	0	1	11.2	40-50	2019	0
5	Bronze	$40,000	36	Jul 23, 2020	1	0	0	1	.416	10.6	30-40	2020	1

FEATURE ENGINEERING TECHNIQUES

*Common **feature engineering** techniques:*

Cust. ID	Status	HH Income	Age	Sign-Up Date	Newsletter	Gold	Silver	Bronze	Scaled Income	Log Income	Age Group	Sign-Up Year	Priority
1	Bronze	$30,000	29	Jan 17, 2019	1	0	0	1	.25	10.3	20-30	2019	1
2	Gold	$60,000	37	Mar 18, 2017	1	1	0	0	.75	11.0	30-40	2017	1
3	Bronze	$15,000	24	Oct 1, 2020	0	0	0	1	0	9.6	20-30	2020	0
4	Silver	$75,000	41	Apr 12, 2019	1	0	1	0	1	11.2	40-50	2019	0
5	Bronze	$40,000	36	Jul 23, 2020	1	0	0	1	.416	10.6	30-40	2020	1

One-Hot Encoding

Splitting categorical fields into separate binary features

Scaling

Standardizing numerical ranges common scales (i.e. 1-10, 0-100%)

Log Transformation

Converting a range of values into a compressed, "less-skewed" distribution

Discretization

Grouping continuous values into discrete segments or bins

Date Extraction

Transforming a date or datetime value into individual components

Boolean Logic

Using and/or logic to encode "interactions" between variables

DATA SPLITTING



Splitting is the process of partitioning data into separate sets of records for the purpose of training and testing machine learning models

- As a rule of thumb, ~70-80% of your data will be used for **Training** *(which is what your model learns from)*, and ~20-30% will be reserved for **Testing** *(to validate the model's accuracy)*

Cust. ID	Gender	Status	HH Income	Age	Sign-Up Date	Newsletter	Churn
1	M	Bronze	$30,000	29	Jan 17, 2019	1	0
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Training *data*

Test *data*



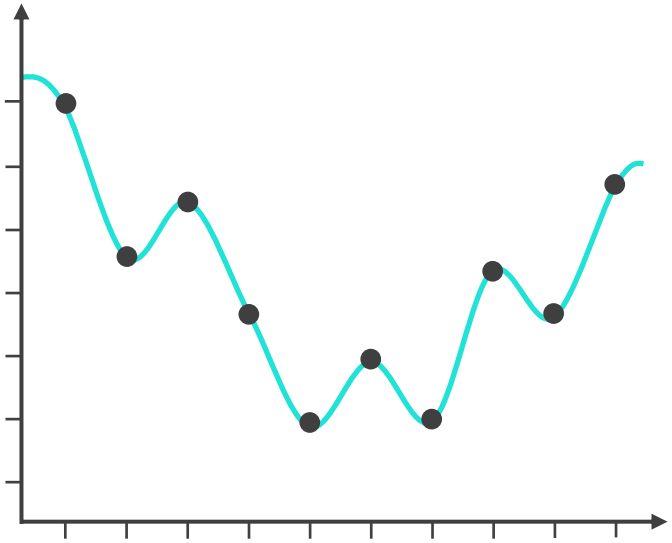
Test data is NOT used to optimize models

Using **Training** data for optimization and **Test** data for validation ensures that your model can accurately predict both *known* and *unknown* values, which helps to prevent **overfitting**

OVERFITTING

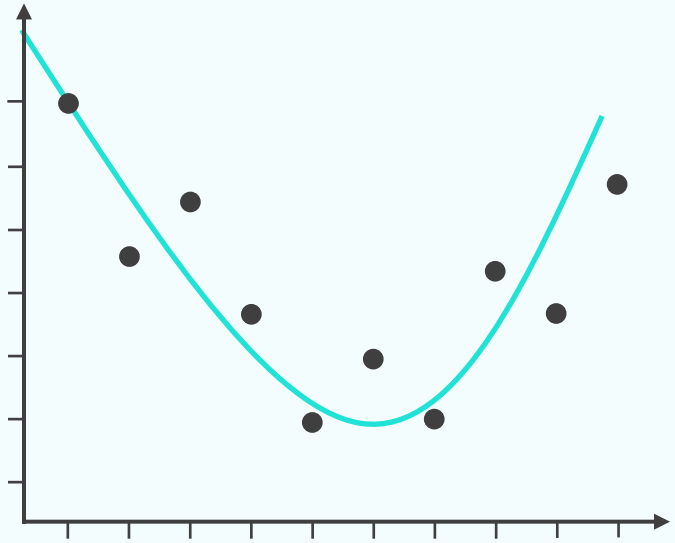
Splitting is primarily used to avoid **overfitting**, which is when a model predicts known *(Training)* data very well but unknown *(Test)* data poorly

- Think of overfitting like memorizing the answers to a test instead of actually *learning* the material; you'll ace the test, but lack the ability to ***generalize*** and apply your knowledge to unfamiliar questions



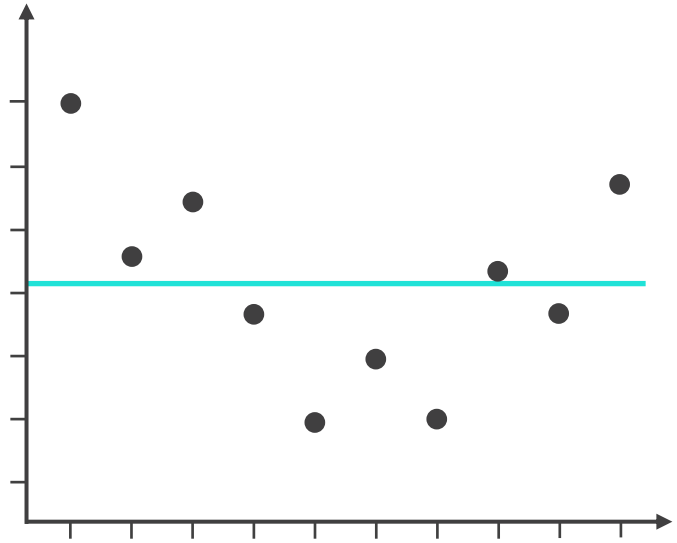
OVERFIT model

- Models the Training data too well
- Doesn't generalize well to Test data *(high variance, low bias)*



WELL-FIT model

- Models the Training data just right
- Generalizes well to Test data *(balance of bias & variance)*



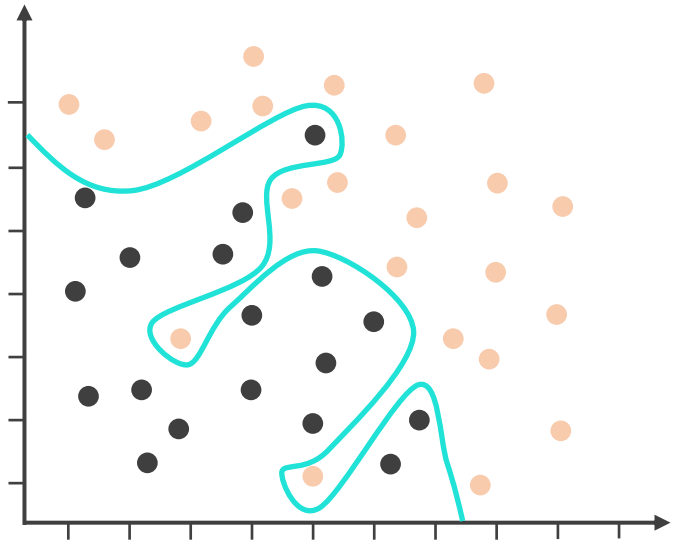
UNDERFIT model

- Doesn't model Training data well enough
- Doesn't generalize well to Test data *(high bias, low variance)*

OVERFITTING

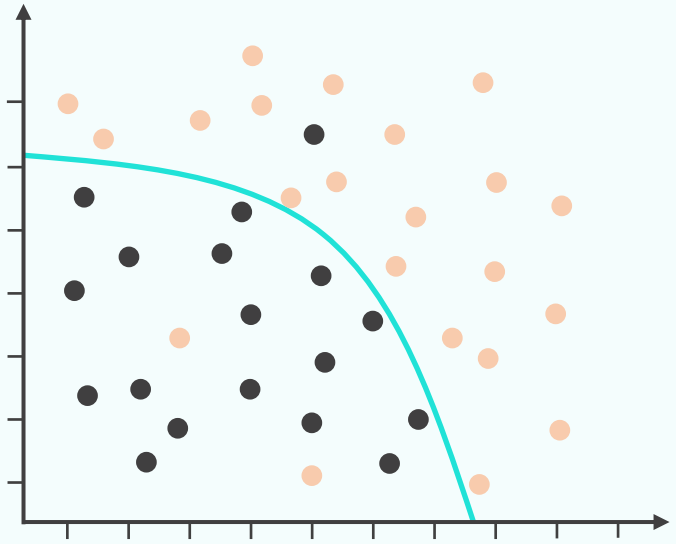
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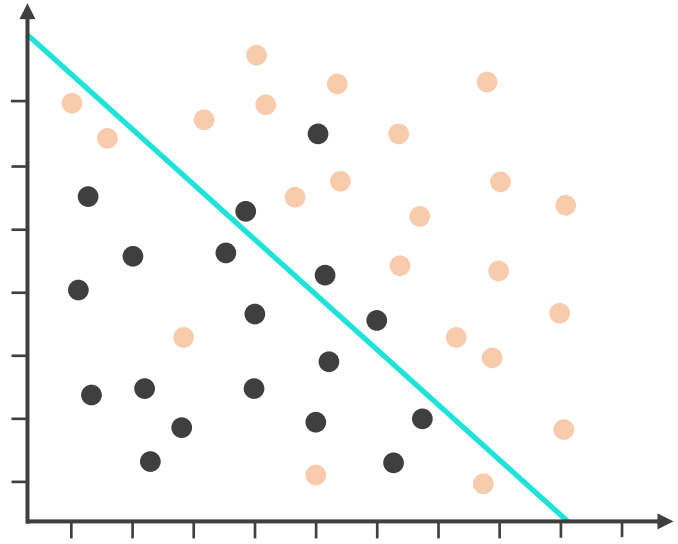
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- Models the Training data just right
- Generalizes well to Test data *(balance of bias & variance)*



UNDERFIT model

- Doesn't model Training data well enough
- Doesn't generalize well to Test data *(high bias, low variance)*

CLASSIFICATION MODELS

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CLASSIFICATION MODELS



In this section we'll introduce common **classification models** used to predict categorical outcomes, including KNN, naïve bayes, decision trees, logistic regression, and more

TOPICS WE'LL COVER:

- K-Nearest Neighbors
- Naïve Bayes
- Decision Trees
- Random Forests
- Logistic Regression
- Sentiment Analysis

COMMON USE CASES:

- *Predicting if a subscriber will stay or churn*
- *Predicting which product a customer will purchase*
- *Predicting tone or sentiment from raw text (product reviews, survey responses, tweets, etc.)*
- *Predicting if an email is spam, or a bank transaction is fraudulent*

K-NEAREST NEIGHBORS

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K-NEAREST NEIGHBORS (KNN)

- **K-Nearest Neighbors**
- Naïve Bayes
- Decision Trees
- Random Forests
- Logistic Regression
- Sentiment Analysis

K-Nearest Neighbors (KNN) is a classification technique designed to predict outcomes based on similar observed values

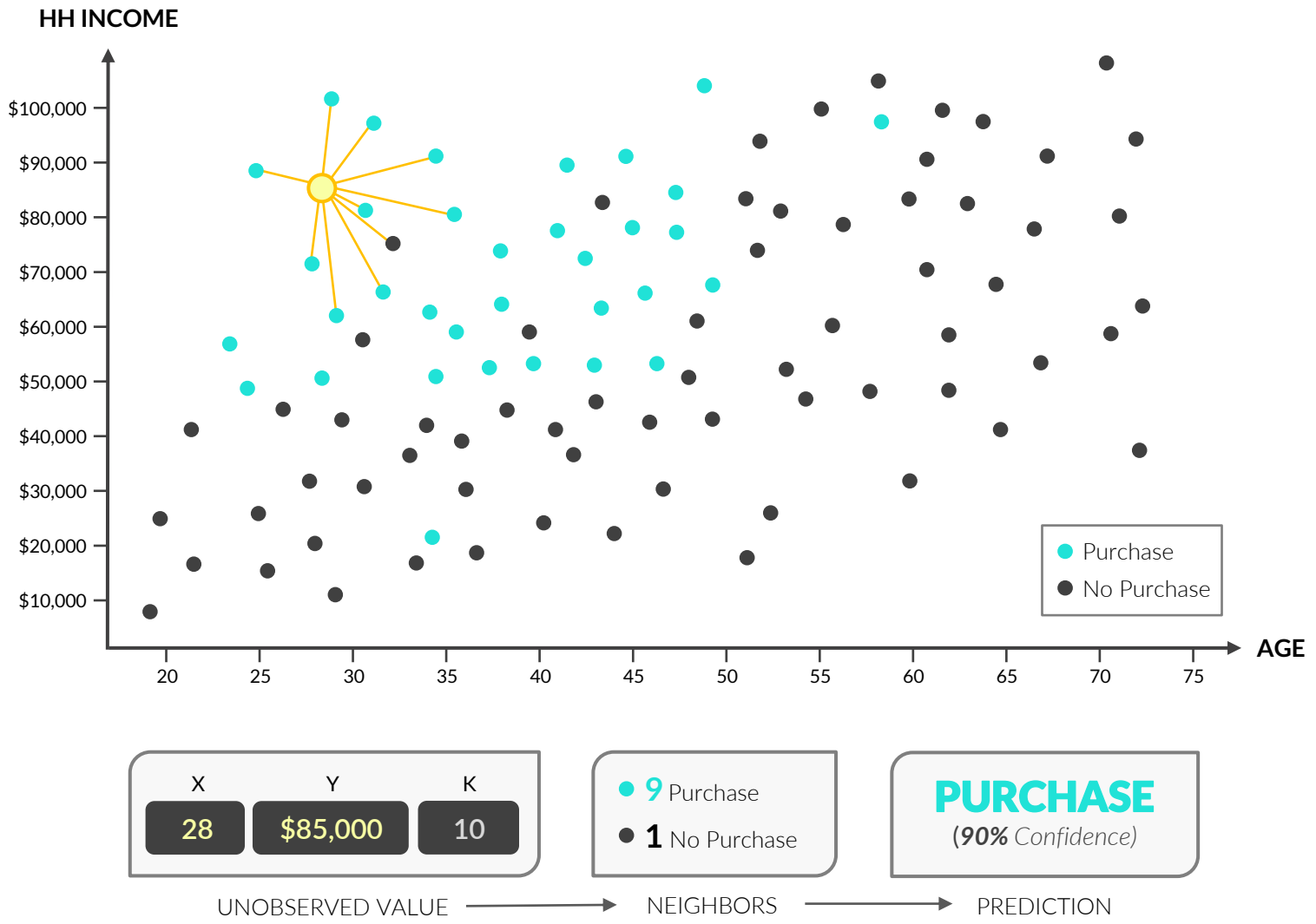
- In its simplest form, KNN creates a scatter plot with training data, plots a new unobserved value, and assigns a class (DV) based on the classes of nearby points
- **K** represents the number of nearby points (or "neighbors") the model will consider when making a prediction

Example use cases:

- Generating product recommendations on a website
- Classifying customers into pre-existing segments

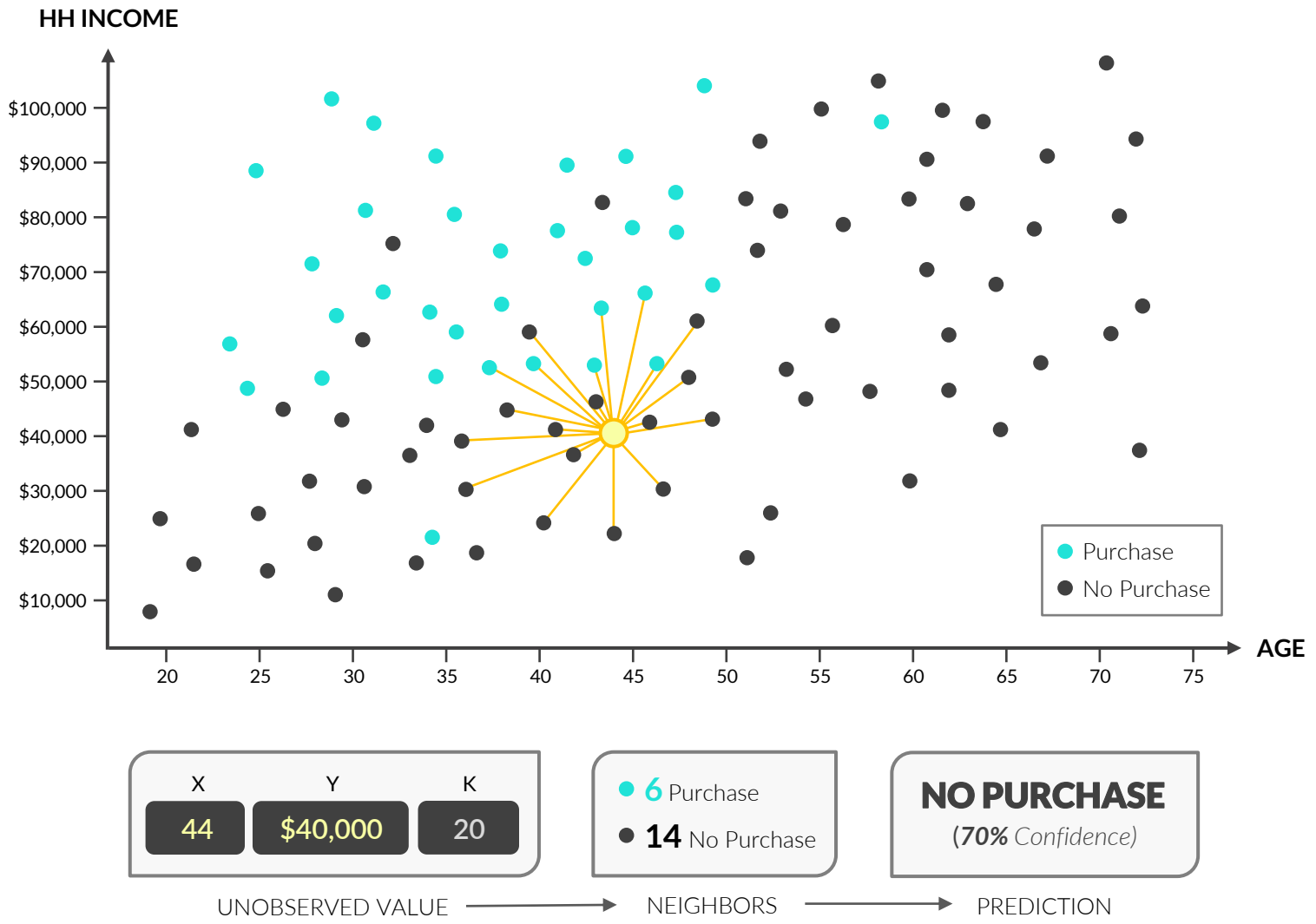
K-NEAREST NEIGHBORS (KNN)

- **K-Nearest Neighbors**
- Naïve Bayes
- Decision Trees
- Random Forests
- Logistic Regression
- Sentiment Analysis



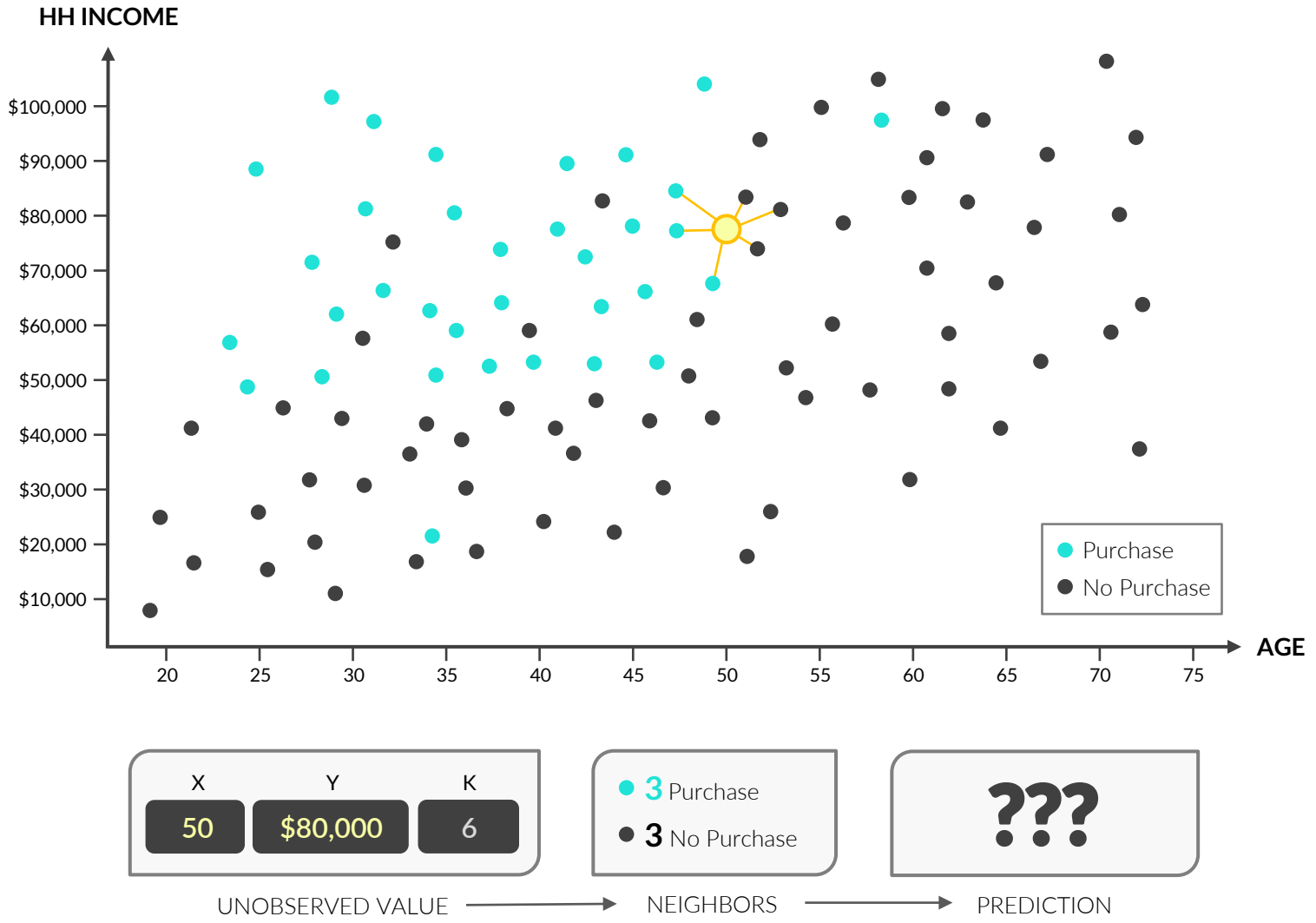
K-NEAREST NEIGHBORS (KNN)

- **K-Nearest Neighbors**
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K-NEAREST NEIGHBORS (KNN)

- **K-Nearest Neighbors**
- Naïve Bayes
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- Sentiment Analysis





K-NEAREST NEIGHBORS (KNN)

- **K-Nearest Neighbors**
- Naïve Bayes
- Decision Trees
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- Sentiment Analysis



What if there's a tie?

- In practice, KNN also factors *distance* between neighbors; if there are multiple modes, the class with the **shortest total distance** to the observation wins



How do we figure out the "right" value for K?

- This is where **Machine Learning** comes in! The model tests multiple K values to see which one is most accurate when applied to your Test data



What if we have more than 2 IVs?

- Scatter plots are great for demonstrating how KNN works for 2 independent variables, but humans can't visualize beyond 3 or 4 dimensions
- Computers can easily apply the same logic and calculations to *many* IVs

CASE STUDY: K-NEAREST NEIGHBORS



THE **SITUATION**

Congratulations! You just landed your dream job as a Machine Learning Engineer at **Spotify**, one of the world's largest music streaming platforms



THE **ASSIGNMENT**

Your first assignment is to **analyze user listening behavior to help improve Spotify's recommendation engine**.

You'll need to use a KNN model to predict if a user will **listen**, **skip** or **favorite** a given track, based on various attributes.



THE **OBJECTIVES**

1. Collect sample data containing listener behaviors and song attributes
2. Visualize outcomes for a given pair of attributes using a Scatter Plot
3. Calculate the predicted outcome *(listen, skip or favorite)* and level of confidence for any unobserved value

NAÏVE BAYES

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NAÏVE BAYES

- K-Nearest Neighbors
- **Naïve Bayes**
- Decision Trees
- Random Forests
- Logistic Regression
- Sentiment Analysis

Naïve Bayes is a classification technique which uses conditional probabilities to predict multi-class or binary outcomes

- Naïve Bayes essentially creates **frequency tables** for each combination of variables, then calculates the **conditional probability** of each IV value for a given outcome
- For new observations, the model looks at the probability that each IV value would be observed *given each outcome*, and compares the results to make a prediction

Example use cases:

- Predicting purchase probability for prospects in a marketing database
- Credit risk scoring in banking or financial industries



NAÏVE BAYES

- K-Nearest Neighbors
- **Naïve Bayes**
- Decision Trees
- Random Forests
- Logistic Regression
- Sentiment Analysis

*Each record represents a **customer***

*These are the **independent variables** (IV's) which will help us make a prediction*

Cust. ID	Subscribed to Newsletter	Followed FB Page	Visited Website	Purchase?
1	0	1	0	1
2	1	0	1	0
3	1	1	0	1
4	0	0	0	0
5	1	0	0	0
6	1	1	1	1
7	1	0	1	0
8	0	1	1	1
9	1	0	1	1
10	1	1	0	0
11	0	1	1	???

*This is the **dependent variable** (DV) we are predicting*

*These are our **observed values**, which we use to train the model*

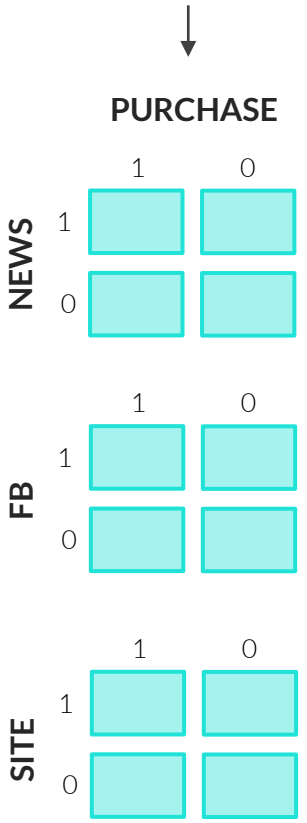
*This is an **unobserved value**; which our model will predict*

NAÏVE BAYES

- K-Nearest Neighbors
- **Naïve Bayes**
- Decision Trees
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Cust. ID	Subscribed to Newsletter	Followed FB Page	Visited Website	Purchase?
1	0	1	0	1
2	1	0	1	0
3	1	1	0	1
4	0	0	0	0
5	1	0	0	0
6	1	1	1	1
7	1	0	1	0
8	0	1	1	1
9	1	0	1	1
10	1	1	0	0
11	0	1	1	???

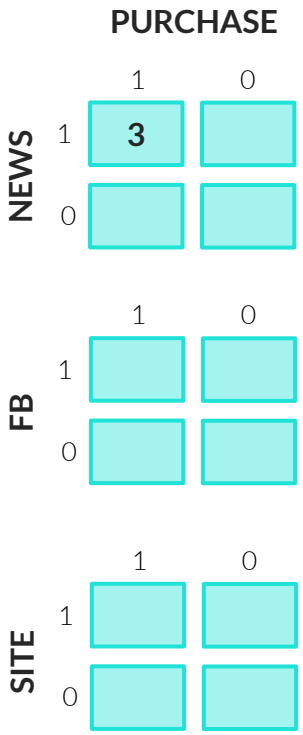
Frequency tables between each IV and the DV



NAÏVE BAYES

- K-Nearest Neighbors
- **Naïve Bayes**
- Decision Trees
- Random Forests
- Logistic Regression
- Sentiment Analysis

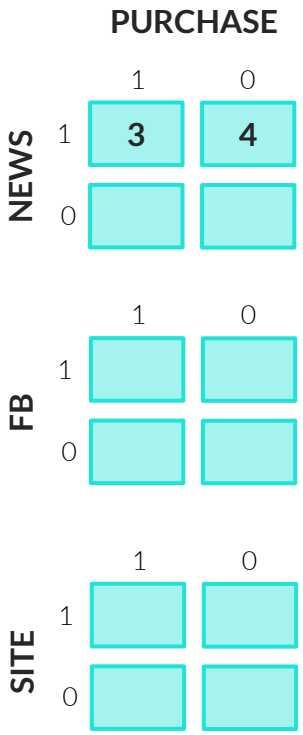
Cust. ID	Subscribed to Newsletter	Followed FB Page	Visited Website	Purchase?
1	0	1	0	1
2	1	0	1	0
3	1	1	0	1
4	0	0	0	0
5	1	0	0	0
6	1	1	1	1
7	1	0	1	0
8	0	1	1	1
9	1	0	1	1
10	1	1	0	0
11	0	1	1	???



NAÏVE BAYES

- K-Nearest Neighbors
- **Naïve Bayes**
- Decision Trees
- Random Forests
- Logistic Regression
- Sentiment Analysis

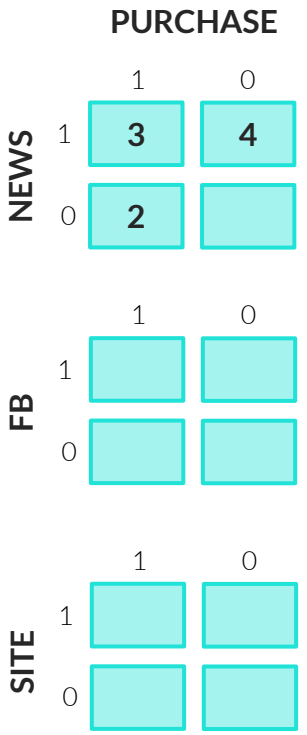
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2	1	0	1	0
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5	1	0	0	0
6	1	1	1	1
7	1	0	1	0
8	0	1	1	1
9	1	0	1	1
10	1	1	0	0
11	0	1	1	???



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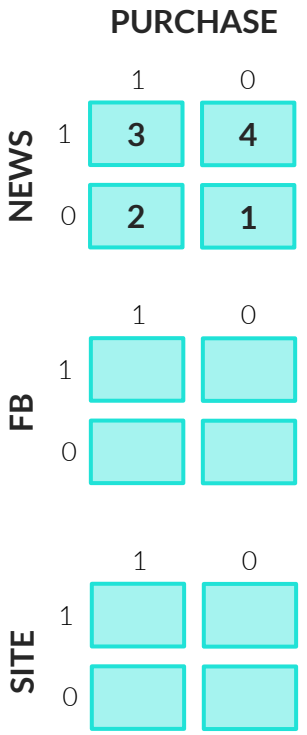
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7	1	0	1	0
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9	1	0	1	1
10	1	1	0	0
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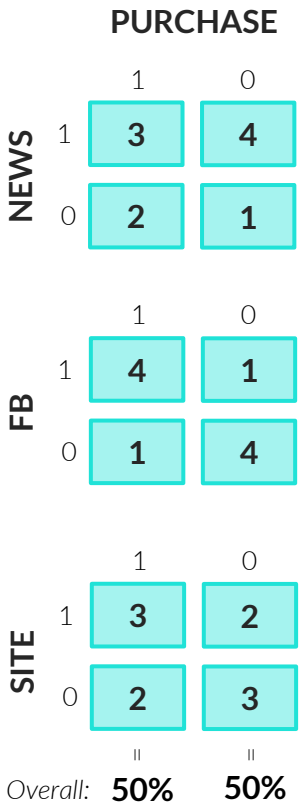
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Frequency tables give us the **conditional probability** of each outcome

- For example, **P (News | Purchase)** tells us the probability that a customer subscribed to the newsletter, given *(or conditioned on)* the fact that they purchased

PURCHASE

NEWS

	1	0
1	3	4
0	2	1

FB

	1	0
1	4	1
0	1	4

SITE

	1	0
1	3	2
0	2	3

P (News | Purchase)	**60%**	*Probability of each independent variable, given that a purchase was made*
P (No News | Purchase)	**40%**	
P (FB | Purchase)	**80%**	
P (No FB | Purchase)	**20%**	
P (Site | Purchase)	**60%**	
P (No Site | Purchase)	**40%**	
P (News | No Purchase)	**80%**	*Probability of each independent variable, given that a purchase was **NOT** made*
P (No News | No Purchase)	**20%**	
P (FB | No Purchase)	**20%**	
P (No FB | No Purchase)	**80%**	
P (Site | No Purchase)	**40%**	
P (No Site | No Purchase)	**60%**	
P (Purchase)	**50%**	*Overall probability of purchase*
P (No Purchase)	**50%**	



NAÏVE BAYES

- K-Nearest Neighbors
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P (News | Purchase)	**60%**
P (No News | Purchase)	**40%**
P (FB | Purchase)	**80%**
P (No FB | Purchase)	**20%**
P (Site | Purchase)	**60%**
P (No Site | Purchase)	**40%**
P (News | No Purchase)	**80%**
P (No News | No Purchase)	**20%**
P (FB | No Purchase)	**20%**
P (No FB | No Purchase)	**80%**
P (Site | No Purchase)	**40%**
P (No Site | No Purchase)	**60%**
P (Purchase)	**50%**
P (No Purchase)	**50%**

Unobserved value:

NEWS	FB	SITE
0	1	1

*Prob. given Purchase = **1***

=

P (No News | Purchase)
x
P (FB | Purchase)
x
P (Site | Purchase)
x
P (Purchase)

=

*Prob. given Purchase = **0***

=

P (No News | No Purchase)
x
P (FB | No Purchase)
x
P (Site | No Purchase)
x
P (No Purchase)

=

NAÏVE BAYES

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P (News | Purchase)	**60%**
P (No News | Purchase)	**40%**
P (FB | Purchase)	**80%**
P (No FB | Purchase)	**20%**
P (Site | Purchase)	**60%**
P (No Site | Purchase)	**40%**
P (News | No Purchase)	**80%**
P (No News | No Purchase)	**20%**
P (FB | No Purchase)	**20%**
P (No FB | No Purchase)	**80%**
P (Site | No Purchase)	**40%**
P (No Site | No Purchase)	**60%**
P (Purchase)	**50%**
P (No Purchase)	**50%**

Unobserved value:

NEWS	FB	SITE
0	1	1

*Prob. given Purchase = **1***

=

40%
x
80%
x
60%
x
50%

=

9.6%

*Prob. given Purchase = **0***

=

P (No News | No Purchase)
x
P (FB | No Purchase)
x
P (Site | No Purchase)
x
P (No Purchase)

=

NAÏVE BAYES

- K-Nearest Neighbors
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P (News | Purchase)	**60%**
P (No News | Purchase)	**40%**
P (FB | Purchase)	**80%**
P (No FB | Purchase)	**20%**
P (Site | Purchase)	**60%**
P (No Site | Purchase)	**40%**
P (News | No Purchase)	**80%**
P (No News | No Purchase)	**20%**
P (FB | No Purchase)	**20%**
P (No FB | No Purchase)	**80%**
P (Site | No Purchase)	**40%**
P (No Site | No Purchase)	**60%**
P (Purchase)	**50%**
P (No Purchase)	**50%**

Unobserved value:

NEWS	FB	SITE
0	1	1

*Prob. given Purchase = **1***

=

40%
x
80%
x
60%
x
50%

=

9.6%

*Prob. given Purchase = **0***

=

20%
x
20%
x
40%
x
50%

=

0.8%

NAÏVE BAYES

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P (News | Purchase)	**60%**
P (No News | Purchase)	**40%**
P (FB | Purchase)	**80%**
P (No FB | Purchase)	**20%**
P (Site | Purchase)	**60%**
P (No Site | Purchase)	**40%**
P (News | No Purchase)	**80%**
P (No News | No Purchase)	**20%**
P (FB | No Purchase)	**20%**
P (No FB | No Purchase)	**80%**
P (Site | No Purchase)	**40%**
P (No Site | No Purchase)	**60%**
P (Purchase)	**50%**
P (No Purchase)	**50%**

Unobserved value:

NEWS	FB	SITE
0	1	1

*Prob. given Purchase = **1***

=

40%
x
80%
x
60%
x
50%

=

9.6%

*Prob. given Purchase = **0***

=

20%
x
20%
x
40%
x
50%

=

0.8%

Overall Purchase Probability: $\frac{9.6\%}{(9.6\% + 0.8\%)}$ = **92.3%** → **PURCHASE**

PREDICTION



NAÏVE BAYES

- K-Nearest Neighbors
- **Naïve Bayes**
- Decision Trees
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- Sentiment Analysis



Who has time to work through all this math?

- No one! These probabilities are all calculated automatically by the model *(no manual effort required!)*



Doesn't multiplying many probabilities lead to very small values?

- Yes, which is why the *relative* probability is so important; even so, Naïve Bayes can struggle with a very large number of IVs

CASE STUDY: NAÏVE BAYES



THE SITUATION

You've just been promoted to Marketing Manager at **Cat Slacks**, a global retail powerhouse specializing in high-quality pants for cats



THE ASSIGNMENT

Your boss wants to understand the impact of key customer interactions on purchase behavior, including **subscribing to the newsletter**, **following the Facebook page**, and **visiting the Cat Slacks website**

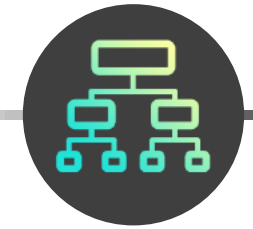


THE OBJECTIVES

1. Collect sample data containing customer interactions and purchase events
2. Compare the purchase probability for specific interactions
3. Calculate the overall purchase probability for any given set of interactions

DECISION TREES

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DECISION TREES

- K-Nearest Neighbors
- Naïve Bayes
- **Decision Trees**
- Random Forests
- Logistic Regression
- Sentiment Analysis

Decision Trees use a series of binary (true/false) rules to predict multi-class or binary outcomes

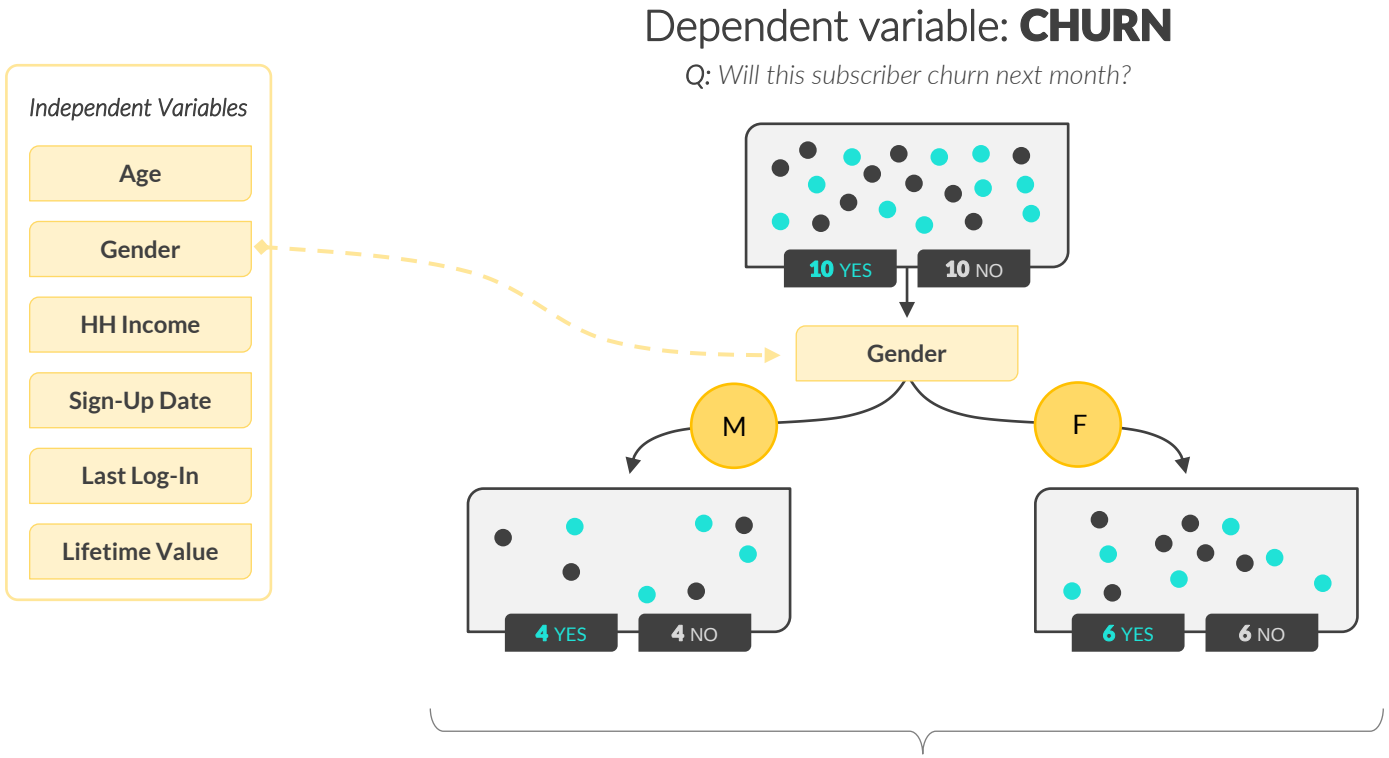
- For each rule, a decision tree selects a single IV and uses it to split observations into two groups, where each group *(ideally)* skews towards one class
- The goal is to determine which rules and IVs do the best job splitting up the classes, which we can measure using a metric known as **entropy**
- This is NOT a manual process; decision trees test *many* variables and select rules based on the change in entropy after the split (known as **information gain**)

Example use cases:

- Predicting if a customer will cancel/churn next month
- Identifying fraudulent bank transactions or insurance claims

DECISION TREES

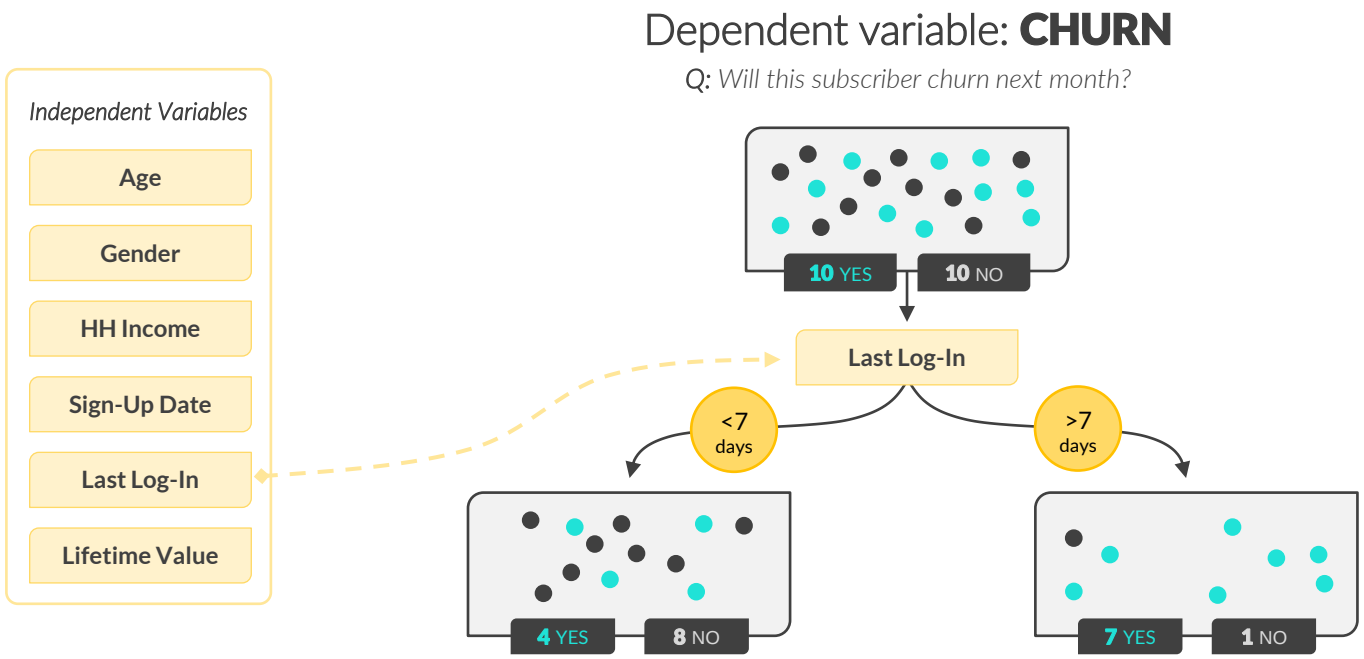
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*Splitting on **Gender** isn't effective, since our classes are still evenly distributed after the split (in other words, Gender isn't a good predictor for churn)*

DECISION TREES

- K-Nearest Neighbors
- Naïve Bayes
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- Sentiment Analysis



How do we know which independent variables to split?

- **Entropy** measures the "evenness" or "uncertainty" between classes, and can be compared before/after each split to measure information gain
- This is where **Machine Learning** comes in; the model tries *all* IVs to determine which splits reduce entropy the most

DECISION TREES

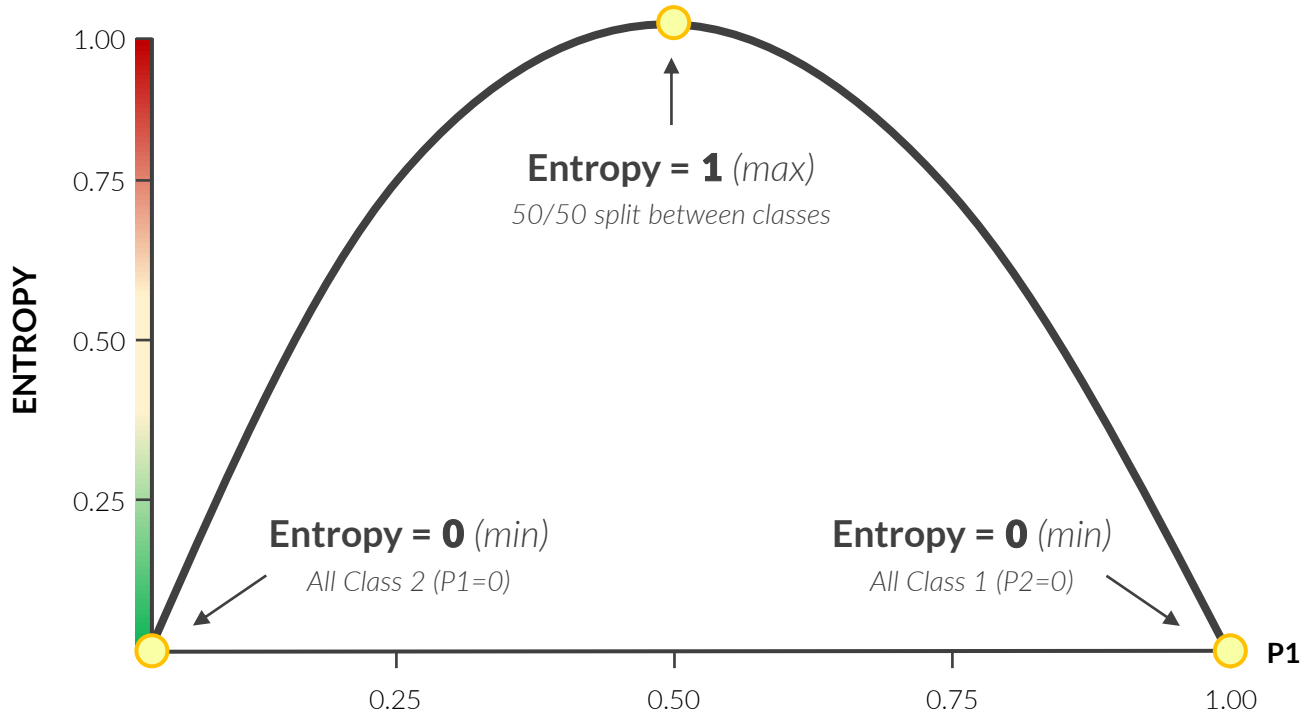
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Probability of **Class 1** (*Count of Class 1 / N*)

Probability of **Class 2** (*Count of Class 2 / N*)

$$\textbf{ENTROPY} = -P_1 * log_2(P_1) - P_2 * log_2(P_2)$$

Curve between **0** - **1**



DECISION TREES

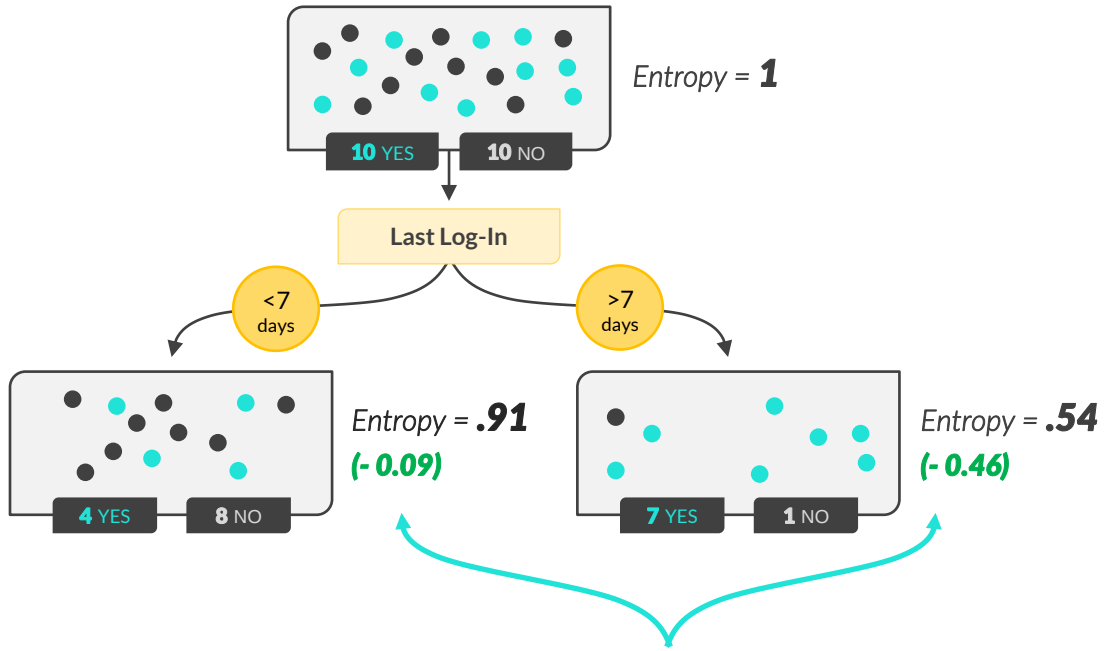
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Independent Variables

- Age
- Gender
- HH Income
- Sign-Up Date
- Last Log-In
- Lifetime Value

Dependent variable: **CHURN**

Q: Will this subscriber churn next month?



*The reduction in entropy after the split tells us that we're **gaining information**, and teasing out differences between those who churn and those who do not*

DECISION TREES

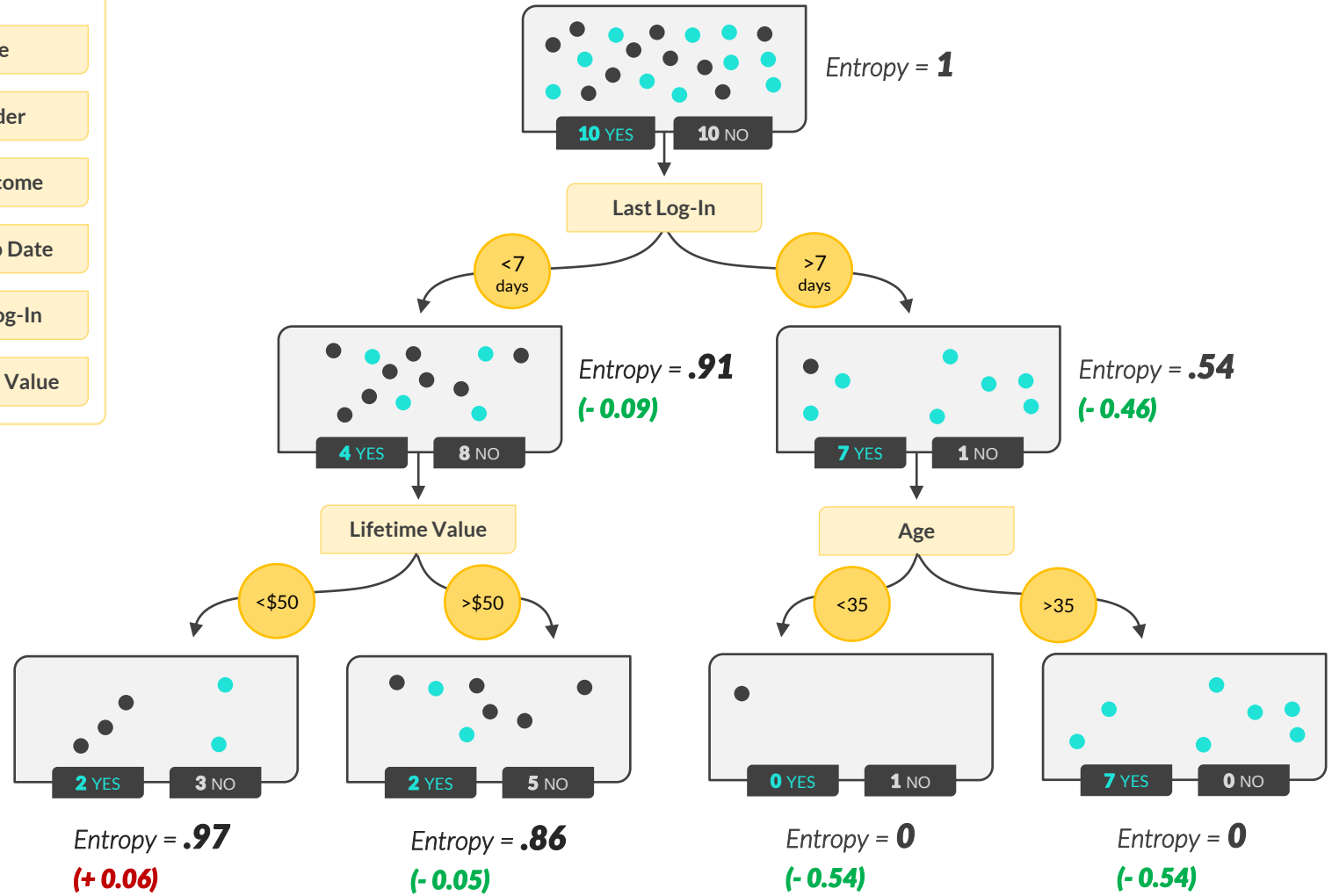
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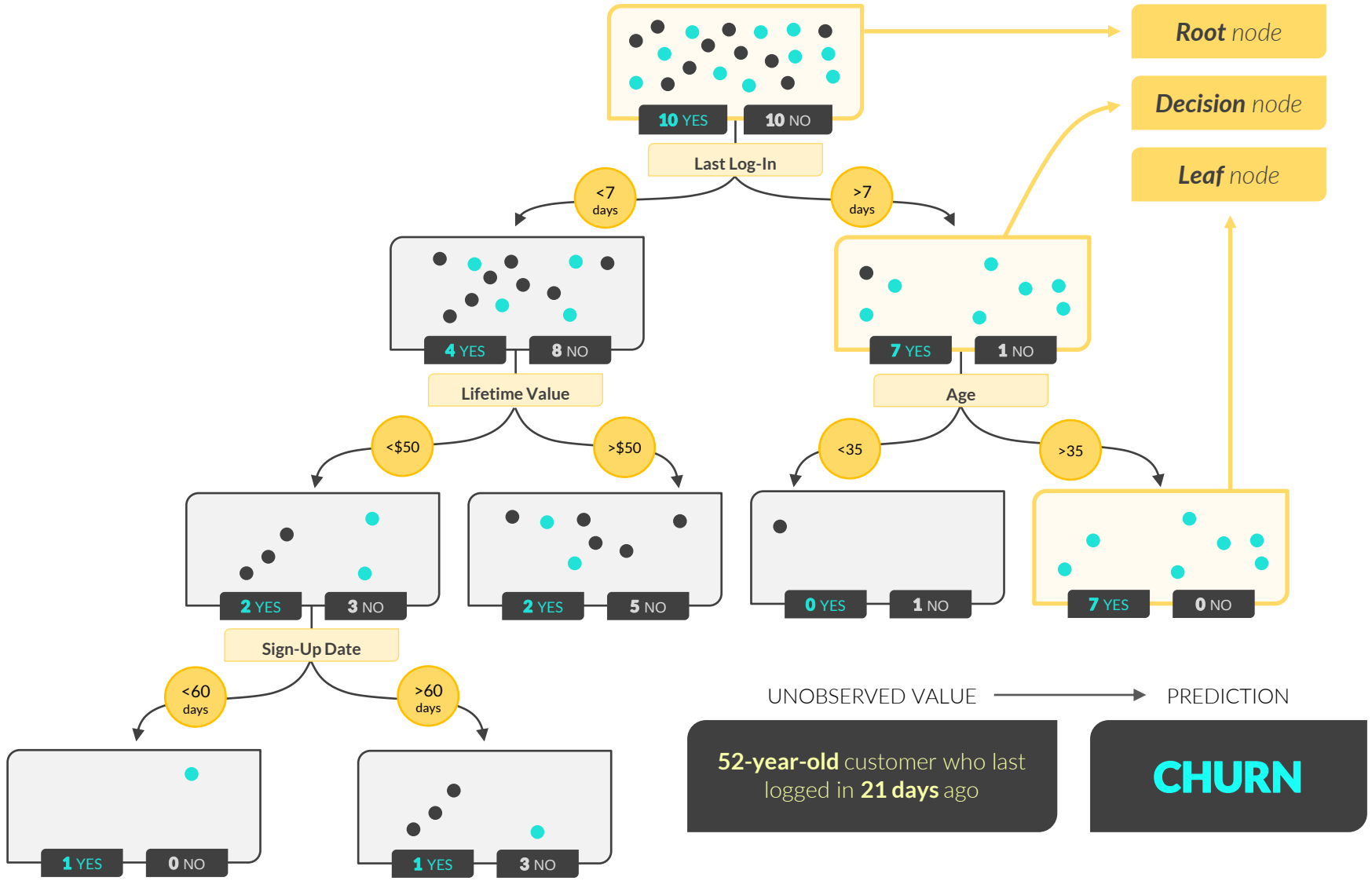
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Q: Will this subscriber churn next month?



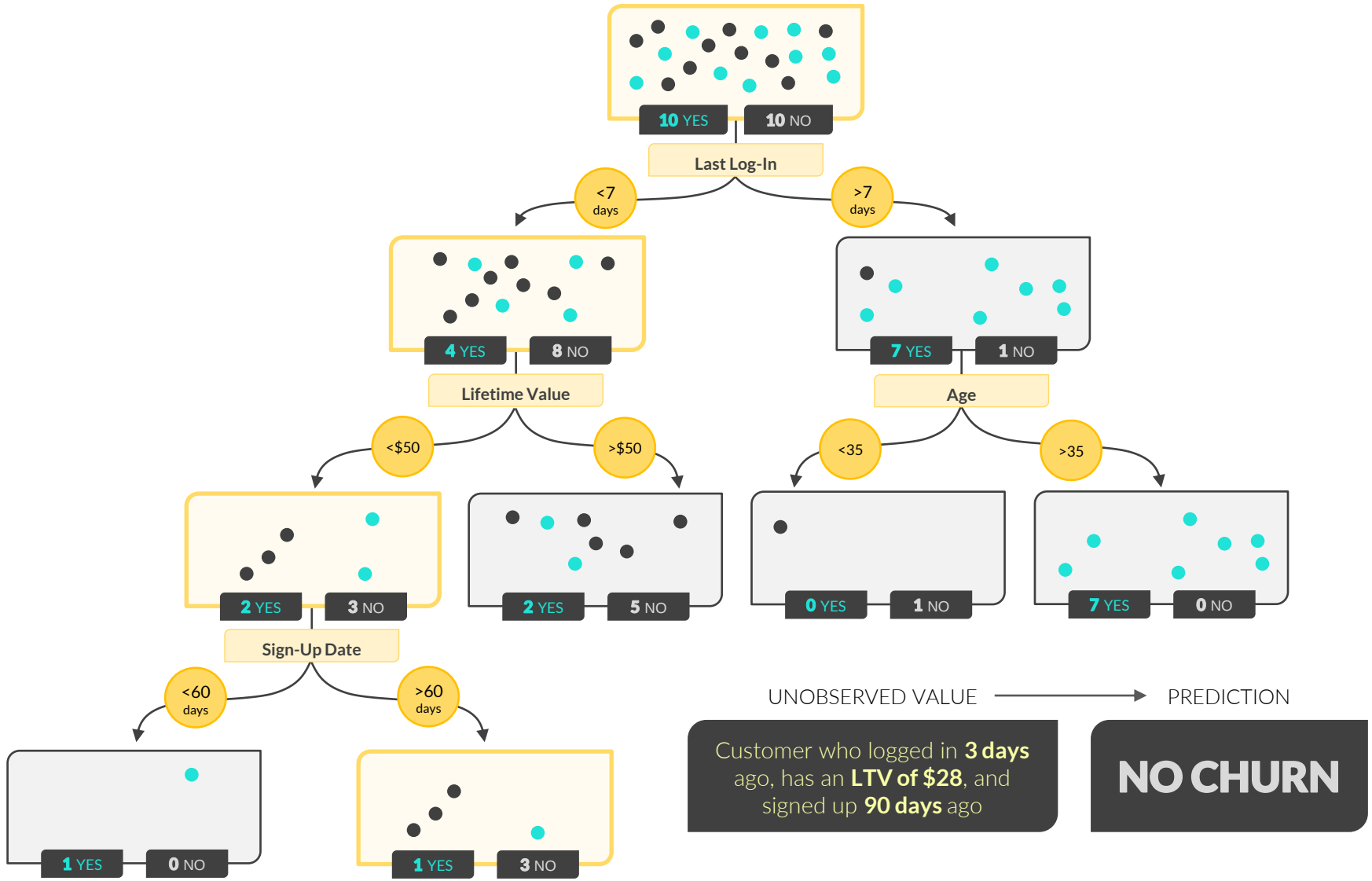
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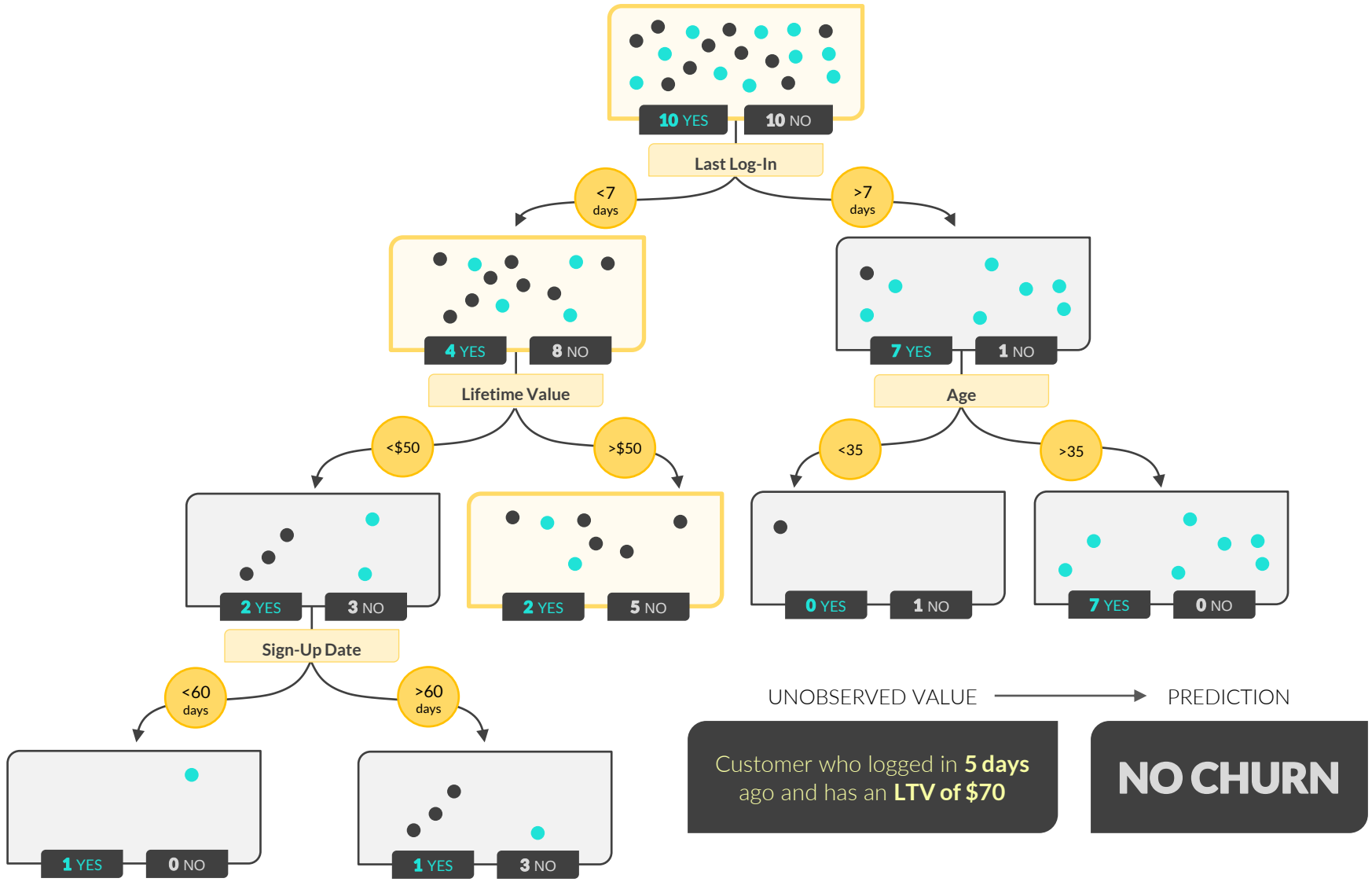
DECISION TREES

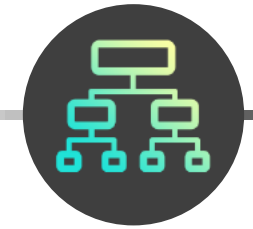
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DECISION TREES

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DECISION TREES

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- Sentiment Analysis



How do we know when to stop splitting?

- Splitting too much can lead to overfitting, so you can adjust model inputs (known as "hyperparameters") to limit things like **tree depth** or **leaf size**



How do we know where to split numerical IVs?

- In practice, decision trees will test different numerical splits and optimize based on information gain (just like testing individual IV splits)



Does the best first split *always* lead to the most accurate model?

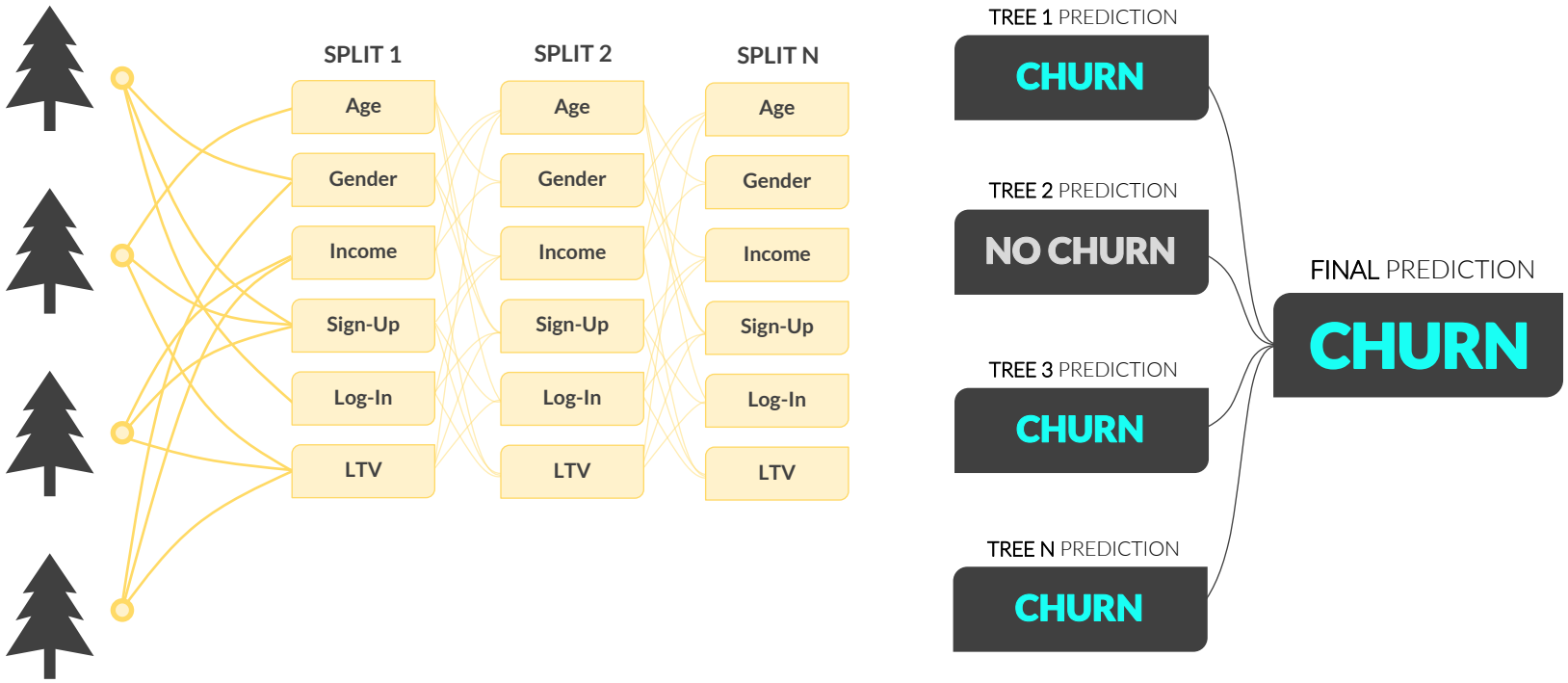
- Not necessarily! That's why we often use a collection of *multiple* decision trees, known as a **random forest**, to maximize accuracy

RANDOM FORESTS

- K-Nearest Neighbors
- Naïve Bayes
- Decision Trees
- **Random Forests**
- Logistic Regression
- Sentiment Analysis

A **Random Forest** is a collection of individual decision trees, each built using a random subset of observations

- Each decision tree randomly selects variables to evaluate at each split
- Each tree produces a prediction, and the **mode**, or most frequent prediction, wins



CASE STUDY: DECISION TREES



THE **SITUATION**

You are the founder and CEO of **Trip Genie**, an online subscription service designed to connect global travelers with local guides.



THE **ASSIGNMENT**

You'd like to better understand your customers and identify which types of behaviors can be used to help predict paid subscriptions.

Your goal is to **build a decision tree** to help you predict subscriptions based on multiple factors (*newsletter*, *Facebook follow*, *time on site*, *sessions*, *etc.*).

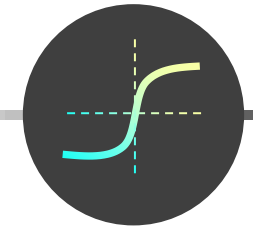


THE **OBJECTIVES**

1. Collect Training data at the customer-level
2. Split on each independent variable and compare changes in entropy
3. Build a decision tree to best predict subscriptions based on available IVs

LOGISTIC REGRESSION

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LOGISTIC REGRESSION

- K-Nearest Neighbors
- Naïve Bayes
- Decision Trees
- Random Forests
- **Logistic Regression**
- Sentiment Analysis

Logistic Regression is a classification technique used to predict the probability of a binary (true/false) outcome

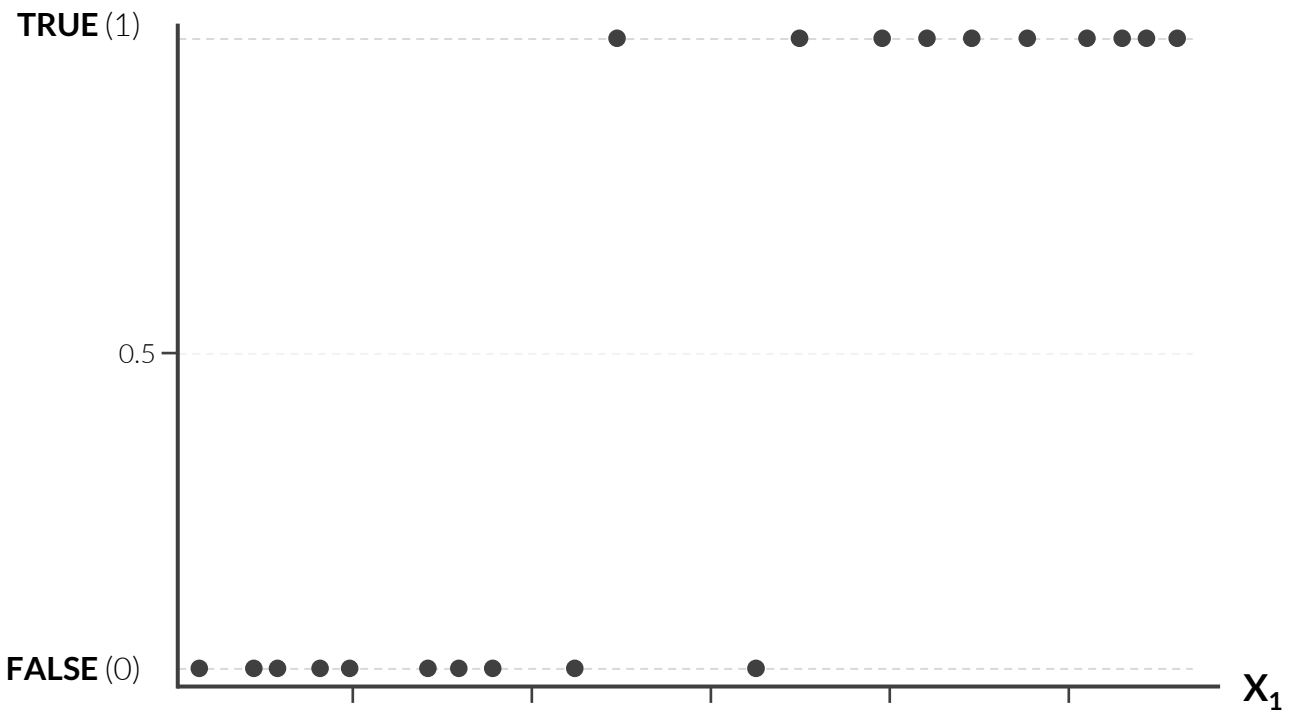
- In its simplest form, logistic regression forms an **S-shaped curve between 0 - 1**, which represents the probability of a TRUE outcome for any given value of X
- The **likelihood function** measures how accurately a model predicts outcomes, and is used to optimize the "shape" of the curve
- Although it has the word *"regression"* in its name, logistic regression is not used for predicting numeric variables

Example use cases:

- Flagging spam emails or fraudulent credit card transactions
- Determining whether to serve a particular ad to a website visitor

LOGISTIC REGRESSION

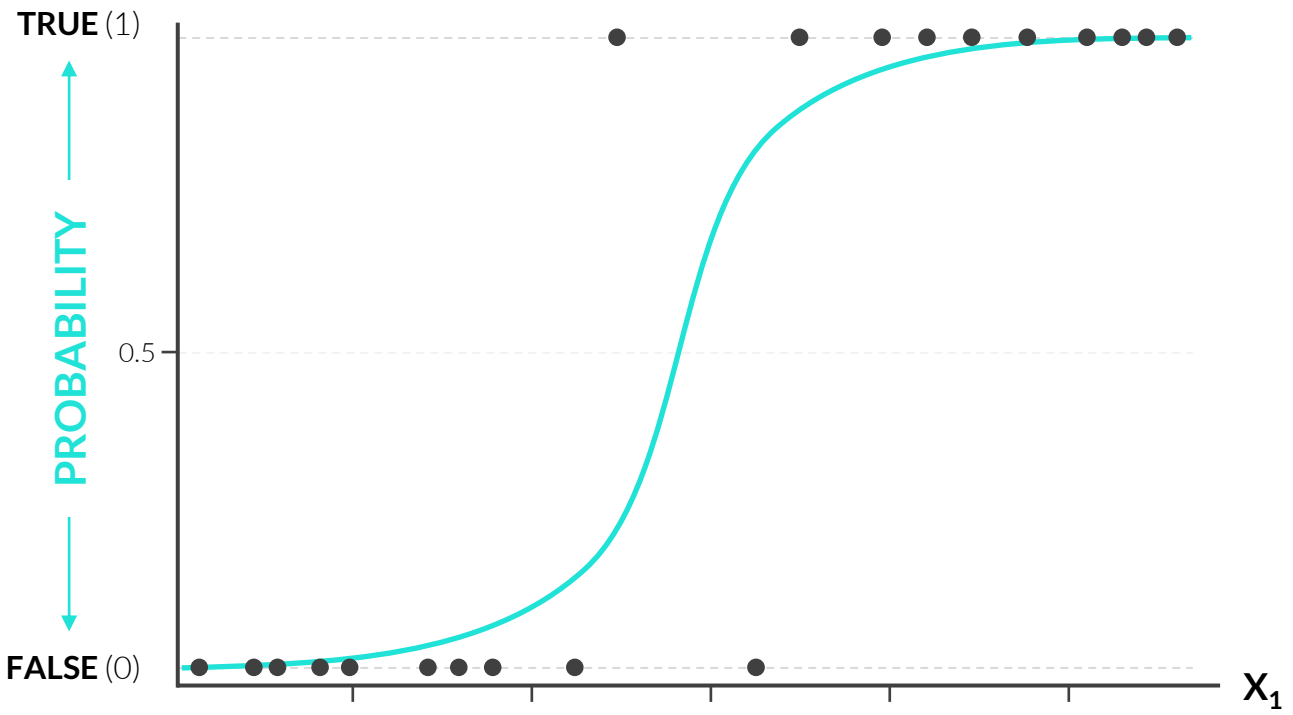
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- Each dot represents an observed value, where **X is a numerical independent variable** and **Y is the binary outcome** *(true/false)* that we want to predict

LOGISTIC REGRESSION

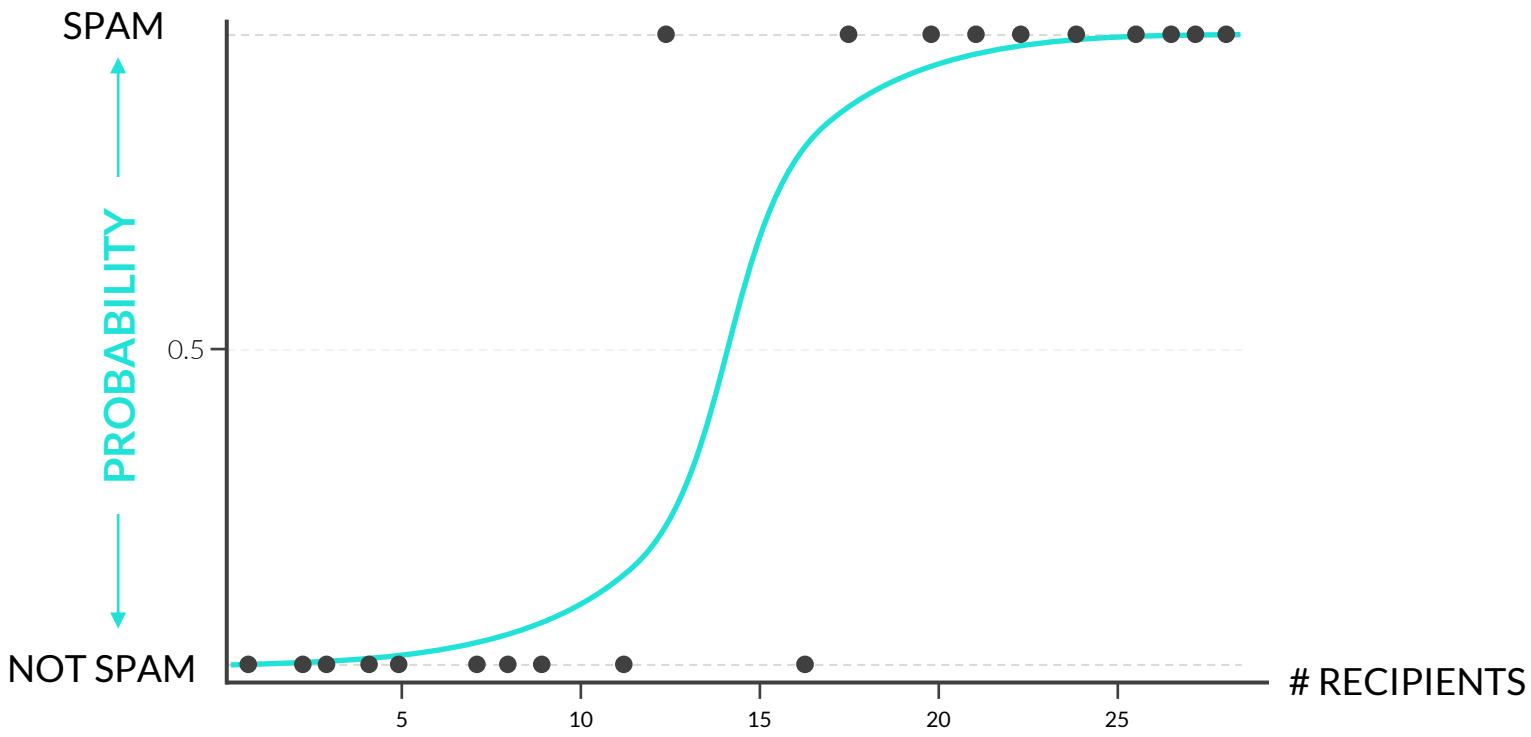
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- Logistic regression plots the **best-fitting curve between 0 and 1**, which tells us the probability of Y being TRUE for any given value of X_1

LOGISTIC REGRESSION

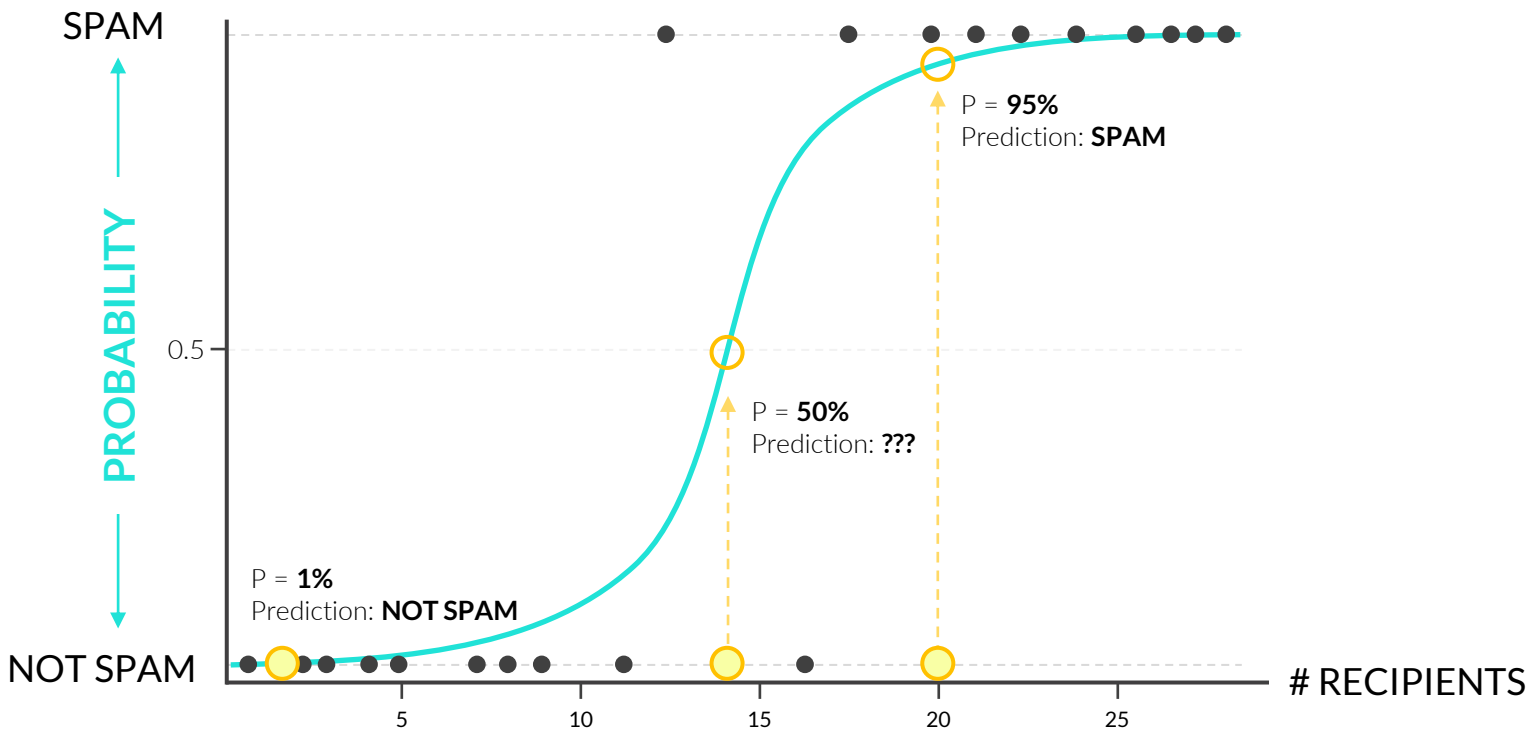
- K-Nearest Neighbors
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- Here we're using logistic regression to **predict if an email will be marked as spam**, based on the number of email recipients (X_1)

LOGISTIC REGRESSION

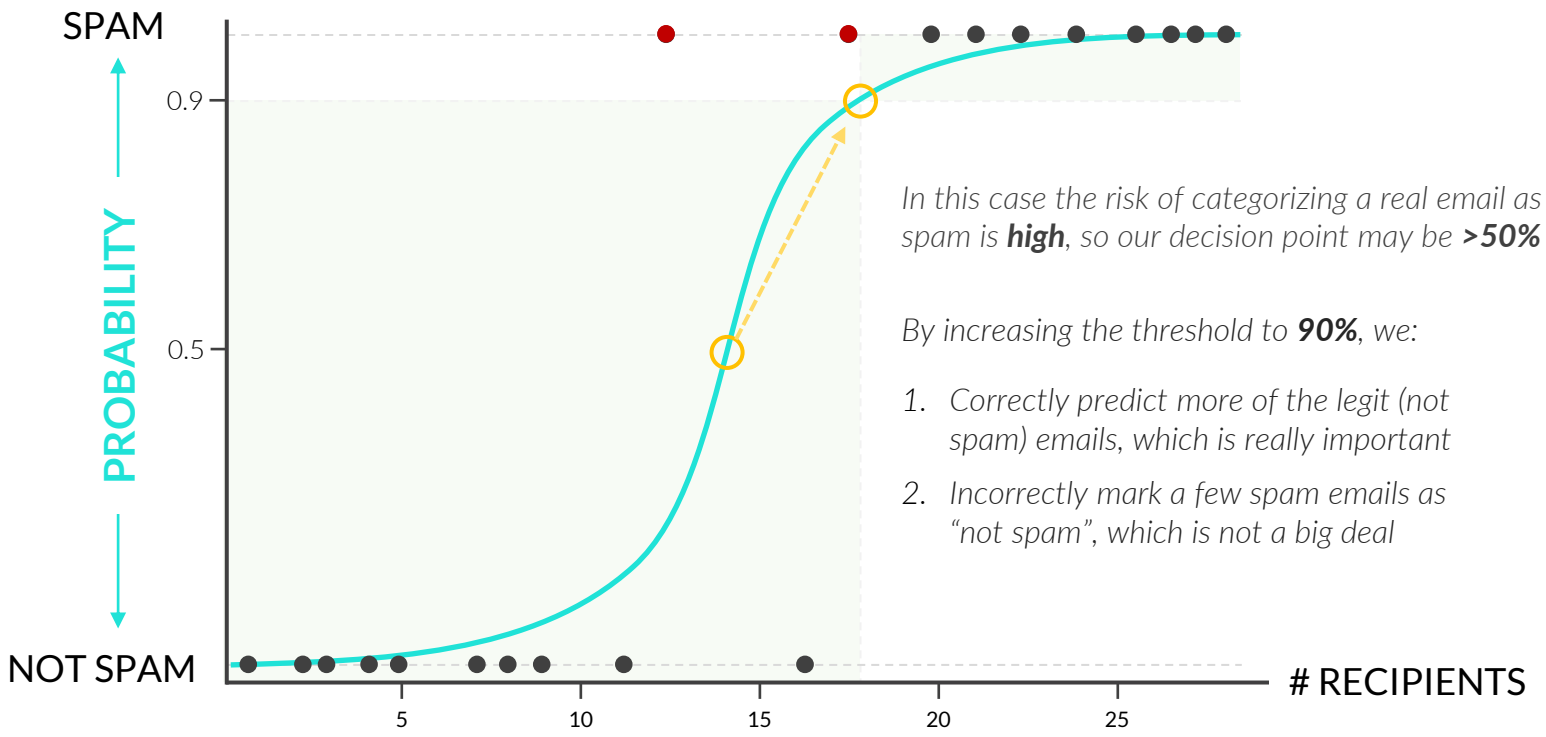
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- Using this model, we can test unobserved values of $\mathbf{X_1}$ to predict the probability that **Y** is true or false *(in this case the probability that an email is marked as spam)*

LOGISTIC REGRESSION

- K-Nearest Neighbors
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Is **50%** *always* the right decision point for logistic models?

- **No.** It depends on the relative risk of a **false positive** *(incorrectly predicting a TRUE outcome)* or **false negative** *(incorrectly predicting a FALSE outcome)*

LOGISTIC REGRESSION

- K-Nearest Neighbors
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- **Logistic Regression**
- Sentiment Analysis



- Now consider a case where we're **predicting if an employee will quit** based on negative feedback from HR
- It's easier to train an employee than hire a new one, so the risk of a false positive is **low** but the risk of a false negative *(incorrectly predicting someone will stay)* is **high**

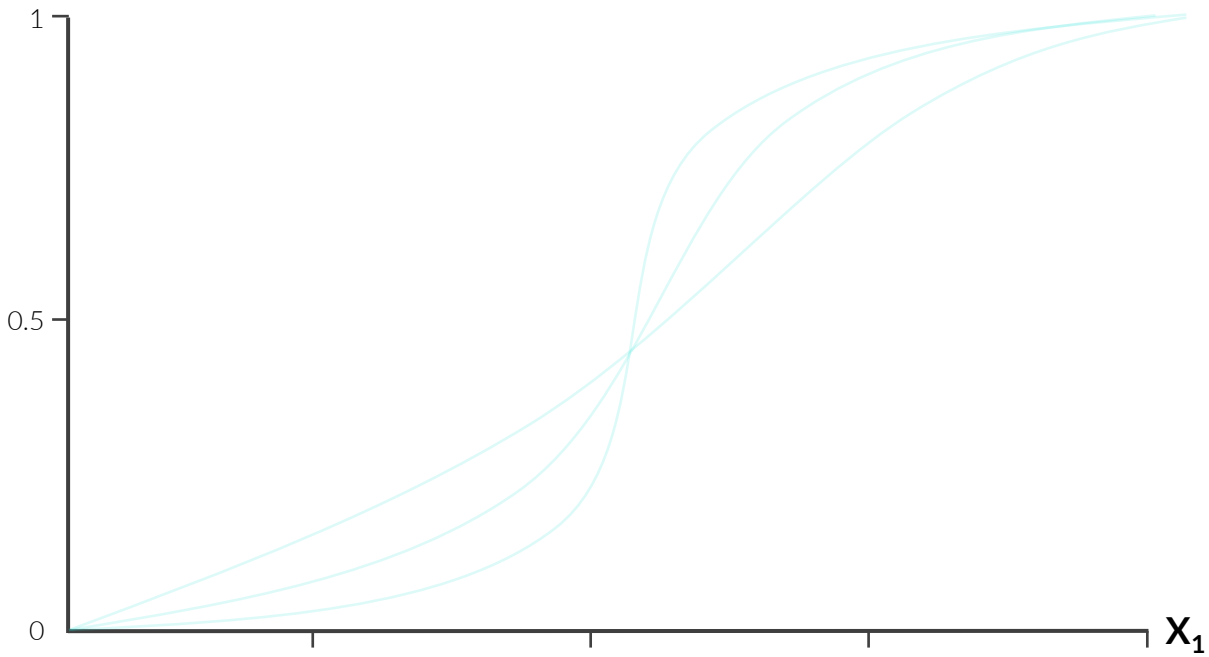
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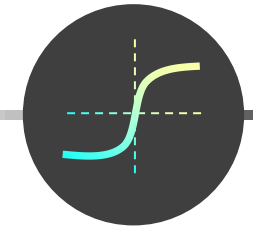
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Makes the output fall between **0** and **1**

$$\frac{1}{1 + e^{-(\beta_0 + \beta_1 x_1)}}$$

Linear equation, where $\boldsymbol{\beta_0}$ is the intercept, **X** is the IV value and $\boldsymbol{\beta_1}$ is the weight *(or slope)*





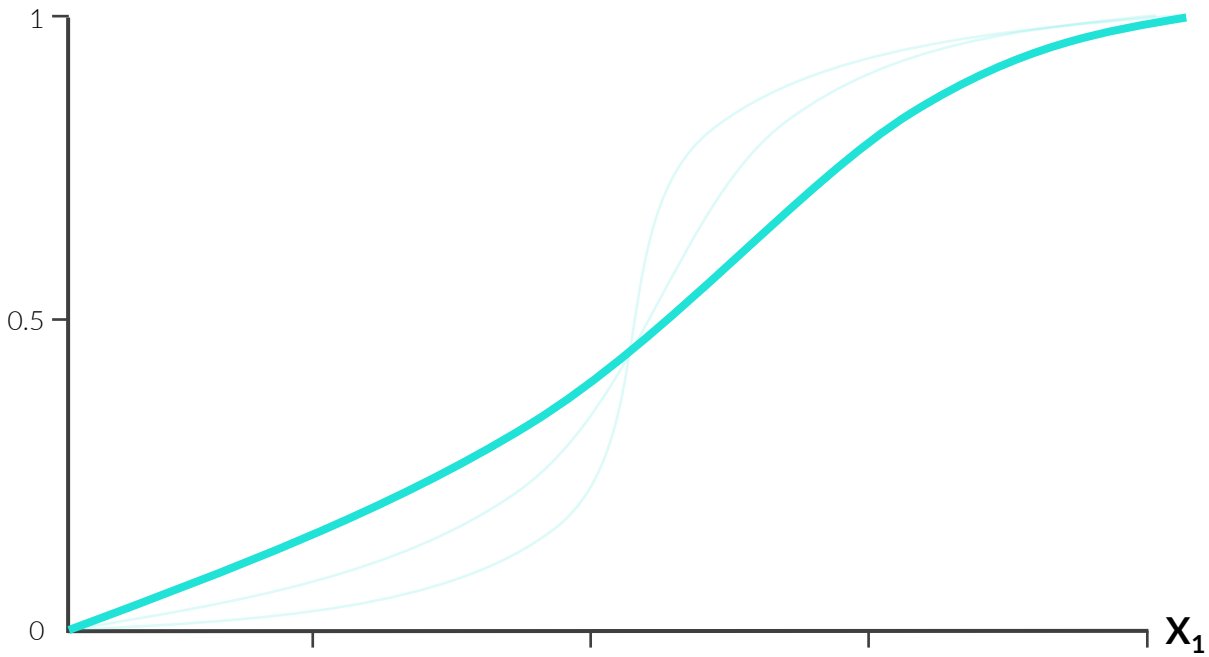
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$\boldsymbol{\beta_1}$ = 1.0

$\boldsymbol{\beta_1}$ = 5.0

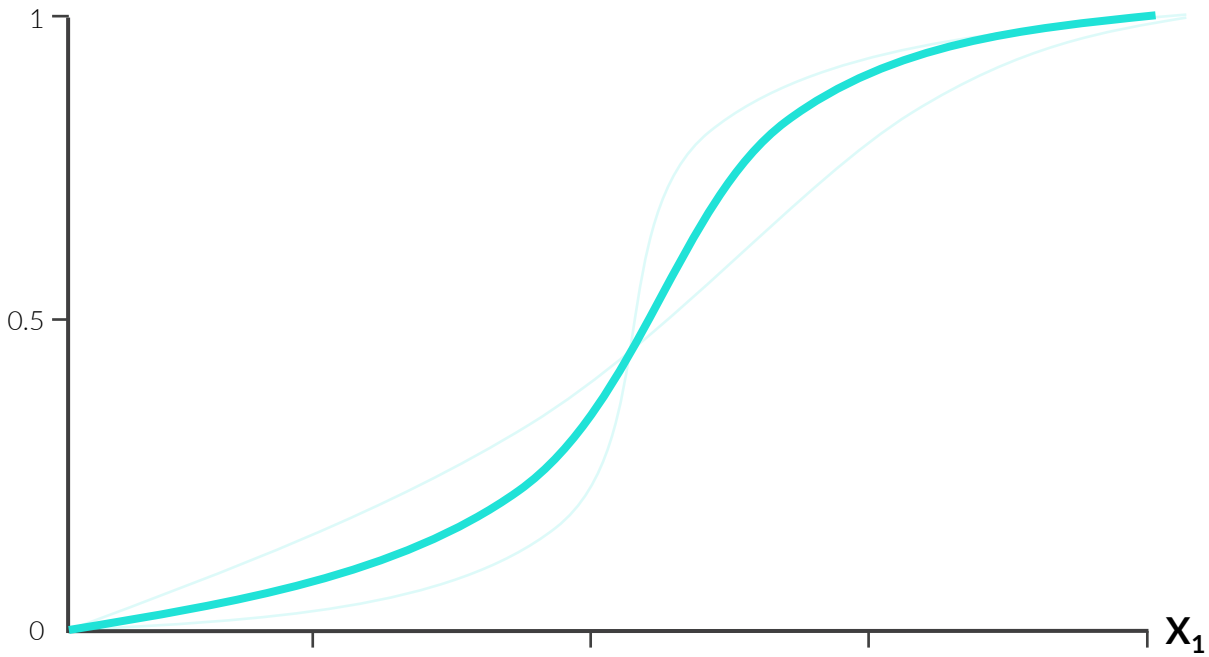
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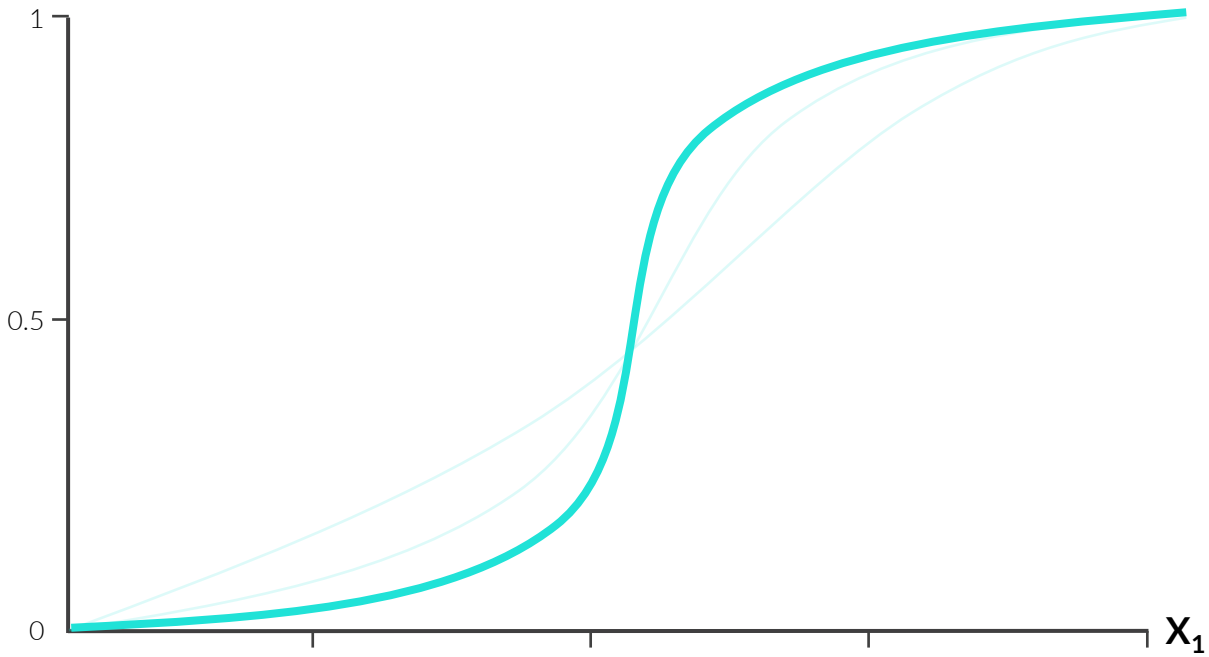
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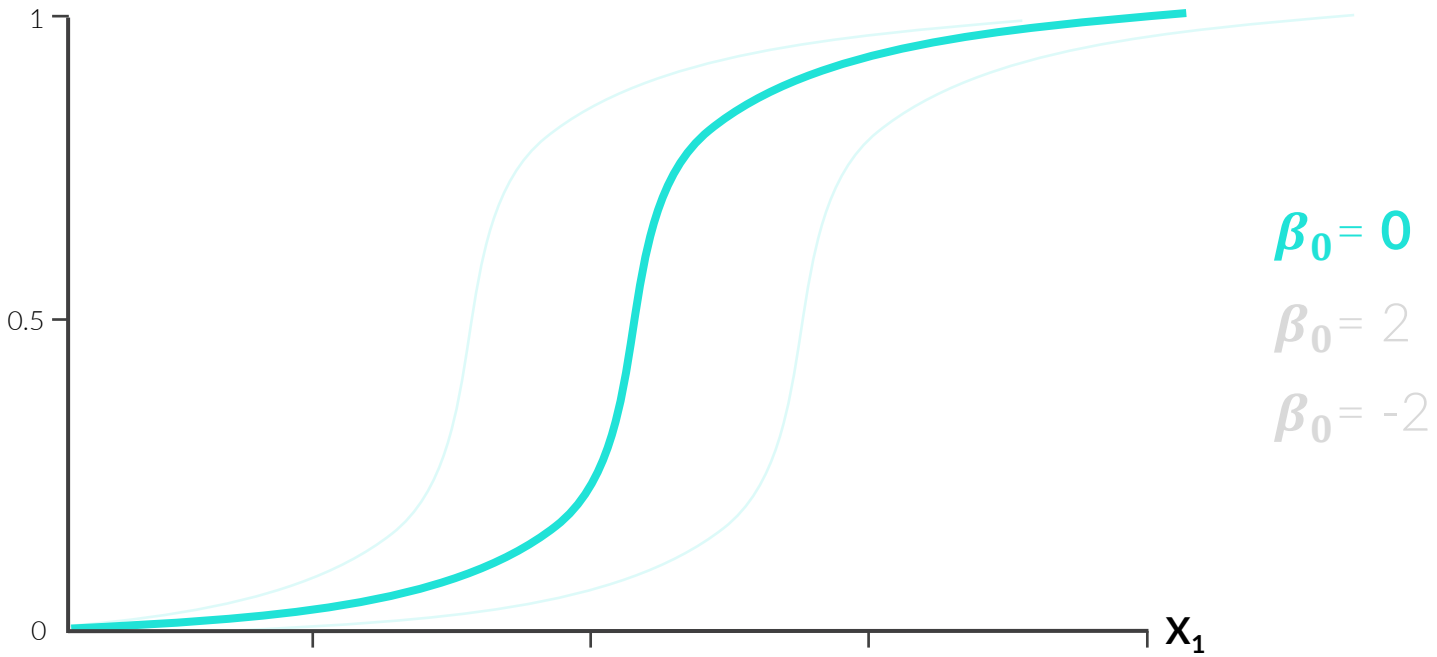
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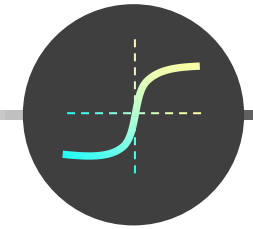
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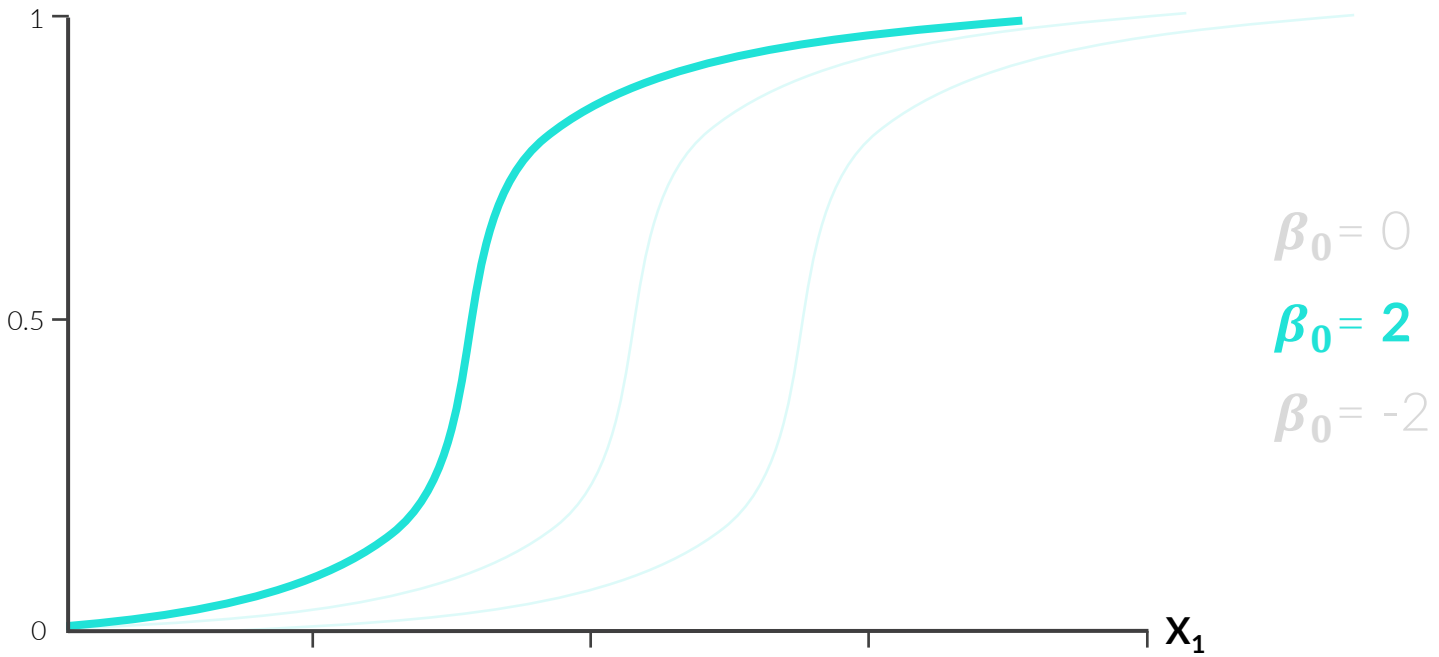
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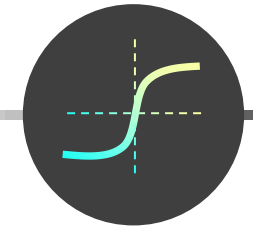
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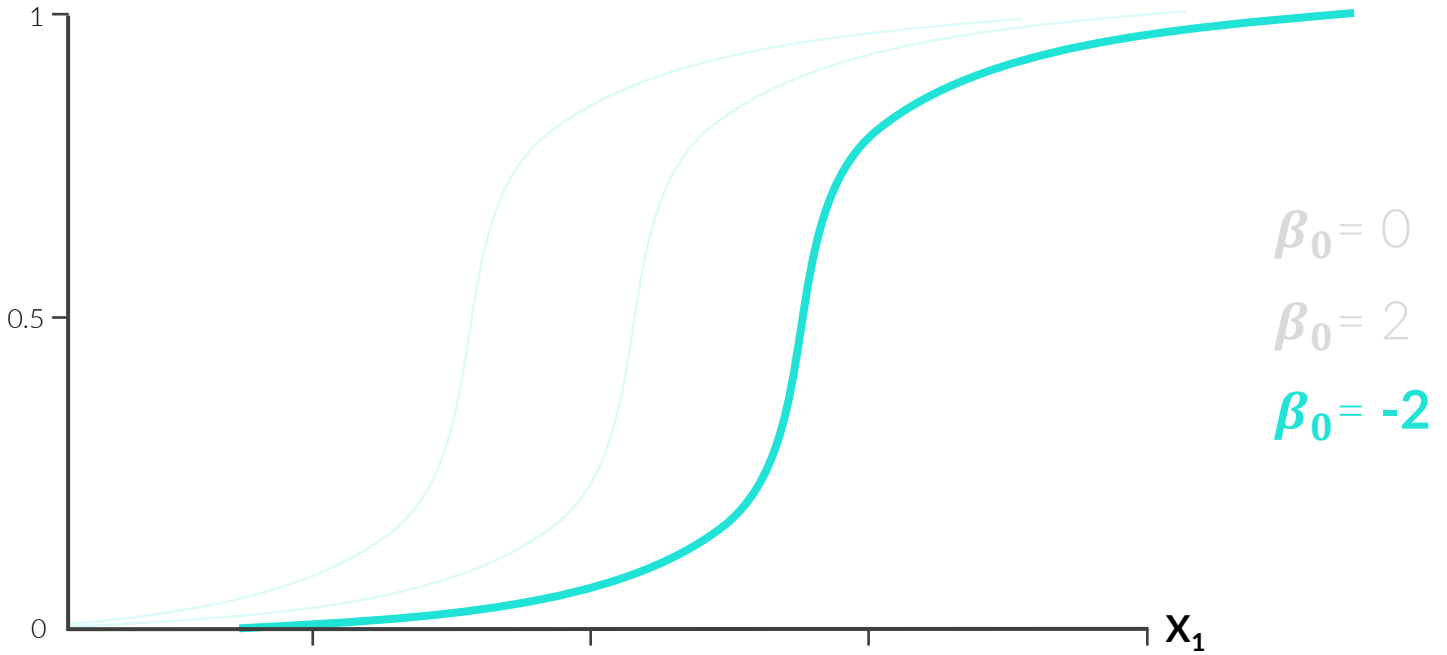
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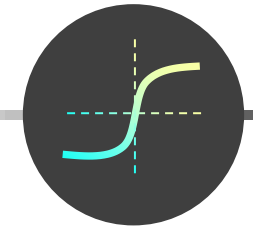
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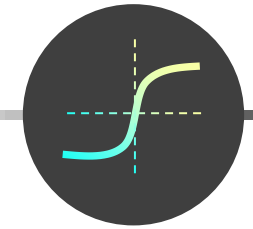
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How do we determine the "right" values for β? **LIKELIHOOD!**

- **Likelihood** is a metric that tells us how good our model is at correctly predicting Y, based on the shape of the curve
- This is where **machine learning** comes in; instead of human trial-and-error, an algorithm determines the best weights to maximize likelihood using observed values

		Actual Observation: Y = **1**	Actual Observation: Y = **0**
Model Output	~**1**	HIGH	LOW
	~**0**	LOW	HIGH

- *When our model output is close to the actual Y, we want likelihood to be **HIGH** (near **1**)*
- *When our model output is far from the actual Y, we want likelihood to be **LOW** (near **0**)*



LOGISTIC REGRESSION

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Model Output	~**1**	HIGH	LOW
Model Output	~**0**	LOW	HIGH

LIKELIHOOD FUNCTION:

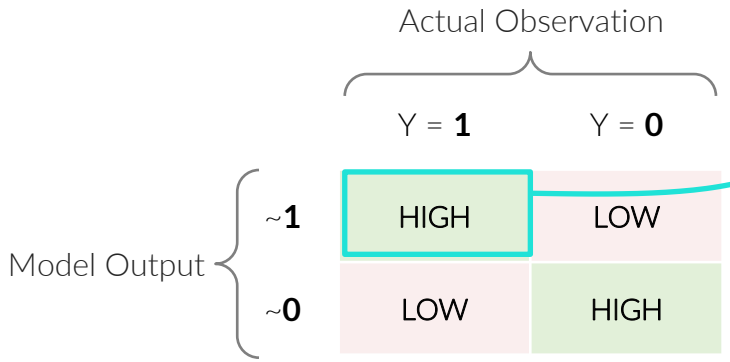
$$\mathcal{L} = (output)^{y} * (1 - output)^{1-y}$$

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$$(output)^{y} * (1 - output)^{1-y}$$

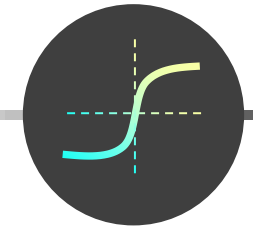
$$(.99)^{1} * (1 - .99)^{1-1}$$

$$(.99)^{1} * (.01)^{0}$$

$$.99 * 1$$

=

.99



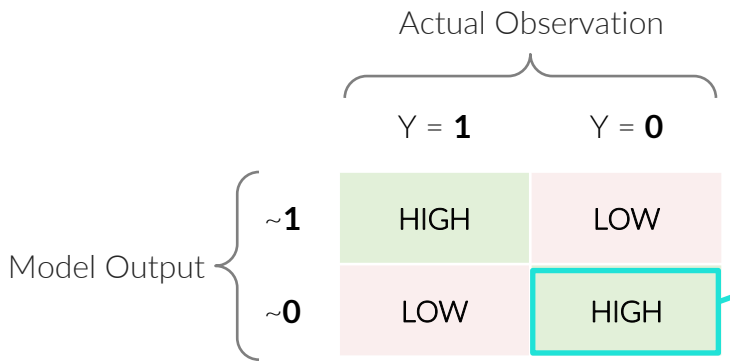
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$$(.01)^{0} * (1 - .01)^{1-0}$$

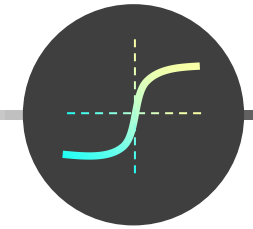
$$1 * (.99)^{1}$$

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=

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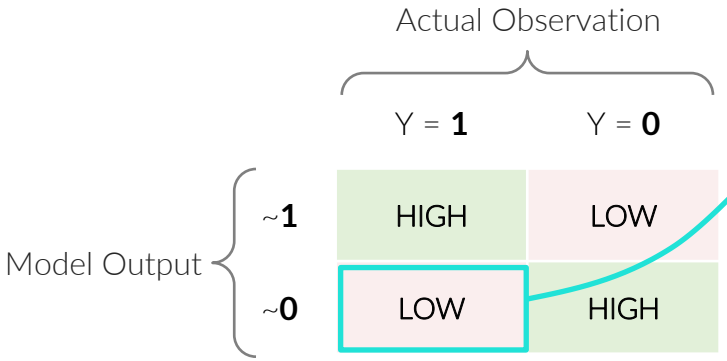


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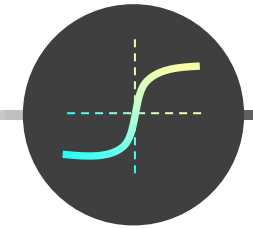
$$(.01)^{1} * (1 - .01)^{1-1}$$

$$.01 * (.99)^{0}$$

$$.01 * 1$$

=

.01



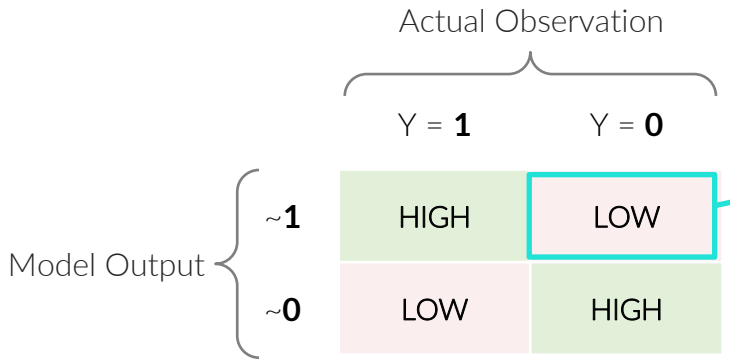
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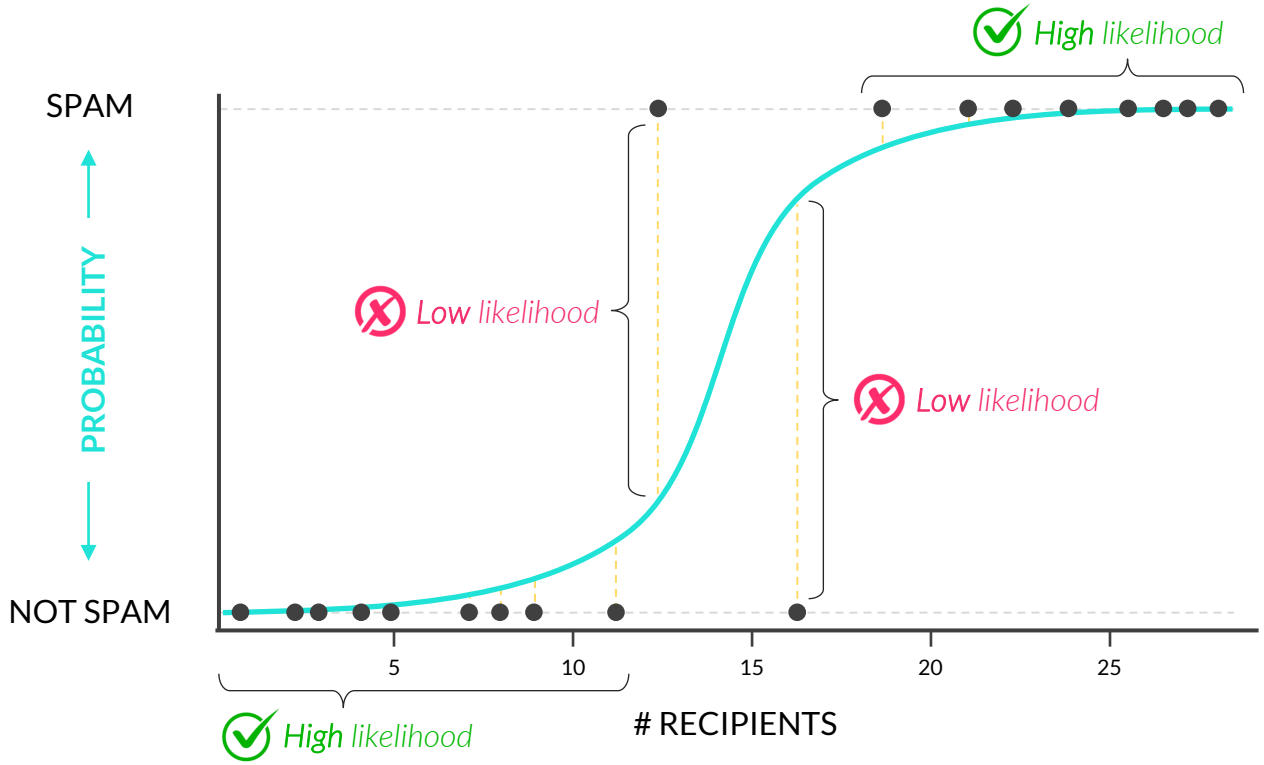
$$=$$

.01



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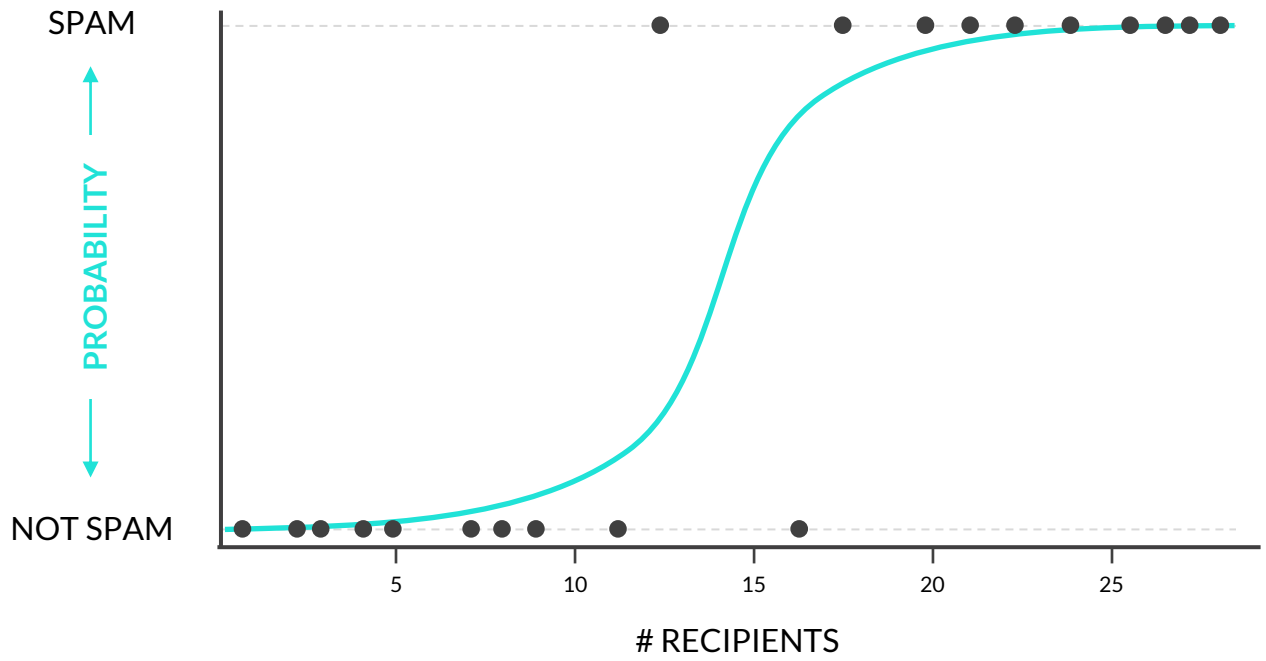


- Observations closest to the curve have the **highest likelihood values** *(and vice versa)*, so maximizing total likelihood allows us to find the curve that fits our data best

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But what if *multiple* X variables could help predict Y?

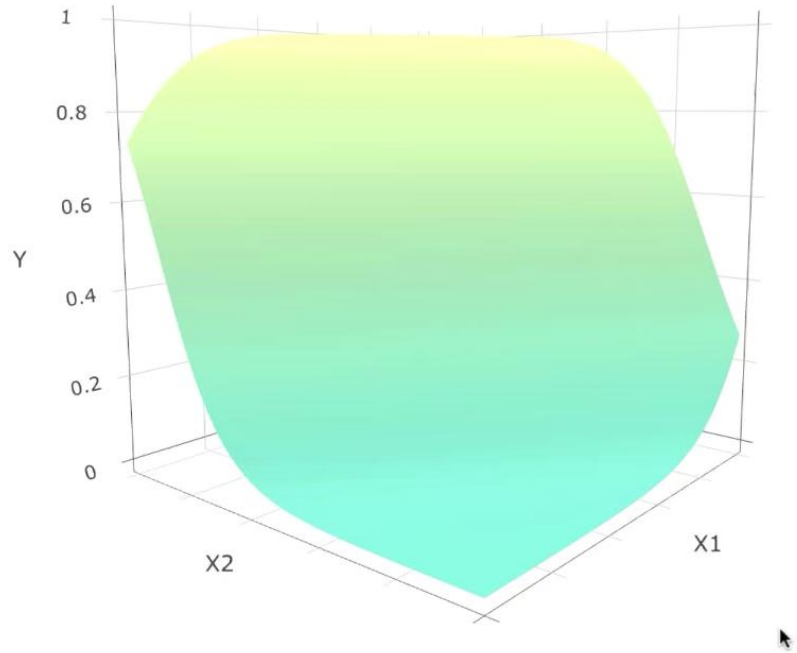


- **# of Recipients** can help us detect spam, but so can other variables like the **number of typos**, **count of words like "free" or "bonus"**, **sender reputation score**, etc.

LOGISTIC REGRESSION

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But what if ***multiple* X variables** could help predict Y?



- Logistic regression can handle multiple independent variables, but the visual interpretation breaks down at >2 IV's *(this is why we need **machine learning**!)*

LOGISTIC REGRESSION

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But what if ***multiple* X variables** could help predict Y?

Makes the output fall between **0** and **1**

$$\frac{1}{1 + e^{-(\beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_n x_n)}}$$

Weighted independent variables (x_1, $x_2 \ldots x_n$)

- Logistic regression is about finding the **best combination of weights** (β_1, $\beta_2 \ldots \beta_n$) for a given set of independent variables (x_1, $x_2 \ldots x_n$) to maximize the likelihood function

CASE STUDY: LOGISTIC REGRESSION



THE **SITUATION**

You've just been promoted to Marketing Manager for **Lux Dining**, a wildly popular international food blog.



THE **ASSIGNMENT**

The CMO is concerned about unsubscribe rates and thinks it may be related to the frequency of emails your team has been sending.

Your job is to use **logistic regression** to plot this relationship and predict if a user will unsubscribe based on the number of weekly emails received.



THE **OBJECTIVES**

1. Collect customer data containing email frequency and subscription activity
2. Plot the logistic regression curve which maximizes likelihood
3. Determine the appropriate email frequency *(based on a 50% decision point)*

SENTIMENT ANALYSIS

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SENTIMENT ANALYSIS

- K-Nearest Neighbors
- Naïve Bayes
- Decision Trees
- Random Forests
- Logistic Regression
- **Sentiment Analysis**

Sentiment analysis is a technique used to determine the emotional tone or "sentiment" behind text-based data

- Sentiment analysis often falls under **Natural Language Processing** (NLP), but is typically applied as a classification technique
- Sentiment models typically use a "**bag of words**" approach, which involves calculating the frequency or presence (1/0) of key words to convert text *(which models struggle with)* into numerical inputs
- Unlike other classification models, you must "hand-score" the sentiment (DV) values for your Training data, which your model will learn from

Example use cases:

- Understanding the tone of product reviews posted by customers
- Analyzing open-ended survey responses

SENTIMENT ANALYSIS

- K-Nearest Neighbors
- Naïve Bayes
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- **Sentiment Analysis**



This **word cloud** is *technically* a very simple version of sentiment analysis

Limitations:

- Based entirely on word count
- Straight-forward but not flexible
- Requires manual interpretation
- Subject to inconsistency/human error



SENTIMENT ANALYSIS

- K-Nearest Neighbors
- Naïve Bayes
- Decision Trees
- Random Forests
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- **Sentiment Analysis**

The first step in any sentiment analysis is to **clean** and **QA** the text to remove noise and isolate the most meaningful information:

- Remove **punctuation**, **capitalization** and **special characters**
- Correct **spelling** and **grammatical errors**
- Use proper **encoding** *(i.e. UTF-8)*
- **Lemmatize** or **stem** *(remove grammar tense, convert to "root" term)*
- Remove **stop words** *("a", "the", "or", "of", "are", etc.)*

The computer is running hot because I'm mining bitcoin!

↓

~~the~~ computer ~~is~~ run~~ning~~ hot ~~because~~ ~~I'm~~ min~~ing~~ bitcoin~~!~~

↓

computer run hot mine bitcoin

NOTE: This process can vary based on the context; for example, you may want to preserve capitalization or punctuation if you care about measuring intensity *(i.e. "GREAT!!" vs. "great")*, or choose to allow specific stop words or special characters



SENTIMENT ANALYSIS

- K-Nearest Neighbors
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- **Sentiment Analysis**

Once the text has been cleaned, we can transform our text into numeric data using a "**bag of words**" approach:

- Split cleaned text into individual words *(this is known as **tokenization**)*
- Create a new column with a **binary flag (1/0)** for each word
- Manually **assign sentiment** for observations in your Training data
- Apply any **classification technique** to predict sentiment for unobserved text

*Each word is an **independent variable***

*Sentiment is our **dependent variable***

	hate	garbage	love	book	**sentiment**	
I HATE this garbage!	1	1	0	0	negative	*Observed values*
I love this book.	0	0	1	1	positive	
This book is garbage	0	1	0	1	**???**	→ *Unobserved value*



SENTIMENT ANALYSIS

- K-Nearest Neighbors
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- **Sentiment Analysis**



How do you address **language nuances** like ambiguity, double negatives, slang or sarcasm?

- Generally speaking, the more observations you score in your Training data, the better your model will be at detecting these types of nuances
- No sentiment model is perfect, but feature engineering and advanced techniques can help improve accuracy for more complex cases

"If you like watching paint dry, you'll love this movie!" → *~~If you~~ like watch~~ing~~ paint dry ~~you'll~~ love ~~this~~ movie* → ***like watch paint dry love movie***

"The new version is awfully good, not as bad as expected!" → *~~The~~ new version ~~is~~ awful~~ly~~ good ~~not~~ ~~as~~ bad ~~as~~ expect~~ed~~* → ***new version awful good bad expect***

CASE STUDY: SENTIMENT ANALYSIS



THE **SITUATION**

You're an accomplished author and creator of the hit series **Bark Twain the Data Dog**, featuring a feisty chihuahua who uses machine learning to solve crimes.



THE **ASSIGNMENT**

Reviews for the latest book in the Bark Twain series are coming in, and they aren't looking great...

To apply a bit more rigor to your analysis and automate scoring for future reviews, you've decided to use this feedback to build a basic **sentiment model**.



THE **OBJECTIVES**

1. Collect raw text reviews and tokenize key words
2. Hand-score the sentiment for observed values in the Training data
3. Apply any classification technique and validate using Test data

MODEL SELECTION & TUNING

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MODEL SELECTION & TUNING



In this section we'll discuss techniques for **selecting & tuning** classification models, including hyperparameter optimization, class balancing, confusion matrices and more

TOPICS WE'LL COVER:

- Hyperparameters
- Imbalanced Classes
- Confusion Matrix
- Selection & Drift

COMMON USE CASES:

- *Adjusting model parameters to optimize performance*
- *Rebalancing classes to reduce bias*
- *Comparing performance across multiple classification models to select the best performing option*
- *Retraining models with fresh data to minimize drift*

HYPERPARAMETERS

- **Hyperparameters**
- Imbalanced Classes
- Confusion Matrix
- Model Selection
- Model Drift

Hyperparameters are inputs/settings you can control while training a model

- Adjusting parameters to maximize accuracy is known as *"hyperparameter optimization"*



K-Nearest Neighbors

Examples:

- *K*
- *Distance definition*
- *NN Algorithm*



Naïve Bayes

Examples:

- *Smoothing parameters*



Decision Trees

Examples:

- *Criterion (entropy, gini)*
- *Tree depth*
- *Leaf size*
- *Max Features*
- *Min gain*



Random Forests

Examples:

- *# Trees*
- *# Samples*
- *# Features*
- *Re-Sampling*



Logistic Regression

Examples:

- *Intercept*
- *Penalty Term*
- *Solver*

IMBALANCED CLASSES

Hyperparameters

Imbalanced Classes

Confusion Matrix

Model Selection

Model Drift

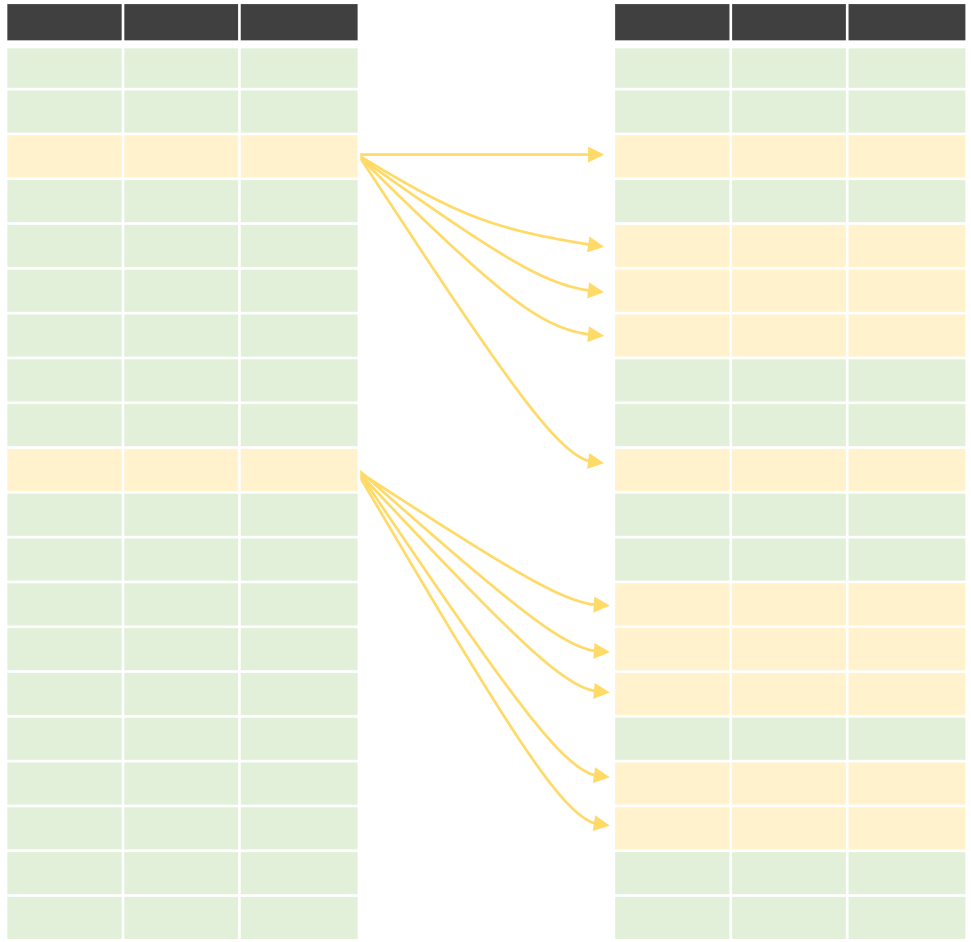
Imbalanced classes occur when you are predicting outcomes with *drastically* different frequencies in the Training data

- The class or outcome which occurs more frequently is known as the **majority class**, while the class which occurs less frequently is the **minority class**
- Imbalanced classes can bias a model towards *always* predicting the majority class, since it often yields the best overall accuracy *(i.e. 99%+)*
- This is a significant issue when predicting very rare and very *important* events, like a nuclear meltdown

There are several ways to balance classes in your Training data, including **up-sampling**, **down-sampling** and **weighting**

IMBALANCED CLASSES

- Hyperparameters
- **Imbalanced Classes**
- Confusion Matrix
- Model Selection
- Model Drift



Up-sampling

Minority class observations are duplicated to balance the data

Down-sampling

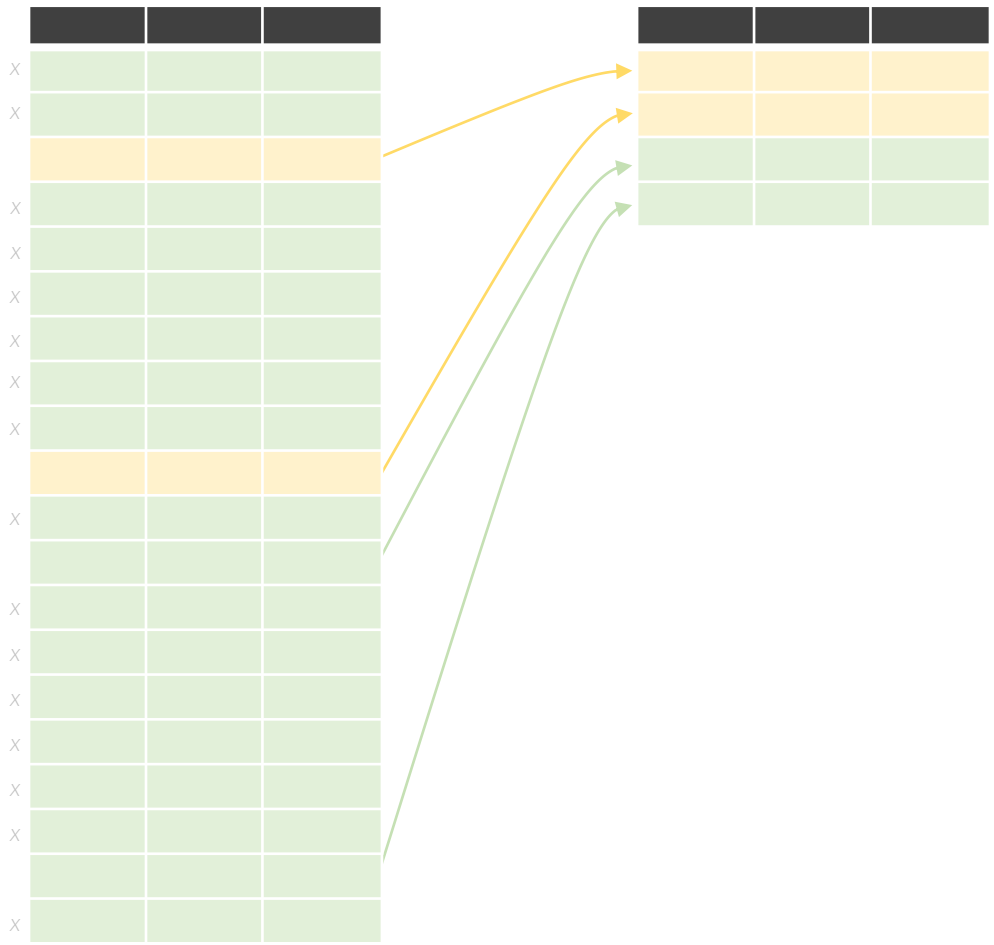
Majority class observations are randomly removed to balance the data

Weighting

For models that randomly sample observations (random forests), increase the probability of selecting the minority class

IMBALANCED CLASSES

- Hyperparameters
- **Imbalanced Classes**
- Confusion Matrix
- Model Selection
- Model Drift



Up-sampling

Minority class observations are duplicated to balance the data

Down-sampling

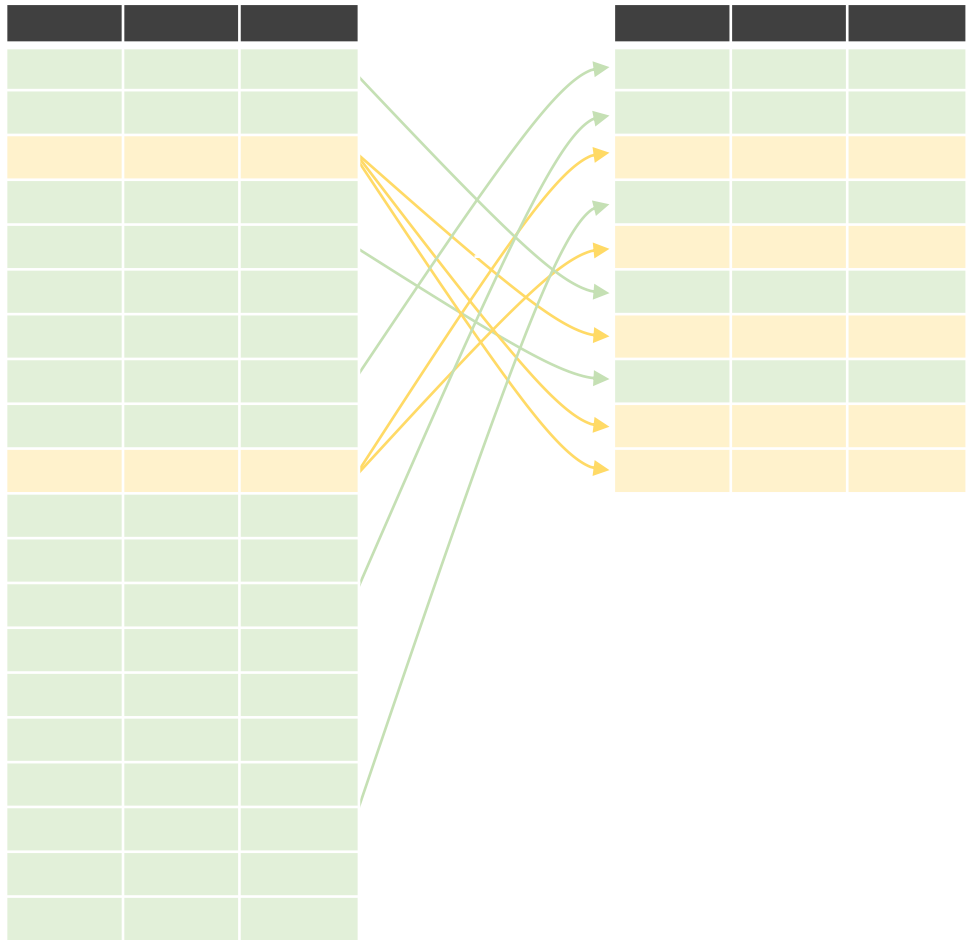
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Weighting

For models that randomly sample observations (random forests), increase the probability of selecting the minority class

IMBALANCED CLASSES

- Hyperparameters
- **Imbalanced Classes**
- Confusion Matrix
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- Model Drift



Up-sampling

Minority class observations are duplicated to balance the data

Down-sampling

Majority class observations are randomly removed to balance the data

Weighting

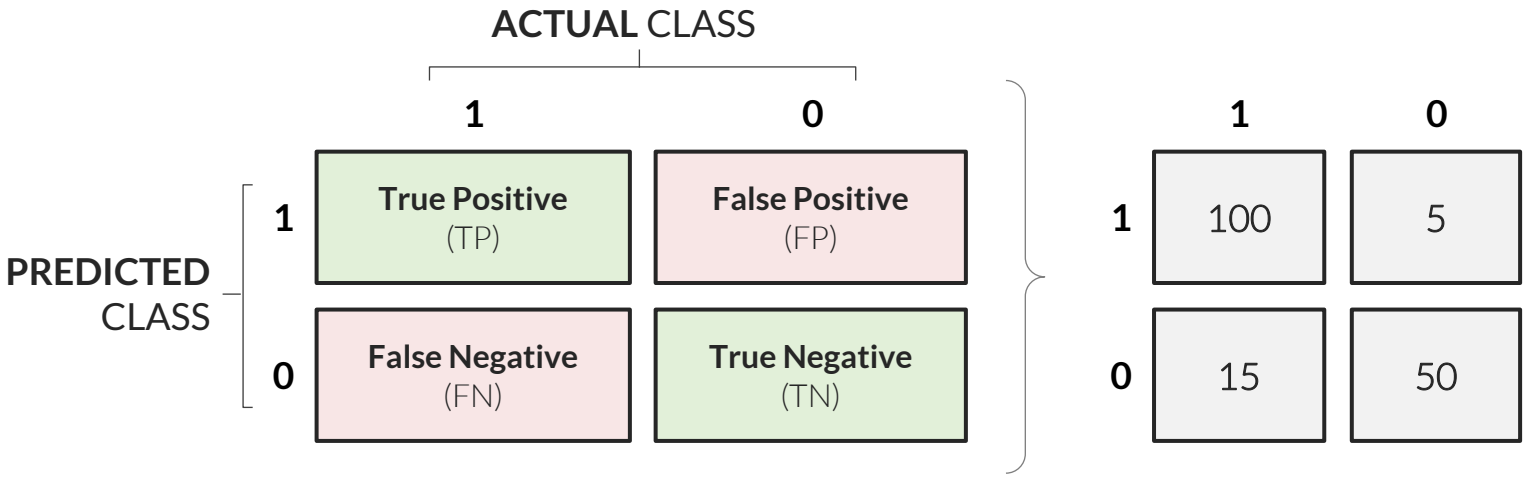
For models that randomly sample observations (i.e. random forests), increase the probability of selecting the minority class

CONFUSION MATRIX

- Hyperparameters
- Imbalanced Classes
- **Confusion Matrix**
- Model Selection
- Model Drift

A **Confusion Matrix** is a table summarizing the frequency of *predicted* vs. *actual* classes in your Test data

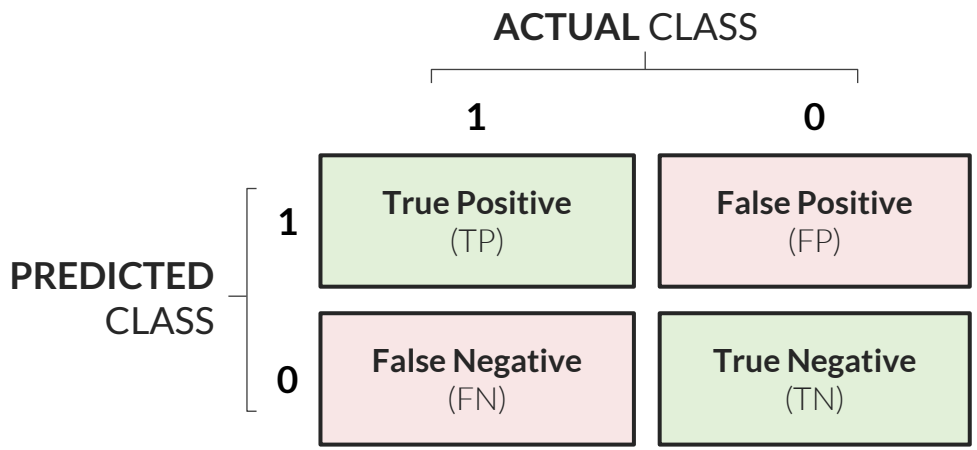
- This is the most common and concise way to evaluate performance and compare classification models against one another
- Confusion matrices can be used to derive several types of model performance metrics, including **accuracy**, **precision** and **recall**



CONFUSION MATRIX

- Hyperparameters
- Imbalanced Classes
- **Confusion Matrix**
- Model Selection
- Model Drift

Confusion matrices are used to derive model **performance metrics**

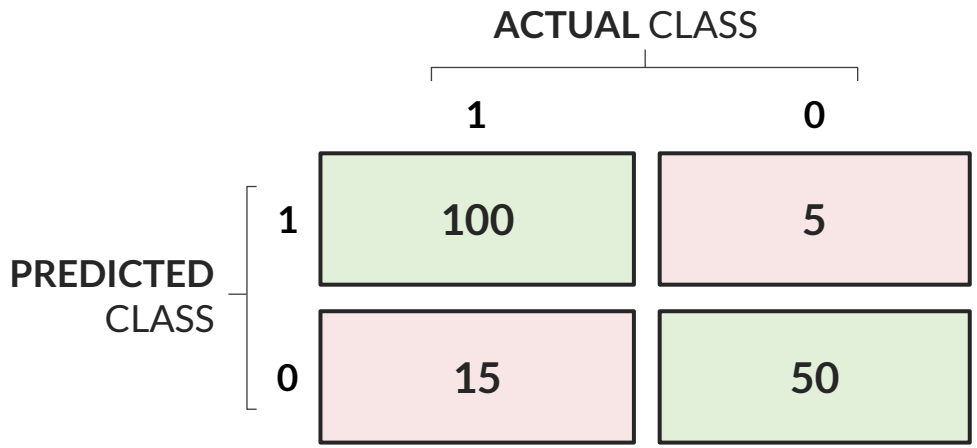


Accuracy	Precision	Recall
(TP+TN) / (TP+TN+FP+FN)	TP / (TP+FP)	TP / (TP+FN)
*Of **all predictions**, what % were correct?*	*Of all **predicted positives**, what % were correct?*	*Of all **actual positives**, what % were predicted correctly?*

CONFUSION MATRIX

- Hyperparameters
- Imbalanced Classes
- **Confusion Matrix**
- Model Selection
- Model Drift

Confusion matrices are used to derive model **performance metrics**



Accuracy

(TP+TN) / (TP+TN+FP+FN)
=
(100+50)/(100+50+5+15)
=
.88

Precision

TP / (TP+FP)
=
100/(100+5)
=
.95

Recall

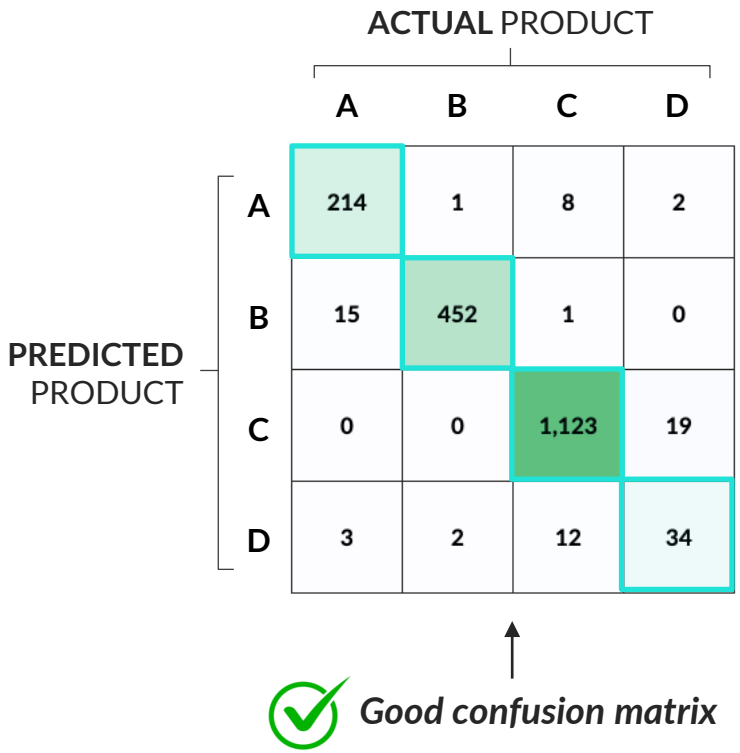
TP / (TP+FN)
=
100/(100+15)
=
.87

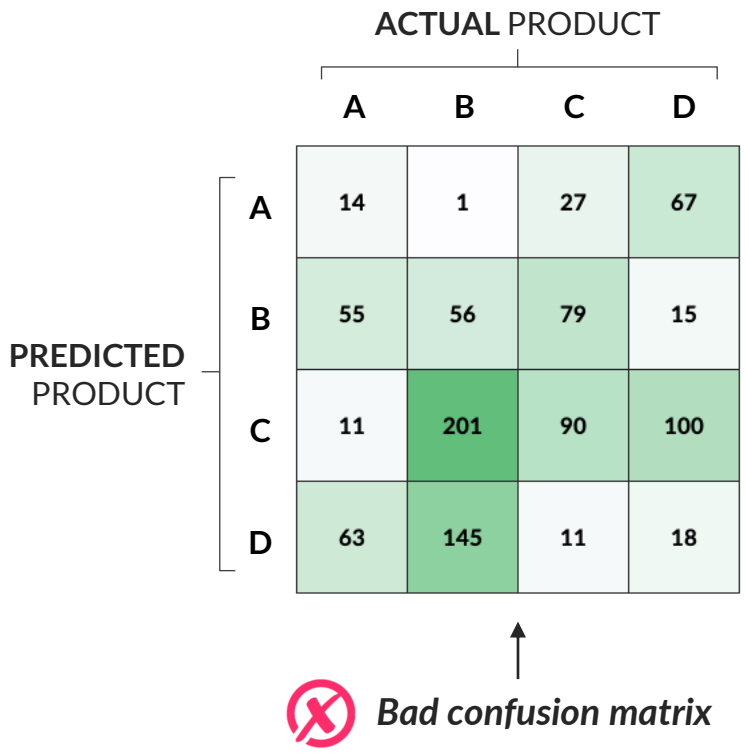
CONFUSION MATRIX

- Hyperparameters
- Imbalanced Classes
- **Confusion Matrix**
- Model Selection
- Model Drift

Will a **Confusion Matrix** work for *multi-class* predictions too?

- **Yes!** Consider a model classifying multiple products (*A*, *B*, *C*, *D*); ideally, we want the highest frequencies along the top-left to bottom-right diagonal

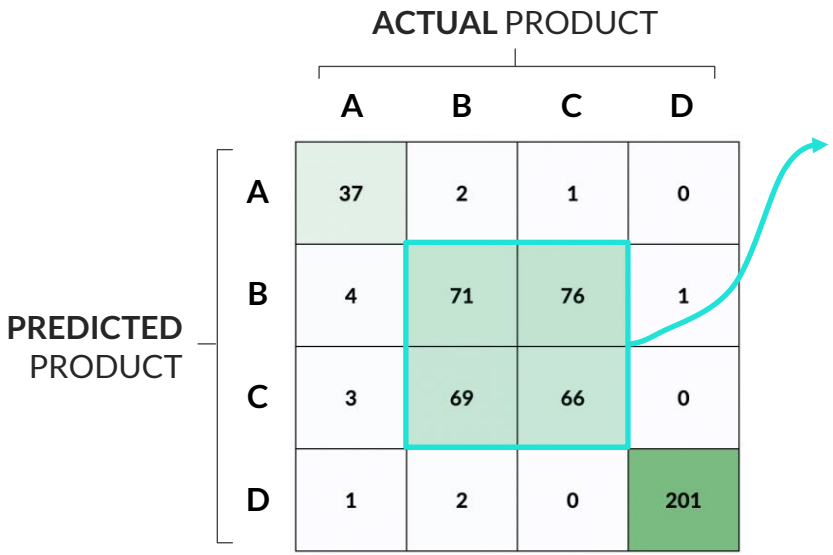




CONFUSION MATRIX

- Hyperparameters
- Imbalanced Classes
- **Confusion Matrix**
- Model Selection
- Model Drift

A **confusion matrix** can also provide insight into relationships and interactions between classes, to help inform model improvements



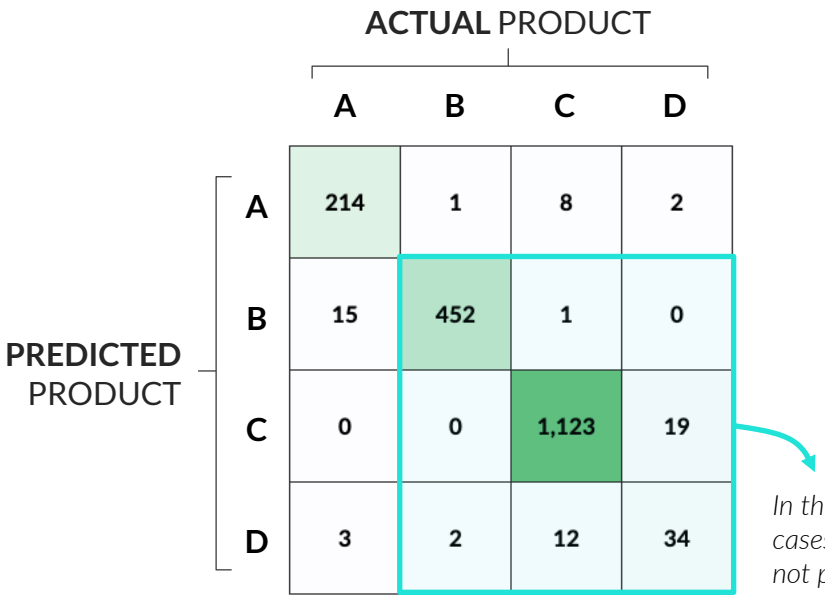
In this case our model has a hard time differentiating products **B** and **C**, often predicting the wrong one

- Are these products very similar? Do we need to predict them separately?
- If so, can we engineer features to help distinguish B from C, and improve model accuracy?

CONFUSION MATRIX

- Hyperparameters
- Imbalanced Classes
- **Confusion Matrix**
- Model Selection
- Model Drift

To score a **multi-class confusion matrix**, calculate metrics for *each* predicted class, then take a weighted average to evaluate the model as a whole



In this case, (TN) includes all cases where Product A was not predicted OR observed

PRODUCT A:

Accuracy

(TP + TN) / (TP + TN + FP + FN)

(214 + 452 + 1 + 1123 + 19 + 2 + 12 + 34) / (1886)

=

.9846

Precision

TP / (TP + FP)

(214) / (214 + 1 + 8 + 2)

=

.9511

Recall

TP / (TP + FN)

(214) / (214 + 15 + 3)

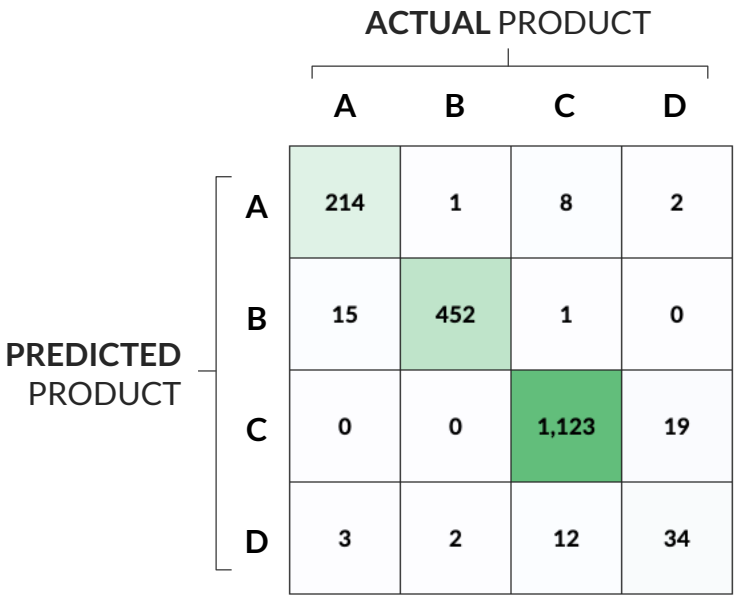
=

.9224

CONFUSION MATRIX

- Hyperparameters
- Imbalanced Classes
- **Confusion Matrix**
- Model Selection
- Model Drift

To score a **multi-class confusion matrix**, calculate metrics for *each* predicted class, then take a weighted average to evaluate the model as a whole



PRODUCT B:

Accuracy

(TP + TN) / (TP + TN + FP + FN)

(452 + 214 + 8 + 2 + 1123 + 19 + 3 + 12 + 34) / (1886)

=

.9899

Precision

TP / (TP + FP)

(452) / (452 + 15 + 1)

=

.9658

Recall

TP / (TP + FN)

(452) / (452 + 1 + 2)

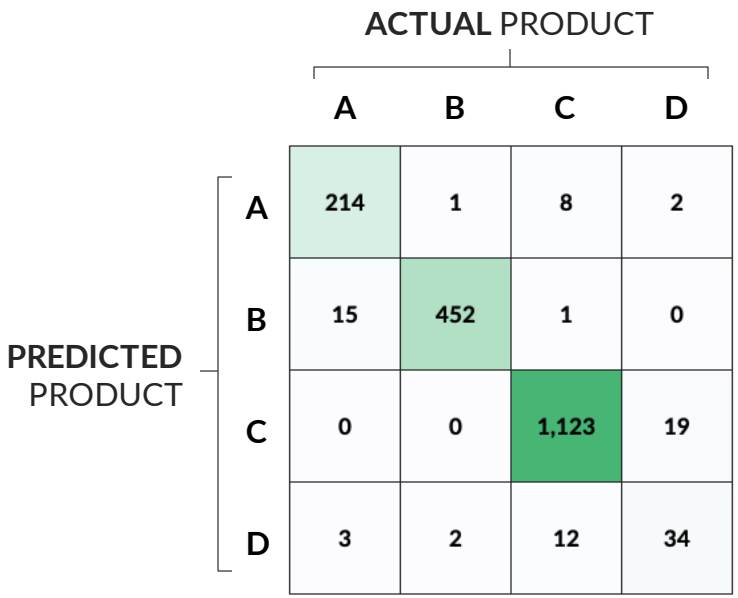
=

.9934

CONFUSION MATRIX

- Hyperparameters
- Imbalanced Classes
- **Confusion Matrix**
- Model Selection
- Model Drift

To score a **multi-class confusion matrix**, calculate metrics for *each* predicted class, then take a weighted average to evaluate the model as a whole



PRODUCT C:

Accuracy

(TP + TN) / (TP + TN + FP + FN)

(1123 + 214 + 1+ 2 + 15 + 452 + 3 + 2 + 34) / (1886)

=

.9788

Precision

TP / (TP + FP)

(1123) / (1123 + 19)

=

.9834

Recall

TP / (TP + FN)

(1123) / (1123 + 8 + 1 + 12)

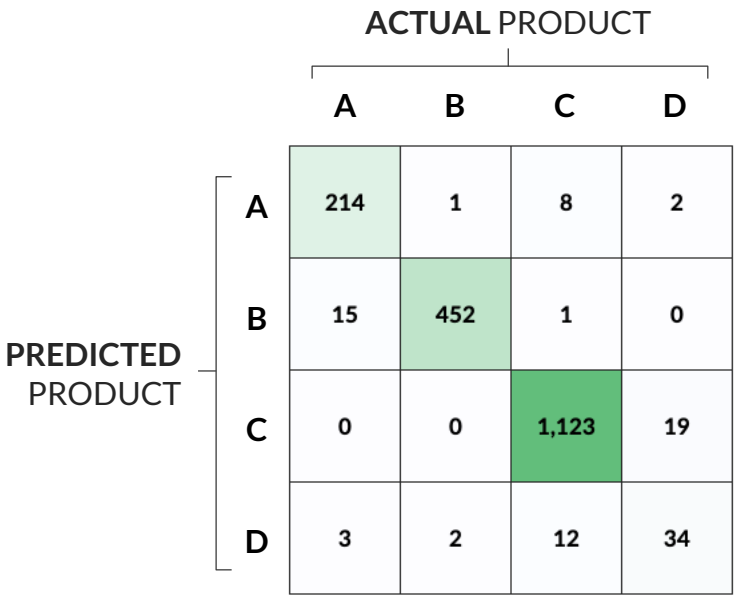
=

.9816

CONFUSION MATRIX

- Hyperparameters
- Imbalanced Classes
- **Confusion Matrix**
- Model Selection
- Model Drift

To score a **multi-class confusion matrix**, calculate metrics for *each* predicted class, then take a weighted average to evaluate the model as a whole



PRODUCT D:

Accuracy

(TP + TN) / (TP + TN + FP + FN)

(34 + 214 + 1 + 8 + 15 + 452 + 1 + 1123) / (1886)

=

.9799

Precision

TP / (TP + FP)

(34) / (34 + 3 + 2 + 12)

=

.6667

Recall

TP / (TP + FN)

(34) / (34 + 2 + 19)

=

.6182

CONFUSION MATRIX

- Hyperparameters
- Imbalanced Classes
- **Confusion Matrix**
- Model Selection
- Model Drift

To score a **multi-class confusion matrix**, calculate metrics for *each* predicted class, then take a weighted average to evaluate the model as a whole

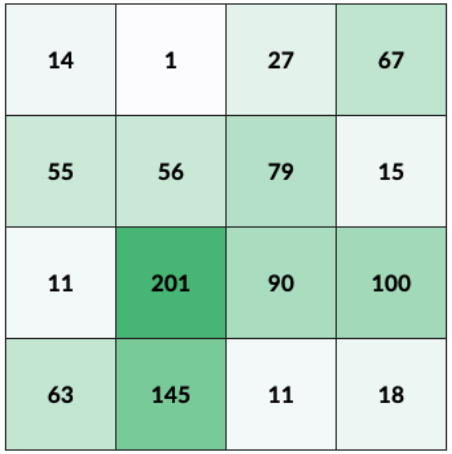
214	1	8	2
15	452	1	0
0	0	1,123	19
3	2	12	34

	# Obs.	Accuracy	Precision	Recall
A	232	**.9846**	**.9511**	**.9224**
B	455	**.9899**	**.9658**	**.9934**
C	1,144	**.9788**	**.9834**	**.9816**
D	55	**.9799**	**.6667**	**.6182**
WEIGHTED AVG:		***.9822***	***.9659***	***.9666***

CONFUSION MATRIX

- Hyperparameters
- Imbalanced Classes
- **Confusion Matrix**
- Model Selection
- Model Drift

To score a **multi-class confusion matrix**, calculate metrics for *each* predicted class, then take a weighted average to evaluate the model as a whole



	# Obs.	Accuracy	Precision	Recall
A	143	.7500	.1284	.0979
B	403	.4795	.2732	.1390
C	207	.5498	.2239	.4348
D	200	.5792	.0759	.0900
WEIGHTED AVG:		.5563	.1994	.1868

MODEL SELECTION

- Hyperparameters
- Imbalanced Classes
- Confusion Matrix
- **Model Selection**
- Model Drift

Model selection describes the process of training multiple models, comparing performance on Test data, and choosing the best option

When comparing performance, it's important to prioritize the *most relevant* metrics based on the context of your model

- **RECALL** may be the best metric if it's critical to predict ALL positive outcomes correctly, and false negatives are a major risk *(i.e. nuclear reactor)*
- **PRECISION** may be the best metric if false negatives aren't a big deal, but false positives are a major risk *(i.e. spam filter or document search)*
- **ACCURACY** may be the best metric if you care about predicting positive and negative outcomes equally, or if the risk of each outcome is comparable

MODEL DRIFT

- Hyperparameters
- Imbalanced Classes
- Confusion Matrix
- Model Selection
- **Model Drift**

Drift is when a trained model gradually becomes less accurate over time, even when all variables and parameters remain the same

- As a best practice, all ML models used for ongoing prediction should be updated or retrained on a regular basis to combat drift

Tips to correct drift:

- Benchmark your model performance *(accuracy, precision, recall)* on Day 1
- Monitor performance against your benchmark over time
- If you notice drift compared to your benchmark, retrain the model using updated Training data and consider discarding old records *(if you have enough volume)*
- Conduct additional feature engineering as necessary

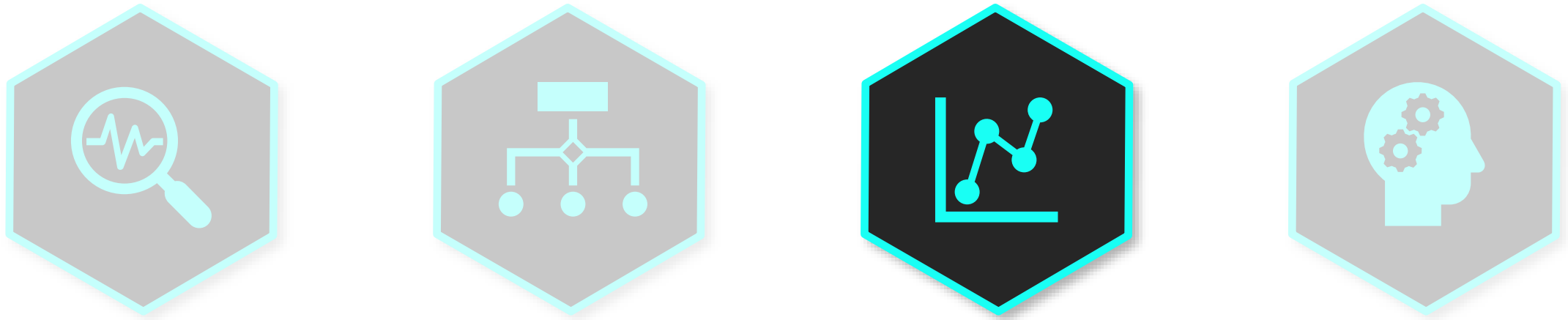
PART 3:

REGRESSION



ABOUT THIS SERIES

This is **Part 3** of a **4-Part series** designed to help you build a deep, foundational understanding of machine learning, including data QA & profiling, classification, forecasting and unsupervised learning



PART 1
QA & Data Profiling

PART 2
Classification

PART 3
Regression & Forecasting

PART 4
Unsupervised Learning

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COURSE OUTLINE

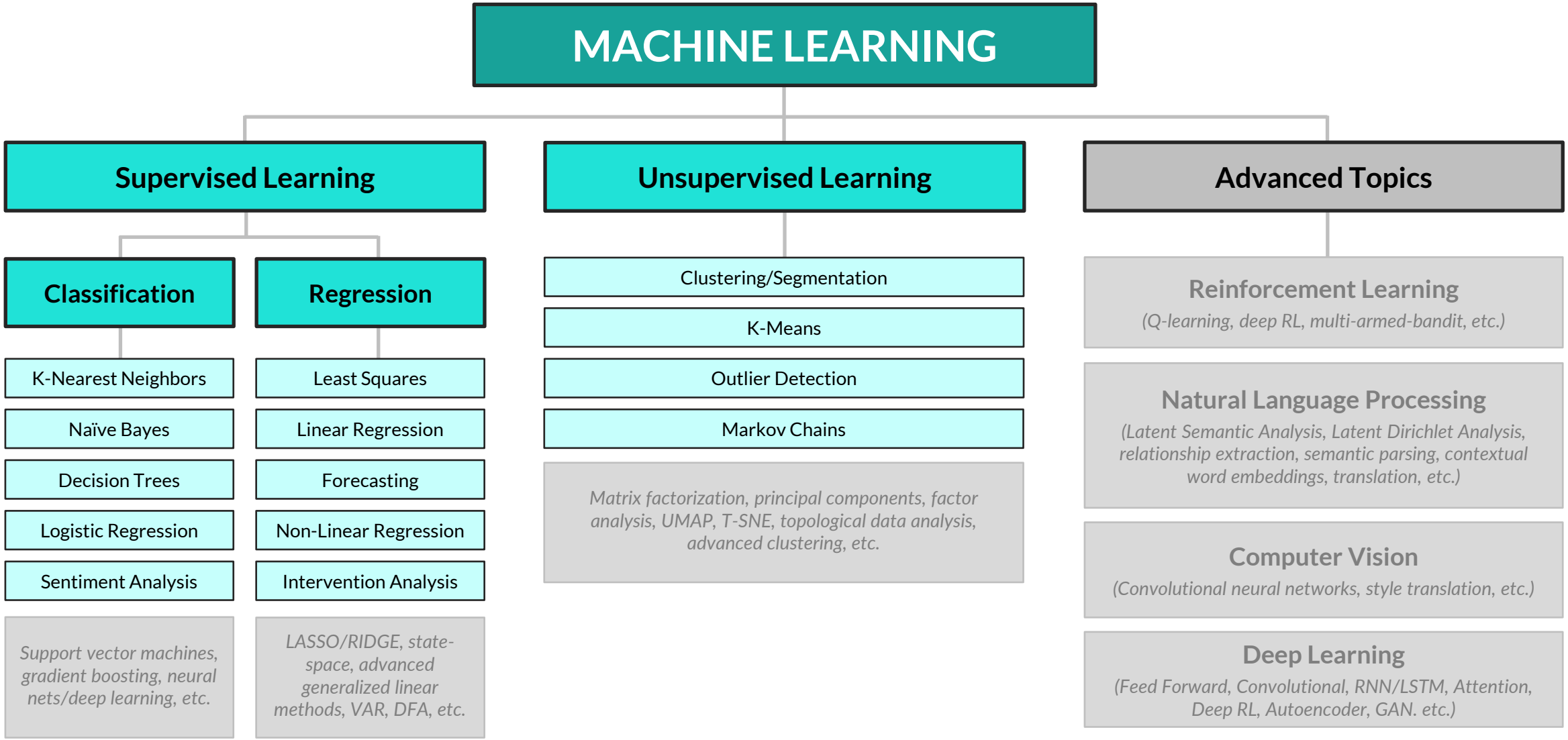
1 Intro to Regression	*Review the ML landscape and key supervised learning concepts, including regression vs. classification, feature engineering, overfitting, etc.*
2 Regression Modeling	*Learn the key building blocks of regression analysis, including linear relationships, least squared error, multivariate and non-linear models*
3 Model Diagnostics	*Understand and interpret model outputs and common diagnostic metrics like R-squared, mean error, F-significance, P-values, and more*
4 Time-Series Forecasting	*Explore powerful tools & techniques for time-series forecasting, including seasonality, linear and non-linear trends, intervention analysis, etc.*

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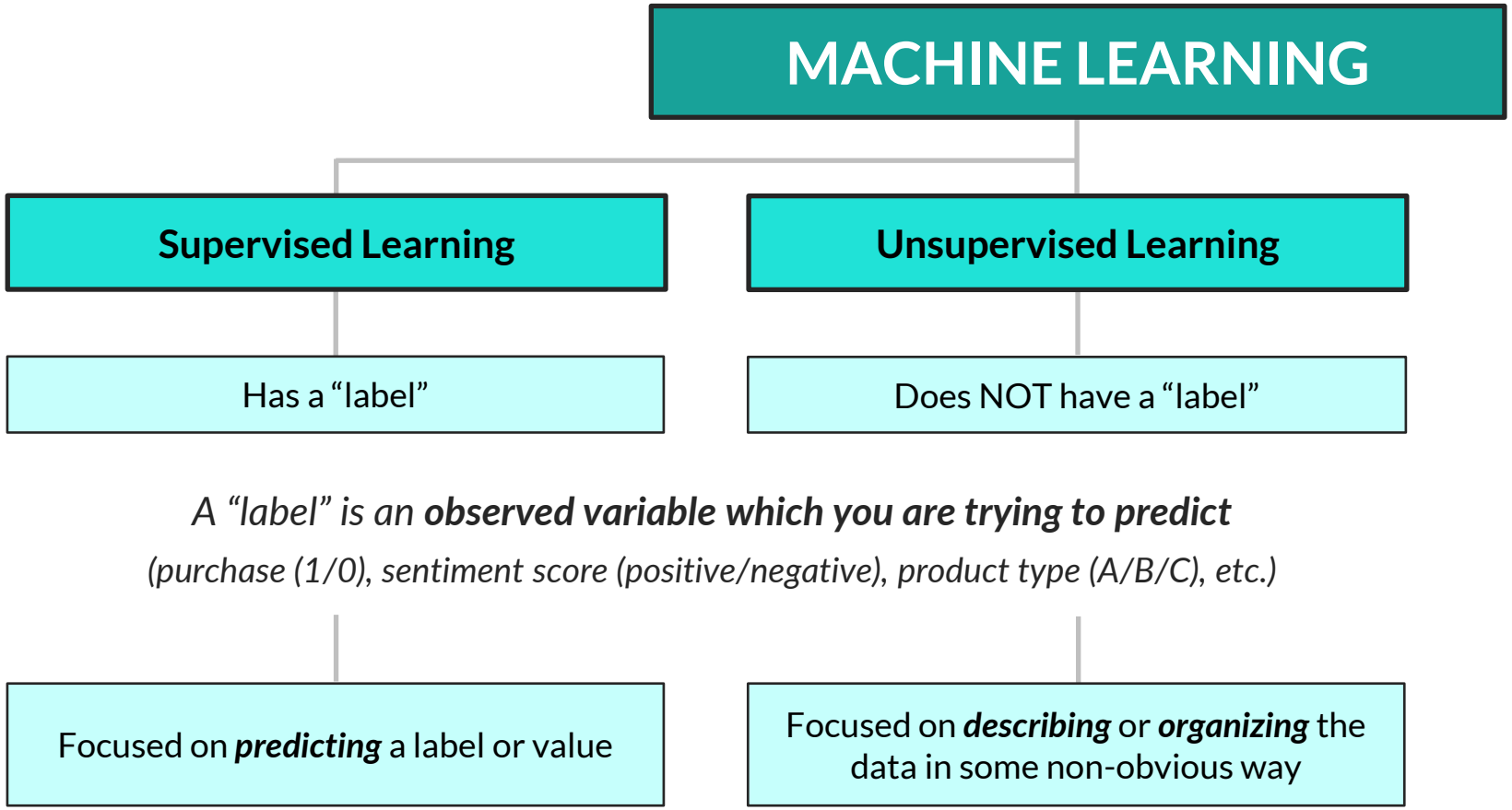
INTRO TO REGRESSION

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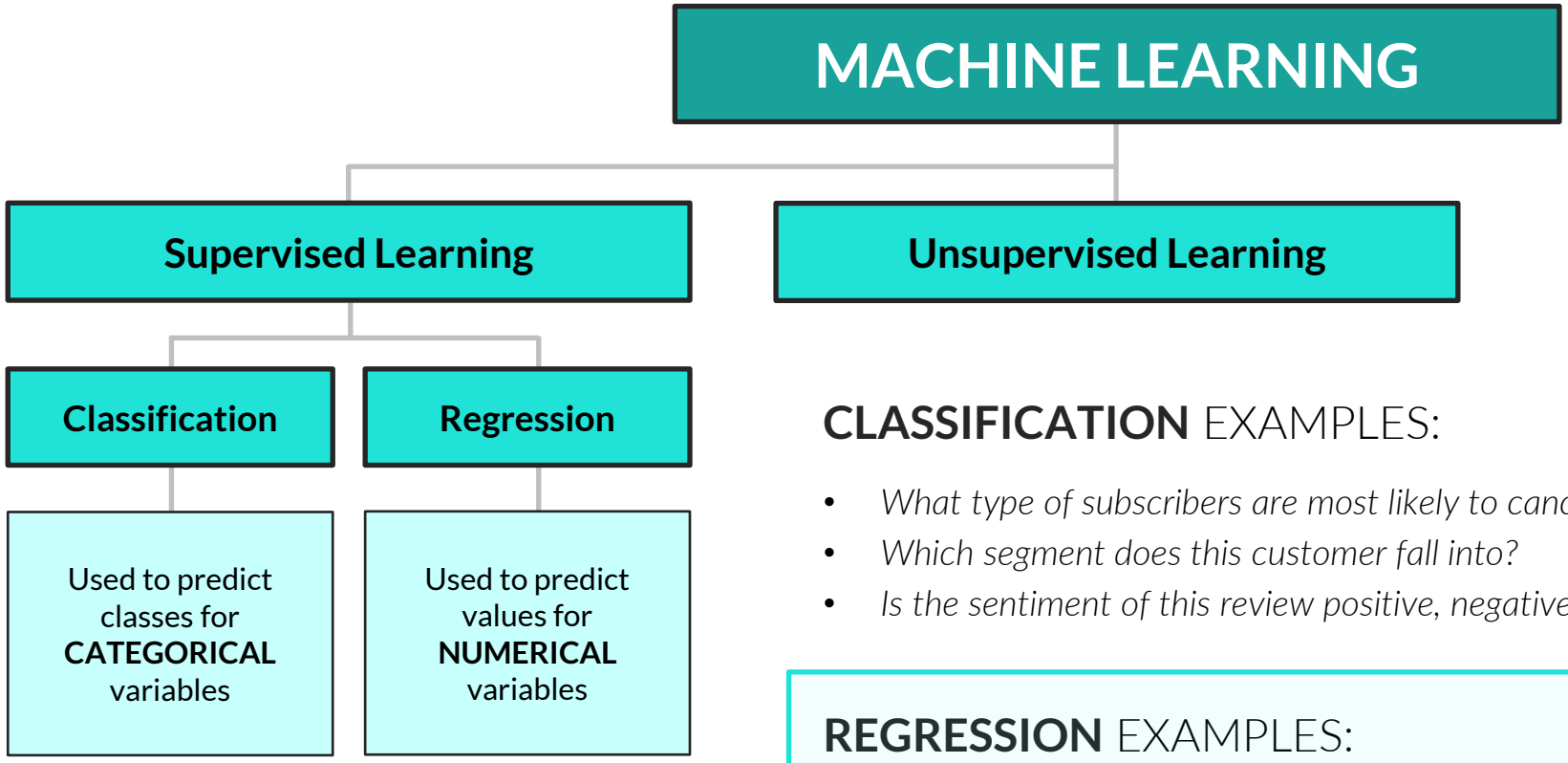
MACHINE LEARNING LANDSCAPE



MACHINE LEARNING LANDSCAPE



MACHINE LEARNING LANDSCAPE



CLASSIFICATION EXAMPLES:

- *What type of subscribers are most likely to cancel or churn?*
- *Which segment does this customer fall into?*
- *Is the sentiment of this review positive, negative or neutral?*

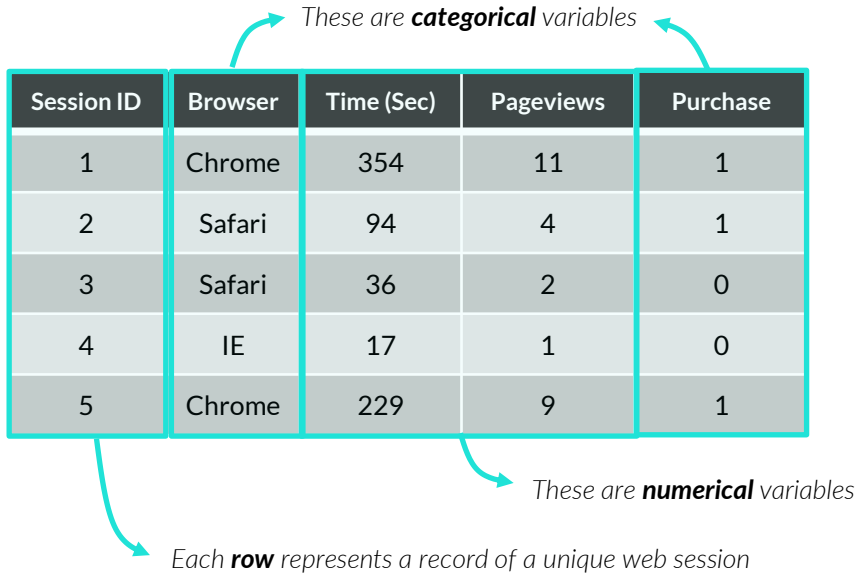
REGRESSION EXAMPLES:

- *How much revenue are we forecasting for the quarter?*
- *How many units do we expect to sell next season?*
- *How will seasonality impact the size of our customer base?*

RECAP: KEY CONCEPTS

All tables contain rows and columns

- Each row represents an individual **record**/**observation**
- Each columns represent an individual **variable**



Variables can be categorical or numerical

- Categorical variables contain **classes** or **categories** *(used for filtering)*
- Numerical variables contain **numbers** *(used for aggregation)*

QA and data profiling comes first, every time

- Without quality data, you can't build quality models *("garbage in, garbage out")*
- Profiling is the first step towards **conditioning** *(or filtering)* on key variables to understand their impact

Machine Learning is needed when visual analysis falls short

- Humans aren't capable of thinking beyond 3+ dimensions, but machines are built for it
- ML is a natural extension of visual analysis and univariate/multivariate profiling

REGRESSION 101

The goal of regression is to predict a ***numeric* dependent variable** using **independent variables**

y **Dependent variable** (DV) – for regression, this must be numerical *(not categorical)*!

- This is the variable **you're trying to predict**
- The dependent variable is commonly referred to as *"Y"*, *"predicted"*, *"output"*, or *"target"* variable
- Regression is about understanding how the numerical DV is impacted by, or *dependent* on, other variables in the model

x **Independent variables** (IVs)

- These are the variables **which help you predict the dependent variable**
- Independent variables are commonly referred to as *"X's"*, *"predictors"*, *"features"*, *"explanatory variables"*, *or "covariates"*
- Regression is about understanding how the IVs impact, or *predict*, the DV

REGRESSION 101

EXAMPLE: Using marketing and sales data *(sample below)* to predict revenue for a given month

Month ID	Social Posts	Competitive Activity	Marketing Spend	Promotion Count	Revenue
1	30	High	$130,000	12	$1,300,050
2	15	Low	$600,000	5	$11,233,310
3	11	Medium	$15,000	10	$1,112,050
4	22	Medium	$705,000	11	$1,582,077
5	41	High	$3,000	3	$1,889,053
6	5	High	$3,500	3	$1,200,089
7	23	Low	$280,000	2	$1,300,080
8	12	Low	$120,000	11	$700,150
9	8	Low	$1,000,000	51	$41,011,150
10	19	Medium	$43,000	10	$1,000,210

Revenue *is our* ***dependent variable****, since it's what we want to predict*

Since Revenue is ***numerical****, we'll use regression (vs. classification) to predict it*

Social Posts, Competitive Activity, Marketing Spend *and* ***Promotion Count*** *are all* ***independent variables****, since they can help us explain, or predict, monthly Revenue*

REGRESSION 101

EXAMPLE: Using marketing and sales data *(sample below)* to predict revenue for a given month

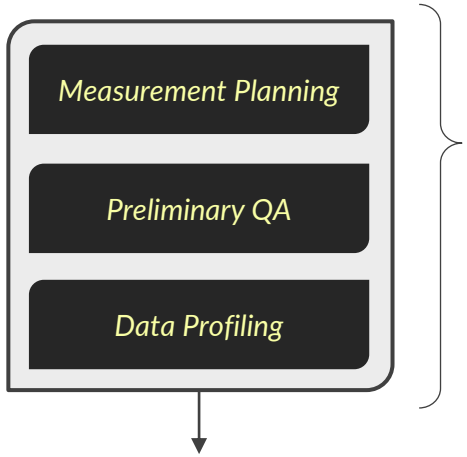
Month ID	Social Posts	Competitive Activity	Marketing Spend	Promotion Count	Revenue
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8	12	Low	$120,000	11	$700,150
9	8	Low	$1,000,000	51	$41,011,150
10	19	Medium	$43,000	10	$1,000,210
11	7	High	$320,112	8	???

We'll use records with **observed values** for both independent and dependent variables to *"train"* our regression model…

…then apply that model to new, **unobserved values** containing IVs but no DV

This is what our regression model will predict!

REGRESSION WORKFLOW



Remember, these steps ALWAYS come first

*Before building a model, you should have a clear **measurement plan** (KPIs, project scope, desired outcome, etc.) and an understanding of the **data at hand** (variable types, table structure, data quality, profiling metrics, etc.)*

Feature Engineering → **Data Splitting** → **Model Training** → **Selection & Tuning**

Adding new, calculated variables (or "features") to a data set based on existing fields

Splitting records into "Training" and "Test" data sets, to validate accuracy and avoid overfitting

Building regression models from Training data and applying to Test data to maximize prediction accuracy

Choosing the best performing model for a given prediction, and tuning it to prevent drift over time

FEATURE ENGINEERING



Feature engineering is the process of enriching a data set by creating additional independent variables based on existing fields

- New features can help improve the accuracy and predictive power of your ML models
- Feature engineering is often used to convert fields into *"model-friendly"* formats; for example, **one-hot encoding** transforms categorical variables into individual, binary (1/0) fields

***Original** features*

***Engineered** features*

Month ID	Social Posts	Competitive Activity	Marketing Spend	Promotion Count	Revenue	Competitive High	Competitive Medium	Competitive Low	Promotion >10 & Social > 25	Log Spend
1	30	High	$130,000	12	$1,300,050	1	0	0	1	11.7
2	15	Low	$600,000	5	$11,233,310	0	0	1	0	13.3
3	8	Medium	$15,000	10	$1,112,050	0	1	0	0	9.6
4	22	Medium	$705,000	11	$1,582,077	0	1	0	0	13.5
5	41	High	$3,000	3	$1,889,053	1	0	0	0	8.0

DATA SPLITTING



Splitting is the process of partitioning data into separate sets of records for the purpose of training and testing machine learning models

- As a rule of thumb, ~70-80% of your data will be used for **Training** *(which is what your model learns from)*, and ~20-30% will be reserved for **Testing** *(to validate the model's accuracy)*

Month ID	Social Posts	Competitive Activity	Marketing Spend	Promotion Count	Revenue
1	30	High	$130,000	12	$1,300,050
2	15	Low	$600,000	5	$11,233,310
3	11	Medium	$15,000	10	$1,112,050
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6	5	High	$3,500	3	$1,200,089
7	23	Low	$280,000	2	$1,300,080
8	12	Low	$120,000	11	$700,150
9	8	Low	$1,000,000	51	$41,011,150
10	19	Medium	$43,000	10	$1,000,210

Training *data*

Test *data*

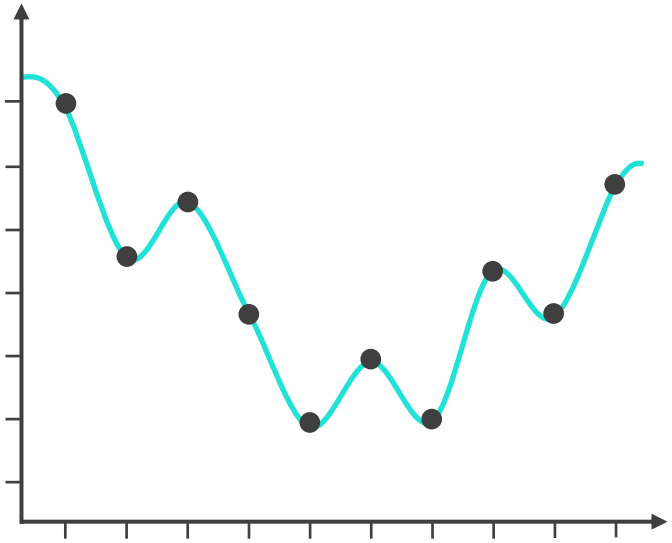
Test data is NOT used to optimize models

Using **Training** data for optimization and **Test** data for validation ensures that your model can accurately predict both *known* and *unknown* values, which helps to prevent **overfitting**

OVERFITTING

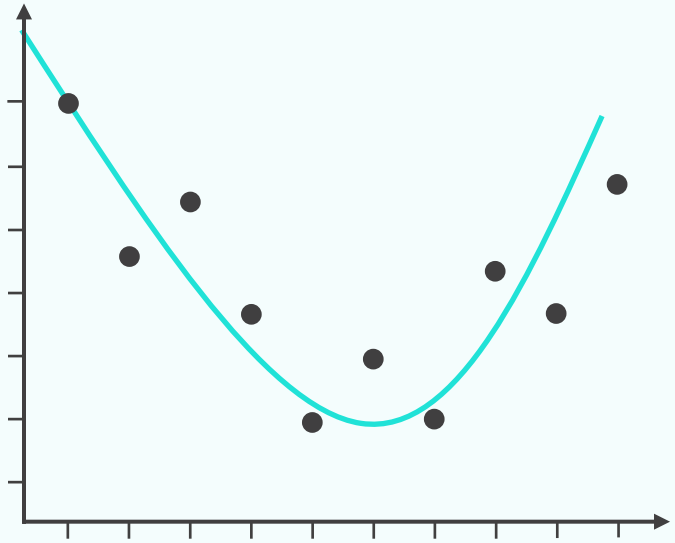
Splitting is primarily used to avoid **overfitting**, which is when a model predicts known *(Training)* data very well but unknown *(Test)* data poorly

- Think of overfitting like memorizing the answers to a test instead of actually *learning* the material; you'll ace the test, but lack the ability to ***generalize*** and apply your knowledge to unfamiliar questions



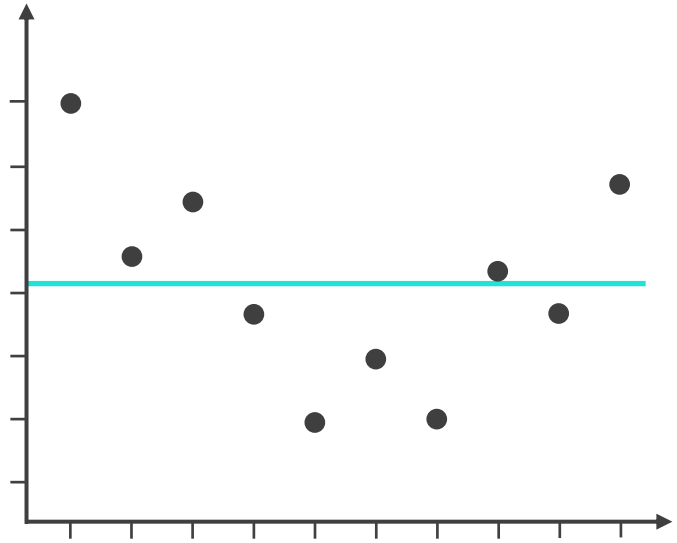
OVERFIT model

- Models the Training data too well
- Doesn't generalize well to Test data *(high variance, low bias)*



WELL-FIT model

- Models the Training data just right
- Generalizes well to Test data *(balance of bias & variance)*



UNDERFIT model

- Doesn't model Training data well enough
- Doesn't generalize well to Test data *(high bias, low variance)*

REGRESSION

There are two common use cases for linear regression: **prediction** and **root-cause analysis**



PREDICTION

- Used to **predict** or **forecast** a numerical dependent variable
- Goal is to make accurate predictions, even if causality cannot necessarily be proven



ROOT-CAUSE ANALYSIS

- Used to determine the **causal impact** of individual model inputs
- Goal is to prove causality by comparing the sensitivity of each IV on the DV

REGRESSION MODELING

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REGRESSION MODELING



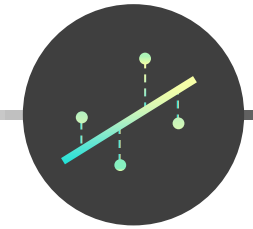
In this section we'll introduce the basics of **regression modeling**, including linear relationships, least squared error, simple and multiple regression and non-linear models

TOPICS WE'LL COVER:

- Linear Relationships
- Least Squared Error
- Univariate Linear Regression
- Multiple Linear Regression
- Non-Linear Regression

GOALS FOR THIS SECTION:

- *Understand basic linear relationships and the concept of least squared errors*
- *Explore the difference between simple (univariate) and multiple linear regression*
- *Demonstrate how regression can be used to model linear or non-linear relationships*

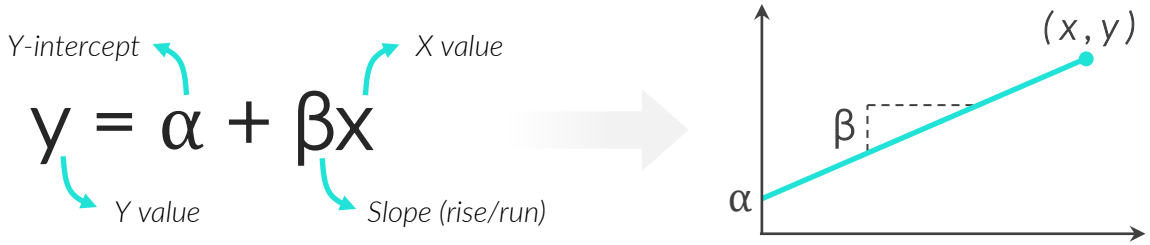


LINEAR RELATIONSHIPS

- Linear Relationships
- Least Squared Error
- Univariate Linear Regression
- Multiple Linear Regression
- Non-Linear Regression

A **linear relationship** means that when you increase or decrease your X variable, your Y variable increases or decreases at a steady rate

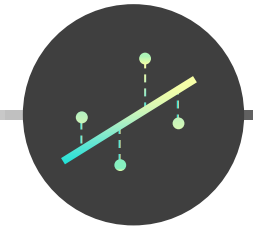
- If you plot a linear relationship over many X/Y values, it will look like a straight line
- Mathematically, linear relationships can be described using the following equation:



- **NOTE:** Not all relationships are linear *(more on that later!)*

Common examples:

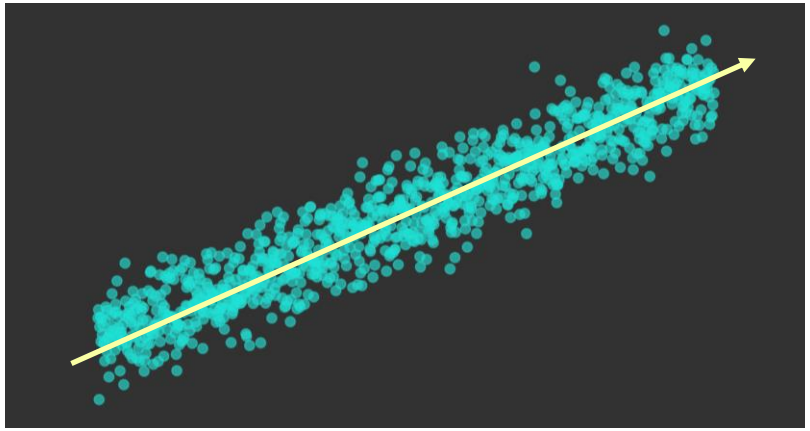
- Taxi Mileage (X) and Total Fare (Y)
- Units Sold (X) and Total Revenue (Y)



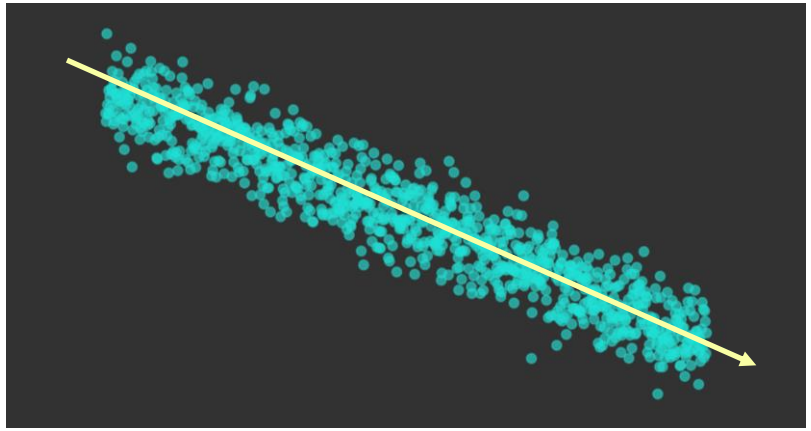
LINEAR RELATIONSHIPS

- Linear Relationships
- Least Squared Error
- Univariate Linear Regression
- Multiple Linear Regression
- Non-Linear Regression

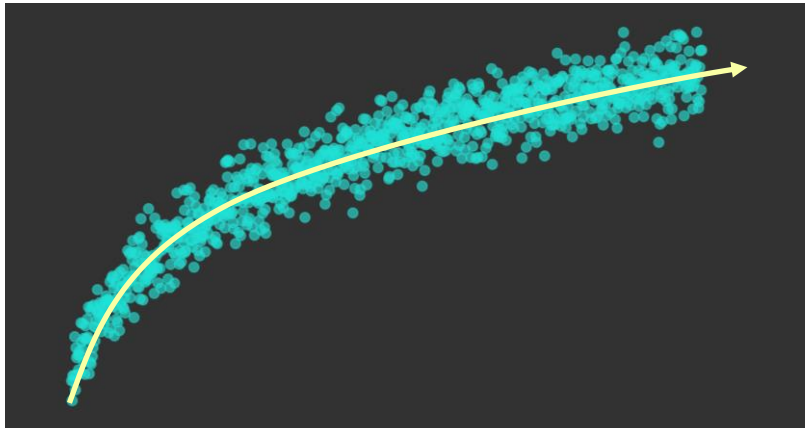
Positive Linear



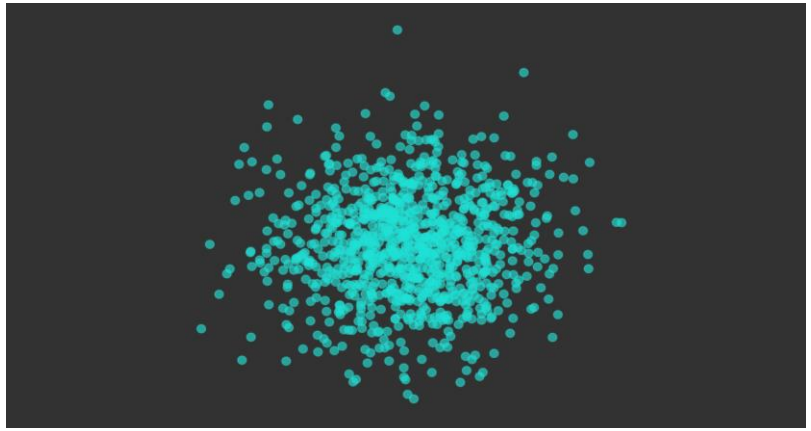
Negative Linear



Non-Linear *(logarithmic)*



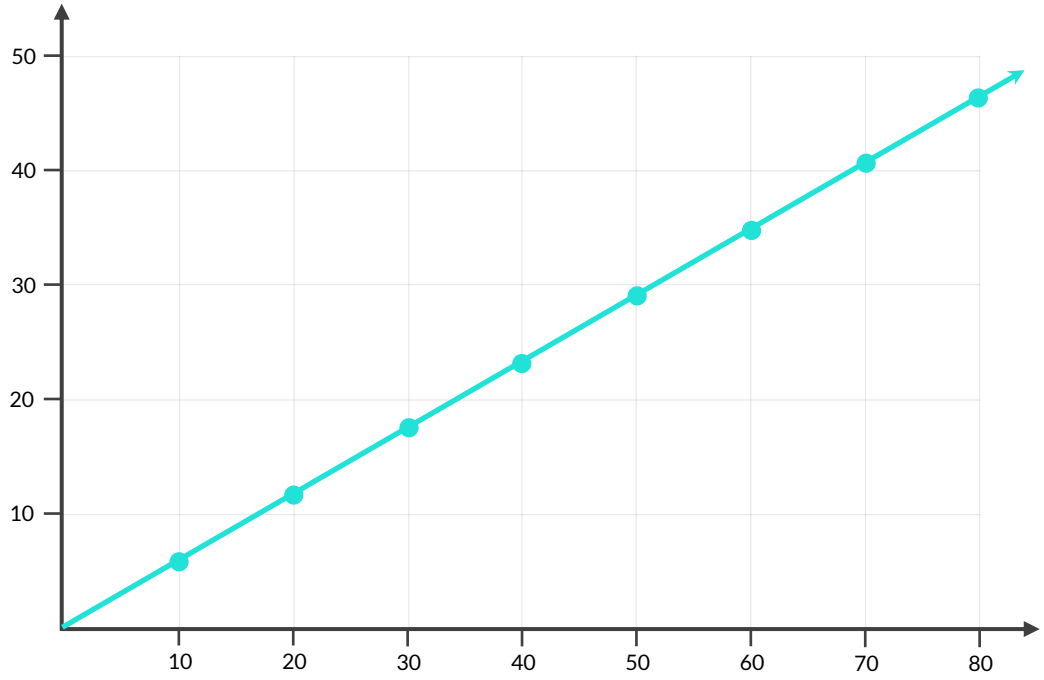
No Relationship



Sometimes there's ***no relationship at all!***

LINEAR RELATIONSHIPS

- Linear Relationships
- Least Squared Error
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- Non-Linear Regression



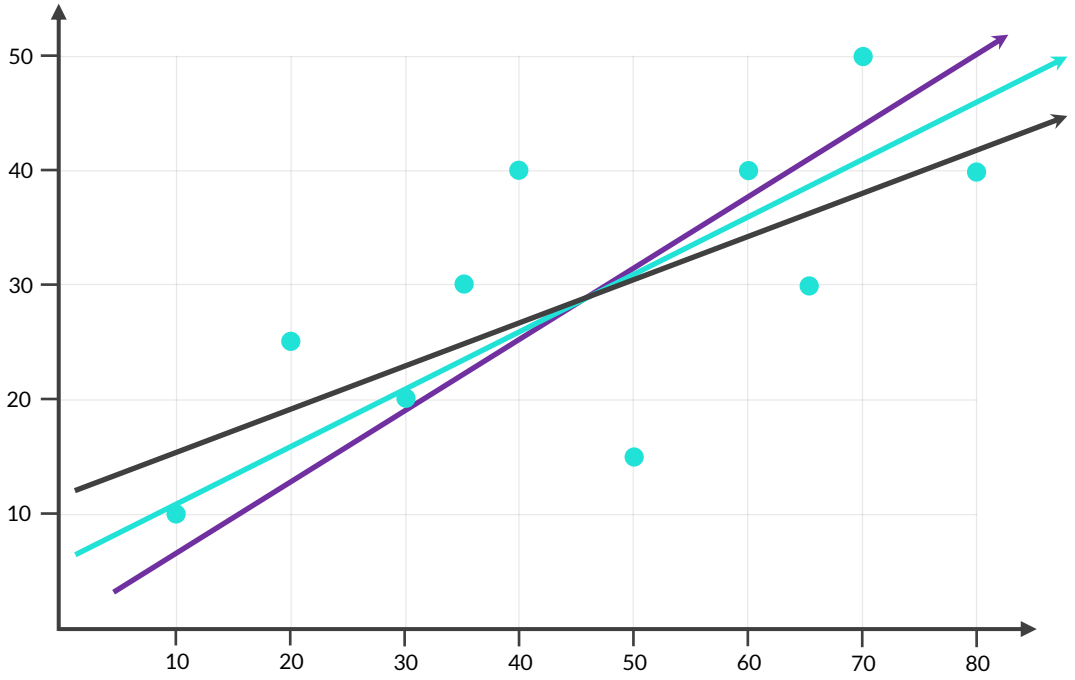
Consider a line that fits *every single point* in the plot

This is known as a **perfectly linear relationship**

In this case you can simply calculate the **exact value of Y** for any given value of X *(no Machine Learning needed, just simple math!)*

LINEAR RELATIONSHIPS

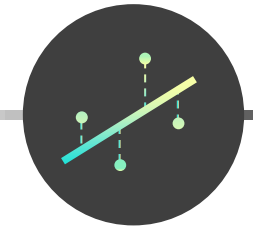
- Linear Relationships
- Least Squared Error
- Univariate Linear Regression
- Multiple Linear Regression
- Non-Linear Regression



In the real world, things aren't quite so simple

When you add **variance**, it means that *many* different lines could potentially fit through the plot

To find the equation of the line with the best possible fit, we can use a technique known as **least squared error** or **"least squares"**



LEAST SQUARED ERROR

Linear Relationships

Least Squared Error

Univariate Linear Regression

Multiple Linear Regression

Non-Linear Regression

Least squared error is used to mathematically determine the line that best fits through a series of data points

- Imagine drawing a line through a scatterplot, and measuring the distance between your line and each point *(these distances are called **errors**, or **residuals**)*
- Now square each of those residuals, add them all up, and adjust your line until you've minimized that sum; **this is how least squares works!**

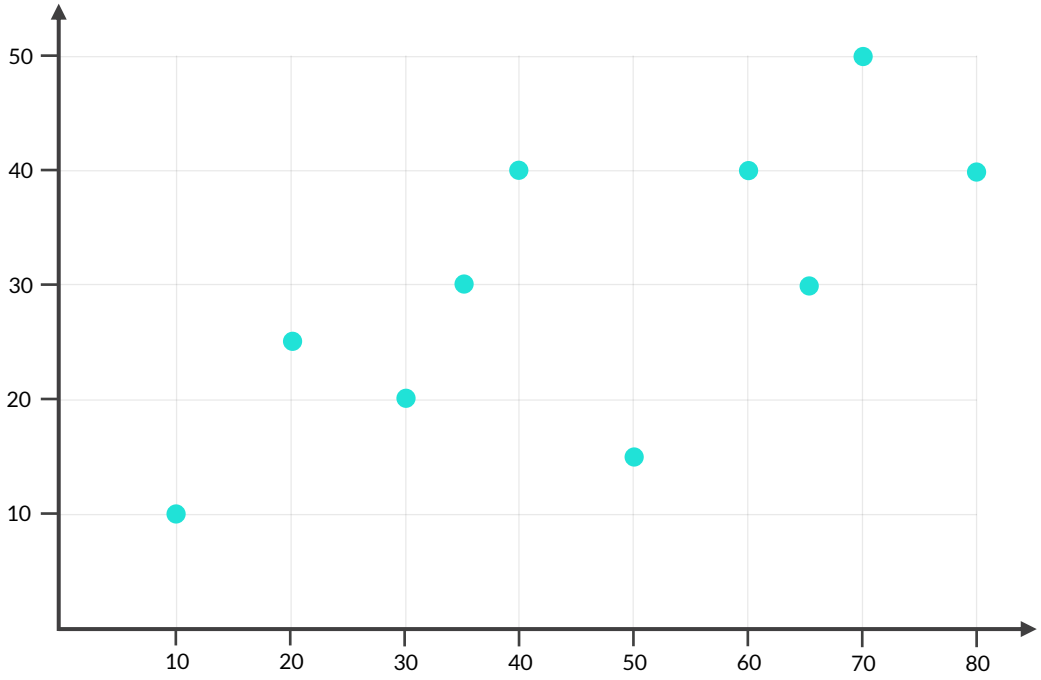


Why "squared" error?

- Squaring the residuals converts them into **positive values**, and prevents positive and negative distances from cancelling each other out *(this helps the model optimize more efficiently, too)*

LEAST SQUARED ERROR

- Linear Relationships
- Least Squared Error
- Univariate Linear Regression
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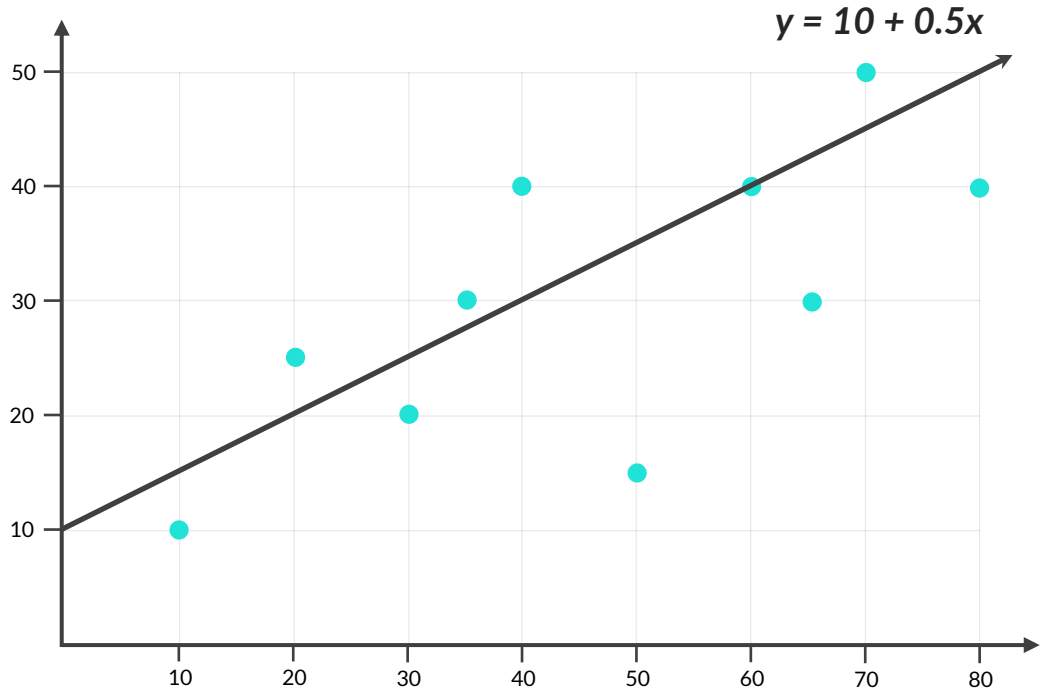


X	Y *(actual)*
10	10
20	25
30	20
35	30
40	40
50	15
60	40
65	30
70	50
80	40

STEP 1: Plot each data point on a scatterplot, and record the X and Y values

LEAST SQUARED ERROR

- Linear Relationships
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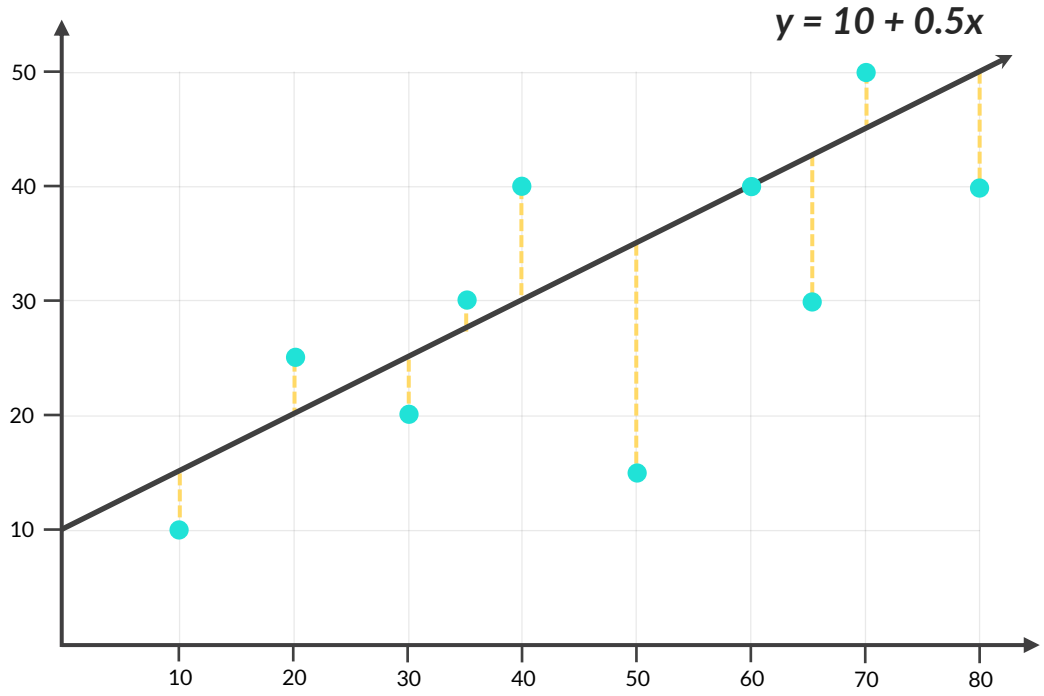


X	Y *(actual)*
10	10
20	25
30	20
35	30
40	40
50	15
60	40
65	30
70	50
80	40

STEP 2: Draw a straight line through the points in the scatterplot, and calculate the Y values derived by your linear equation

LEAST SQUARED ERROR

- Linear Relationships
- Least Squared Error
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- Multiple Linear Regression
- Non-Linear Regression

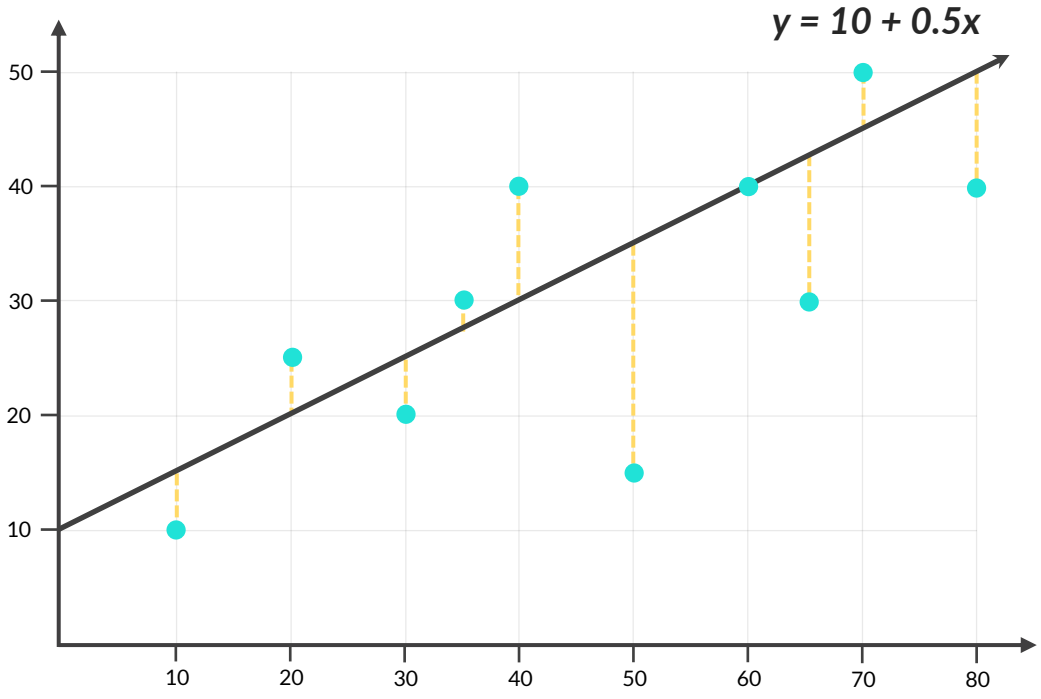


X	Y *(actual)*	Y *(line)*
10	10	15
20	25	20
30	20	25
35	30	27.5
40	40	30
50	15	35
60	40	40
65	30	42.5
70	50	45
80	40	50

STEP 3: For each value of X, calculate the error *(or residual)* by comparing the actual Y value against the Y value produced by your linear equation

LEAST SQUARED ERROR

- Linear Relationships
- Least Squared Error
- Univariate Linear Regression
- Multiple Linear Regression
- Non-Linear Regression



X	Y (*actual*)	Y (*line*)	*Error*
10	10	15	5
20	25	20	-5
30	20	25	5
35	30	27.5	-2.5
40	40	30	-10
50	15	35	20
60	40	40	0
65	30	42.5	12.5
70	50	45	-5
80	40	50	10

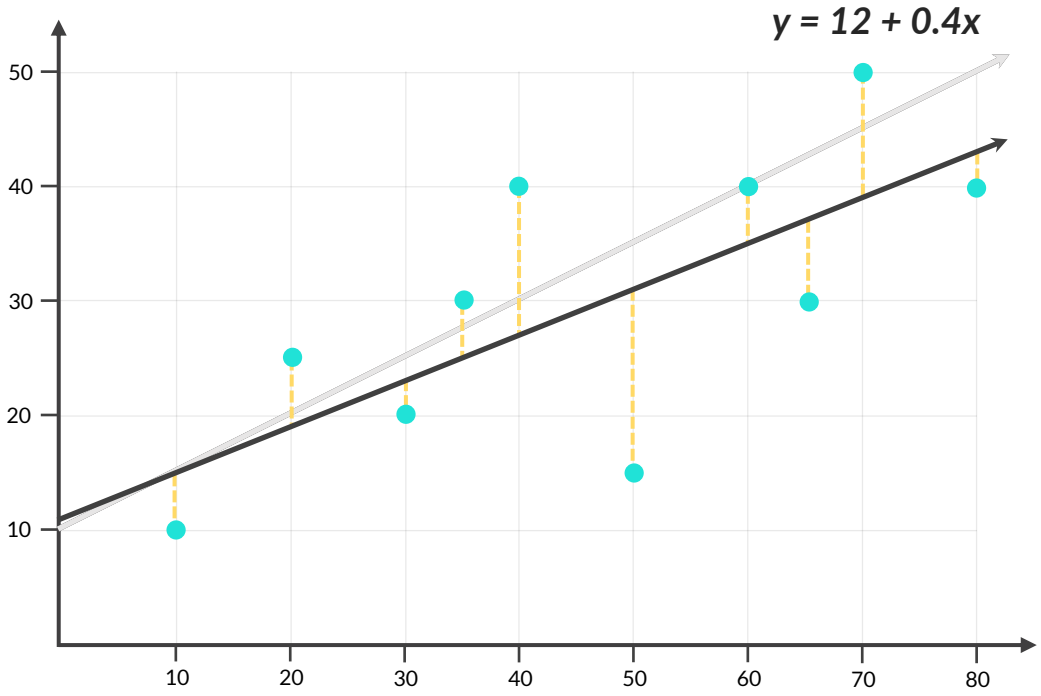
SUM OF SQUARED ERROR: 862.5

STEP 4: Square each individual residual and add them up to determine the **sum of squared error (SSE)**

- This defines exactly how well your line "fits" the plot *(or in other words, how well the linear equation describes the relationship between X and Y)*

LEAST SQUARED ERROR

- Linear Relationships
- Least Squared Error
- Univariate Linear Regression
- Multiple Linear Regression
- Non-Linear Regression

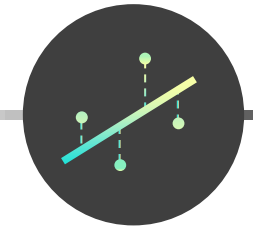


X	Y (*actual*)	Y (*line*)	*Error*	*Sq. Error*
10	10	16	6	**36**
20	25	20	-5	**25**
30	20	24	4	**16**
35	30	26	-4	**16**
40	40	28	-12	**144**
50	15	32	17	**289**
60	40	36	-4	**16**
65	30	38	8	**64**
70	50	40	-10	**100**
80	40	44	4	**16**
				II

SUM OF SQUARED ERROR: 722

STEP 5: Plot a new line, repeat Steps 1-4, and continue the process until you've found the line that **minimizes the sum of squared error**

- This is where **Machine Learning** comes in; human trial-and-error is completely impractical, but machines can find an optimal linear equation in seconds



UNIVARIATE LINEAR REGRESSION

- Linear Relationships
- Least Squared Error
- Univariate Linear Regression
- Multiple Linear Regression
- Non-Linear Regression

Univariate ("simple") linear regression is used for predicting a numerical output (DV) based on a *single* independent variable

- Univariate linear regression is simply an extension of least squares; you use the linear equation that minimizes SSE to predict an output (**Y**) for any given input (**X**)

Dependent variable (DV)

Coefficient/parameter (sensitivity of Y to X)

$$y = \alpha + \beta x + \varepsilon$$

Error/residual

Y-intercept

Independent variable (IV)

This is just the equation of a line, plus an error term

Simple linear regression is rarely used on its own; think of it as a primer for understanding more complex topics like non-linear and multiple regression

CASE STUDY: UNIVARIATE LINEAR REGRESSION



THE SITUATION

You are the proud owner of **The Cone Zone**, a mobile ice cream cart operating on the Atlantic City boardwalk.



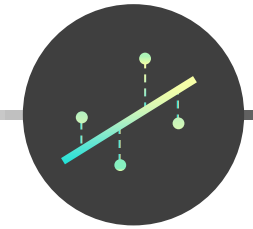
THE ASSIGNMENT

You've noticed that you tend to sell more ice cream on hot days, and want to understand how temperature and sales relate. Your goal is to build a simple linear regression model that you can use to **predict sales based on the weather forecast**.



THE OBJECTIVES

1. Use a scatterplot to visualize a sample of daily temperatures *(X)* and sales *(Y)*
2. Do you notice a clear pattern or trend? How might you interpret this?
3. Find the line that best fits the data by minimizing SSE, then confirm by plotting a linear trendline on the scatter plot



MULTIPLE LINEAR REGRESSION

- Linear Relationships
- Least Squared Error
- Univariate Linear Regression
- Multiple Linear Regression
- Non-Linear Regression



Can we use ***more than one IV*** to predict the DV?

Multiple linear regression is used for predicting a numerical output (DV) based on *multiple* independent variables

- In its simplest form, multiple linear regression is simply univariate linear regression with additional x variables:

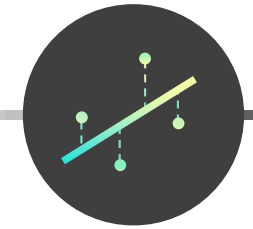
DV

$$y = \alpha + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \ldots + \beta_n x_n + \varepsilon$$

Y-intercept

Error/ residual

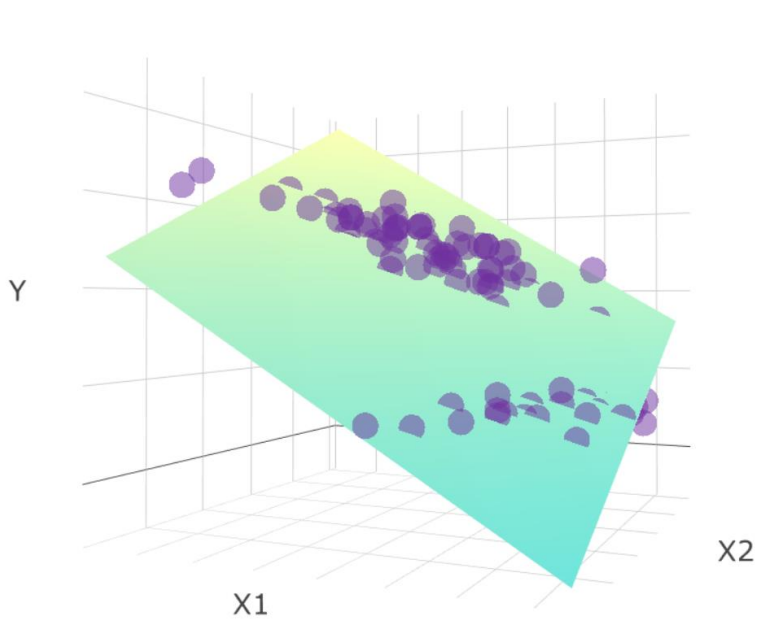
*Instead of just 1 IV, we have a **whole set of independent variables** (and associated coefficients/weights) to help explain our DV*

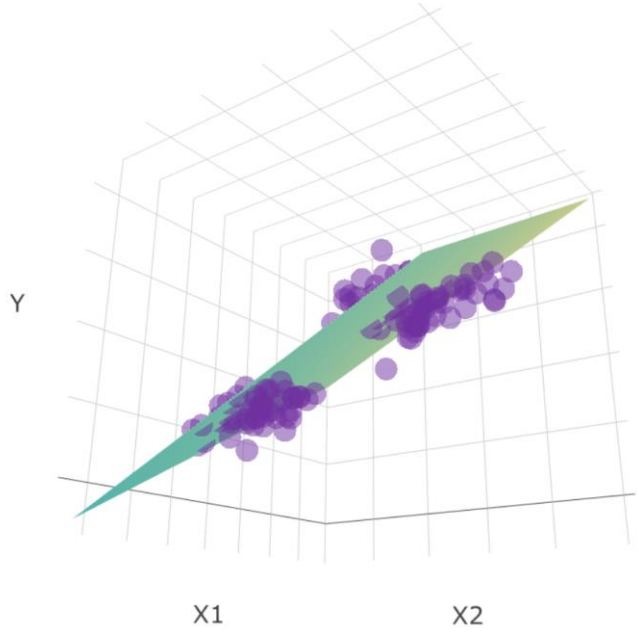


MULTIPLE LINEAR REGRESSION

- Linear Relationships
- Least Squared Error
- Univariate Linear Regression
- **Multiple Linear Regression**
- Non-Linear Regression

To visualize how multiple regression works with 2 independent variables, imagine fitting a plane *(vs. a line)* through a 3D scatterplot:





Multiple regression can scale well beyond 2 variables, but this is where visual analysis breaks down *(and why we need **machine learning**!)*

MULTIPLE LINEAR REGRESSION

- Linear Relationships
- Least Squared Error
- Univariate Linear Regression
- **Multiple Linear Regression**
- Non-Linear Regression

EXAMPLE

*You are preparing to list a new property on **AirBnB**, and want to estimate (or predict) an appropriate price using the listing data below*

list_id	city	district	room_type	accommodates	bedrooms	entire.place	hotel.room	private.room	bronx	brooklyn	manhattan	queens	price
3831	New York	Brooklyn	Entire place	3	1	1	0	0	0	1	0	0	73
5121	New York	Brooklyn	Private room	2	1	0	0	1	0	1	0	0	60
5178	New York	Manhattan	Private room	2	1	0	0	1	0	0	1	0	79
5203	New York	Manhattan	Private room	1	1	0	0	1	0	0	1	0	75
5803	New York	Brooklyn	Private room	2	1	0	0	1	0	1	0	0	83
6872	New York	Manhattan	Private room	1	1	0	0	1	0	0	1	0	65
6990	New York	Manhattan	Private room	1	1	0	0	1	0	0	1	0	62
7097	New York	Brooklyn	Entire place	4	1	1	0	0	0	1	0	0	199
7750	New York	Manhattan	Private room	1	2	0	0	1	0	0	1	0	96
8490	New York	Brooklyn	Entire place	5	1	1	0	0	0	1	0	0	120
9657	New York	Manhattan	Entire place	3	1	1	0	0	0	0	1	0	150
9704	New York	Manhattan	Private room	2	1	0	0	1	0	0	1	0	55
10452	New York	Brooklyn	Private room	3	1	0	0	1	0	1	0	0	70
10962	New York	Brooklyn	Private room	2	1	0	0	1	0	1	0	0	83
11943	New York	Brooklyn	Private room	1	1	0	0	1	0	1	0	0	150
12192	New York	Manhattan	Private room	2	1	0	0	1	0	0	1	0	40
12343	New York	Manhattan	Entire place	3	1	1	0	0	0	0	1	0	150
12937	New York	Queens	Private room	4	1	0	0	1	0	0	0	1	130
12940	New York	Brooklyn	Entire place	2	1	1	0	0	0	1	0	0	99
13121	New York	Queens	Private room	4	1	0	0	1	0	0	0	1	75
13394	New York	Brooklyn	Private room	2	1	0	0	1	0	1	0	0	80
13808	New York	Brooklyn	Private room	2	1	0	0	1	0	1	0	0	81
14290	New York	Brooklyn	Entire place	2	2	1	0	0	0	1	0	0	157
14314	New York	Brooklyn	Entire place	2	1	1	0	0	0	1	0	0	99
14322	New York	Manhattan	Entire place	1	1	1	0	0	0	0	1	0	200
15385	New York	Brooklyn	Private room	1	1	0	0	1	0	1	0	0	80
15711	New York	Manhattan	Entire place	6	2	1	0	0	0	0	1	0	206
16326	New York	Brooklyn	Entire place	8	4	1	0	0	0	1	0	0	200
1633	New York	Brooklyn	Private room	2	1	0	0	1	0	1	0	0	45

MULTIPLE LINEAR REGRESSION

- Linear Relationships
- Least Squared Error
- Univariate Linear Regression
- Multiple Linear Regression
- Non-Linear Regression

EXAMPLE

*You are preparing to list a new property on **AirBnB**, and want to estimate (or predict) an appropriate price using the listing data below*

MODEL 1: Predict **price** (Y) based on **accommodation** (X_1):

$Y = 55.71 + (16.6 * X_1)$

MODEL 2: Predict **price** (Y) based on **accommodation** (X_1) and number of **bedrooms** (X_2):

$Y = 52.59 + (15.4 * X_1) + (5.1 * X_2)$

MODEL 3: Predict **price** (Y) based on **accommodation** (X_1), number of **bedrooms** (X_2), and **room type** (entire.place (X_3), hotel.room (X_4), private.room (X_5)):

$Y = 43.82 + (5.7 * X_1) + (5.1 * X_2) + (63.7 * X_3) + (65.4 * X_4) + (9.8 * X_5)$

MODEL 4: Predict **price** (Y) based on **accommodation** (X_1), number of **bedrooms** (X_2), **room type** (entire.place (X_3), hotel.room (X_4), private.room (X_5)), and **district** (manhattan (X_6), Brooklyn (X_7)):

$Y = 26.1 + (6.3 * X_1) + (6.7 * X_2) + (60.5 * X_3) + (54.5 * X_4) + (10.6 * X_5) + (28.5 * X_6) + 9.8 * X_7)$

MULTIPLE LINEAR REGRESSION

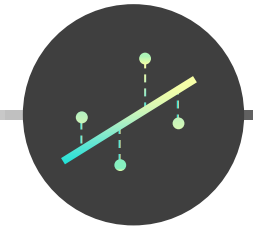
- Linear Relationships
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- Multiple Linear Regression
- Non-Linear Regression

EXAMPLE

*You are preparing to list a new property on **AirBnB**, and want to estimate (or predict) an appropriate price using the listing data below*

list_id	accommodates	bedrooms	entire.place	hotel.room	private.room	brooklyn	manhattan	price	Model 1	Model 2	Model 3	Model 4
3831	3	1	1	0	0	1	0	**$73**	$106	$104	$130	$122
5121	2	1	0	0	1	1	0	**$60**	$89	$88	$70	$66
5178	2	1	0	0	1	0	1	**$79**	$89	$88	$70	$84
5203	1	1	0	0	1	0	1	**$75**	$72	$73	$64	$78
5803	2	1	0	0	1	1	0	**$83**	$89	$88	$70	$66
6872	1	1	0	0	1	0	1	**$65**	$72	$73	$64	$78
6990	1	1	0	0	1	0	1	**$62**	$72	$73	$64	$78
7097	4	1	1	0	0	1	0	**$199**	$122	$119	$135	$128
7750	1	2	0	0	1	0	1	**$96**	$72	$78	$69	$85
8490	5	1	1	0	0	1	0	**$120**	$139	$135	$141	$134
9657	3	1	1	0	0	0	1	**$150**	$106	$104	$130	$140
9704	2	1	0	0	1	0	1	**$55**	$89	$88	$70	$84
10452	3	1	0	0	1	1	0	**$70**	$106	$104	$76	$72
10962	2	1	0	0	1	1	0	**$83**	$89	$88	$70	$66
11943	1	1	0	0	1	1	0	**$150**	$72	$73	$64	$59
12192	2	1	0	0	1	0	1	**$40**	$89	$88	$70	$84
12343	3	1	1	0	0	0	1	**$150**	$106	$104	$130	$140
12937	4	1	0	0	1	0	0	**$130**	$122	$119	$82	$68
12940	2	1	1	0	0	1	0	**$99**	$89	$88	$124	$116
13121	4	1	0	0	1	0	0	**$75**	$122	$119	$82	$68
13394	2	1	0	0	1	1	0	**$80**	$89	$88	$70	$66
13808	2	1	0	0	1	1	0	**$81**	$89	$88	$70	$66
14290	2	2	1	0	0	1	0	**$[illegible]**	[illegible]	$94	$1[illegible]	$17[illegible]

	Model 1	Model 2	Model 3	Model 4
Sum of Squared Error (SSE):	226,577	224,259	172,201	158,591



NON-LINEAR REGRESSION

- Linear Relationships
- Least Squared Error
- Univariate Linear Regression
- Multiple Linear Regression
- Non-Linear Regression

What if the relationship between variables ***isn't linear***?

Non-linear regression is used when variables have a non-linear relationship, but can be *transformed* to create a linear one

- This works exactly like linear regression, except you use *transformed* versions of your dependent or independent variables:

Dependent variable (DV) — *Coefficient/parameter*

$$y = \alpha + \beta * \ln(x) + \varepsilon$$

Error/residual

Y-intercept

Log-transformed *independent variable (IV)*

All we're really doing is transforming the data to create linear relationships between each IV and the DV, then applying a **standard linear regression model** using those transformed values

NON-LINEAR REGRESSION

- Linear Relationships
- Least Squared Error
- Univariate Linear Regression
- Multiple Linear Regression
- Non-Linear Regression

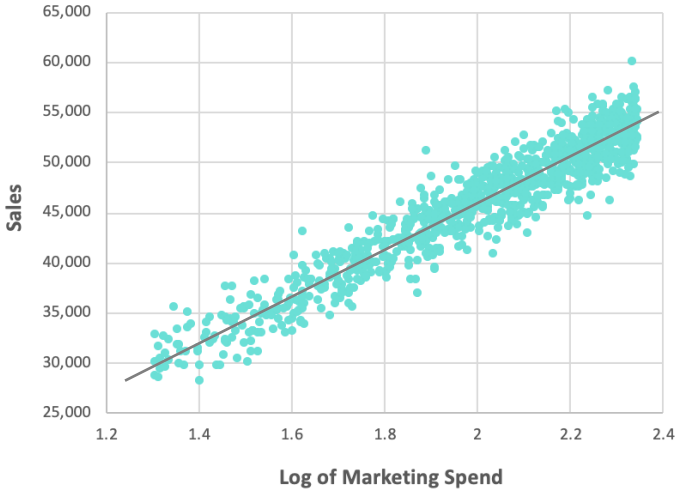
EXAMPLE #1

*You are predicting **Sales** (Y) using **Marketing Spend** (X). As you spend more on marketing, the impact on sales eventually begins to diminish.*



*The relationship between Sales and Marketing Spend is **non-linear (logarithmic)**...*

$$y = \alpha + \beta x + \varepsilon$$



*...but the relationship between Sales and the **log of Marketing Spend** is **linear!***

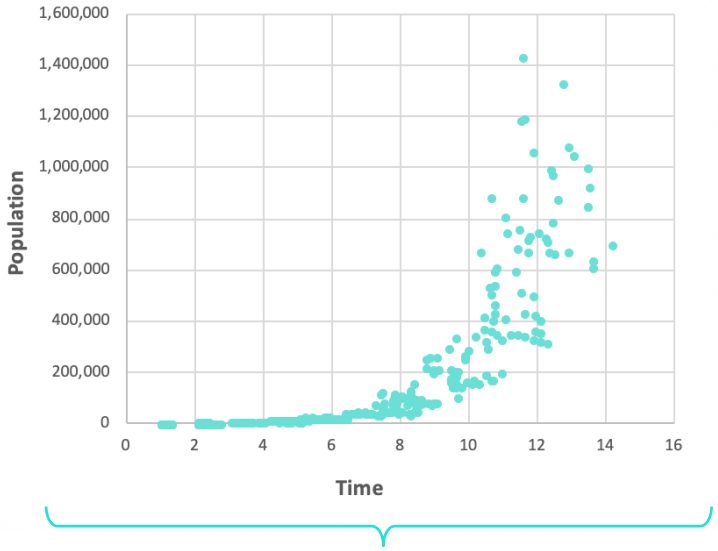
$$y = \alpha + \beta * \ln(x) + \varepsilon$$

NON-LINEAR REGRESSION

- Linear Relationships
- Least Squared Error
- Univariate Linear Regression
- Multiple Linear Regression
- Non-Linear Regression

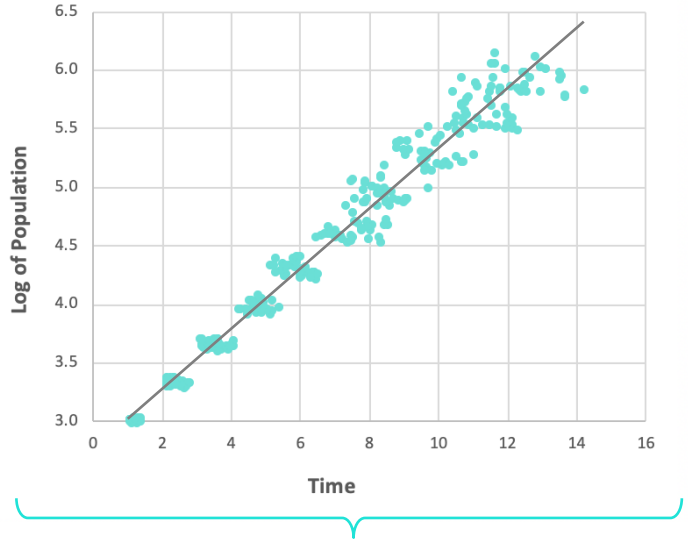
EXAMPLE #2

*You are predicting **population growth** (Y) over **time** (X) and notice an increasing rate of growth as the population size increases.*



*The relationship between Time and Population is **non-linear (exponential)**...*

$$y = \alpha + \beta x + \varepsilon$$



*...but the relationship between Time and the **log of Population** is **linear!***

$$\ln(y) = \alpha + \beta x + \varepsilon$$

NOTE: There are multiple ways to transform variables based on the type of relationship *(log, exponential, cubic, etc.)*, and multiple techniques to model them *(more on that later!)*

CASE STUDY: NON-LINEAR REGRESSION



THE SITUATION

You work as a Marketing Analyst for **Maven Marketing**, an international advertising agency based in London.



THE ASSIGNMENT

Your client has asked you to help set media budgets and estimate sales for an upcoming campaign. Using historical ad spend and revenue, your goal is to **build a regression model to help predict campaign performance**.



THE OBJECTIVES

1. Collect historical data for daily ad spend and revenue
2. Create a linear regression, and gauge the fit. Does it look accurate?
3. Transform the ad spend values to fit a logarithmic relationship, and re-run the model. How does this compare to the linear version?

MODEL DIAGNOSTICS

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MODEL DIAGNOSTICS



In this section we'll explore **common diagnostic metrics** used to evaluate regression models and ensure that predictions are stable and accurate

TOPICS WE'LL COVER:

- Sample Model Output
- R-Squared
- Mean Error Metrics
- Homoskedasticity
- F-Significance
- P-Values & T-Statistics
- Multicollinearity
- Variance Inflation Factor

GOALS FOR THIS SECTION:

- *Understand how to interpret regression model outputs*
- *Define common diagnostic metrics like R^2, MSE/MAE/MAPE, P-Values, F-statistics, etc.*
- *Explore the difference between homoskedasticity and heteroskedasticity in a linear regression model*
- *Understand how to identify and measure multicollinearity using variance inflation factor (VIF)*

SAMPLE MODEL OUTPUT

- Sample Model Output
- R-Squared
- Mean Error Metrics
- Homoskedasticity
- F-Significance
- P-Values & T-Statistics
- Multicollinearity
- Variance Inflation

```
Call:
lm(formula = price ~ accommodates + bedrooms + Entire.place +
    host_has_profile_pic + Hotel.room + Private.room, data = d.2)

Residuals:
     Min       1Q   Median       3Q      Max
-135.057  -27.091   -8.091   20.909  194.388

Coefficients:
                    Estimate Std. Error t value P
(Intercept)          47.9501     4.4393  10.801
accommodates          5.7166     0.2238  25.540
bedrooms              5.0933     0.5478   9.298
Entire.place         63.6845     1.6472  38.663
host_has_profile_pic -4.1485     4.1430  -1.001
Hotel.room           65.4399     3.5595  18.385
Private.room          9.7627     1.6065   6.077 1
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘

Residual stand
Multiple R-squ
F-statistic:
```

	A	B	C	D	E	F	G	H	I
1	SUMMARY OUTPUT								
2									
3	*Regression Statistics*								
4	Multiple R	0.645315							
5	R Square	0.416431							
6	Adjusted R Square	0.416314							
7	Standard Error	40.31352							
8	Observations	29797							
9									
10	ANOVA								
11		*df*	*SS*	*MS*	*F*	*Significance F*			
12	Regression	6	34548002	5758000	3542.993	0			

```
                            OLS Regression Results
==============================================================================
Dep. Variable:                      y   R-squared:                       0.933
Model:                            OLS   Adj. R-squared:                  0.928
Method:                 Least Squares   F-statistic:                     211.8
Date:                Tue, 02 Feb 2021   Prob (F-statistic):           6.30e-27
Time:                        06:52:01   Log-Likelihood:                -34.438
No. Observations:                  50   AIC:                             76.88
Df Residuals:                      46   BIC:                             84.52
Df Model:                           3
Covariance Type:            nonrobust
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
x1             0.4687      0.026     17.751      0.000       0.416       0.522
x2             0.4836      0.104      4.659      0.000       0.275       0.693
x3            -0.0174      0.002     -7.507      0.000      -0.022      -0.013
const          5.2058      0.171     30.405      0.000       4.861       5.550
```

Lower 95%	*Upper 95%*	*ower 95.0%*	*Upper 95.0%*
39.24885206	56.65141	39.24885	56.65141
6.613795082	12.91157	6.613795	12.91157
58.46311957	72.41661	58.46312	72.41661
-12.26887582	3.971917	-12.2689	3.971917
60.45596468	66.91296	60.45596	66.91296
4.019682825	6.166946	4.019683	6.166946
5.277919726	6.155365	5.27792	6.155365



R-SQUARED

Sample Model Output

R-Squared

Mean Error Metrics

Homoskedasticity

F-Significance

P-Values & T-Statistics

Multicollinearity

Variance Inflation

R-Squared measures how well your model explains the variance in the dependent variable you are predicting

- The higher the R-Squared, the "better" your model predicts variance in the DV and the more confident you can be in the accuracy of your predictions
- **Adjusted R-Squared** is often used as it "penalizes" the R-squared value based on the number of variables included in the model

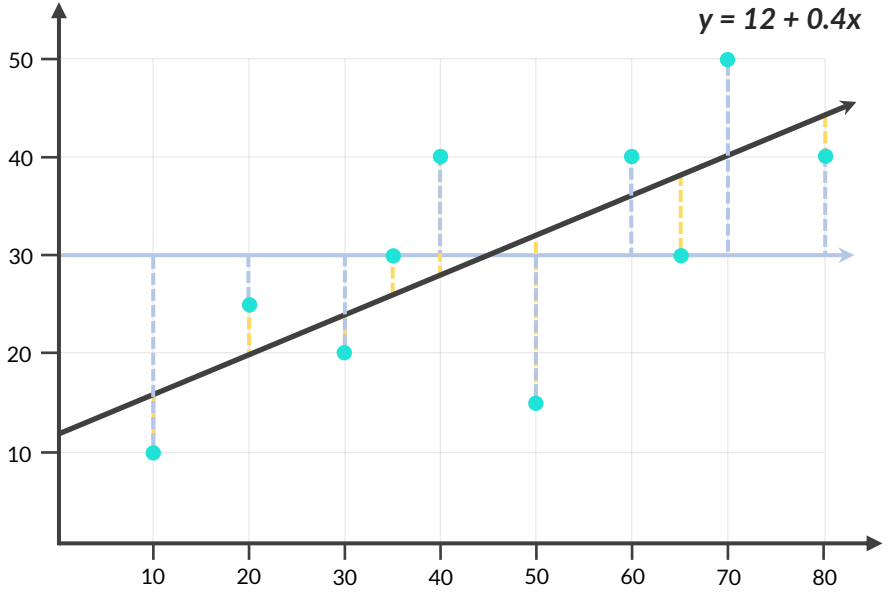
$$R^2 = 1 - \frac{SSE}{TSS} = \frac{\textit{Sum of Squared Error}}{\textit{Total Sum of Squares}} = \frac{\sum_i (y_i - prediction_i)^2}{\sum_i (y_i - \bar{y})^2}$$

*Total distance between predicted and actual values, squared (aka **squared error**)*

*Total distance between each y value and the mean, squared (basically **variance**, without dividing by n)*

R-SQUARED EXAMPLE

- Sample Model Output
- R-Squared
- Mean Error Metrics
- Homoskedasticity
- F-Significance
- P-Values & T-Statistics
- Multicollinearity
- Variance Inflation



x	y	prediction	(y–prediction)²	$\bar{y}$	(y–$\bar{y}$)²
10	10	16	36	30	400
20	25	20	25	30	25
30	20	24	16	30	100
35	30	26	16	30	0
40	40	28	144	30	100
50	15	32	289	30	225
60	40	36	16	30	100
65	30	38	64	30	0
70	50	40	100	30	400
80	40	44	16	30	100
			= 722 SSE		= 1,450 TSS

$$R^2 = 1 - \frac{SSE}{TSS} = \frac{\sum_i (y_i - prediction_i)^2}{\sum_i (y_i - \bar{y})^2}$$

$$1 - \frac{722}{1{,}450} = 0.502$$



MEAN ERROR METRICS

Sample Model Output

R-Squared

Mean Error Metrics

Homoskedasticity

F-Significance

P-Values & T-Statistics

Multicollinearity

Variance Inflation

Mean error metrics measure how well your regression model *predicts*, as opposed to how well it explains variance (like R-Squared)

- There are many variations, but the most common ones are **Mean Squared Error** (MSE), **Mean Absolute Error** (MAE) and **Mean Absolute Percentage Error** (MAPE)
- These metrics provide "standards" which can be used to compare predictive accuracy across multiple regression models

MSE

$$\frac{\sum_i (y_i - prediction_i)^2}{n}$$

*Average of the **squared** distance between actual & predicted values*

MAE

$$\frac{\sum_i |y_i - prediction_i|}{n}$$

*Average of the **absolute** distance between actual & predicted values*

MAPE

$$\frac{\sum_i \frac{|y_i - prediction_i|}{y_i}}{n}$$

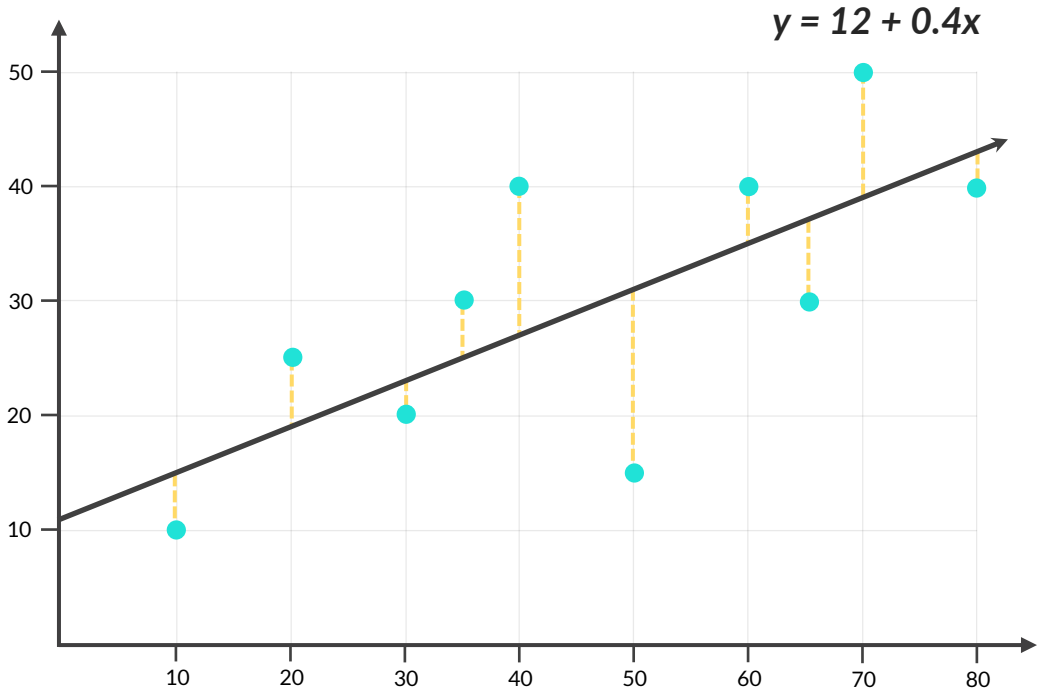
*Mean Absolute Error, converted to a **percentage***



Mean error metrics can be used to evaluate regression models just like performance metrics like **accuracy, precision and recall** can be used to evaluate classification models

MSE EXAMPLE

- Sample Model Output
- R-Squared
- **Mean Error Metrics**
- Homoskedasticity
- F-Significance
- P-Values & T-Statistics
- Multicollinearity
- Variance Inflation



X	Y (actual)	Y (line)	Error	Sq. Error
10	10	16	6	**36**
20	25	20	-5	**25**
30	20	24	4	**16**
35	30	26	-4	**16**
40	40	28	-12	**144**
50	15	32	17	**289**
60	40	36	-4	**16**
65	30	38	8	**64**
70	50	40	-10	**100**
80	40	44	4	**16**

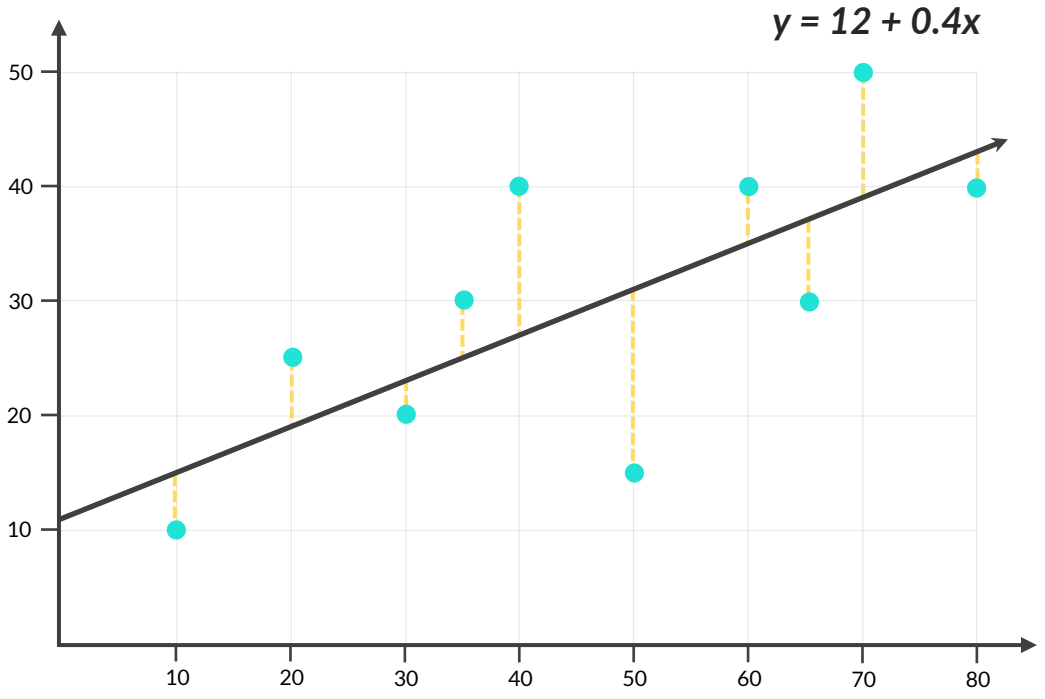
SUM OF SQUARED ERROR: = 722

MSE

$$\frac{\sum_i (y_i - prediction_i)^2}{n} = \frac{722}{10} = 72.2$$

MAE EXAMPLE

- Sample Model Output
- R-Squared
- **Mean Error Metrics**
- Homoskedasticity
- F-Significance
- P-Values & T-Statistics
- Multicollinearity
- Variance Inflation



X	Y (actual)	Y (line)	Error	Abs Error
10	10	16	6	**6**
20	25	20	-5	**5**
30	20	24	4	**4**
35	30	26	-4	**4**
40	40	28	-12	**12**
50	15	32	17	**17**
60	40	36	-4	**4**
65	30	38	8	**8**
70	50	40	-10	**10**
80	40	44	4	**4**
				=

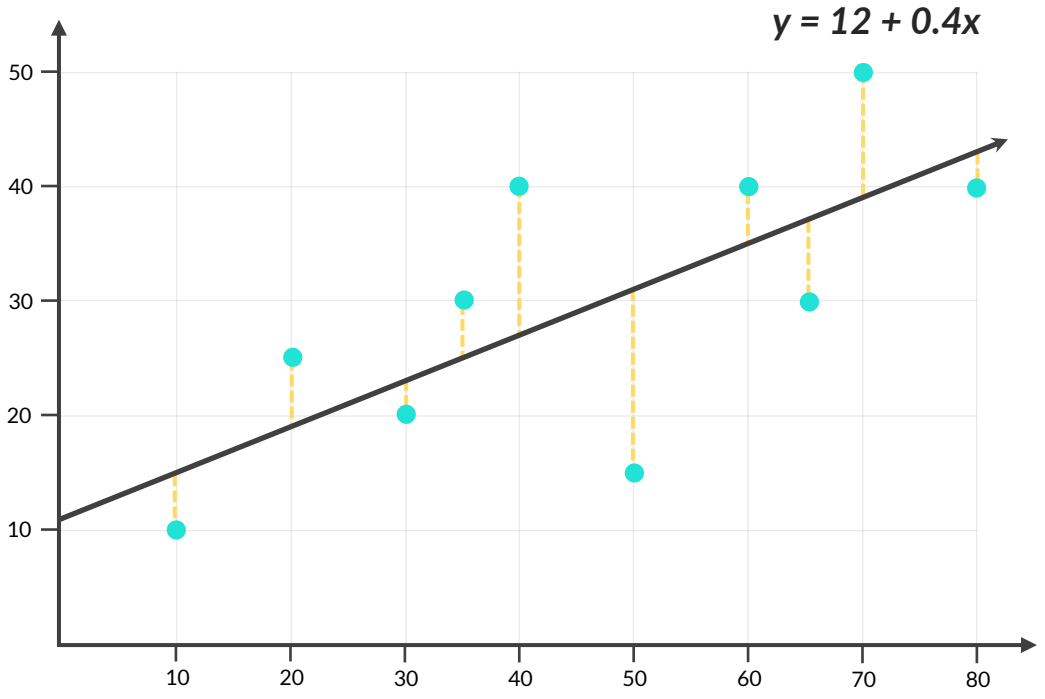
SUM OF ABSOLUTE ERROR: 74

MAE

$$\frac{\sum_i |y_i - prediction_i|}{n} = \frac{74}{10} = 7.4$$

MAPE EXAMPLE

- Sample Model Output
- R-Squared
- **Mean Error Metrics**
- Homoskedasticity
- F-Significance
- P-Values & T-Statistics
- Multicollinearity
- Variance Inflation



X	Y (actual)	Y (line)	Error	Abs Error	Abs % Error
10	10	16	6	6	0.6
20	25	20	-5	5	0.2
30	20	24	4	4	0.2
35	30	26	-4	4	0.133
40	40	28	-12	12	0.3
50	15	32	17	17	1.133
60	40	36	-4	4	0.1
65	30	38	8	8	0.267
70	50	40	-10	10	0.2
80	40	44	4	4	0.1

SUM OF ABSOLUTE % ERROR: 3.233

MAPE

$$\frac{\sum_i \frac{|y_i - prediction_i|}{y_i}}{n} = \frac{3.233}{10} = 32.33\%$$

MEAN ERROR METRICS

Sample Model Output

R-Squared

Mean Error Metrics

Homoskedasticity

F-Significance

P-Values & T-Statistics

Multicollinearity

Variance Inflation

When should I use each type of error metric?

- **Mean Squared Error** (MSE) is particularly useful when outliers or extreme values are important to predict
- **Mean Absolute Error** (MAE) is useful if you want to minimize the impact of outliers on model selection
- **Mean Absolute Percentage Error** (MAPE) is useful when your DV is on a very large scale, or if you want to compare models on a more intuitive scale



PRO TIP: In general we recommend considering **all of them**, since they can be calculated instantly and each provide helpful context into model performance

HOMOSKEDASTICITY

Sample Model Output

R-Squared

Mean Error Metrics

Homoskedasticity

F-Significance

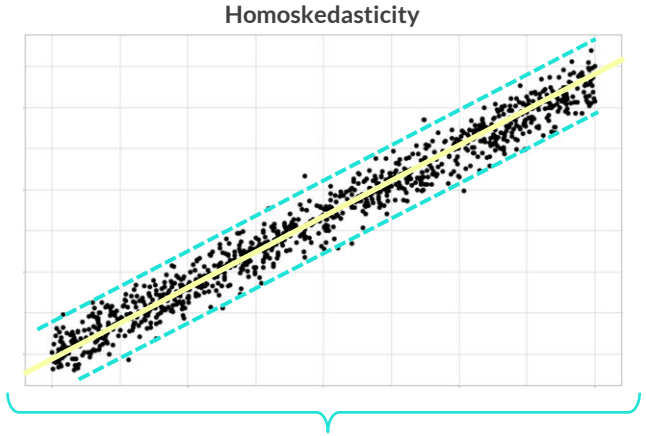
P-Values & T-Statistics

Multicollinearity

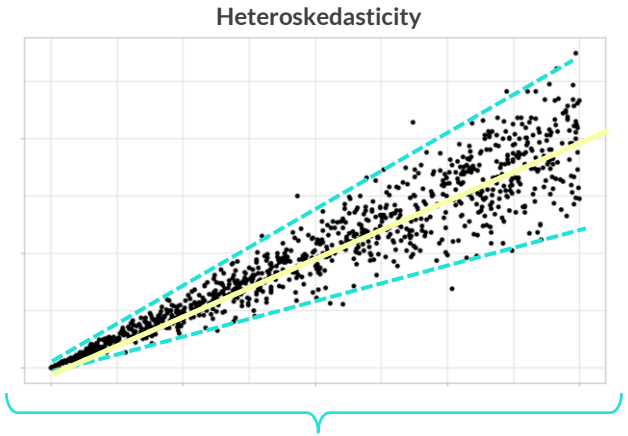
Variance Inflation

Homoskedasticity is a term used to describe a model with consistent "scatter", or variance in residuals, across all IV values

- In order to make accurate predictions across the full range of IV values, models should have normally distributed residuals (errors) with a zero mean
- **Heteroskedasticity** describes a model with *inconsistent* residual variance, meaning that it predicts poorly for certain IV values and will fail to generalize



Residuals are consistent over the entire IV range



Residuals increase at higher IV values, ***indicating that there is some variance that the IVs are unable to explain***

Breusch-Pagan tests can report a formal calculation of Heteroskedasticity, but usually a simple visual check is enough

NULL HYPOTHESIS



NULL HYPOTHESIS (H_0) [null hy·poth·e·sis]

noun

1. ~~In a statistical test, the hypothesis that there is no significant difference between specified populations, any observed difference being due to sampling or experimental error *~~

The hypothesis that your model is garbage

Our goal is to **reject the null hypothesis** and prove *(with a high level of confidence)* that our model can produce accurate, statistically significant predictions and not just random outputs

*Oxford Dictionary



F-STATISTIC & F-SIGNIFICANCE

Sample Model Output

R-Squared

Mean Error Metrics

Homoskedasticity

F-Significance

P-Values & T-Statistics

Multicollinearity

Variance Inflation

The **F-Statistic** and associated **P-Value** (aka **F-Significance**) help us understand the predictive power of the model *as a whole*

- **F-Significance** is technically defined as *"the probability that the null hypothesis cannot be rejected"*, which can be interpreted as **the probability that your model predicts poorly**
- The smaller the F-Significance, the more useful your regression is for prediction
- **NOTE:** It's common practice to use a P-Value of **.05** (aka **95%**) as a threshold to determine if a model is "statistically significant", or valid for prediction



PRO TIP: F-Significance should be the ***first thing*** you check when you evaluate a regression model; if it's above your threshold, you may need more training, if it's below your threshold, move on to coefficient-level significance *(up next!)*

T-STATISTICS & P-VALUES

- Sample Model Output
- R-Squared
- Mean Error Metrics
- Homoskedasticity
- F-Significance
- **P-Values & T-Statistics**
- Multicollinearity
- Variance Inflation

T-Statistics and their associated **P-Values** help us understand the predictive power of the *individual model coefficients*

```
Call:
lm(formula = price ~ accommodates + bedrooms + Entire.place +
    host_has_profile_pic + Hotel.room + Private.room, data = d.2)

Residuals:
     Min       1Q   Median       3Q      Max
-135.057  -27.091   -8.091   20.909  194.388

Coefficients:
                   Estimate Std. Error t value Pr(>|t|)
(Intercept)         47.9501     4.4393  10.801  < 2e-16 ***
accommodates         5.7166     0.2238  25.540  < 2e-16 ***
bedrooms             5.0933     0.5478   9.298  < 2e-16 ***
Entire.place        63.6845     1.6472  38.663  < 2e-16 ***
host_has_profile_pic -4.1485     4.1430  -1.001    0.317
Hotel.room          65.4399     3.5595  18.385  < 2e-16 ***
Private.room         9.7627     1.6065   6.077 1.24e-09 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 40.31 on 29790 degrees of freedom
Multiple R-squared:  0.4164,    Adjusted R-squared:  0.4163
F-statistic:  3543 on 6 and 29790 DF,  p-value: < 2.2e-16
```

T-Statistics tell us the degree to which we can “trust” the coefficient estimates

- Calculated by dividing the coefficient by its standard error
- Primarily used as a stepping-stone to calculate P-Values, which are easier to interpret and more commonly used for diagnostics

P-Values tell us the probability that the coefficient is meaningless, statistically speaking

- The smaller the P-value, the more confident you can be that the coefficient is valid *(not 0)*
- **Statistical significance** is calculated as **(1 – P)**

P-Value	Statistical Significance
0.001	99.9%***
0.01	99%**
0.05	95%*



MULTICOLLINEARITY

- Sample Model Output
- R-Squared
- Mean Error Metrics
- Homoskedasticity
- F-Significance
- P-Values & T-Statistics
- Multicollinearity
- Variance Inflation

Multicollinearity occurs when two or more independent variables are highly correlated, leading to untrustworthy model coefficients

- Correlation means that one IV can be used to predict another *(i.e.* ***height*** *and* ***weight****)*, leading to *many* combinations of coefficients that predict equally well
- This leads to unreliable coefficient estimates, and means that your model will fail to generalize when applied to non-training data



How do I measure multicollinearity, and what can I do about it?

- **Variance Inflation Factor** (VIF) can help you quantify the degree of multicollinearity, and determine which IVs to exclude from the model

VARIANCE INFLATION FACTOR

Sample Model Output

R-Squared

Mean Error Metrics

Homoskedasticity

F-Significance

P-Values & T-Statistics

Multicollinearity

Variance Inflation

To calculate **Variance Inflation Factor** (VIF) you treat each individual IV as the *dependent* variable, and use the R^2 value to measure how well you can predict them using the other IVs in the model

$$Y = \alpha + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \ldots + \beta_n x_n + \varepsilon$$

$$x_1 = \alpha + \beta_2 x_2 + \beta_3 x_3 + \ldots + \beta_n x_n + \varepsilon \longrightarrow \frac{1}{1-R^2} \longrightarrow \text{VIF for } X_1$$

$$x_2 = \alpha + \beta_1 x_1 + \beta_3 x_3 + \ldots + \beta_n x_n + \varepsilon \longrightarrow \frac{1}{1-R^2} \longrightarrow \text{VIF for } X_2$$



PRO TIP: As a rule of thumb, a **VIF >10** indicates that multicollinearity is a problem, and that one or more IVs is redundant and should be removed from the model

VARIANCE INFLATION FACTOR

- Sample Model Output
- R-Squared
- Mean Error Metrics
- Homoskedasticity
- F-Significance
- P-Values & T-Statistics
- Multicollinearity
- Variance Inflation

EXAMPLE

*You are predicting the **price of an AirBnB listing**, and notice the following results after calculating the VIF for each independent variable:*

accommodates	bedrooms	Entire.place	Hotel.room	Private.room	Manhattan	Brooklyn
2.144259	1.720523	12.304416	1.246123	11.791492	1.988467	1.949623

***Entire.place** and **Private.room** produce high VIF values, since they essentially measure the same thing; if a listing isn't a private room, there's a high probability that it's an entire place (and vice versa)*

SOLUTION: Pick one high-VIF variable to remove *(arbitrarily)*, re-run the model, and recalculate the VIF values to see if multicollinearity is gone

accommodates	bedrooms	Entire.place	Hotel.room	Manhattan	Brooklyn
2.143976	1.720513	1.473047	1.013757	1.988459	1.949182

Adios multicollinearity!



PRO TIP: Use a frequency table to confirm correlation!

Entire.place \ Private.room	NO	YES
NO	816	15,815
YES	13,166	0

RECAP: SAMPLE MODEL OUTPUT

- Sample Model Output
- R-Squared
- Mean Error Metrics
- Homoskedasticity
- F-Significance
- P-Values & T-Statistics
- Multicollinearity
- Variance Inflation

Formula for the regression (variables & data set)

```
Call:
lm(formula = price ~ accommodates + bedrooms + Entire.place +
    host_has_profile_pic + Hotel.room + Private.room, data = d.2)

Residuals:
     Min       1Q   Median       3Q      Max
-135.057  -27.091   -8.091   20.909  194.388

Coefficients:
                    Estimate Std. Error t value Pr(>|t|)
(Intercept)          47.9501     4.4393  10.801  < 2e-16 ***
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bedrooms              5.0933     0.5478   9.298  < 2e-16 ***
Entire.place         63.6845     1.6472  38.663  < 2e-16 ***
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Hotel.room           65.4399     3.5595  18.385  < 2e-16 ***
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---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 40.31 on 29790 degrees of freedom
Multiple R-squared:  0.4164,    Adjusted R-squared:  0.4163
F-statistic:  3543 on 6 and 29790 DF,  p-value: < 2.2e-16
```

Profile of ***Residuals/Errors***

Y-Intercept *&* ***IV coefficients***

R² *and* ***Adjusted R²***

Coefficient ***P-Values***

Coefficient Standard Errors & ***T-Values***

F-Statistic *and* ***P-Value*** *(aka* ***F-Significance****)*

TIME-SERIES FORECASTING

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TIME-SERIES FORECASTING



In this section we'll explore common **time-series forecasting** techniques, which use regression models to predict future values based on seasonality and trends

TOPICS WE'LL COVER:

- Forecasting 101
- Seasonality
- Linear Trending
- Smoothing
- Non-Linear Trends
- Intervention Analysis

GOALS FOR THIS SECTION:

- *Understand when, why and how to use common time-series forecasting techniques*
- *Learn how to identify and quantify seasonality using auto correlation and one-hot encoding*
- *Explore common models for non-linear forecasting (like ADBUDG and Gompertz)*
- *Apply forecasting techniques to analyze the impact of key business decisions (aka "interventions")*

FORECASTING 101

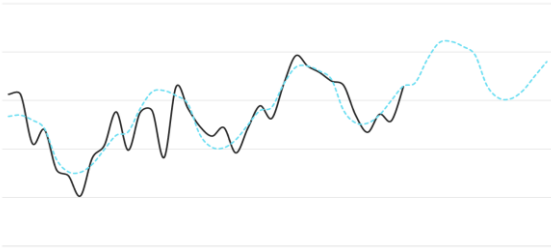
- Forecasting 101
- Seasonality
- Linear Trending
- Smoothing
- Non-Linear Trends
- Intervention Analysis

Time-series forecasting is all about predicting *future values* of a single, numeric dependent variable

- Forecasting works just like any other regression model, except your data must contain multiple observations of your DV over time (aka **time-series** data)
- Time-series forecast models look for patterns in the observed data – like **seasonality** or **linear/non-linear trends** – to accurately predict future values

Common Examples:

- Forecasting revenue for the next fiscal year
- Predicting website traffic growth over time
- Estimating sales for an new product launch



SEASONALITY

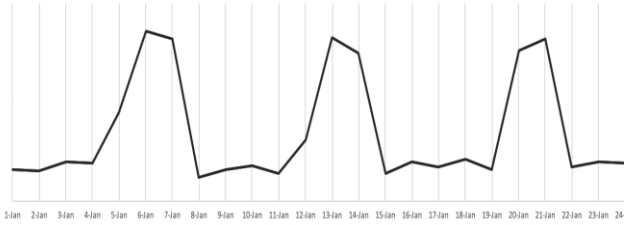
- Forecasting 101
- Seasonality
- Linear Trending
- Smoothing
- Non-Linear Trends
- Intervention Analysis

Seasonality is a repeatable, predictable pattern that a dependent variable may follow over time

- Seasonality often aligns with specific time periods *(calendar months, fiscal periods, days of the week, hours of the day, etc.)* but that isn't always the case
- We can identify seasonal patterns using an **Auto Correlation Function (ACF)**, then apply that seasonality to forecasts using techniques like **one-hot encoding** or **moving averages** *(more on that soon!)*

Common examples:

- Website traffic by hour of the day
- Seasonal product sales
- Airline ticket prices around major holidays

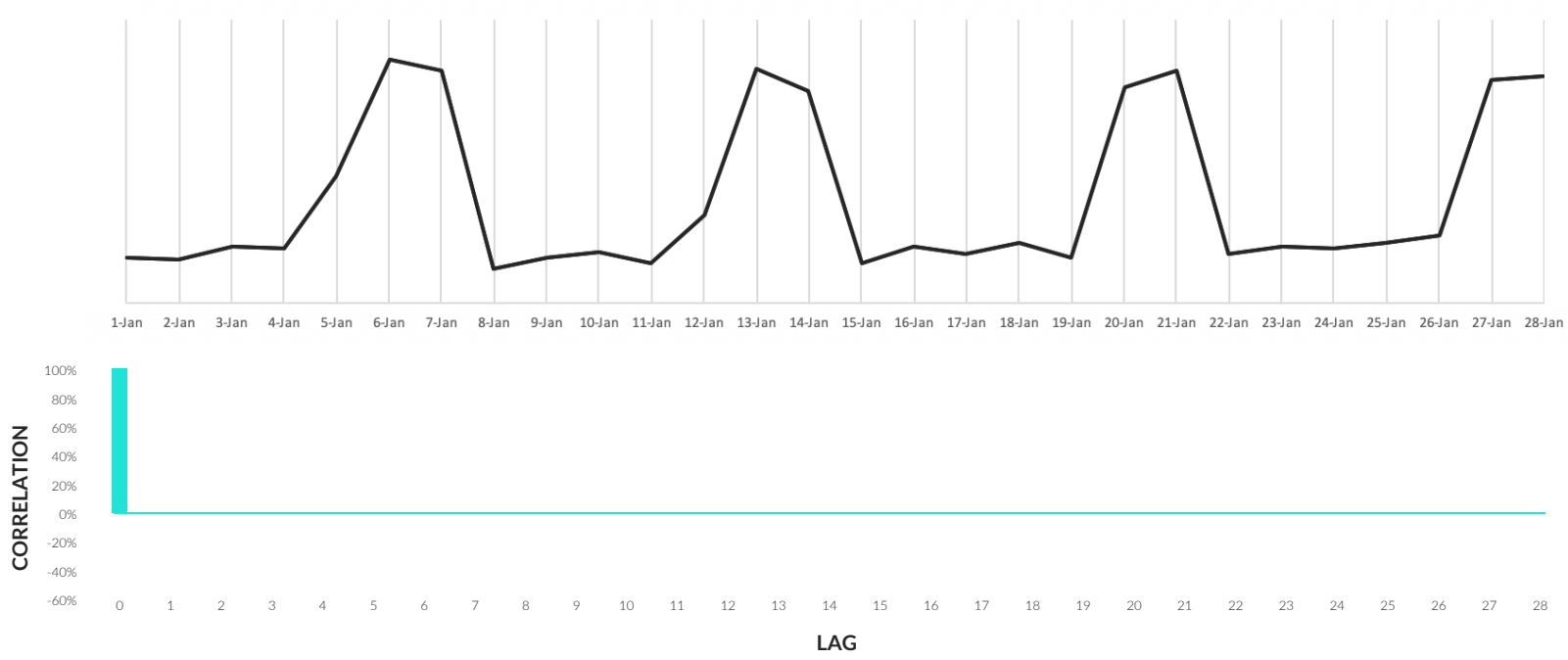


AUTO CORRELATION FUNCTION

- Forecasting 101
- Seasonality
- Linear Trending
- Smoothing
- Non-Linear Trends
- Intervention Analysis

ACF essentially involves calculating the correlation between time-series data and lagged versions of itself, then plotting those correlations

- This allows you to visualize which lags are highly correlated with the original data, and reveals the length (or period) of the seasonal trend

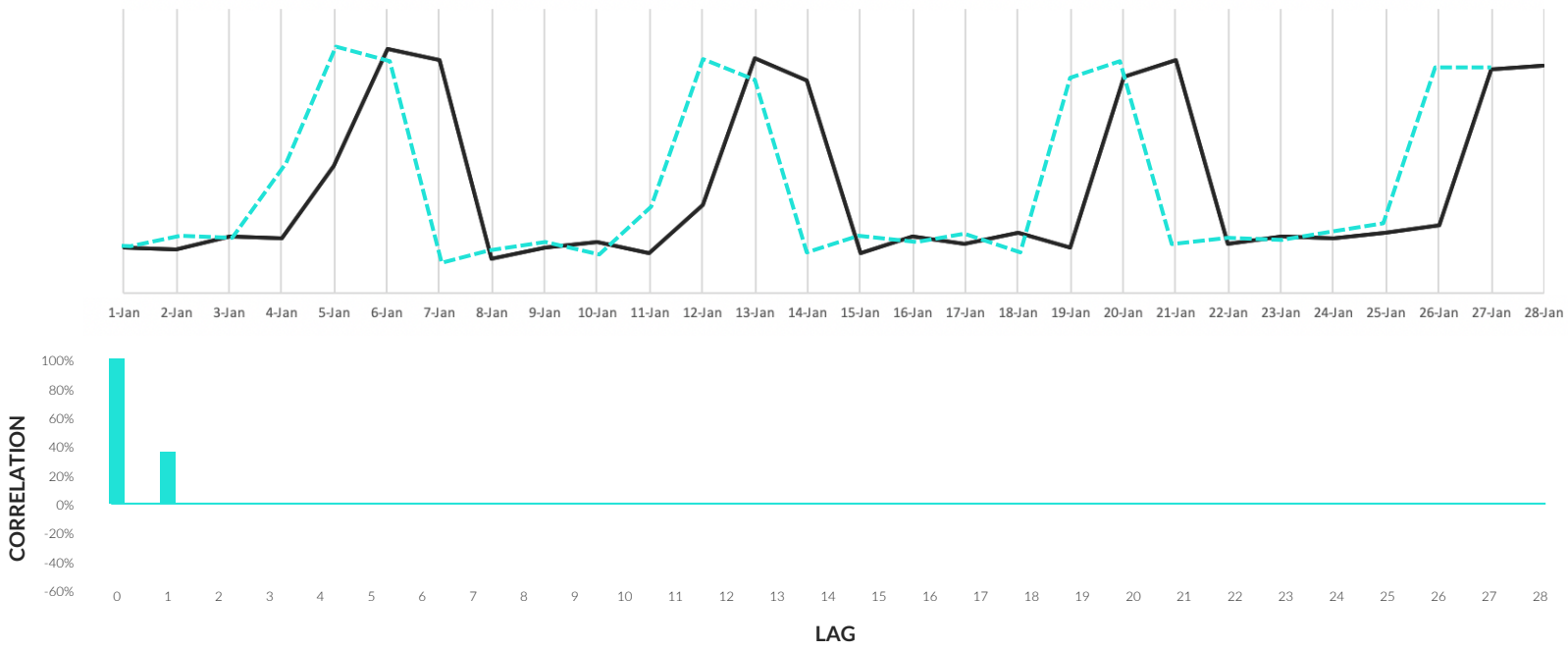


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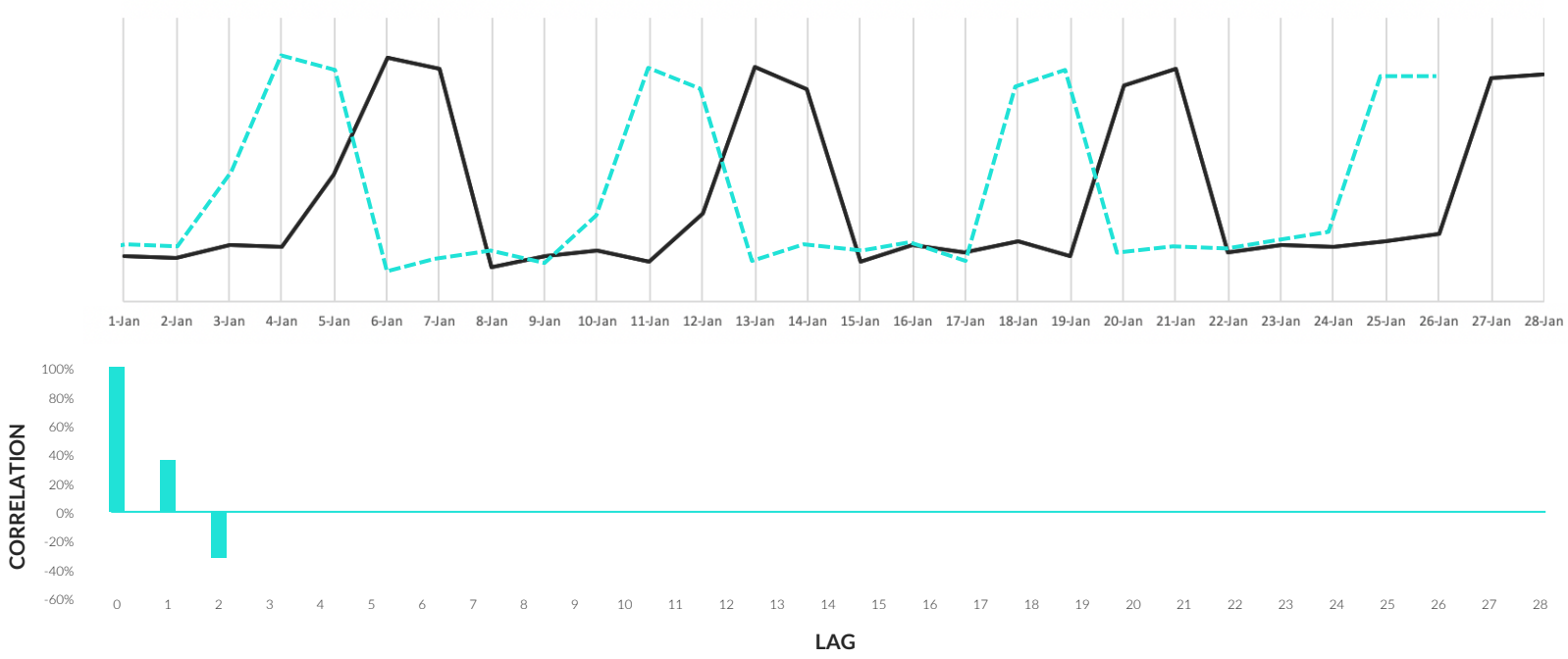


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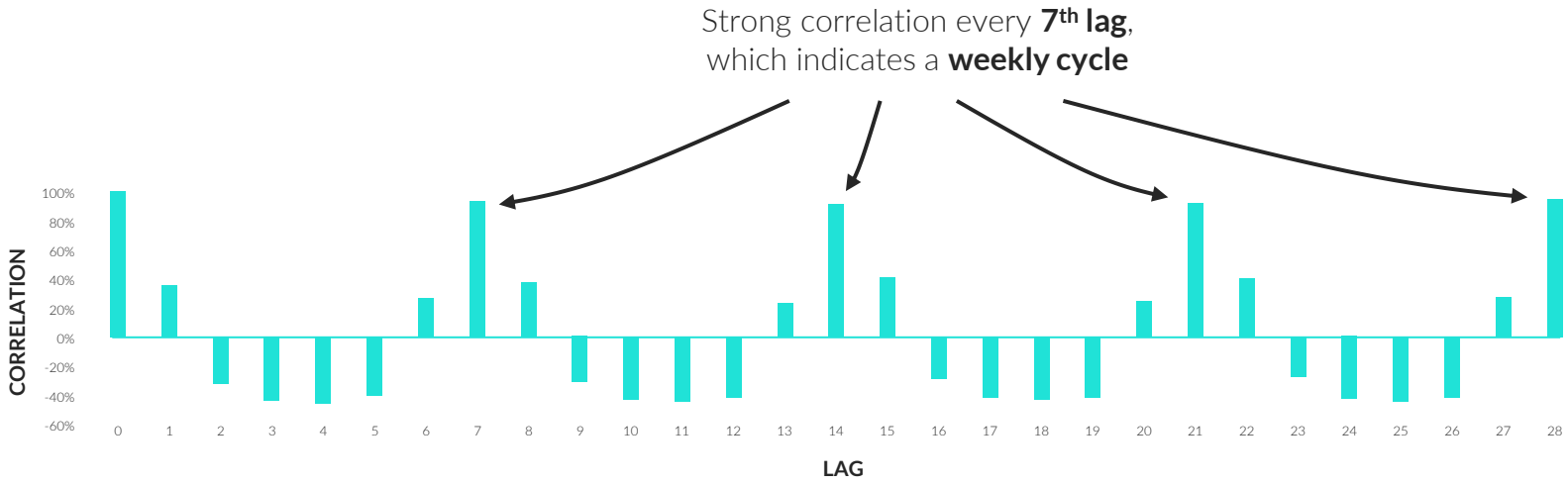


AUTO CORRELATION FUNCTION

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CASE STUDY: AUTO CORRELATION



THE **SITUATION**

You are a Business Intelligence Analyst for **Accucorp Accounting**, a national firm specializing in tax preparation services.



THE **ASSIGNMENT**

The marketing team is hoping to better understand customer behavior, and would like to know if online searches for "tax"-related keywords follow a seasonal pattern. Your goal is to **analyze monthly search data and use auto correlation to test for seasonality**.



THE **OBJECTIVES**

1. Collect monthly search volume for relevant tax keywords
2. Create 36 monthly lags, and calculate the correlation between each lag and the original values
3. Plot the correlations and determine the period of seasonality, if it exists *(monthly, quarterly, annual, etc.)*

SEASONALITY: ONE-HOT ENCODING

Forecasting 101

Seasonality

Linear Trending

Smoothing

Non-Linear Trends

Intervention Analysis

Once you've identified a seasonal pattern, you can use **one-hot encoding** to create independent variables which capture the effect of those time periods *(days, weeks, months, quarters, etc.)*

- **NOTE:** When you use one-hot encoding, you **must exclude one of the options** rather than encoding all of them *(it doesn't matter which one you exclude)*
- Consider the equation **A+B=5**; there are an *infinite* combination of A & B values that can solve it. One-hot encoding all options creates a similar problem for regression

Quarter ID	Revenue	Q1	Q2	Q3	Q4
1	$1,300,050	1	0	0	0
2	$11,233,310	0	1	0	0
3	$1,112,050	0	0	1	0
4	$1,582,077	0	0	0	1



PRO TIP: If your data contains multiple seasonal patterns *(i.e. hour of day + day of week)*, include both dimensions in the model as one-hot encoded independent variables

CASE STUDY: ONE-HOT ENCODING



THE **SITUATION**

You're are a Senior Analyst for **Weather Trends**, a Brazilian weather station boasting the longest and most accurate forecasts in the biz.



THE **ASSIGNMENT**

You've been asked to help prepare temperature forecasts for the upcoming year. To do this, you'll need to analyze ~5 years of historical data from Rio de Janeiro, and use regression to **predict monthly average temperatures**.



THE **OBJECTIVES**

1. Collect historical average monthly temperatures from Rio de Janeiro
2. Create independent variables for each month using one-hot encoding
3. Build a regression model to produce a forecast for the following year



LINEAR TRENDING

Forecasting 101

Seasonality

Linear Trending

Smoothing

Non-Linear Trends

Intervention Analysis



What if the data includes both seasonality and a linear trend?

Trend describes an overarching direction or movement in a time series, not counting seasonality

- Trends are often linear (up/down), but can be non-linear as well *(more on that later!)*
- To account for linear trending in a regression, you can include a **time step** IV; this is simply an index value that starts at 1 and increments with each time period
- If the time step coefficient isn't statistically significant, it means you don't have a meaningful linear trend



PRO TIP: It's common for time-series models to include trending AND seasonality; in this case, use a combination of **one-hot encoding** and **time step variables** to account for both!

LINEAR TRENDING

- Forecasting 101
- Seasonality
- **Linear Trending**
- Smoothing
- Non-Linear Trends
- Intervention Analysis

What if the data includes both seasonality and a linear trend?

Trend describes an overarching direction or movement in a time series, not counting seasonality

Month	Revenue	T-Step	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov
Jan	$1,300,050	1	1	0	0	0	0	0	0	0	0	0	0
Feb	$1,233,310	2	0	1	0	0	0	0	0	0	0	0	0
Mar	$1,112,050	3	0	0	1	0	0	0	0	0	0	0	0
Apr	$1,582,077	4	0	0	0	1	0	0	0	0	0	0	0
May	$1,776,392	5	0	0	0	0	1	0	0	0	0	0	0
Jun	$2,110,201	6	0	0	0	0	0	1	0	0	0	0	0
Jul	$1,928,290	7	0	0	0	0	0	0	1	0	0	0	0
Aug	$2,250,293	8	0	0	0	0	0	0	0	1	0	0	0
Sep	$2,120,050	9	0	0	0	0	0	0	0	0	1	0	0
Oct	$2,479,293	10	0	0	0	0	0	0	0	0	0	1	0
Nov	$2,560,203	11	0	0	0	0	0	0	0	0	0	0	1
Dec	$2,739,022	12	0	0	0	0	0	0	0	0	0	0	0

CASE STUDY: SEASONALITY + TREND



THE SITUATION

You are a Business Intelligence Analyst for **Maven Muscles**, a large national chain of fitness centers.



THE ASSIGNMENT

The Analytics Director needs your help **building a monthly revenue forecast for the upcoming year**. Memberships follow a clear seasonal pattern, and revenue has been steadily rising as the gym continues to open new locations.



THE OBJECTIVES

1. Collect historical monthly revenue data
2. Use one-hot encoding and a time-step variable to account for both seasonality and linear trending
3. Fit a regression model to forecast revenue for the upcoming year

SMOOTHING

- Forecasting 101
- Seasonality
- Linear Trending
- Smoothing
- Non-Linear Trends
- Intervention Analysis

What if my data is so noisy that I can't tell if a trend exists?

If your data is highly volatile, **smoothing techniques** can be used to reveal underlying patterns or trends

- Common techniques like **moving averages** or **weighted smoothing** remove random variation and "noise" from the data to help expose seasonality or trends
- If the volatility is truly just noise *(and something we want our model to ignore)*, averaging or weighting the values around each time step can help us "smooth" the data and produce more accurate forecasts



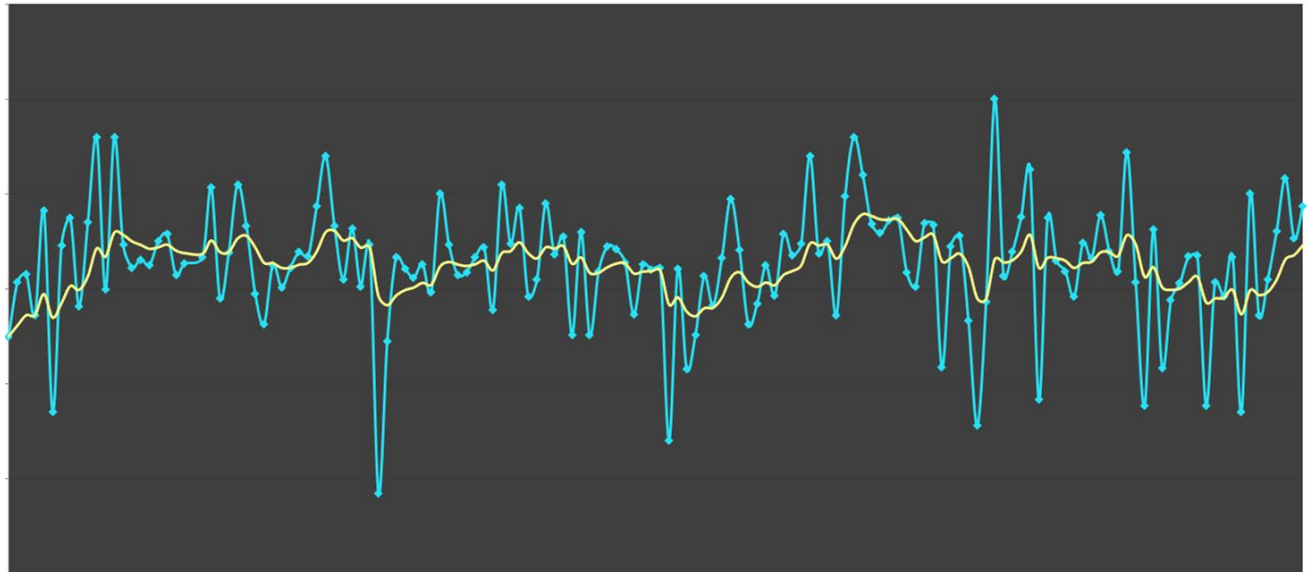
PRO TIP: Smoothing is a great way to expose patterns and trends that otherwise might be tough to see; make this part of your data profiling process!

SMOOTHING

- Forecasting 101
- Seasonality
- Linear Trending
- Smoothing
- Non-Linear Trends
- Intervention Analysis

What if my data is so noisy that I can't tell if a trend exists?

If your data is highly volatile, **smoothing techniques** can be used to reveal underlying patterns or trends



CASE STUDY: SMOOTHING



THE SITUATION

You've just been hired as an analytics consultant for **Maven Motel**, a struggling national motel chain.



THE ASSIGNMENT

Management has been taking steps to improve guest satisfaction, and has asked you to analyze daily data to determine how ratings are trending. Your task is to **use a moving average calculation to discern if an underlying trend is present**.



THE OBJECTIVES

1. Collect daily average guest ratings for the motel. Do you see any clear patterns or trends?
2. Calculate a moving average, and compare various windows from 1-12 weeks
3. Determine if an underlying trend is present. How would you describe it?

NON-LINEAR TRENDS

Forecasting 101

Seasonality

Linear Trending

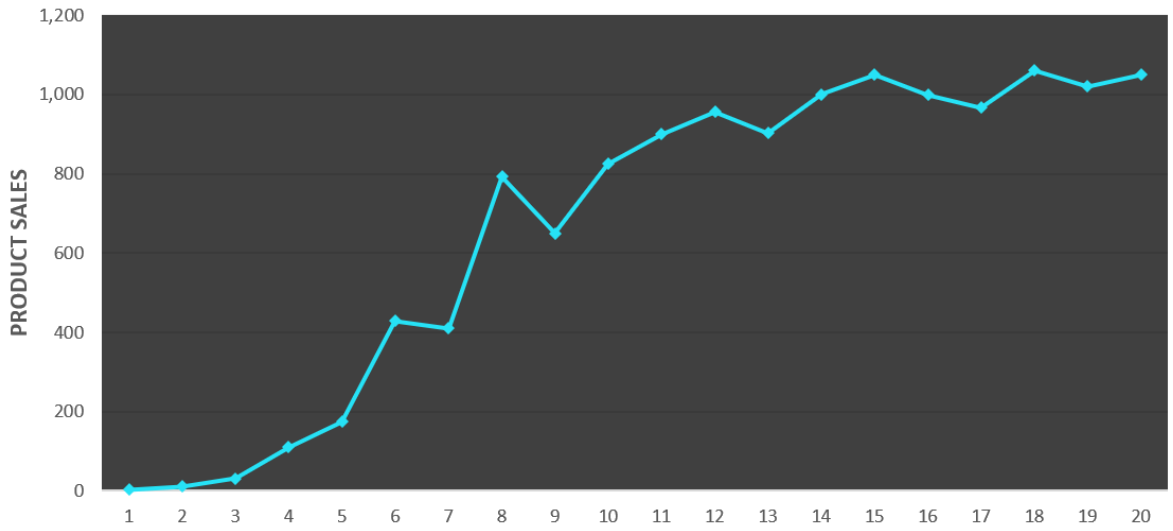
Smoothing

Non-Linear Trends

Intervention Analysis

Time-series data won't always follow a seasonal pattern or linear trend; it may follow a **non-linear trend**, or have no predictable trend at all

- There are many formulas designed to forecast common non-linear trends:



- Exponential
- Cubiq
- Step-Functions
- ADBUDG
- Gompertz

NON-LINEAR TRENDS

Forecasting 101

Seasonality

Linear Trending

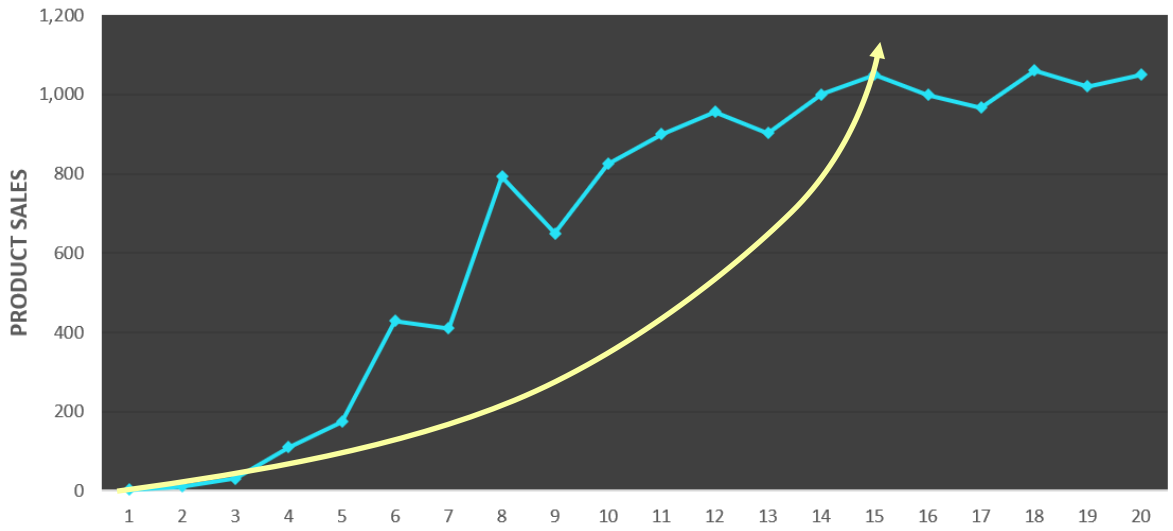
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NON-LINEAR TRENDS

Forecasting 101

Seasonality

Linear Trending

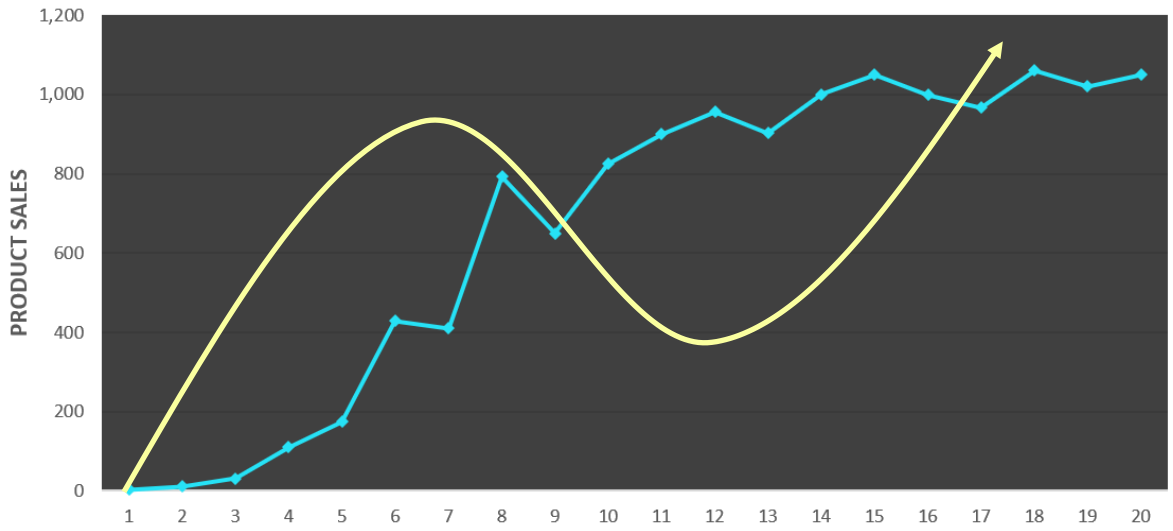
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NON-LINEAR TRENDS

Forecasting 101

Seasonality

Linear Trending

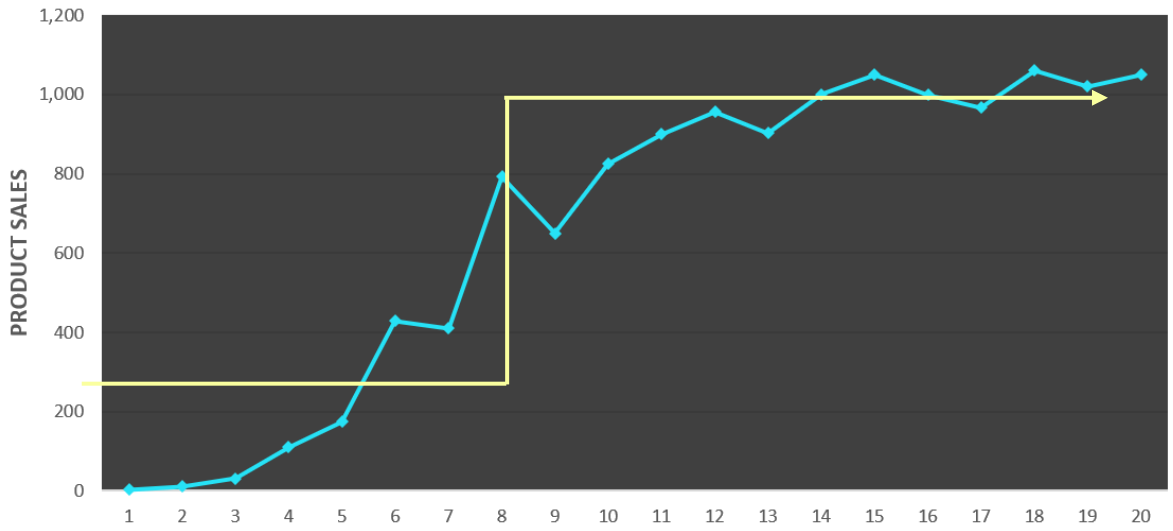
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NON-LINEAR TRENDS

Forecasting 101

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Linear Trending

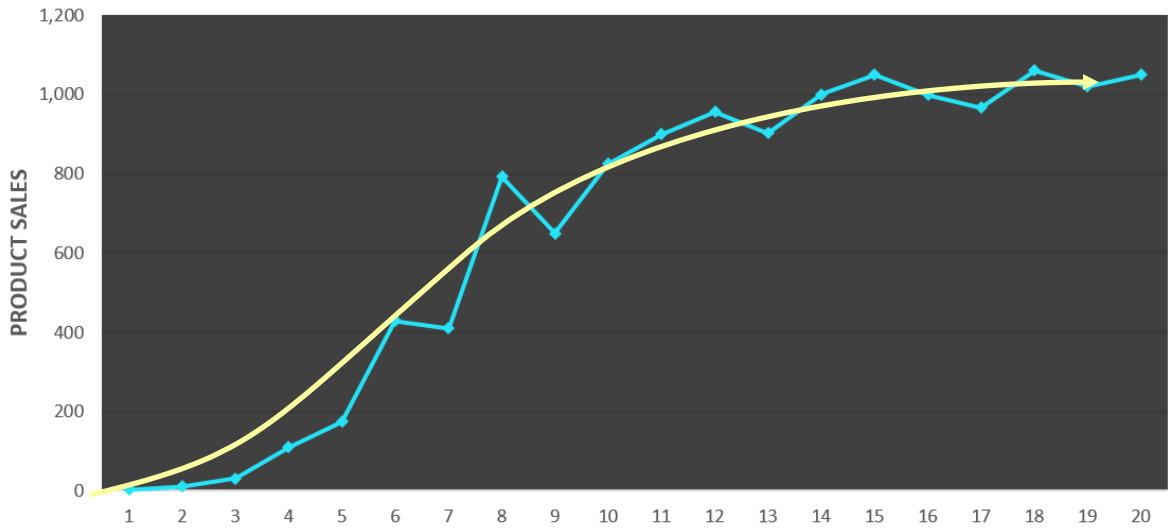
Smoothing

Non-Linear Trends

Intervention Analysis

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- There are many formulas designed to forecast common non-linear trends:



- Exponential
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- Step-Functions
- **ADBUDG**
- **Gompertz**



PRO TIP: ADBUDG and Gompertz are more flexible versions of a logistic curve, and are commonly seen in BI use cases *(product launches, diminishing returns, etc.)*

CASE STUDY: NON-LINEAR TREND



THE **SITUATION**

The team at **Cat Slacks** just launched a new product poised to revolutionize the world of feline fashion: a lightweight, breatheable jogging short designed for active cats who refuse to compromise on quality.



THE **ASSIGNMENT**

You've been asked to **provide a weekly sales forecast to help the manufacturing and warehouse teams with capacity planning**. You only have 8 weeks of data to work with, but expect the launch to follow a logistic curve.



THE **OBJECTIVES**

1. Collect sales data for the first 8 weeks since the launch
2. Apply a Gompertz curve to fit a logistic trend
3. Adjust parameters to compare various capacity limits and growth rates



INTERVENTION ANALYSIS

- Forecasting 101
- Seasonality
- Linear Trending
- Smoothing
- Non-Linear Trends
- Intervention Analysis

Intervention analysis is a technique used to estimate the impact of a specific change (or "intervention") on the dependent variable

- Simply put, intervention analysis is about predicting what ***would have happened*** if the change or intervention never took place
- By fitting a model to the "pre-intervention" data (up to the date of the change), you can compare predicted vs. actual values after that date to estimate the impact of the intervention

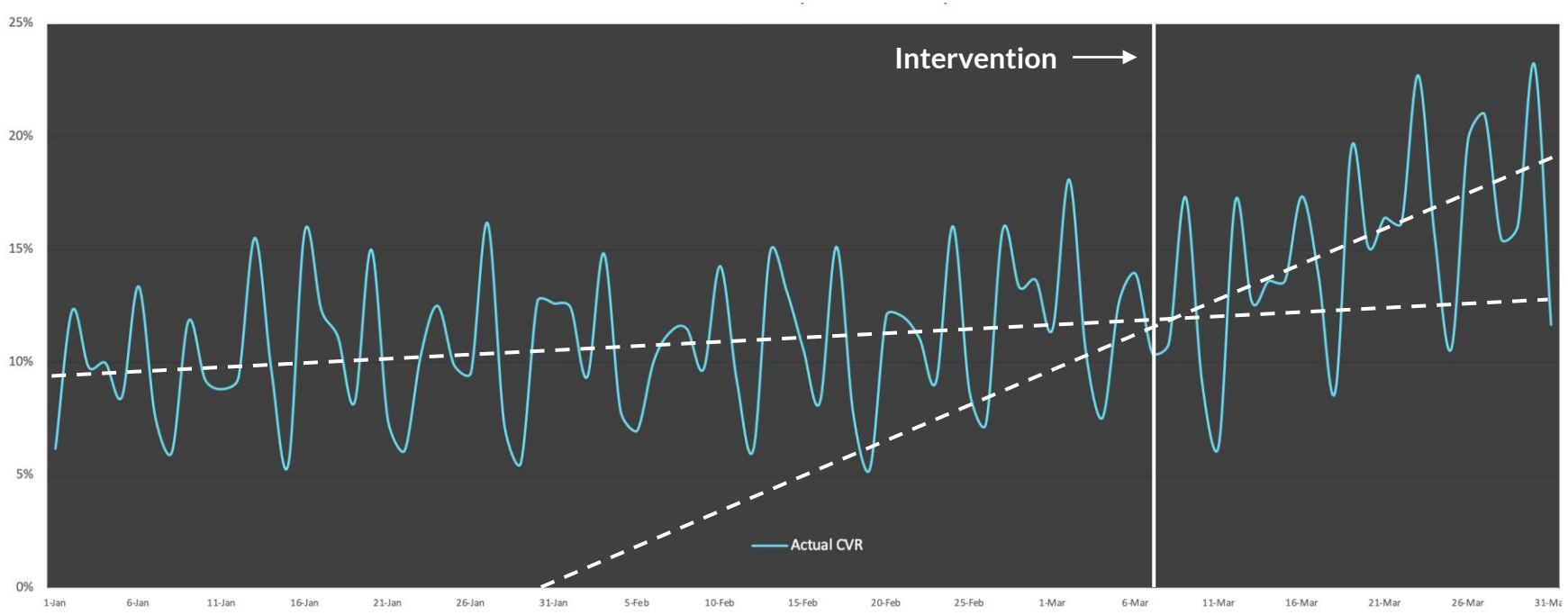
Common examples:

- Measuring the impact of a new website or check-out page on conversion rates
- Quantifying the impact of a new HR program to reduce employee churn

INTERVENTION ANALYSIS

- Forecasting 101
- Seasonality
- Linear Trending
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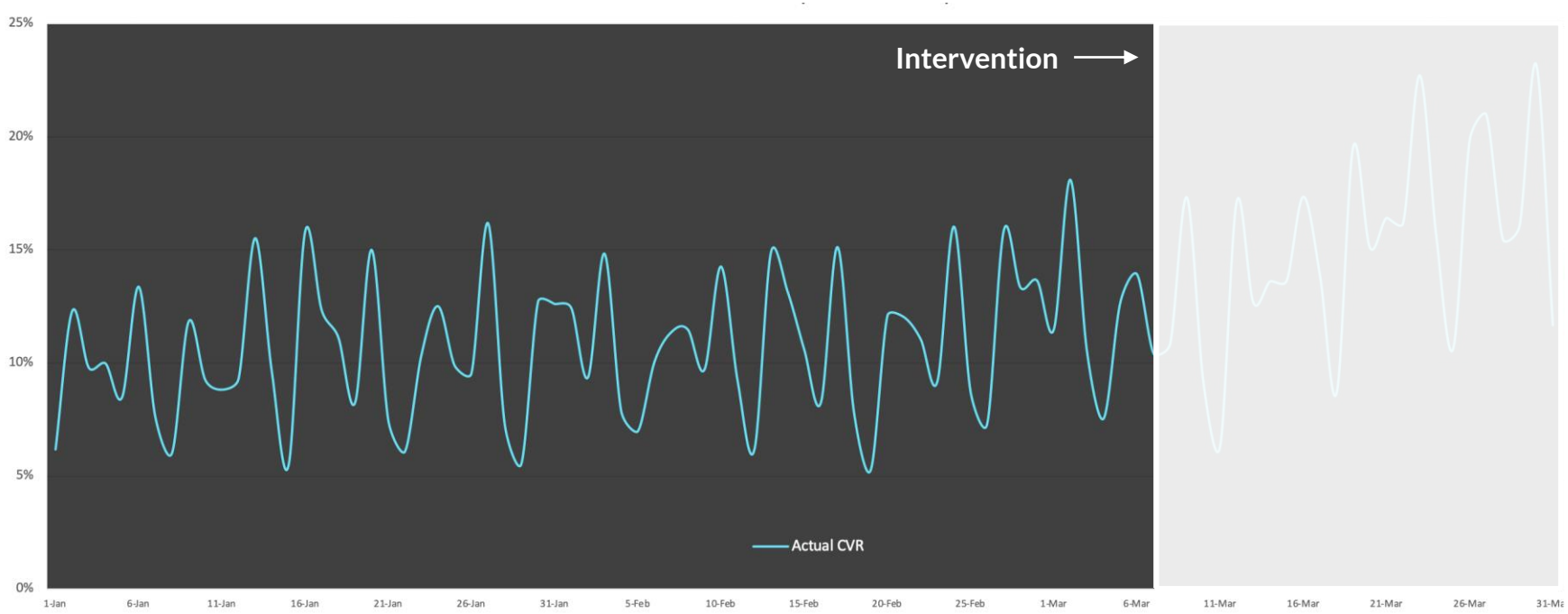
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INTERVENTION ANALYSIS

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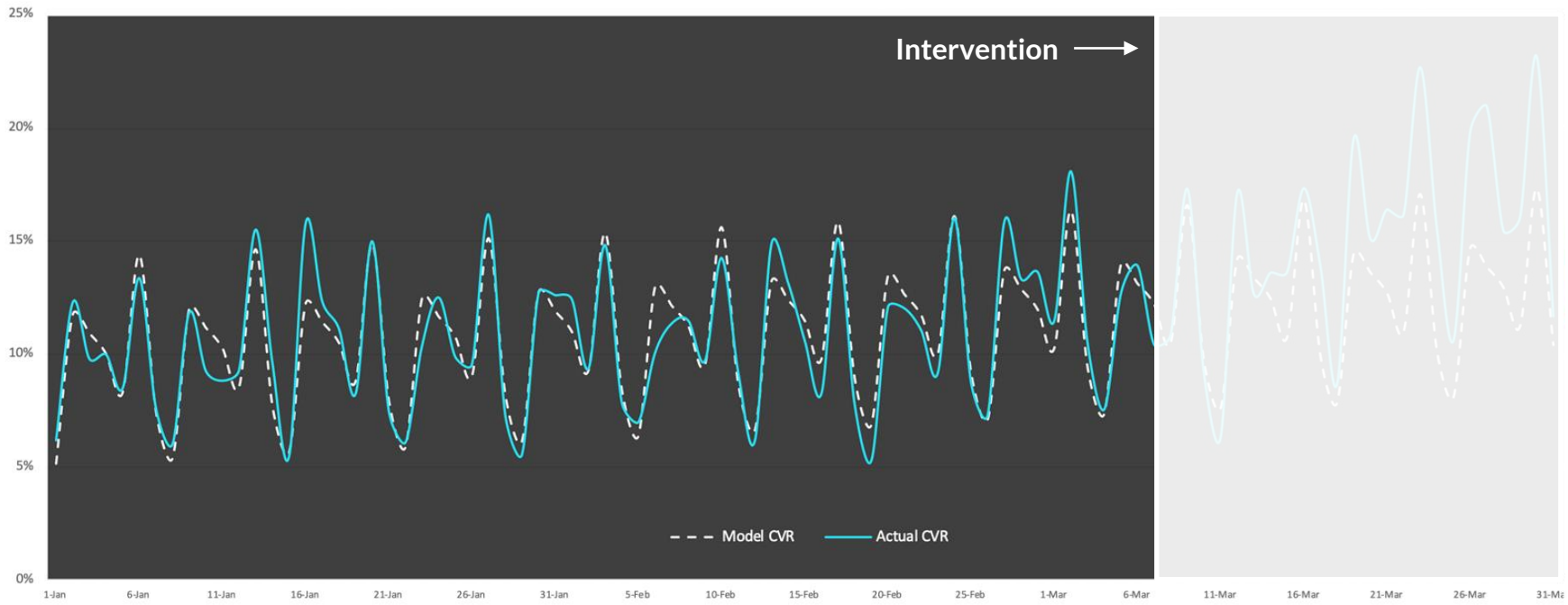
STEP 1: Fit a regression model to the data, using only observations from the **pre-intervention** period:



INTERVENTION ANALYSIS

- Forecasting 101
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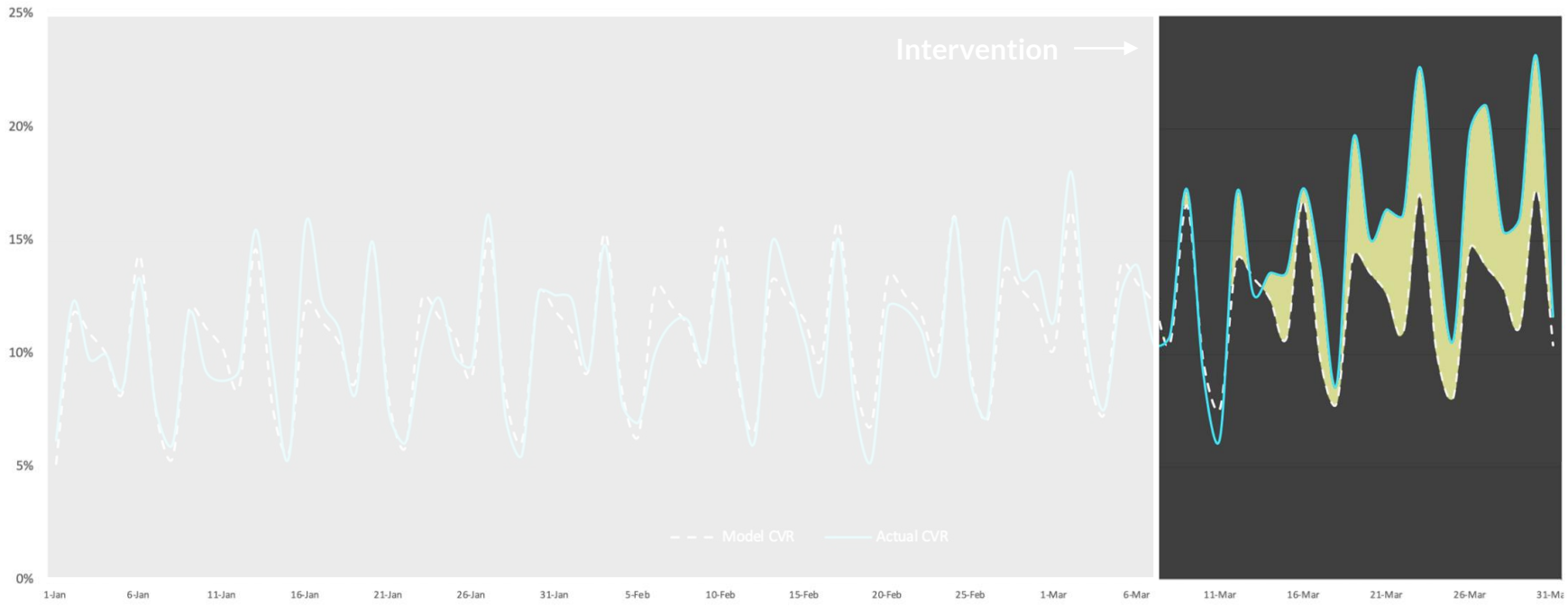
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INTERVENTION ANALYSIS

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- Linear Trending
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- Intervention Analysis

STEP 2: Compare the predicted and observed values in the **post-intervention** period, and sum the daily residuals to estimate the impact of the change:



CASE STUDY: INTERVENTION ANALYSIS



THE SITUATION

You are a Web Analyst for **Alpine Supplies**, an online retailer specializing in high-end camping and hiking gear.



THE ASSIGNMENT

The company recently rolled out a new product landing page, and the CMO has asked you to help **quantify the impact on conversion rate (CVR) and sales.** You'll need to conduct an intervention analysis, and make sure to capture day of week seasonality and trending in your model.



THE OBJECTIVES

1. Collect data to track sessions and CVR before and after the change
2. Fit a regression model to predict CVR using data from before the redesign
3. Use the model to forecast "baseline" CVR after the change, and calculate both incremental daily sales and the total cumulative impact

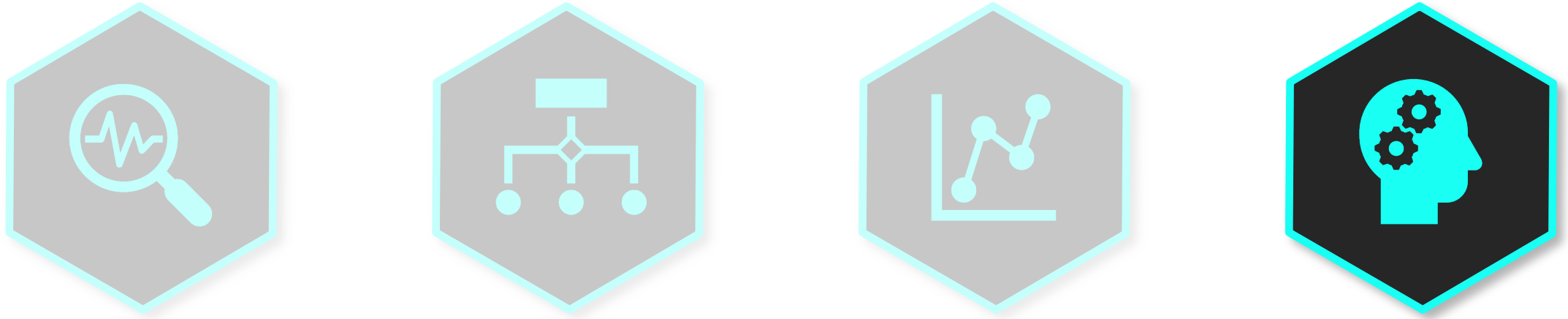
PART 4:

UNSUPERVISED



ABOUT THIS SERIES

This is **Part 4** of a **4-Part series** designed to help you build a deep, foundational understanding of machine learning, including data QA & profiling, classification, forecasting and unsupervised learning



PART 1
QA & Data Profiling

PART 2
Classification

PART 3
Regression & Forecasting

PART 4
Unsupervised Learning

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COURSE OUTLINE

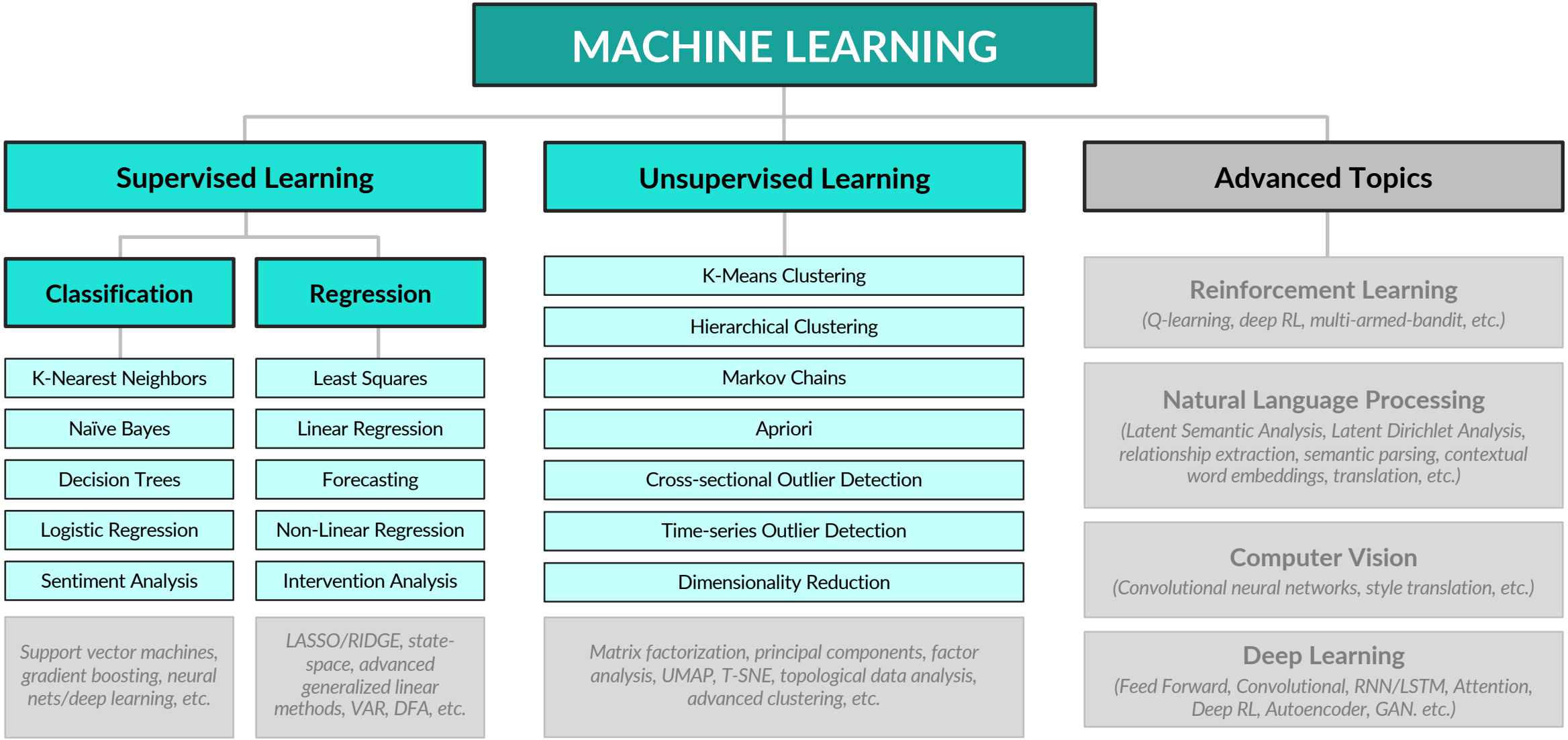
1	**Intro to Unsupervised ML**	*Review the ML landscape and key unsupervised learning concepts, including unsupervised vs. supervised, feature engineering, workflow, etc.*
2	**Segmentation & Clustering**	*Learn the key building blocks of segmentation and clustering, and review two of the most-used algorithms: K-Means and Hierarchical Clustering*
3	**Association Mining**	*Explore common techniques for association mining and when to use each of them, including A Priori and Markov models*
4	**Outlier Detection**	*Learn how to detect cross-sectional and time-series outliers using techniques like Nearest Neighbor Distance and Time-Series models*
5	**Dimensionality Reduction**	*Understand the core foundations of dimensionality reduction and how it can be used in the context of BI and unsupervised learning*

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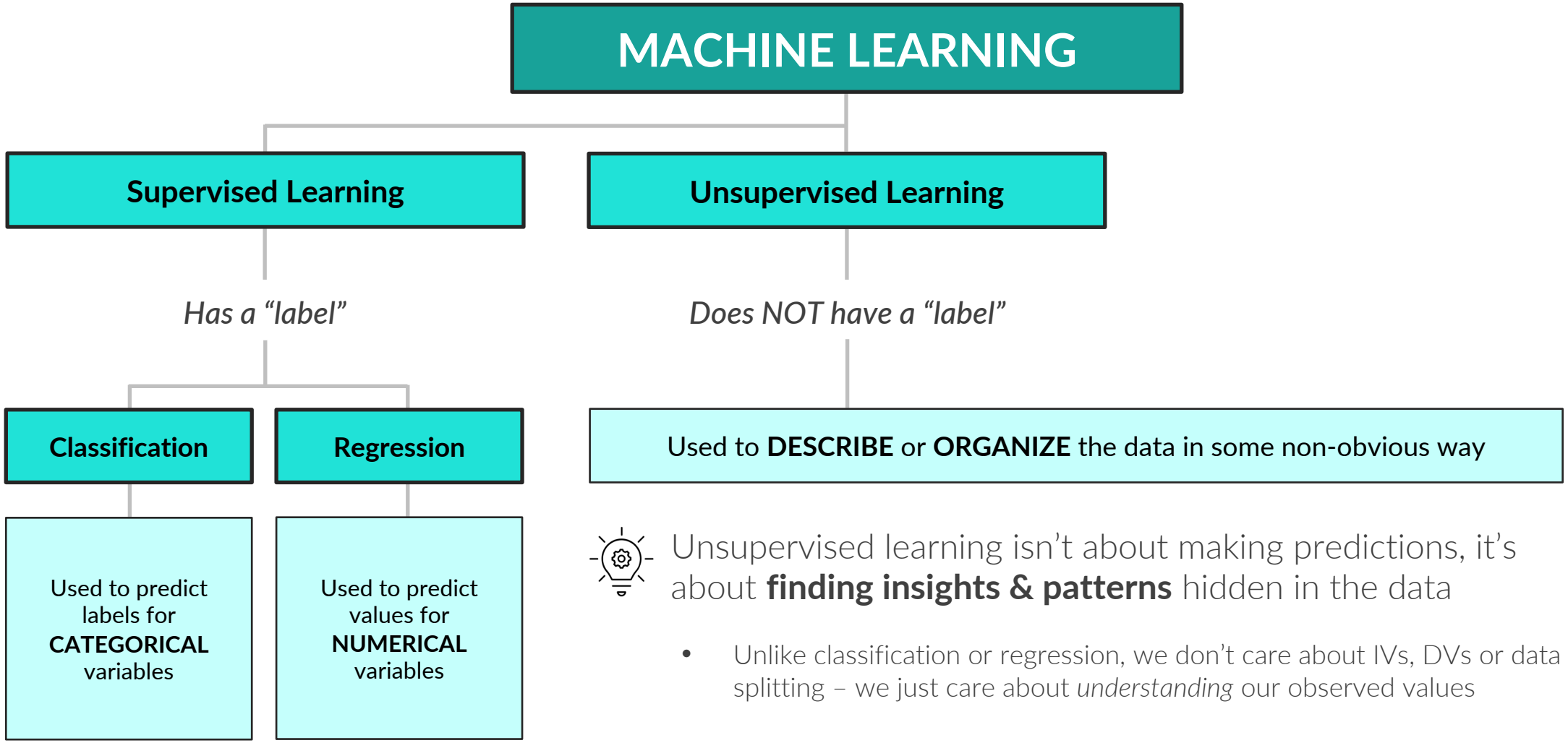
INTRO TO UNSUPERVISED ML

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MACHINE LEARNING LANDSCAPE



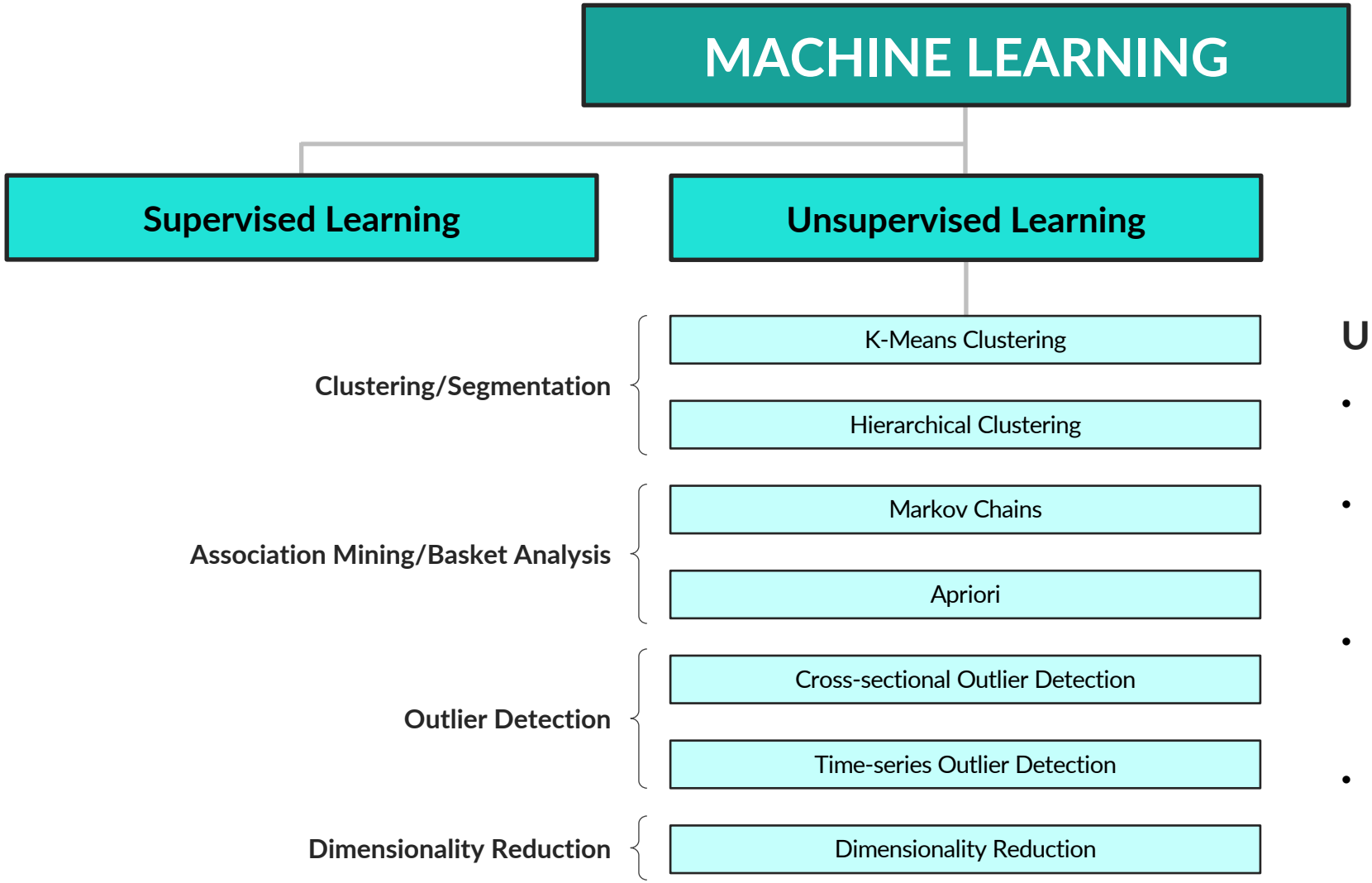
MACHINE LEARNING LANDSCAPE



Unsupervised learning isn't about making predictions, it's about **finding insights & patterns** hidden in the data

- Unlike classification or regression, we don't care about IVs, DVs or data splitting – we just care about *understanding* our observed values

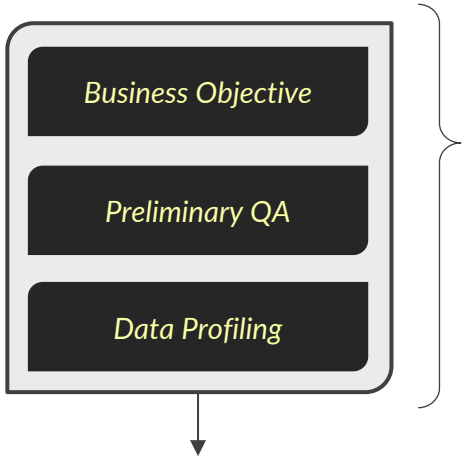
MACHINE LEARNING LANDSCAPE



UNSUPERVISED EXAMPLES:

- *Are there specific groups or "clusters" of customers who behave in similar ways?*
- *When customers purchase a particular item, which other items tend to be purchased as well?*
- *Are there any anomalies in the data that we wouldn't expect to see under normal conditions?*
- *Can we combine or reduce the number of dimensions in our data to better understand it?*

UNSUPERVISED LEARNING WORKFLOW



Remember, these steps ALWAYS come first

*Before building a model, you should have a clear understanding of the **business objective** (identifying clusters, detecting anomalies, etc.) and the **data at hand** (variable types, table structure, data quality, profiling metrics, etc.)*

Feature Engineering → **Model Application** → **Hyperparameter Tuning** → **Model Selection**

Add new, calculated variables (or "features") to the data set based on existing fields

Apply relevant unsupervised ML techniques, based on the objective (you will typically test multiple models)

Adjust and tune model parameters (this is typically an iterative process)

Select the model that yields the most useful or insightful results, based on the objective at hand

There often are **no strict rules** to determine which model is best; it's about which one helps you **best answer the question at hand**

RECAP: FEATURE ENGINEERING



Feature engineering is the process of enriching a data set by creating additional variables *(or dimensions)* based on existing fields

- When preparing data for segmentation/clustering, features should be **intuitive** and **actionable**; they should help you clearly describe each cluster and make practical business recommendations

Original features						*Engineered* features				
Month ID	Social Posts	Competitive Activity	Marketing Spend	Promotion Count	Revenue	Competitive High	Competitive Medium	Competitive Low	Promotion >10 & Social > 25	Log Spend
1	30	High	$130,000	12	$1,300,050	1	0	0	1	11.7
2	15	Low	$600,000	5	$11,233,310	0	0	1	0	13.3
3	8	Medium	$15,000	10	$1,112,050	0	1	0	0	9.6
4	22	Medium	$705,000	11	$1,582,077	0	1	0	1	13.5
5	41	High	$3,000	3	$1,889,053	1	0	0	0	8.0

KEY TAKEAWAYS



Unsupervised learning is about **finding patterns**, not making predictions

- *Unlike classification or regression, it's not about labels, dependent/independent variables or splitting test and training data; it's about better understanding the observed values in a data set*



For many unsupervised techniques, there's no **"right" or "wrong"** solution

- *Unsupervised learning is often less about optimizing specific metrics or parameters, and more about using Machine Learning to clearly answer the business question at hand*



Always focus on the desired **business outcome**

- *Make sure that you can clearly interpret your model output and translate it into intuitive, practical insights and recommendations (otherwise what's the point?)*

CLUSTERING & SEGMENTATION

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CLUSTERING & SEGMENTATION



In this section we'll introduce the fundamentals of **clustering** & **segmentation** and compare two common unsupervised models: K-means and hierarchical clustering

TOPICS WE'LL COVER:

- Clustering Basics
- K-Means
- Hierarchical Clustering
- Key Takeaways

GOALS FOR THIS SECTION:

- Understand how segmentation and clustering models fundamentally work
- Explore common models including K-Means and hierarchical clustering
- Compare and contrast common clustering models



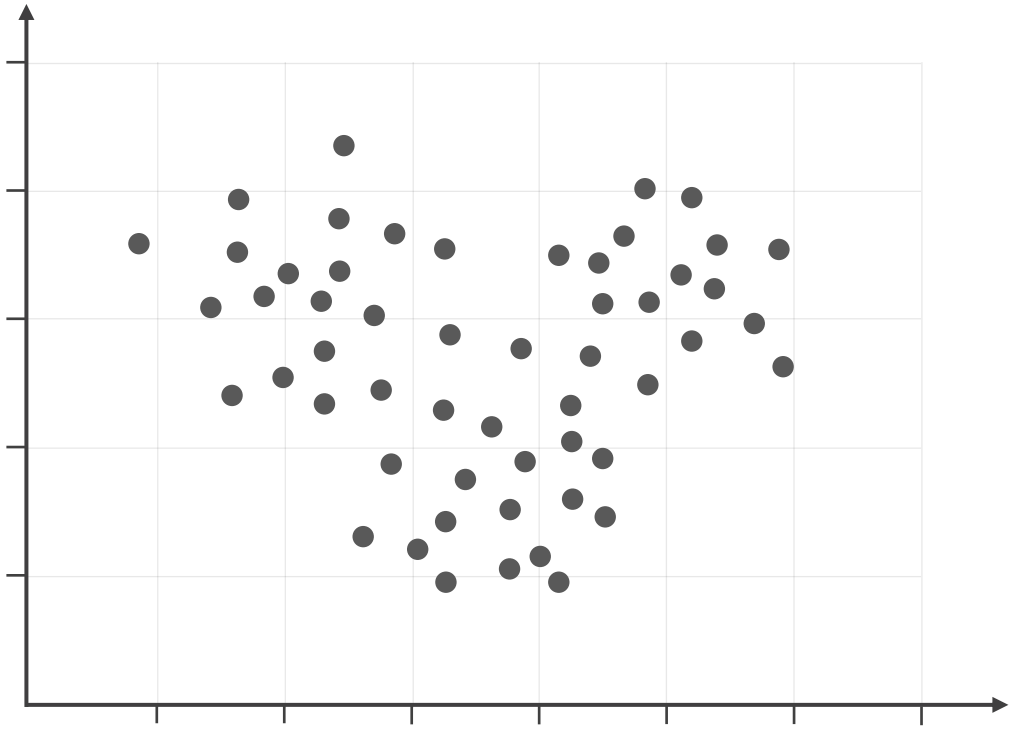
Remember, there's no "right" answer or single optimization metric when it comes to clustering and segmentation; the best outputs are the ones which help you answer the question at hand and make practical, data-driven business decisions



CLUSTERING BASICS

- Clustering Basics
- K-Means
- Hierarchical Clustering
- Key Takeaways

Clustering allows you to find concentrations or groups of observations which are similar to one another but distinct from other groups

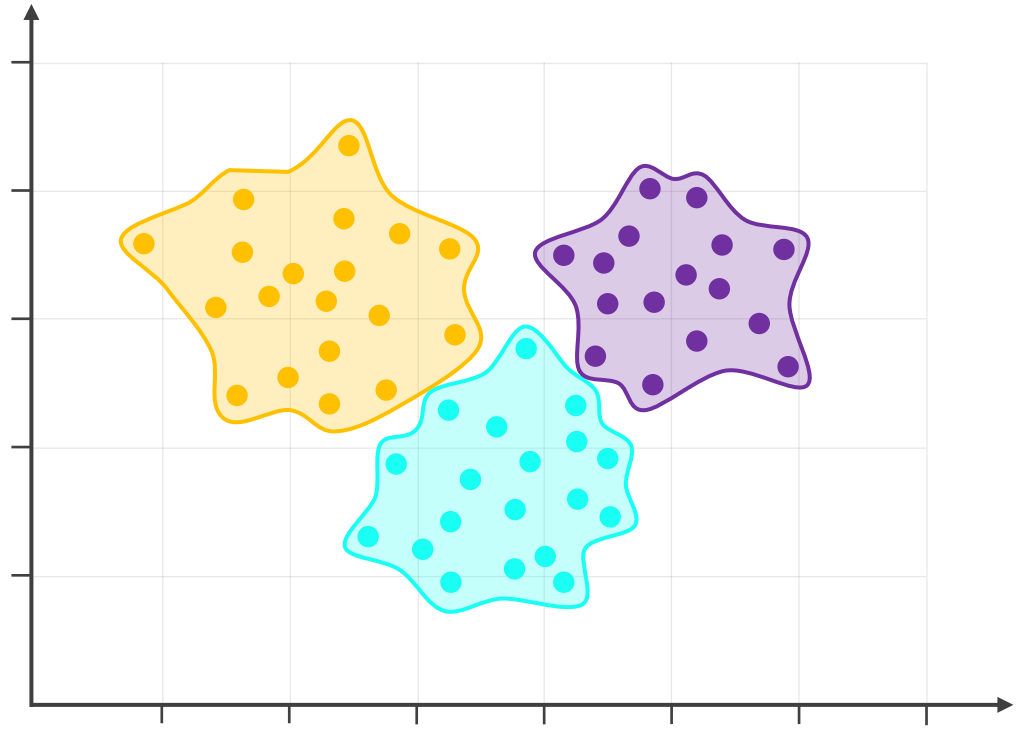




CLUSTERING BASICS

- Clustering Basics
- K-Means
- Hierarchical Clustering
- Key Takeaways

Clustering allows you to find concentrations or groups of observations which are similar to one another but distinct from other groups





CLUSTERING BASICS

- Clustering Basics
- K-Means
- Hierarchical Clustering
- Key Takeaways

Clustering allows you to find concentrations or groups of observations which are similar to one another but distinct from other groups

Multiple models can be used for clustering, but the general workflow consists of the following steps:

EXTRACT → **TRANSFORM** → **MODEL** → **INTERPRET**

Collect the raw data you need for analysis *(i.e. transactions, customer records, survey results, website pathing, etc.)*

Use **feature engineering** to create clear, intuitive dimensions which can help you interpret each cluster and make actionable recommendations

Apply one or more **clustering models** and determine an appropriate number of clusters based on the model output and business context

Use **multivariate analysis** to understand the characterics of each cluster *(profiling metrics, distributions, etc.)*. **The model won't do this for you!**



K-MEANS

Clustering Basics

K-Means

Hierarchical Clustering

Key Takeaways

K-Means is a common algorithm which assigns each observation in a data set to a specific cluster, where **K** represents the number of clusters

In it's simplest form *(2 dimensions)*, here's how it works:

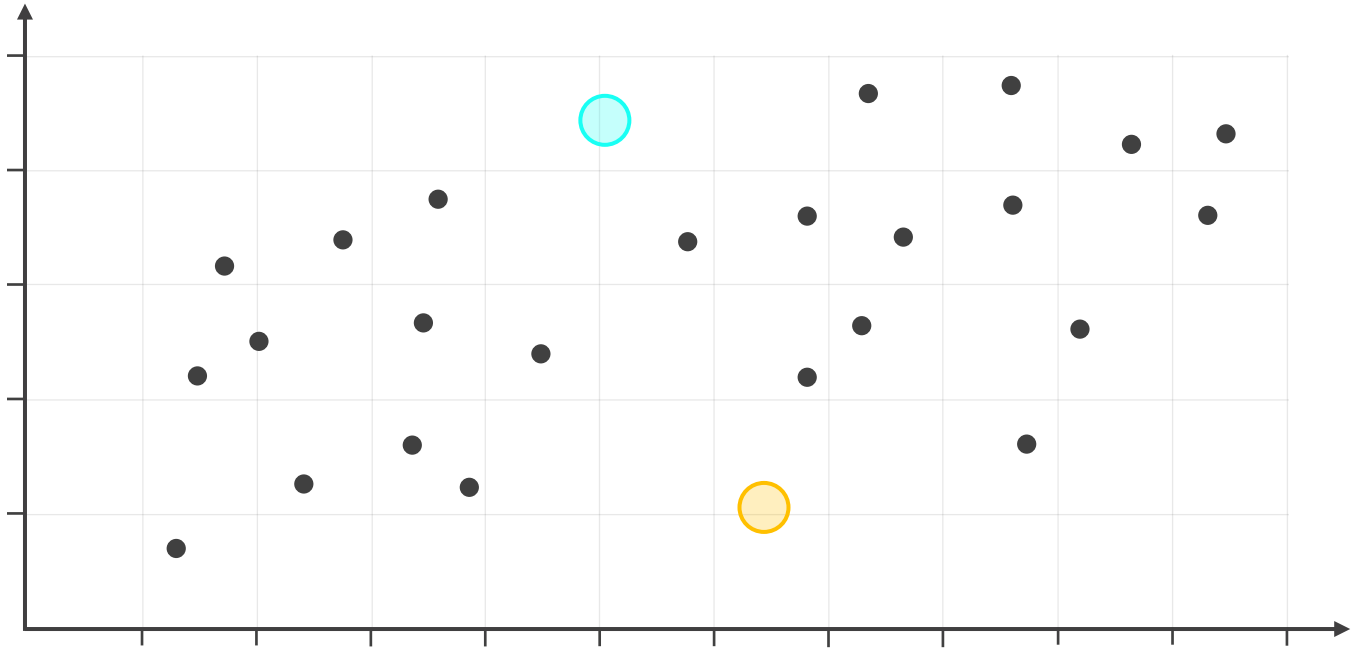
1. Select K arbitrary locations in a scatterplot as cluster centers (or **centroids**), and assign each observation to a cluster based on the closest centroid
2. Recalculate and relocate each centroid to the mean of the observations assigned to it, then reassign each observation to its *new* closest centroid
3. Repeat the process until observations no longer change clusters

Example use cases:

- Identifying customer segments for targeted marketing campaigns
- Clustering store locations based on factors like sales, ratings, size, etc.

K-MEANS

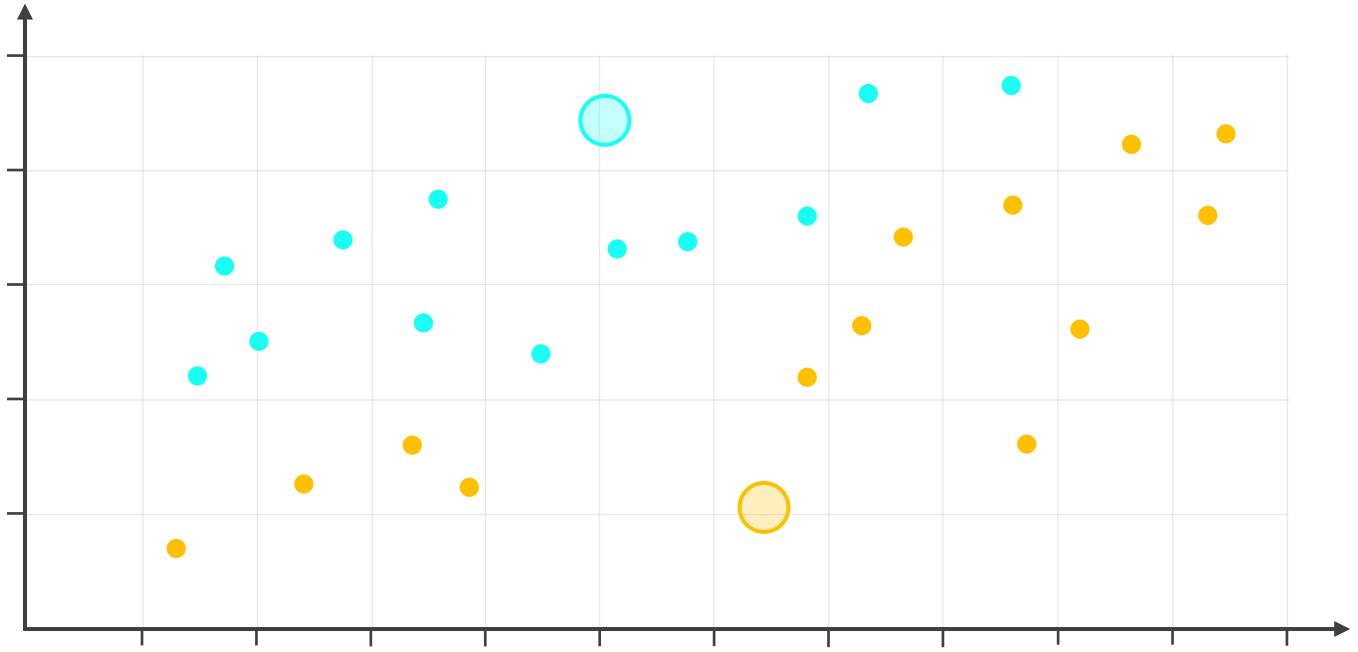
- Clustering Basics
- K-Means
- Hierarchical Clustering
- Key Takeaways



STEP 1: Determine what you think might be an appropriate number of clusters *(in this case 2)*, and select arbitrary locations as initial centroids

K-MEANS

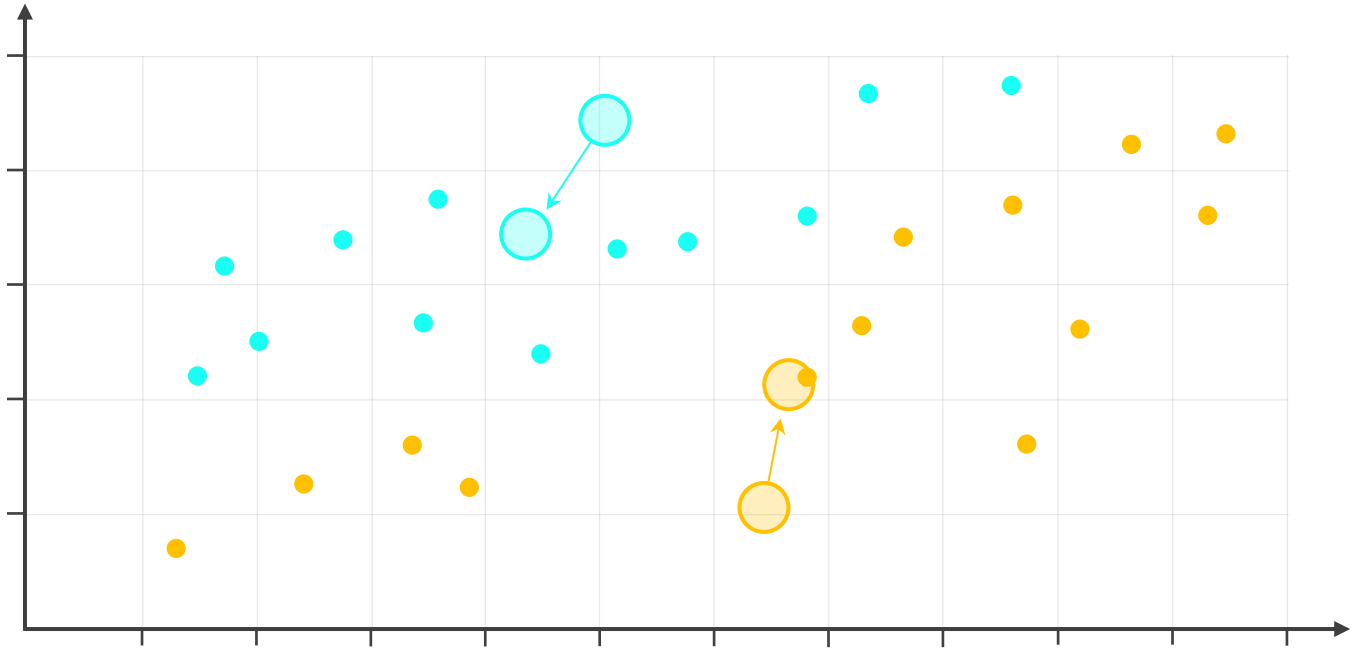
- Clustering Basics
- K-Means
- Hierarchical Clustering
- Key Takeaways



STEP 2: Assign each observation to a cluster, based on the closest centroid

K-MEANS

- Clustering Basics
- K-Means
- Hierarchical Clustering
- Key Takeaways

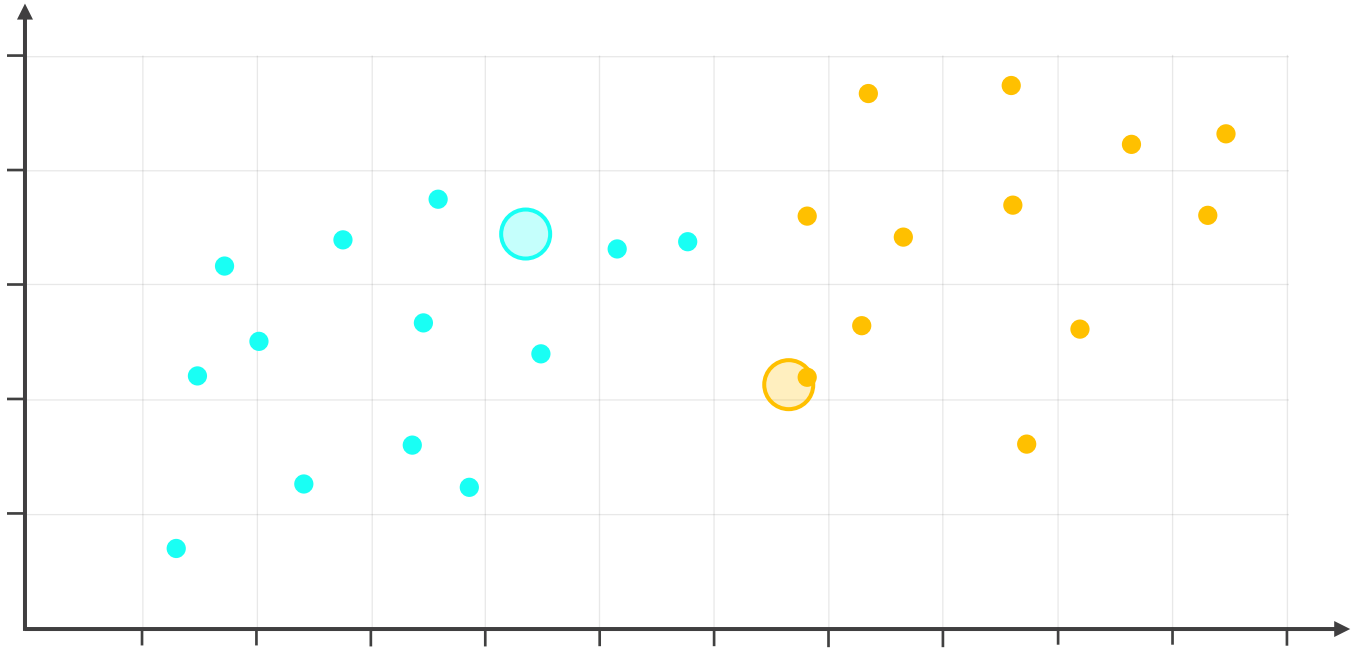


STEP 3: Relocate each centroid to the mean of its assigned observations, and reassign each observation to the new closest centroid

K-MEANS



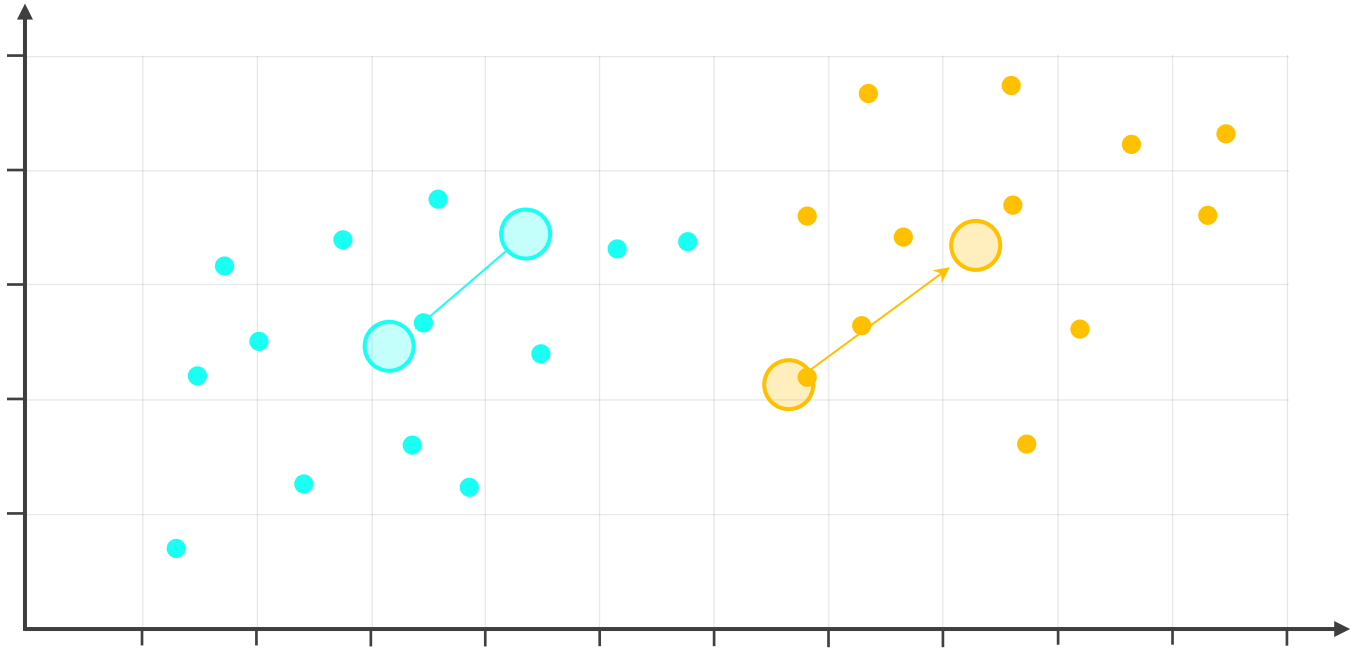
- Clustering Basics
- K-Means
- Hierarchical Clustering
- Key Takeaways



STEP 3: Relocate each centroid to the mean of its assigned observations, and reassign each observation to the new closest centroid

K-MEANS

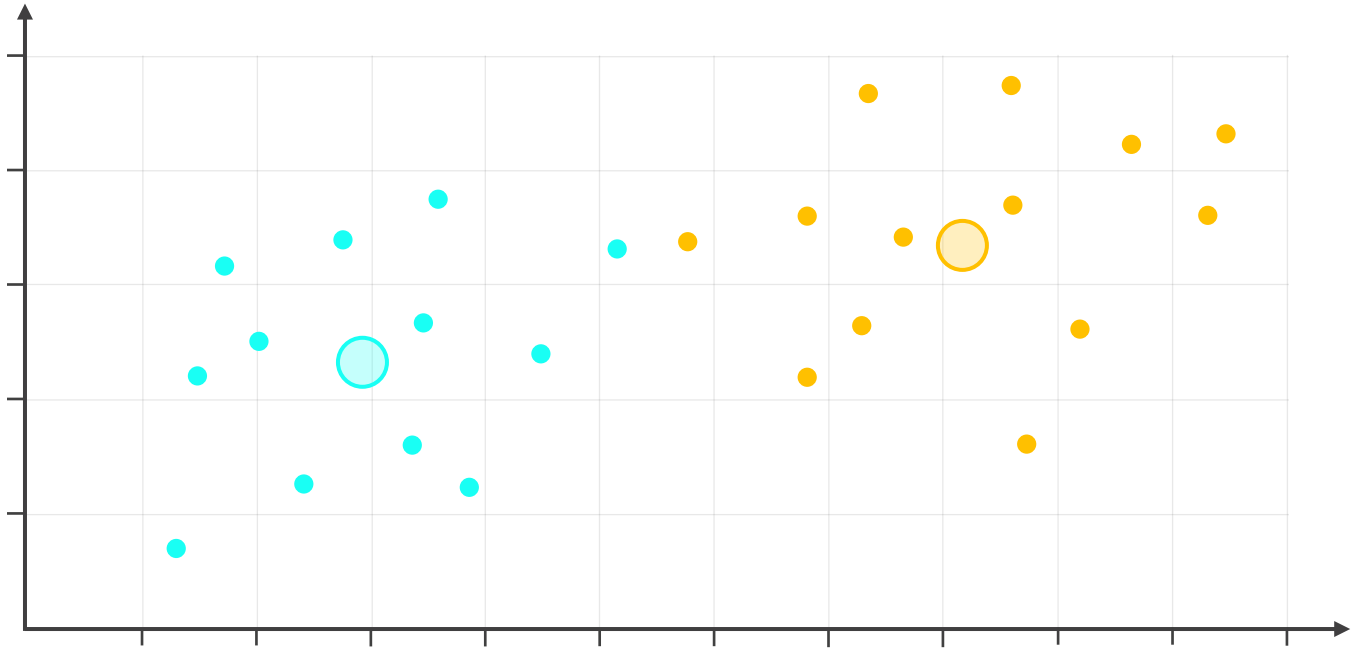
- Clustering Basics
- K-Means
- Hierarchical Clustering
- Key Takeaways



STEP 4: Continue to relocate each centroid to the mean of its assigned observations, until the clusters no longer change

K-MEANS

- Clustering Basics
- K-Means
- Hierarchical Clustering
- Key Takeaways



STEP 4: Continue to relocate each centroid to the mean of its assigned observations, until the clusters no longer change

K-MEANS

Clustering Basics

K-Means

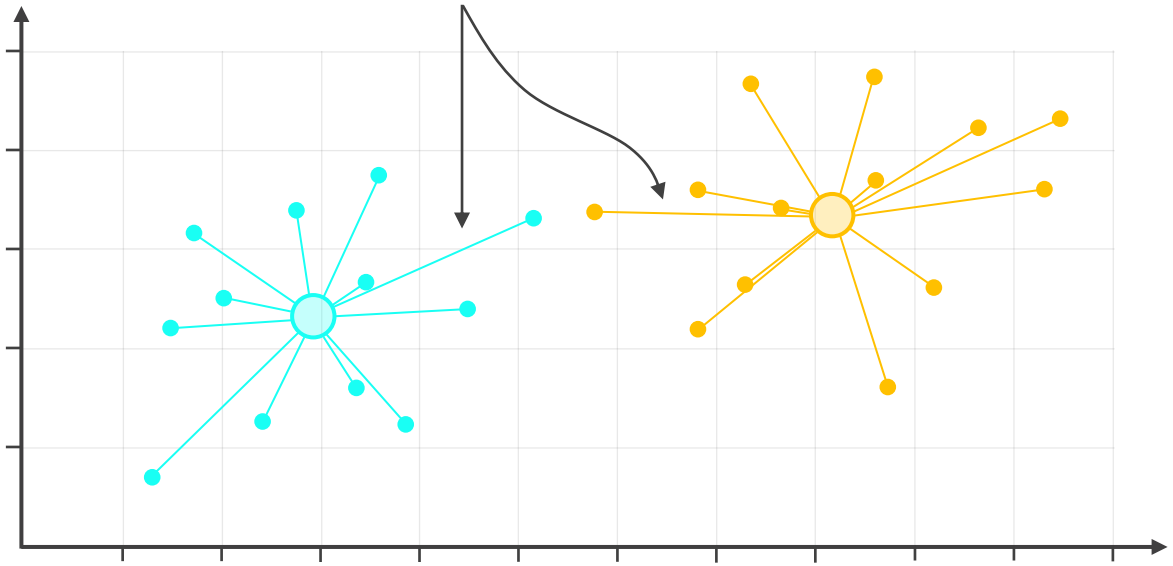
Hierarchical Clustering

Key Takeaways

How do we know what's the "right" number of clusters (K)?

- While there is no "right" or "wrong" number of clusters, you can use the **within-cluster sum of squares (WSS)** to help inform your decision

Square these distances and sum them to calculate WSS for two clusters (K=2)



K-MEANS

Clustering Basics

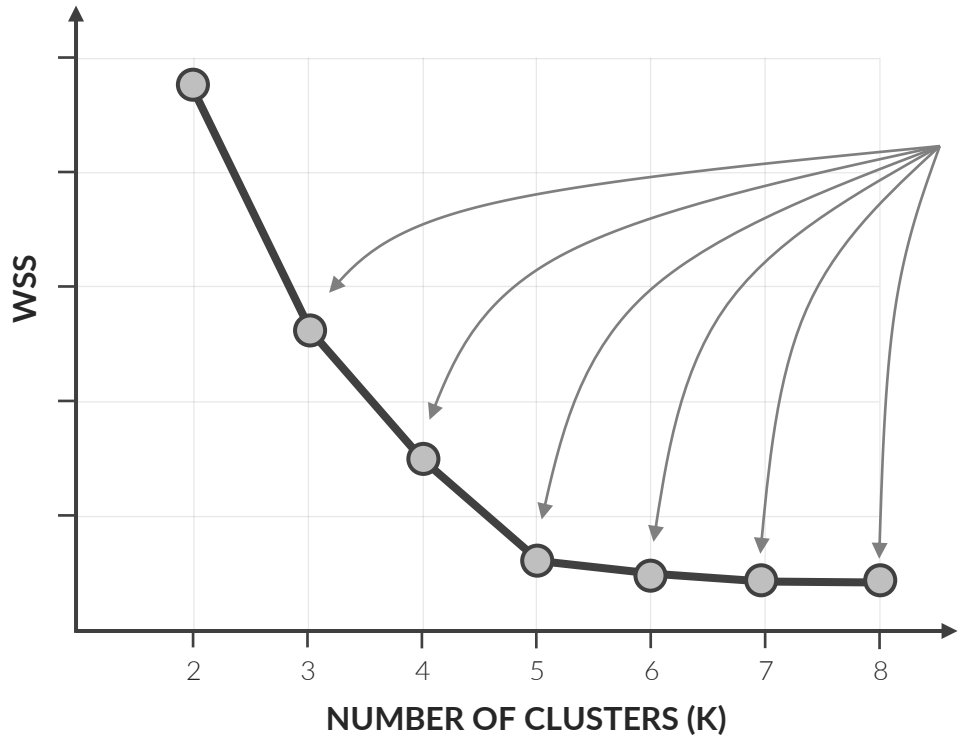
K-Means

Hierarchical Clustering

Key Takeaways

How do we know what's the "right" number of clusters (K)?

- While there is no "right" or "wrong" number of clusters, you can use the **within-cluster sum of squares (WSS)** to help inform your decision



Rerun the model with additional clusters, and plot the WSS for each value of K

K-MEANS

Clustering Basics

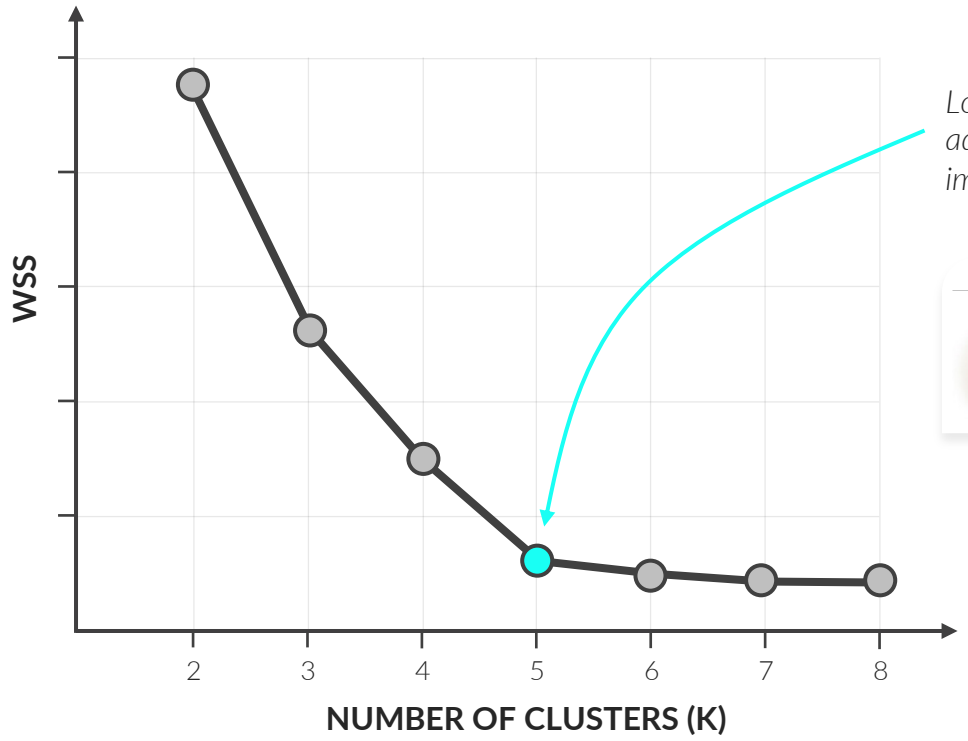
K-Means

Hierarchical Clustering

Key Takeaways

How do we know what's the "right" number of clusters (K)?

- While there is no "right" or "wrong" number of clusters, you can use the **within-cluster sum of squares (WSS)** to help inform your decision



*Look for an "elbow" or inflection point, where adding another cluster has a relatively small impact on WSS (in this case where **K=5**)*



PRO TIP: Think of this as a guideline, not a strict rule



K-MEANS

Clustering Basics

K-Means

Hierarchical Clustering

Key Takeaways



Can we create clusters based on **more than 2 dimensions**?

- Yes! K-Means and WSS plots can scale to *any* number of dimensions
- At higher dimensions, profiling metrics are critical for understanding what each cluster represents, since you can't compare them visually



Does the **shape** of the clusters matter?

- Yes, K-Means works best when the clusters are mostly circular in shape; but other tools like Hierarchical Clustering *(up next!)* can address this



Does the **initial centroid location** make a difference?

- It can, so a common best practice is to re-run the model K times to ensure that you are minimizing WSS for the selected number of clusters

CASE STUDY: K-MEANS



THE SITUATION

You've just been hired as a Data Science Intern for **Pet Palz**, an ecommerce start-up selling custom 3-D printed models of household pets.



THE ASSIGNMENT

You've been asked to analyze customer-level data to help **identify meaningful audience segments and refine the company's marketing strategy**.

As a first step, you'll be exploring the relationship between how often customers engage with promotional emails and how much they actually spend, using a K-means model for clustering.



THE OBJECTIVES

1. Collect sample data containing email engagement and spend by customer
2. Use K-means to identify clusters
3. Interpret each cluster and suggest next steps based on your analysis



HIERARCHICAL CLUSTERING

Clustering Basics

K-Means

Hierarchical Clustering

Key Takeaways

Hierarchical clustering is an unsupervised learning technique that creates clusters by grouping similar data points together*

In it's simplest form *(2 dimensions)*, here's how it works:

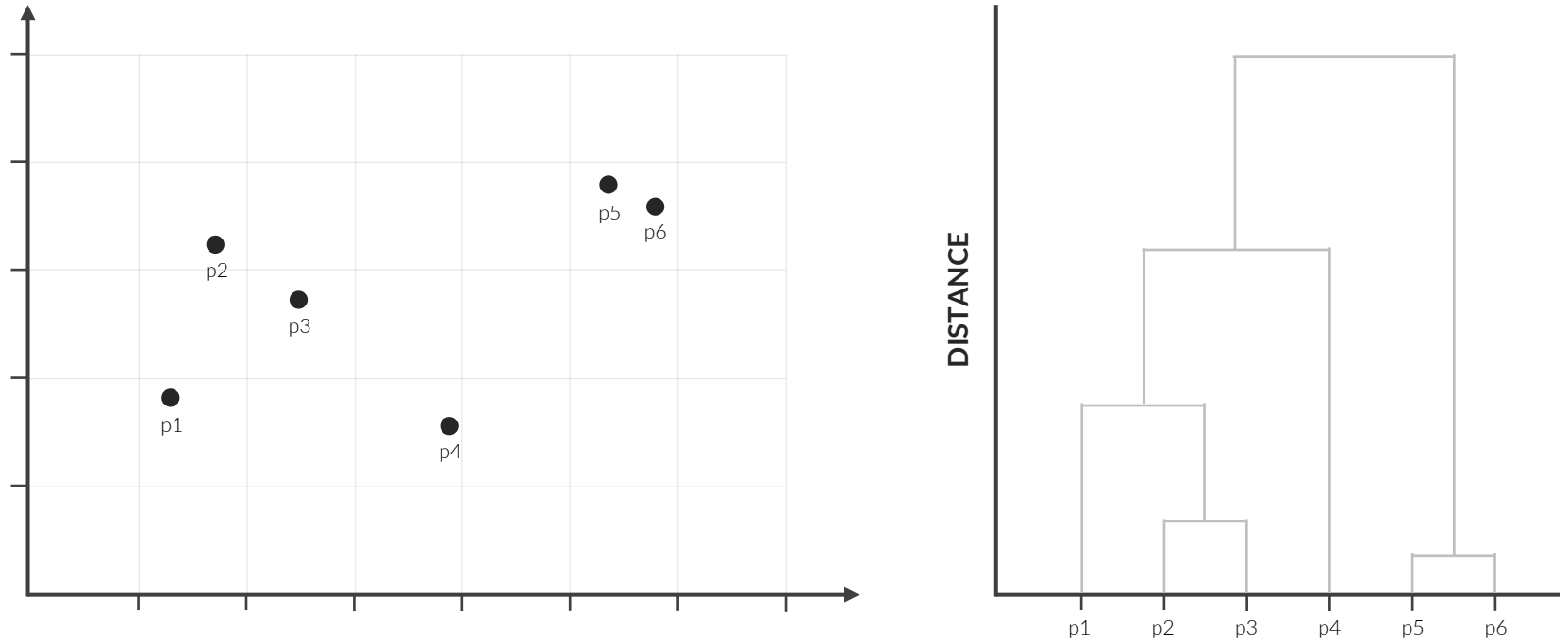
1. Create a scatterplot, find the 2 closest points, and group them into a cluster
2. Then find the *next* two closest points or clusters, and group them to a cluster
3. Repeat the process of combining the closest pairs of points or clusters until you eventually end up with one single cluster

This process is visualized using a tree diagram called a **dendrogram**, which shows the hierarchical relationship between clusters

**This is known as agglomerative or "bottom-up" clustering (vs. divisive or "top-down" clustering, which is much less common)*

HIERARCHICAL CLUSTERING

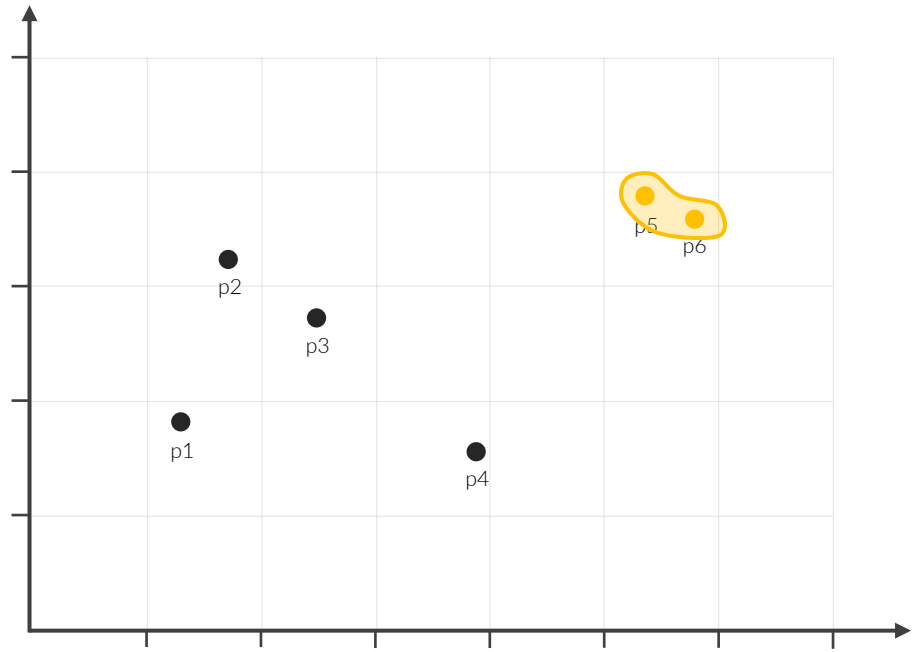
- Clustering Basics
- K-Means
- Hierarchical Clustering
- Key Takeaways

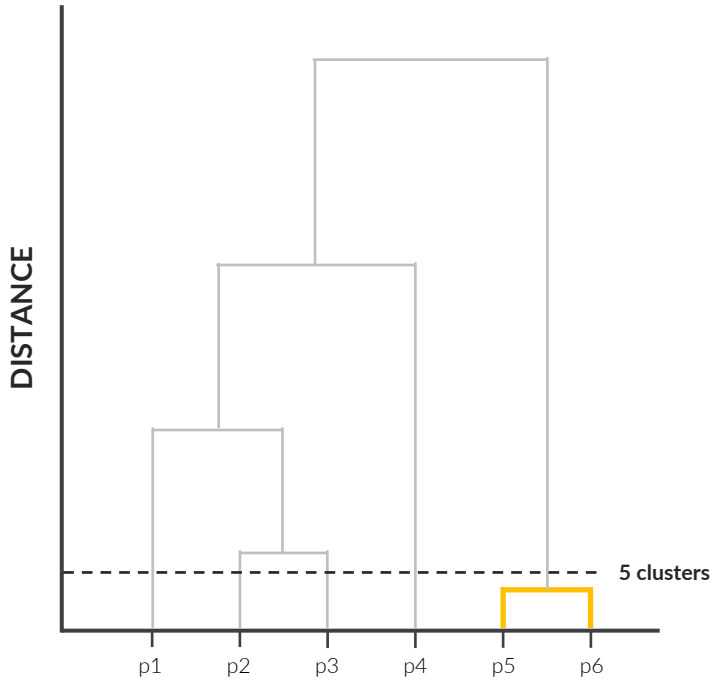


STEP 1: Find the two closest points, and group them into a cluster

HIERARCHICAL CLUSTERING

- Clustering Basics
- K-Means
- Hierarchical Clustering
- Key Takeaways

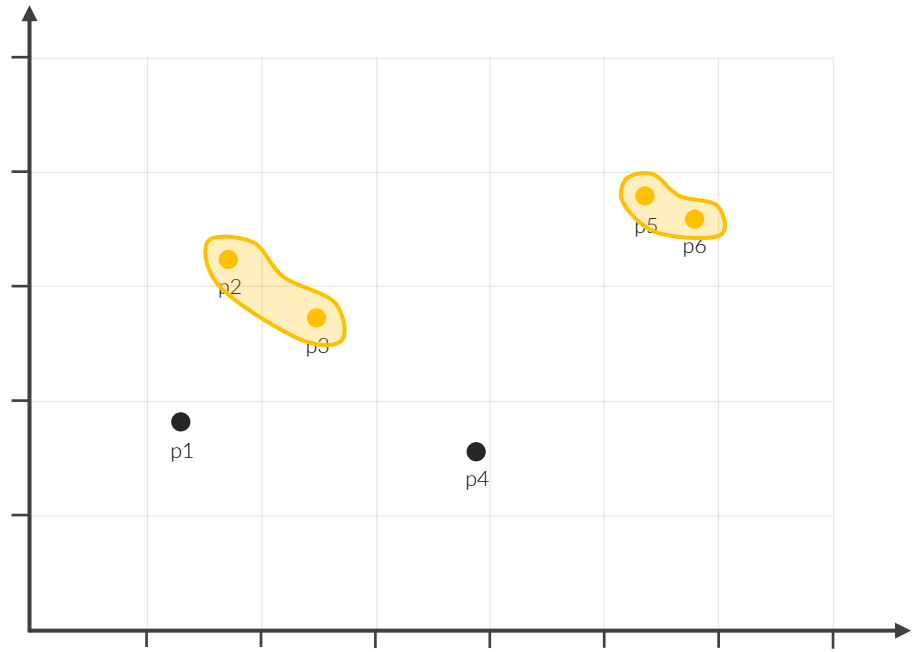


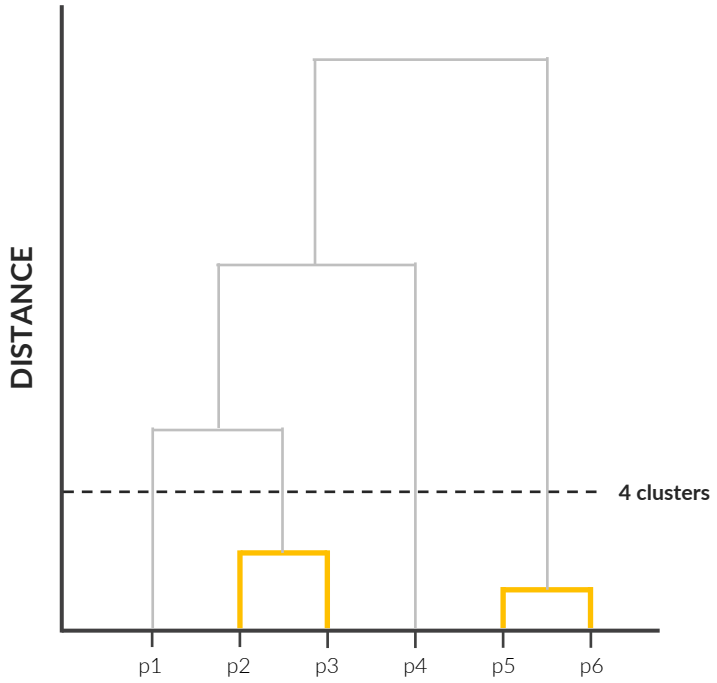


STEP 1: Find the two closest points, and group them into a cluster

HIERARCHICAL CLUSTERING

- Clustering Basics
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- Key Takeaways

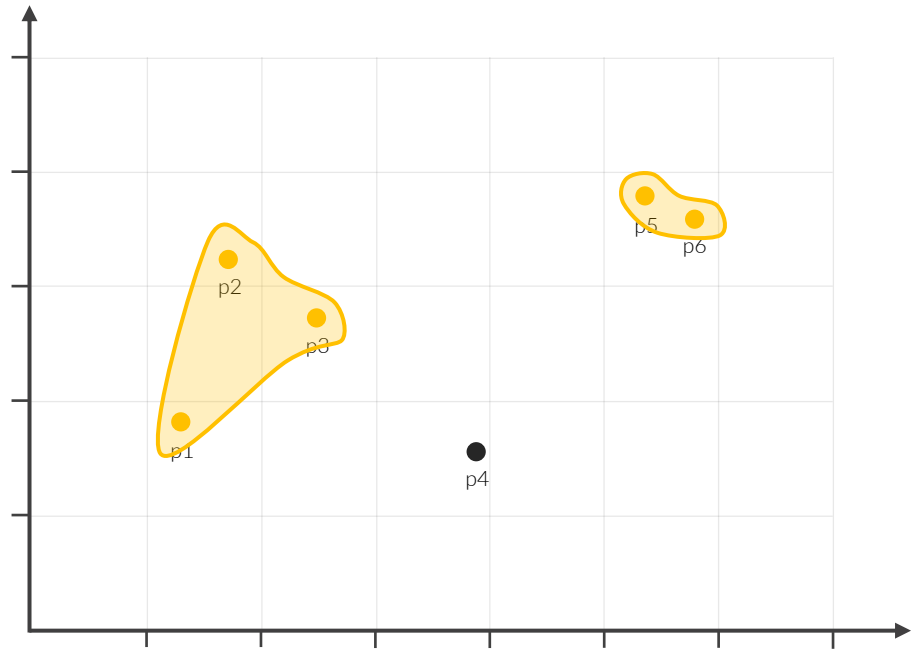


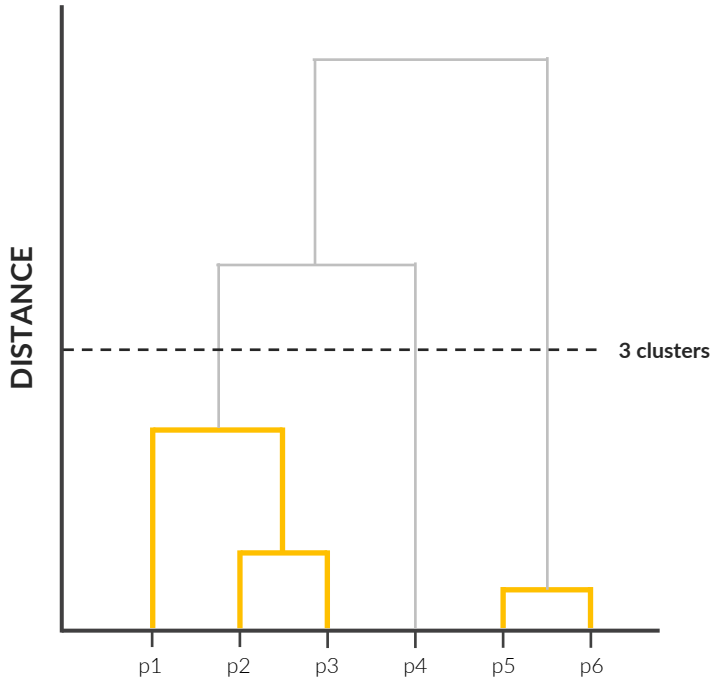


STEP 2: Find the *next* two closest points/clusters, and group them together

HIERARCHICAL CLUSTERING

- Clustering Basics
- K-Means
- Hierarchical Clustering
- Key Takeaways

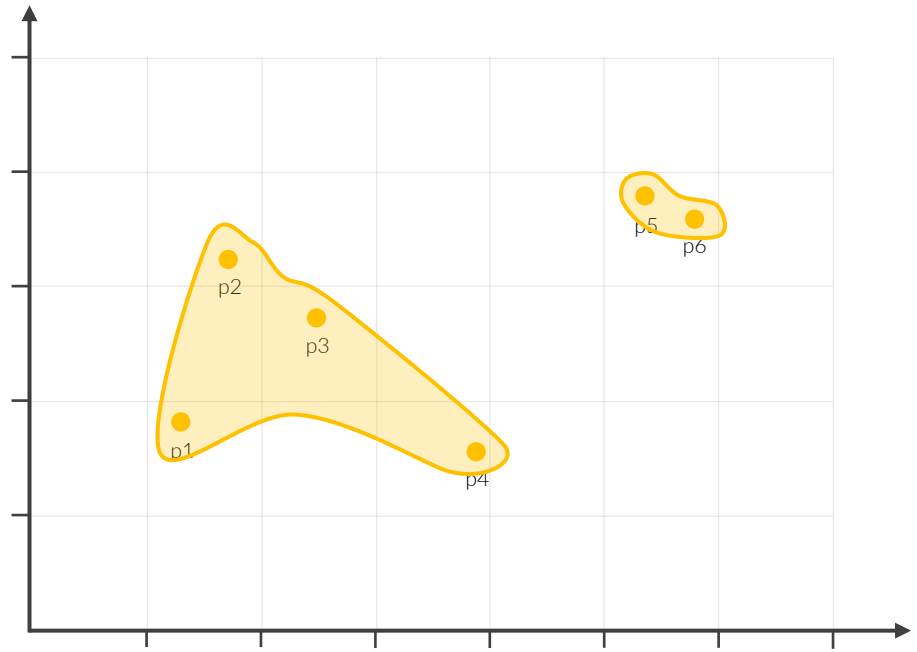


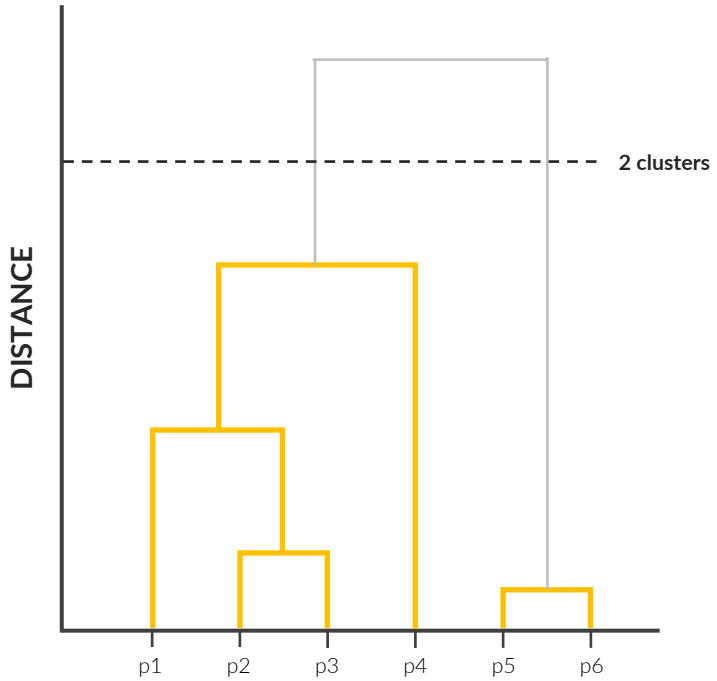


STEP 3: Repeat the process until all points are part of the same cluster

HIERARCHICAL CLUSTERING

- Clustering Basics
- K-Means
- Hierarchical Clustering
- Key Takeaways

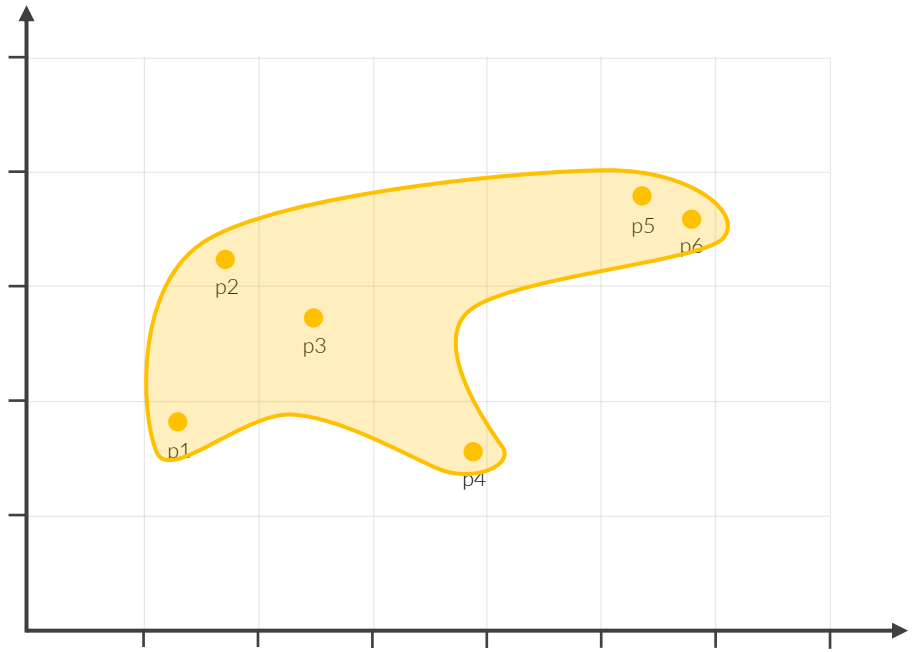


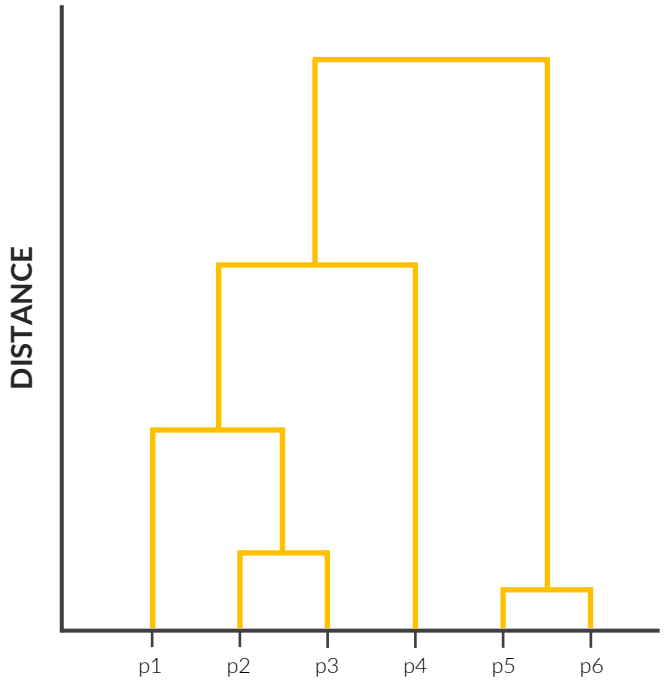


STEP 3: Repeat the process until all points are part of the same cluster

HIERARCHICAL CLUSTERING

- Clustering Basics
- K-Means
- Hierarchical Clustering
- Key Takeaways





STEP 3: Repeat the process until all points are part of the same cluster

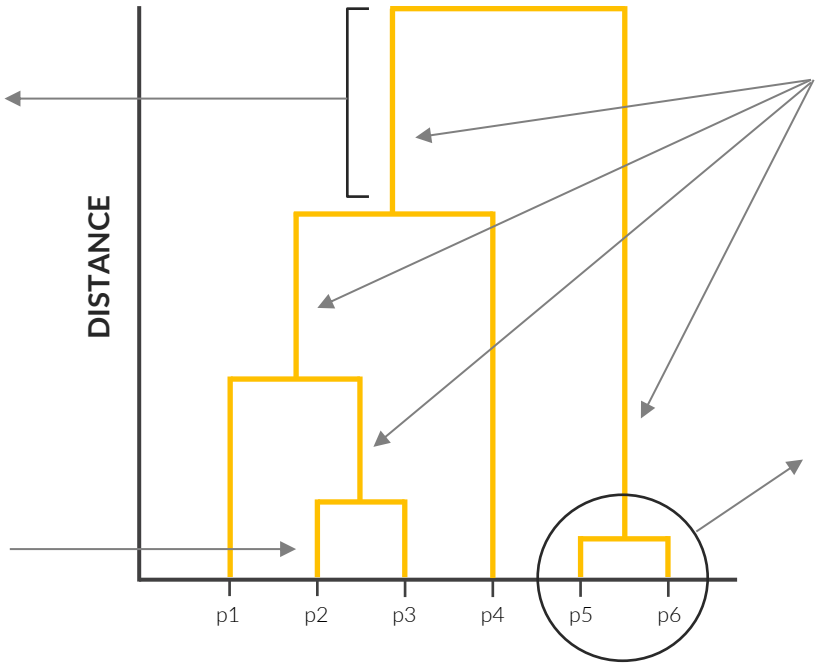
HIERARCHICAL CLUSTERING

- Clustering Basics
- K-Means
- Hierarchical Clustering
- Key Takeaways

How do you know when to stop creating new clusters?

*The **height** of each branch tells us how close the clusters/data points are to each other **(taller = longer distance)***

*At the end of each clade is a **leaf**, representing a single data point*



*Vertical branches that lead to splits are called **clades***

*Clades help us understand **similarity**; two leaves in the same clade are more similar to each other than they are to the leaves in another clade*



HIERARCHICAL CLUSTERING

- Clustering Basics
- K-Means
- Hierarchical Clustering
- Key Takeaways



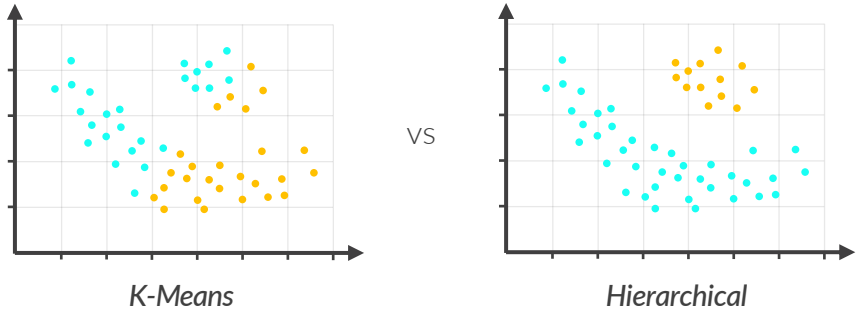
How exactly do you define the **distance** between clusters?

- There are a number of valid ways to measure the distance between clusters, which are often referred to as "**linkage methods**"
- Common methods include measuring the closest *min/max* distance between clusters, the lowest *average* distance, or the distance between cluster centroids



When might you use **hierarchical clustering** over **K-Means**?

- Run both models and compare outputs; Hierarchical clustering may produce more meaningful results if clusters are not circular or uniform in shape



KEY TAKEAWAYS

- Cluster analysis is about finding concentrations of observations which are **similar to one another** and **distinct from other groups**
  - *Common use cases include identifying key customer segments, clustering stores based on performance, etc.*
- Two common techniques are **K-means** and **Hierarchical clustering**
  - *In practice, we recommend testing multiple models since each technique has strengths and weaknesses*
- Models **won't tell you how to interpret** what each cluster represents
  - *Use multivariate analysis and profiling techniques to understand the characteristics of each cluster*

ASSOCIATION MINING

*Copyright Maven Analytics, LLC

ASSOCIATION MINING



In this section we'll introduce the fundamentals of **association mining** and **basket analysis**, and compare common techniques including Apriori and Markov Chains

TOPICS WE'LL COVER:

- Association Mining Basics
- Apriori
- Markov Chains
- Key Takeaways

GOALS FOR THIS SECTION:

- Understand the basics of association mining and introduce several real-world use cases
- Explore common association mining techniques like Apriori and Markov Chains
- Compare and contrast common unsupervised models for basket analysis



ASSOCIATION MINING BASICS

Association Mining Basics

Apriori

Markov Chains

Key Takeaways

Association Mining is used to reveal underlying patterns, relationships and correlations in the data, and translate them into IF/THEN association rules *(i.e. "**If** a customer buys product A, **then** they will likely buy product B")*

- Association Mining is often applied to transactional data, to analyze which products tend to be purchased together (known as "**basket analysis**")

Common use cases:

- Building product recommendation systems *(i.e. Amazon, Netflix)*
- Sending targeted product offers based on purchase history
- Optimizing physical store layouts to maximize cross-selling



Association mining is NOT about trying to prove or establish causation; it's just about identifying frequently occurring patterns and correlations in large datasets

APRIORI

Association Mining Basics

Apriori

Markov Chains

Key Takeaways

The **Apriori** algorithm is commonly used to analyze how frequently items are purchased together in a single transaction *(i.e. beer and diapers, peanut butter and jelly, etc.)*

- In its simplest form *(2-item sets)*, Apriori models compare the frequency of transactions containing **item A, item B**, and both **items A and B**
- This allows you to understand how often these items tend to be purchased together, and calculate the strength of the association between them

Apriori models typically include **three key metrics**:

- **Support** *(frequency of transactions containing a given item or set of items) / total transactions*
- **Confidence** *(conditional probability of item B occurring, given item A)*
- **Lift** *(measures the importance or "strength" of the association between items)*

$$\frac{\text{Support } \mathbf{(A,B)}}{\text{Support } \mathbf{(A)} \times \text{Support } \mathbf{(B)}}$$

APRIORI



- Association Mining Basics
- **Apriori**
- Markov Chains
- Key Takeaways

TRANS.	ITEM 1	ITEM 2
1	🥓	🥚
2	🍌	🌿
3	🥓	🥚
4	🍞	🥛
5	🥓	🌿
6	🍓	🥦
7	🥓	🥚
8	☕	🥛
9	🥓	🥚
10	🥛	🥛
11	🥓	🥚
12	🍌	🥛
13	🥓	🥚
14	🥦	🌿
15	🥓	🧀
16	🥦	🥚
17	🥓	🍼
18	🍌	🌿
19	🥓	🥛
20	🍌	🌿

EXAMPLE #1

*Calculating the association between **bacon** & **eggs***

1) **Support** (🥓) = 10/20 = **0.5**

Support (🥚) = 7/20 = **0.35**

Support (🥓 , 🥚) = 6/20 = **0.3**

2) **Confidence** (🥓 → 🥚) = $\frac{\text{Support}\ (🥓, 🥚)}{\text{Support}\ (🥓)} = \frac{0.3}{0.5} = 60\%$

3) **Lift** (🥓 → 🥚) = $\frac{\text{Support}\ (🥓, 🥚)}{\text{Support}\ (🥓) \times \text{Support}\ (🥚)} = \frac{0.3}{0.5 \times 0.35} = 1.7$

Since **Lift > 1**, we can interpret the association between bacon and eggs as real and informative *(eggs are likely to be purchased with bacon)*

**Inspired by KDNuggets (www.kdnuggets.com)*

APRIORI

- Association Mining Basics
- Apriori
- Markov Chains
- Key Takeaways

TRANS.	ITEM 1	ITEM 2
1	🥓	🥚
2	🍌	🌿
3	🥓	🥚
4	🍞	🥛
5	🥓	🌿
6	🍓	🥦
7	🥓	🥚
8	☕	🥛
9	🥓	🥚
10	🥛	🥛
11	🥓	🥚
12	🍌	🥛
13	🥓	🥚
14	🥦	🌿
15	🥓	🧀
16	🥦	🥚
17	🥓	🥛
18	🍌	🌿
19	🥓	🥛
20	🍌	🌿

EXAMPLE #2

*Calculating the association between **bacon** & **basil***

1) **Support** (🥓) = 10/20 = **0.5**

Support (🌿) = 5/20 = **0.25**

Support (🥓 , 🌿) = 1/20 = **0.05**

2) **Confidence** (🥓 → 🌿) = $\frac{\text{Support}(🥓, 🌿)}{\text{Support}(🥓)} = \frac{0.05}{0.5} = 10\%$

3) **Lift** (🥓 → 🌿) = $\frac{\text{Support}(🥓, 🌿)}{\text{Support}(🥓) \times \text{Support}(🌿)} = \frac{0.05}{0.5 \times 0.25} = 0.4$

Since **Lift < 1**, we can conclude that there is no positive association between bacon and basil *(basil is unlikely to be purchased with bacon)*

*Inspired by KDNuggets (www.kdnuggets.com)

APRIORI

- Association Mining Basics
- **Apriori**
- Markov Chains
- Key Takeaways

TRANS.	ITEM 1	ITEM 2
1	🥓	🥚
2	🍌	🌿
3	🥓	🥚
4	🍞	🥛
5	🥓	🌿
6	🍓	🥦
7	🥓	🥚
8	☕	🥛
9	🥓	🥚
10	🥛	🥛
11	🥓	🥚
12	🍌	🥛
13	🥓	🥚
14	🥦	🌿
15	🥓	🧀
16	🥦	🥚
17	🥓	🥛
18	🍌	🌿
19	🥓	🥛
20	🍌	🌿

EXAMPLE #3

*Calculating the association between **bacon** & **water***

1) **Support** (🥓) = 10/20 = **0.5**

Support (🥛) = 1/20 = **0.05**

Support (🥓 , 🥛) = 1/20 = **0.05**

2) **Confidence** (🥓 → 🥛) = $\frac{\text{Support}(🥓, 🥛)}{\text{Support}(🥓)} = \frac{0.05}{0.5} = 10\%$

3) **Lift** (🥓 → 🥛) = $\frac{\text{Support}(🥓, 🥛)}{\text{Support}(🥓) \times \text{Support}(🥛)} = \frac{0.05}{0.5 \times 0.05} = 2$

Since **Lift = 2** we might assume a strong association between bacon and water, but this is skewed since water only appears in one transaction

**Inspired by KDNuggets (www.kdnuggets.com)*

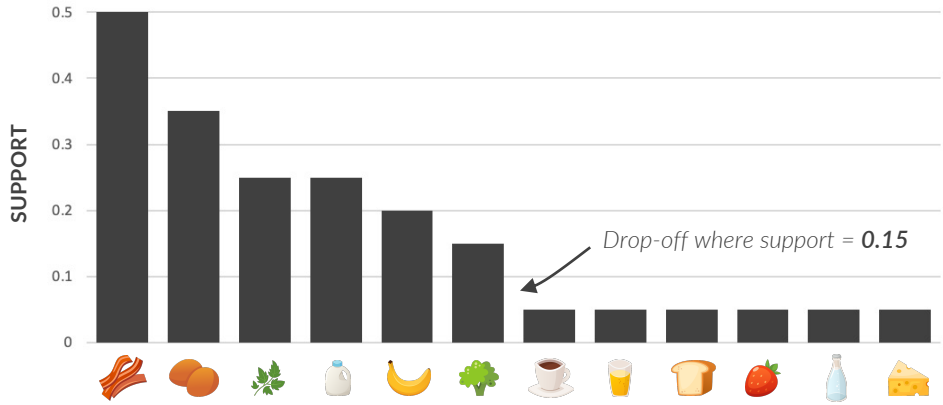
APRIORI

- Association Mining Basics
- Apriori
- Markov Chains
- Key Takeaways

TRANS.	ITEM 1	ITEM 2
1		
2		
3		
4		
5		
6		
7		
8		
9		
10		
11		
12		
13		
14		
15		
16		
17		
18		
19		
20		

How do we account for **infrequently purchased items**?

To filter low-volume purchases, you can plot support for each item and determine a threshold or cutoff value:



In this case we might filter out any transactions containing items with support **<0.15** *(transactions 4, 6, 8, 10, 15, 17)*

- This helps us avoid misleading confidence and lift calculations, and reduces the number of transactions we need to analyze

**Inspired by KDNuggets (www.kdnuggets.com)*

APRIORI

Association Mining Basics

Apriori

Markov Chains

Key Takeaways

Wouldn't you want to analyze *all* possible item combinations, instead of calculating individual associations?

- Yes! In reality that's how Apriori works. But since the number of configurations increases exponentially with each item, this is impractical for humans to do
- To reduce or "prune" the number of itemsets to analyze, we can use the **apriori principle**, which basically states that if an item is infrequent, any combinations containing that item must also be infrequent

Consider a shop that sells **4 items** *(chocolate, cheese, bananas & bread)*

There are **15 possible combinations** or itemsets *(ignoring duplicates)*



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STEP 1:

Calculate support for each **1-item** set, and filter out *all* transactions containing items below the threshold *(in this case **cheese**)*



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STEP 2:

Based on the remaining itemsets, calculate support for each **2-item** set, and filter out transactions containing any pairs below the threshold *(in this case **chocolate & bread**)*



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APRIORI

Association Mining Basics

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STEP 3:

Repeat until all infrequent itemsets have been eliminated, and filter transactions accordingly



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Association Mining Basics

Apriori

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STEP 4:

Based on the filtered transactions, you can use an apriori model to calculate confidence and lift and identify the strongest associations



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APRIORI

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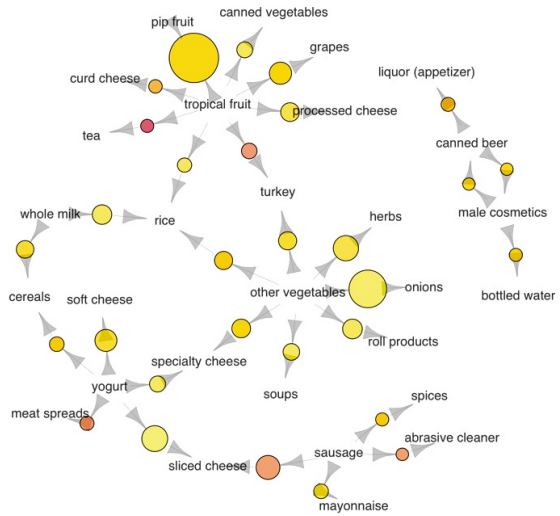
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Associations can be visualized using **network graphs** like this one



*Inspired by KDNuggets (www.kdnuggets.com)

APRIORI



- Association Mining Basics
- Apriori
- Markov Chains
- Key Takeaways



Can you calculate associations between **multiple items**, like coffee being purchased with bacon and eggs?

- Yes, you can calculate support, confidence and lift using the same exact logic as you would with individual items

ID	ITEM 1	ITEM 2	ITEM 3
1			
2			
3			
4			
5			
6			
7			
8			
9			
10			

Support (,) = 3/10 = **0.3**

Support ( ) = 4/10 = **0.4**

Support (, ,) = 2/10 = **0.2**

$$\textbf{Confidence}\left(\& \rightarrow \right) = \frac{\text{Support}\left(, , \right)}{\text{Support}\left(, \right)} = \frac{0.2}{0.3} = \textbf{67\%}$$

$$\textbf{Lift}\left(\& \rightarrow \right) = \frac{\text{Support}\left(, , \right)}{\text{Support}\left(, \right) \times \text{Support}\left(\right)} = \frac{0.2}{0.12} = \textbf{1.7}$$



Calculating all possible associations would be impossible for a human; this is where Machine Learning comes in!

**Inspired by KDNuggets (www.kdnuggets.com)*

CASE STUDY: APRIORI



THE SITUATION

You are the proud owner of **Coastal Roasters**, a local coffee shop and bakery based in the Pacific Northwest.



THE ASSIGNMENT

You'd like to better understand **which items customers tend to purchase together**, to help inform product placement and promotional strategies.

As a first step, you've decided to analyze a sample of ~10,000 transactions over a 6-month period, and conduct a simple basket analysis using an apriori model.



THE OBJECTIVES

1. Collect transaction-level data, including date, time, and items purchased
2. Determine an appropriate support threshold, based on product frequency
3. Calculate confidence and lift to measure association between popular items

MARKOV CHAINS

Association Mining Basics

Apriori

Markov Chains

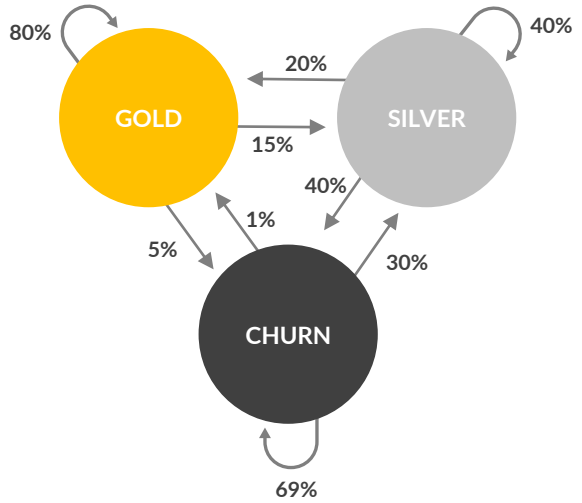
Key Takeaways

Markov Chains are used to describe the flow or transition between "states" of a categorical variable

- Markov Chains **calculate the probability of moving between states** *(subscribers renewing or churning, customers purchasing one product after another, etc.)*
- Observations must be tracked over time and tied to a unique ID, so that all state transitions can be recorded as probabilities in a matrix *(where all rows = 100%)*

EXAMPLE: Monthly subscribers transitioning between **Gold**, **Silver** and **Churn** states each month:

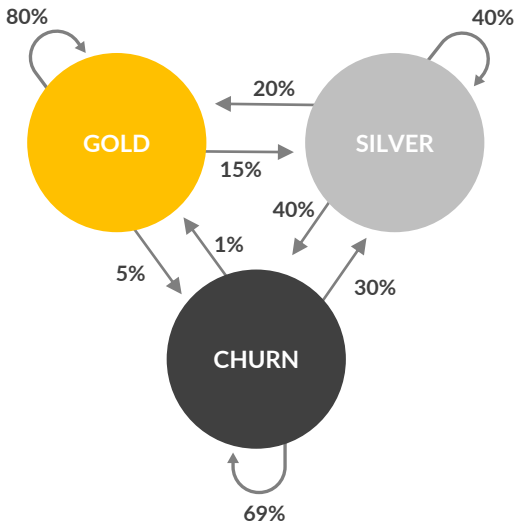
FROM STATE	**TO** STATE: Gold	Silver	Churn
Gold	**0.8**	**0.15**	**0.05**
Silver	**0.2**	**0.4**	**0.4**
Churn	**0.01**	**0.3**	**0.69**



MARKOV CHAINS

- Association Mining Basics
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Example insights & recommendations:

- Most customers who churn stayed churned, but **31%** do come back; of those who return, nearly all of them re-subscribe to a Silver plan *(vs. Gold)*
  - ✓ **RECOMMENDATION:** Launch targeted marketing to recently churned customers, offering a discount to resubscribe to a Silver membership plan
- Once customers upgrade to a Gold membership, the majority **80%** renew each month
  - ✓ **RECOMMENDATION:** Offer a one-time discount for Silver customers to upgrade to Gold; while you may sacrifice some short-term revenue, it will likely be profitable in the long term

To account for prior transitions *(vs. just the previous)* you can use more complex "higher-order" Markov Chains

CASE STUDY: MARKOV CHAINS



THE **SITUATION**

You've just been promoted to Senior Web Analyst at **Alpine Supplies**, an online retailer specializing in equipment and supplies for outdoor enthusiasts.



THE **ASSIGNMENT**

The VP of Sales just shared a sample of ~15,000 customer purchase paths, and would like you to analyze the data to **help inform a new cross-sell sales strategy**.

Your goal is to explore the data and build a simple Markov model to predict which product an existing customer is most likely to purchase next.



THE **OBJECTIVES**

1. Collect sample data containing customer-level purchase paths
2. Calculate a frequency and probability matrix for popular products
3. For any given product, predict the most likely future purchase

KEY TAKEAWAYS

- Association mining reveals **patterns, relationships & correlations** in the data, and translates them into IF/THEN association rules
    - *Common use cases include building recommendation systems (i.e. Amazon, Netflix), targeting product offers based on purchase history, optimizing physical store layouts to maximize cross-selling, etc.*
- **Apriori models** measure how frequently specific items tend to be purchased together *(known as "basket analysis")*
    - *Rather than analzying every possible combination of itemsets, the apriori principle can be used to speed up calculations by filtering out infrequent transactions*
- **Markov Chains** can be used to measure the probability of transitioning between "states" of a categorical variable
    - *Markov models analyze sequences of events over time to determine what is most likely to happen next*

OUTLIER DETECTION

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OUTLIER DETECTION



In this section we'll introduce the concept of **statistical outliers**, and review common methods for detecting both cross-sectional and time-series outliers and anomalies

TOPICS WE'LL COVER:

- Outlier Detection Basics
- Cross-sectional Outliers
- Time-Series Outliers
- Key Takeaways

GOALS FOR THIS SECTION:

- Understand different types of outliers, and how they can impact an analysis
- Learn how to identify outliers in a cross-sectional dataset using a "nearest neighbors" approach
- Learn how to use regression analysis to detect outliers in a time-series dataset



The terms **outlier detection, anomaly detection, and rare-event detection** are often used interchangeably; they all focus on finding observations which are materially different than the others *(outliers are also sometimes called "pathological" data)*



OUTLIER DETECTION BASICS

Outlier Detection Basics

Cross-Sectional Outliers

Time-Series Outliers

Key Takeaways

Outlier detection is used to identify observations in a dataset which are either unexpected or statistically different from the others

There are **two general types** of outliers you may encounter:

- **Data/recording issues:** Values which were captured or categorized incorrectly, and which must be removed or excluded to avoid skewing the analysis
- **Business outliers:** Legitimate values which provide real and meaningful information, such as a fraudulent transaction or an unexpected spike in sales

Common use cases:

- Identifying store locations which are behaving abnormally compared to others
- Detecting web traffic anomalies which would be impossible to detect otherwise
- Flagging potentially fraudulent credit card transactions

CROSS-SECTIONAL OUTLIERS

Outlier Detection Basics

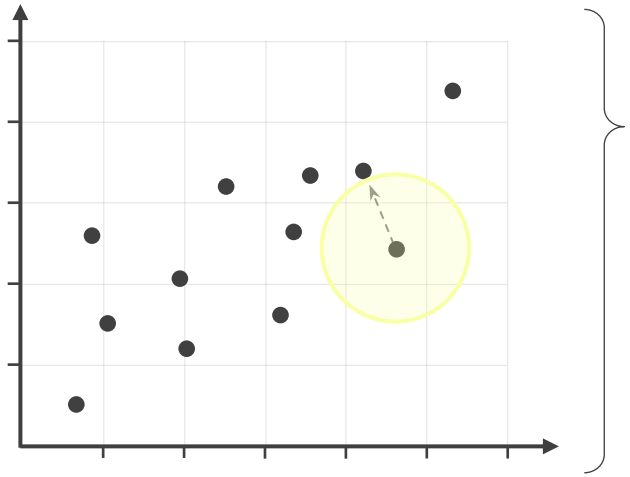
Cross-Sectional Outliers

Time-Series Outliers

Key Takeaways

Cross-sectional outlier detection is used to measure the similarity between observations in a dataset, and identify observations which are unusually different or dissimilar

- Detecting outliers in one or two dimensions is often trivial, using basic profiling metrics *(i.e. interquartile range)* or visual analysis *(scatterplots, box plots, etc.)*
- Detecting outliers 3+ dimensions is tricker, and often requires more sophisticated techniques; **this is where machine learning comes in!**



A common technique is to calculate the distance between each observation and its **nearest neighbor**, and plot those distances to reveal outliers

NOTE: Raw data should first be **scaled** to produce consistent and meaningful distance calculations

CROSS-SECTIONAL OUTLIERS

Outlier Detection Basics

Cross-Sectional Outliers

Time-Series Outliers

Key Takeaways

EXAMPLE | *Normalized store sales by product category (n=101)*

	A	B	C	D
1	**id**	**x_1**	**x_2**	**x_3**
2	1	0.0187	-0.7806	-0.1540
3	2	-0.1843	0.6036	0.5848
4	3	-1.3713	0.3314	-0.3448
5	4	-0.5992	1.3107	1.1476
6	5	0.2945	-0.9278	-1.1736
7	6	0.3898	0.1734	0.8111
8	7	-1.2081	1.8691	1.1873
9	8	-0.3637	-1.2944	-2.2954
10	9	-1.6267	2.6548	0.9313
11	10	-0.2565	1.3844	0.4804
12	11	1.1018	-2.3819	-1.2092
13	12	0.7558	0.3731	1.7597
14	13	-0.2382	-0.2259	-0.6799
15	14	0.9874	-1.3032	-1.2271
16	15	0.7414	0.1829	1.1006
17	16	0.0893	-0.0122	-0.5971
18	17	-0.9549	1.9949	1.3938
19	18	-0.1952	0.9370	0.5364
20	19	0.9255	0.3300	1.0325
21	20	0.4830	0.4679	0.4303
22	21	-0.5963	0.1149	-0.6460
23	22	-2.1853	2.3882	0.0615

In this example we're looking at normalized sales for 3 product categories (**x_1**, **x_2** and **x_3**) across **101** different store locations

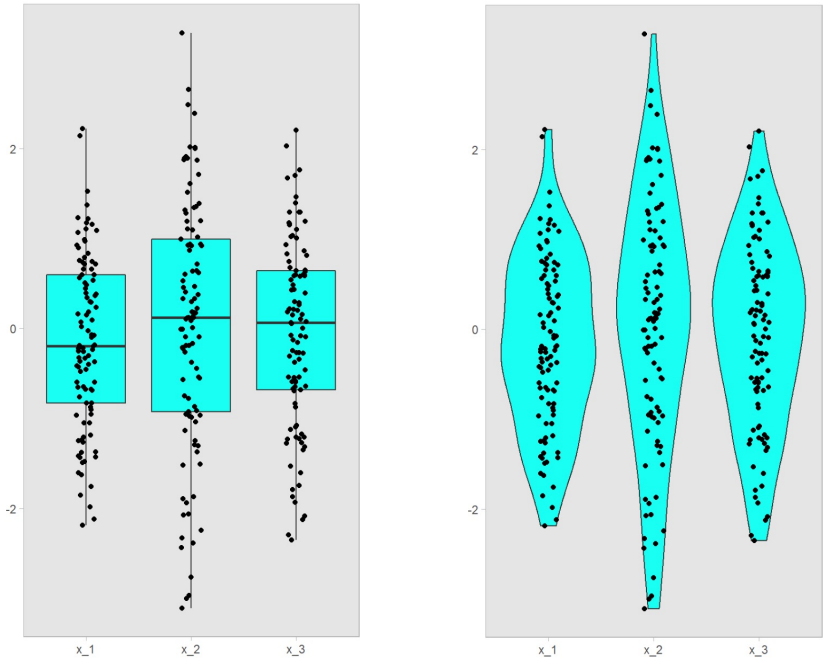
- In cases like these, detecting outliers by visualizing distributions or scatterplots can be difficult *(and often impossible)*

CROSS-SECTIONAL OUTLIERS

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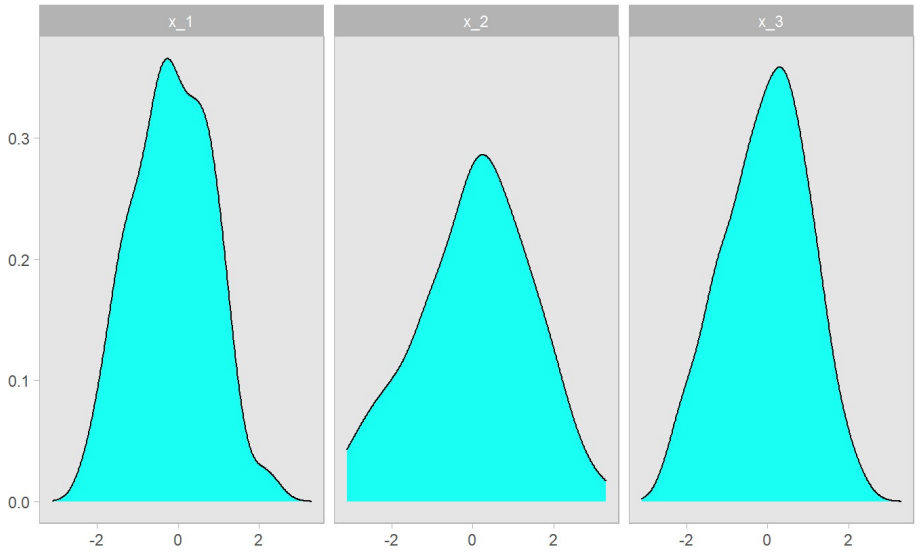
In this case, **box and violin plots** don't reveal any obvious outliers…

CROSS-SECTIONAL OUTLIERS

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EXAMPLE | *Normalized store sales by product category (n=101)*

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Kernel density plots show relatively normal distributions for each variable…

CROSS-SECTIONAL OUTLIERS

Outlier Detection Basics

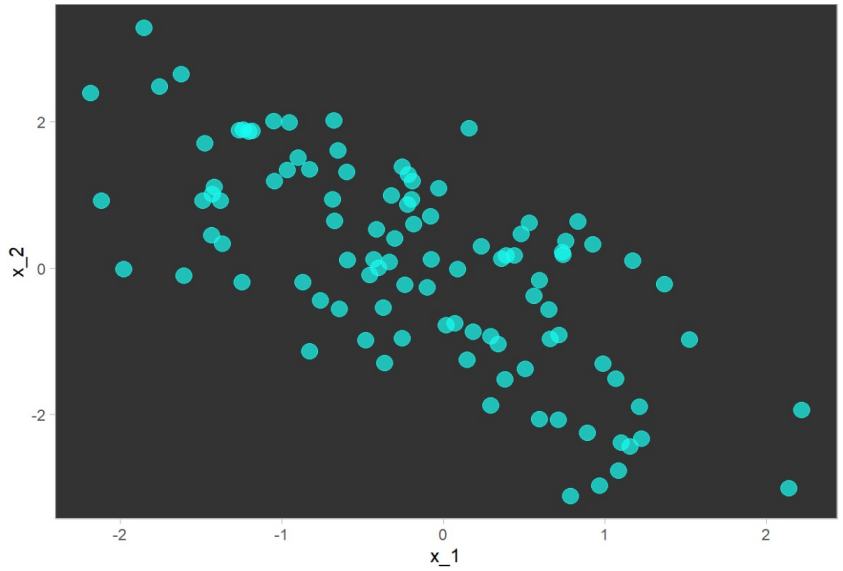
Cross-Sectional Outliers

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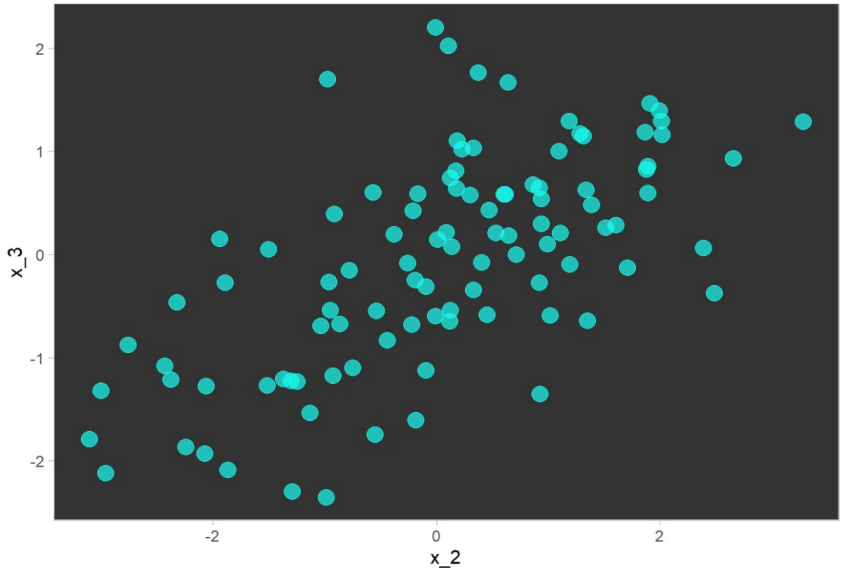
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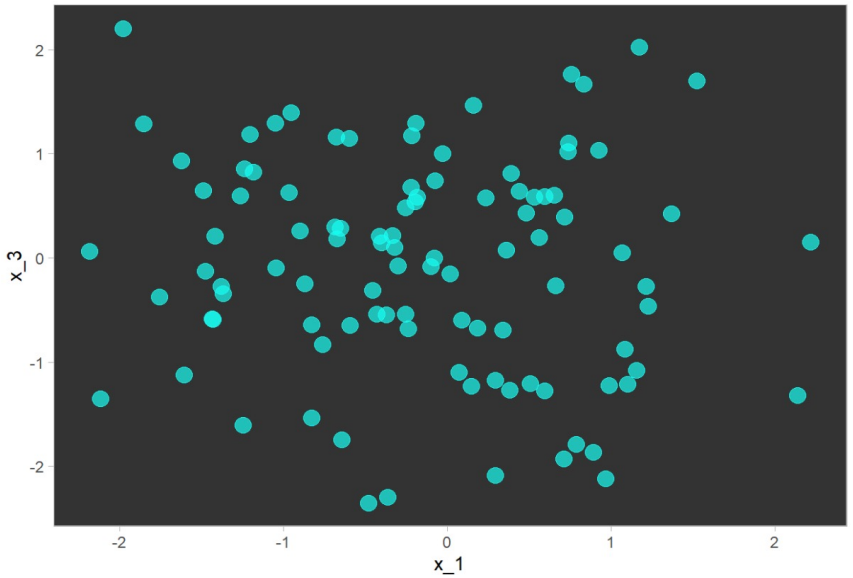
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20	19	0.9255	0.3300	1.0325
21	20	0.4830	0.4679	0.4303
22	21	-0.5963	0.1149	-0.6460
23	22	-2.1853	2.3882	0.0615



And **scatter plots** fail to expose any anomalies in the data

CROSS-SECTIONAL OUTLIERS

Outlier Detection Basics

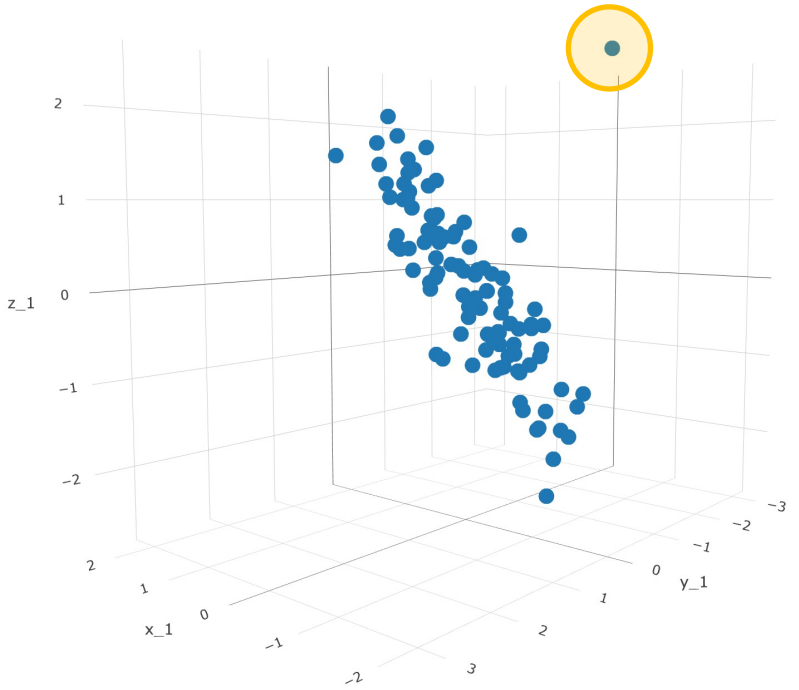
Cross-Sectional Outliers

Time-Series Outliers

Key Takeaways

EXAMPLE | *Normalized store sales by product category (n=101)*

	A	B	C	D
1	**id**	**x_1**	**x_2**	**x_3**
2	1	0.0187	-0.7806	-0.1540
3	2	-0.1843	0.6036	0.5848
4	3	-1.3713	0.3314	-0.3448
5	4	-0.5992	1.3107	1.1476
6	5	0.2945	-0.9278	-1.1736
7	6	0.3898	0.1734	0.8111
8	7	-1.2081	1.8691	1.1873
9	8	-0.3637	-1.2944	-2.2954
10	9	-1.6267	2.6548	0.9313
11	10	-0.2565	1.3844	0.4804
12	11	1.1018	-2.3819	-1.2092
13	12	0.7558	0.3731	1.7597
14	13	-0.2382	-0.2259	-0.6799
15	14	0.9874	-1.3032	-1.2271
16	15	0.7414	0.1829	1.1006
17	16	0.0893	-0.0122	-0.5971
18	17	-0.9549	1.9949	1.3938
19	18	-0.1952	0.9370	0.5364
20	19	0.9255	0.3300	1.0325
21	20	0.4830	0.4679	0.4303
22	21	-0.5963	0.1149	-0.6460
23	22	-2.1853	2.3882	0.0615



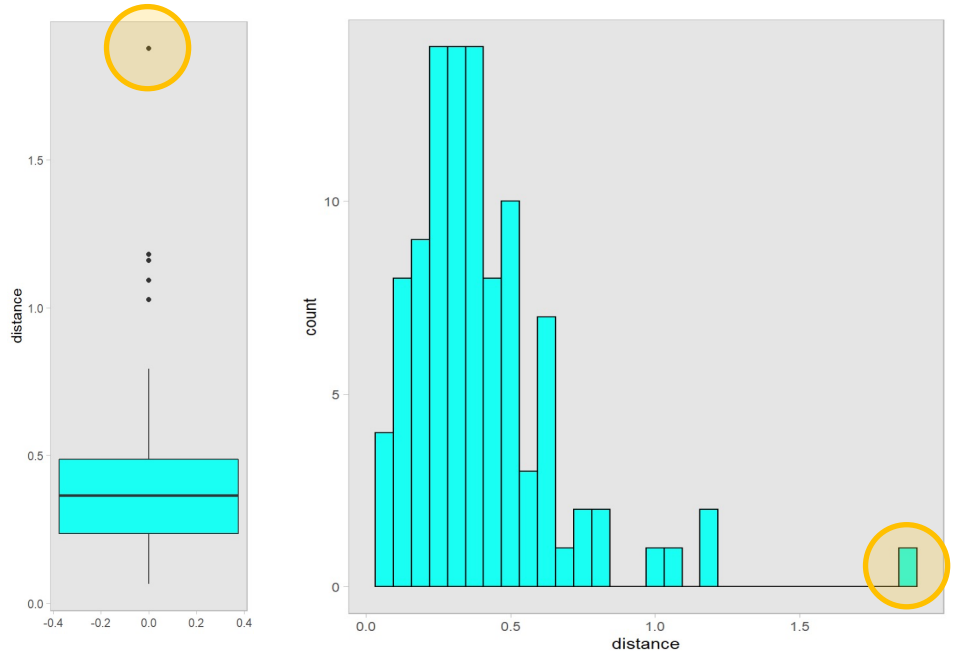
But there *is* an outlier in the sample, which we can clearly see when we visualize a **3rd dimension**

CROSS-SECTIONAL OUTLIERS

- Outlier Detection Basics
- Cross-Sectional Outliers
- Time-Series Outliers
- Key Takeaways

For cross-sectional outlier detection, techniques like **nearest neighbors** are more efficient and can scale to *any* number of dimensions

	A	B	C	D	E
1	id	x_1	x_2	x_3	NN
2	1	0.0187	-0.7806	-0.1540	0.505
3	2	-0.1843	0.6036	0.5848	0.280
4	3	-1.3713	0.3314	-0.3448	0.277
5	4	-0.5992	1.3107	1.1476	0.383
6	5	0.2945	-0.9278	-1.1736	0.297
7	6	0.3898	0.1734	0.8111	0.178
8	7	-1.2081	1.8691	1.1873	0.236
9	8	-0.3637	-1.2944	-2.2954	0.511
10	9	-1.6267	2.6548	0.9313	0.758
11	10	-0.2565	1.3844	0.4804	0.455
12	11	1.1018	-2.3819	-1.2092	0.148
13	12	0.7558	0.3731	1.7597	0.290
14	13	-0.2382	-0.2259	-0.6799	0.368
15	14	0.9874	-1.3032	-1.2271	0.487
16	15	0.7414	0.1829	1.1006	0.091
17	16	0.0893	-0.0122	-0.5971	0.400
18	17	-0.9549	1.9949	1.3938	0.142
19	18	-0.1952	0.9370	0.5364	0.158
20	19	0.9255	0.3300	1.0325	0.218
21	20	0.4830	0.4679	0.4303	0.219
22	21	-0.5963	0.1149	-0.6460	0.196
23	22	-2.1853	2.3882	0.0615	0.619
24	23	-0.6749	0.6431	0.1845	0.284
25	24	-1.119	0.9135	-1.3479	1.027



By calculating the distance from each point to its nearest neighbor and plotting the distribution, we can detect a **clear outlier** in the data

CASE STUDY: OUTLIER DETECTION



THE SITUATION

You are a Senior Data Analyst for **Brain Games**, a large chain of retail shops specializing in educational toys, puzzles and board games.



THE ASSIGNMENT

You'd like to analyze store-level sales by product category, to identify patterns and **see if any locations show an unusual composition of revenue**.

To do this, you'll be exploring revenue data across 100 individual store locations, and detecting outliers using "nearest neighbor" calculations.



THE OBJECTIVES

1. Collect category-level sales data by store location
2. Use scatterplots to visualize relationships and scan for outliers
3. Calculate the nearest neighbor for each observation and visualize the data using a box plot and histogram. Do you see any outliers now?

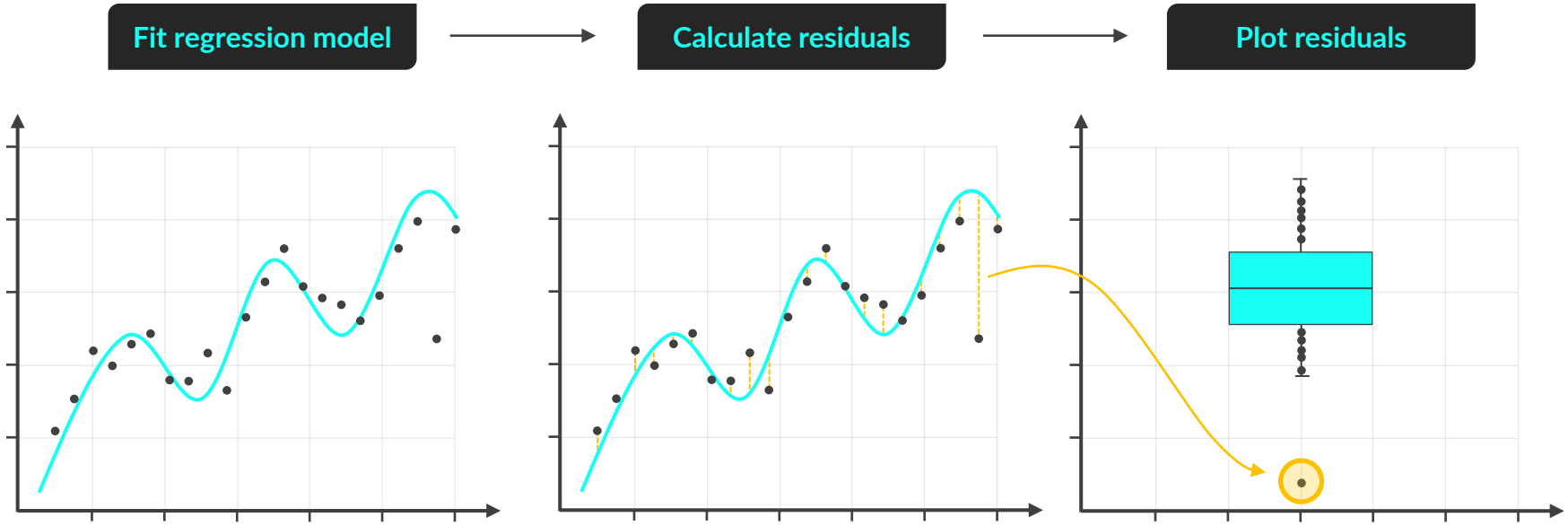
TIME-SERIES OUTLIERS



- Outlier Detection Basics
- Cross-Sectional Outliers
- Time-Series Outliers
- Key Takeaways

Time-series outlier detection is used to identify observations which fall well outside expectations for a particular point in time, after accounting for seasonality and trending

- In practice, this involves building a **time-series regression**, comparing each observation to its predicted value, and plotting the residuals to detect anomalies



TIME-SERIES OUTLIERS

Outlier Detection Basics

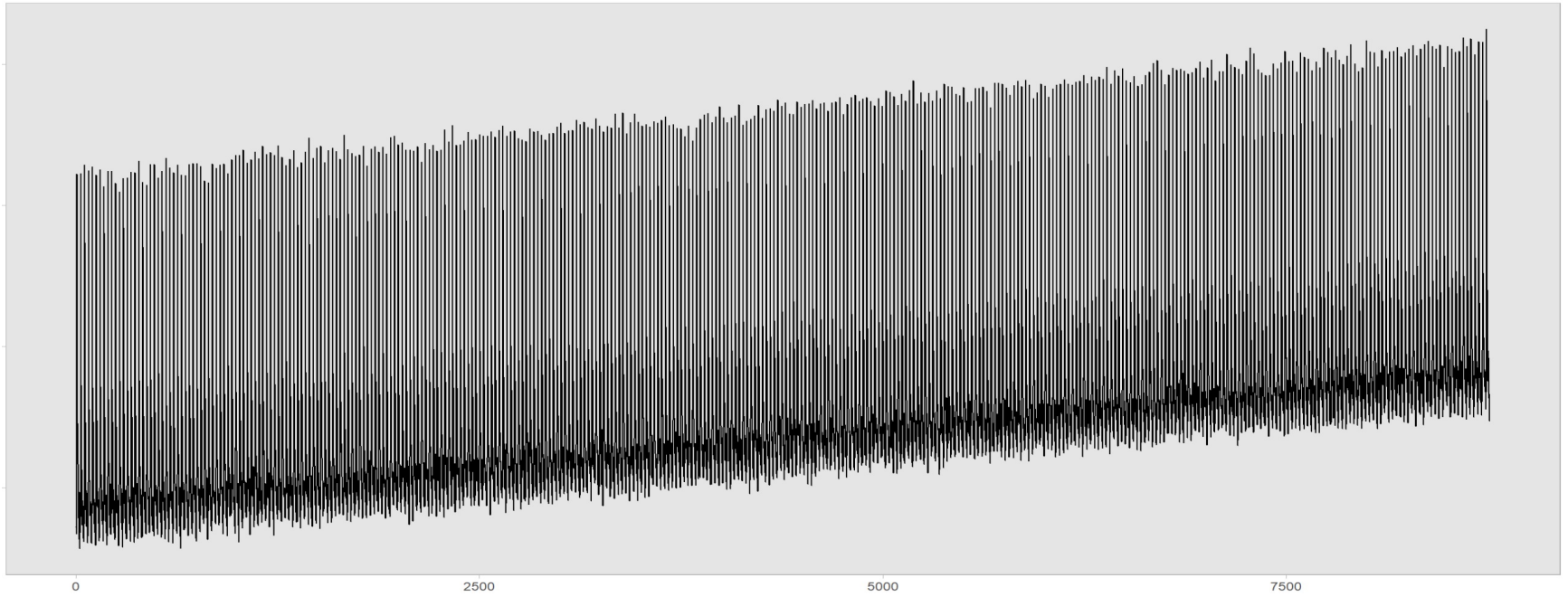
Cross-Sectional Outliers

Time-Series Outliers

Key Takeaways

Real-world time-series data is often **extremely granular**, with recurring patterns and trends which make visual outlier detection virtually impossible

EXAMPLE *Hourly website sessions (n=8,760)*



TIME-SERIES OUTLIERS

Outlier Detection Basics

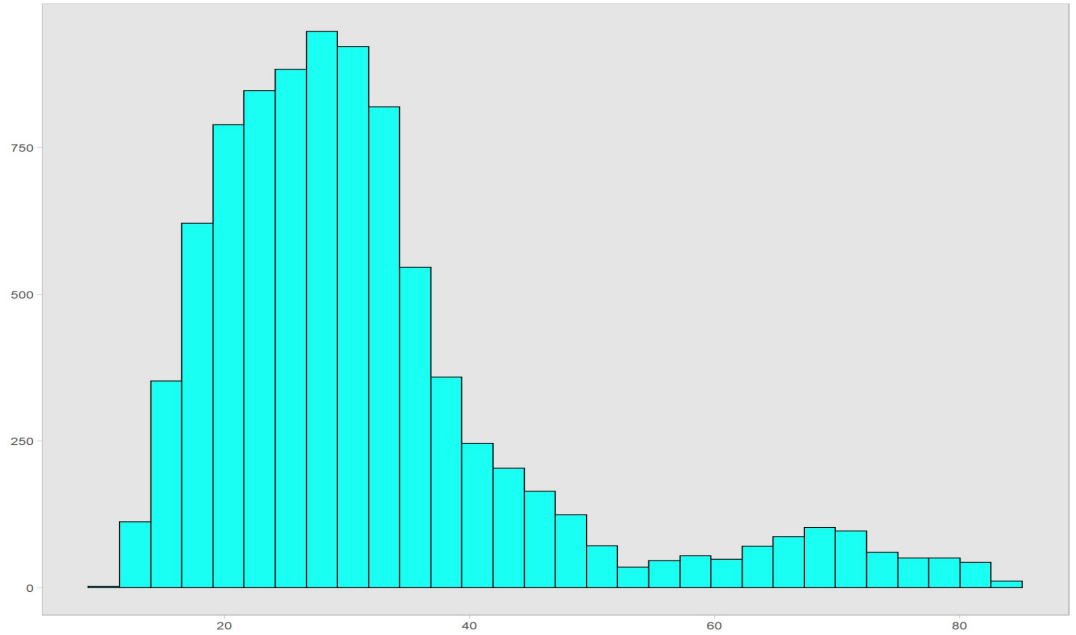
Cross-Sectional Outliers

Time-Series Outliers

Key Takeaways

EXAMPLE *Hourly website sessions (n=8,760)*

	A	B
1	index	value
2	1	14.3710
3	2	13.4375
4	3	16.3675
5	4	20.6394
6	5	64.4129
7	6	53.9047
8	7	35.5245
9	8	31.9205
10	9	29.0358
11	10	28.9568
12	11	23.3266
13	12	23.3105
14	13	15.6372
15	14	17.7494
16	15	16.8970
17	16	18.6685
18	17	14.7505
19	18	13.3804
20	19	12.5986
21	20	19.3613
22	21	13.7367
23	22	11.2642
24	23	12.8758



Just like our cross-sectional example, univariate distributions don't reveal any outliers or anomalies

TIME-SERIES OUTLIERS



- Outlier Detection Basics
- Cross-Sectional Outliers
- Time-Series Outliers
- Key Takeaways

Instead of simple visual analysis, we can fit a **linear regression model** to account for seasonality (hour of day) and trending

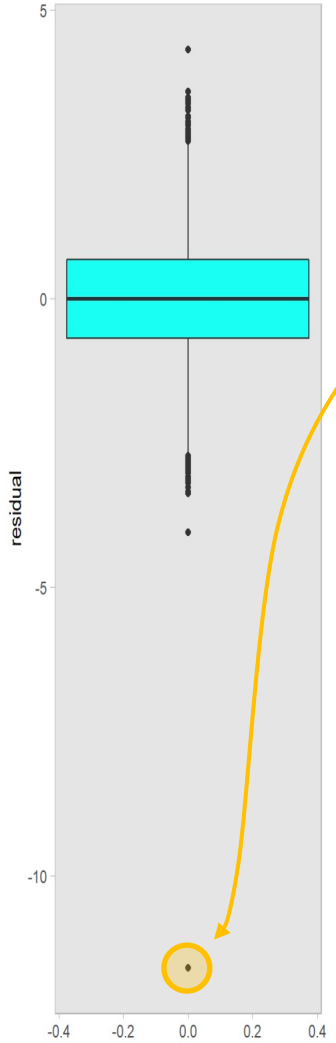
- From there, we can **plot the distribution of residuals** in order to quickly identify any observations which significantly deviated from the model's prediction

web_visits	t-step	hour_1	hour_2	hour_3	hour_4	hour_5	hour_6	hour_7	hour_8	hour_9	hour_10	hour_11	hour_12	hour_13
14.3710	1	1	0	0	0	0	0	0	0	0	0	0	0	0
13.4375	2	0	1	0	0	0	0	0	0	0	0	0	0	0
16.3675	3	0	0	1	0	0	0	0	0	0	0	0	0	0
20.6394	4	0	0	0	1	0	0	0	0	0	0	0	0	0
64.4129	5	0	0	0	0	1	0	0	0	0	0	0	0	0
53.9047	6	0	0	0	0	0	1	0	0	0	0	0	0	0
35.5245	7	0	0	0	0	0	0	1	0	0	0	0	0	0
31.9205	8	0	0	0	0	0	0	0	1	0	0	0	0	0
29.0358	9	0	0	0	0	0	0	0	0	1	0	0	0	0
28.9568	10	0	0	0	0	0	0	0	0	0	1	0	0	0
23.3266	11	0	0	0	0	0	0	0	0	0	0	1	0	0
23.3105	12	0	0	0	0	0	0	0	0	0	0	0	1	0
15.6372	13	0	0	0	0	0	0	0	0	0	0	0	0	1
17.7494	14	0	0	0	0	0	0	0	0	0	0	0	0	0
16.8970	15	0	0	0	0	0	0	0	0	0	0	0	0	0
18.6685	16	0	0	0	0	0	0	0	0	0	0	0	0	0
14.7505	17	0	0	0	0	0	0	0	0	0	0	0	0	0
13.[illegible]4		0		0	0					0	0			

TIME-SERIES OUTLIERS

- Outlier Detection Basics
- Cross-Sectional Outliers
- Time-Series Outliers
- Key Takeaways

index	web_visits	prediction	residual
1	14.3710	12.9582	1.4127
2	13.4375	13.9417	-0.5042
3	16.3675	15.9503	0.4172
4	20.6394	20.0701	0.5693
5	64.4129	64.0274	0.3855
6	53.9047	53.9894	-0.0847
7	35.5245	34.0143	1.5102
8	31.9205	31.9487	-0.0281
9	29.0358	26.9367	2.0991
10	28.9568	28.9841	-0.0273
11	23.3266	21.9954	1.3312
12	23.3105	21.0162	2.2943
13	15.6372	17.0982	-1.4611
14	17.7494	18.0255	-0.2761
15	16.8970	17.0690	-0.1719
16	18.6685	18.0356	0.6329
17	14.7505	15.0293	-0.2788
18	13.3804	16.0404	-2.6599
19	12.5986	14.9461	-2.3475
20	19.3613	18.0447	1.3166
21	13.7367	14.0055	-0.2687
22	11.2642	13.0380	-1.7737
23	12.8758	12.9748	-0.0990
24	14.2646	12.9861	1.2785
25	14.9473	13.0103	1.9369



When we plot the distribution of model residuals, we detect a clear outlier in the data



PRO TIP: Not all outliers are bad! If you find an anomaly, understand what happened and how you can learn from it

KEY TAKEAWAYS

- **Outlier detection** is used to identify observations in a dataset which are either unexpected or statistically different from the others
  - *Common use cases include detecting web traffic anomalies, flagging fraudulent credit card transactions, identifying unusually high or low-performing store locations, etc.*

- For cross-sectional analysis, plotting **"nearest neighbor" distances** can help detect anomalies when visual profiling falls short
  - *This approach is ideal for exposing outliers in large, complex or highly-dimensional datasets*

- For time-series analysis, you can fit a **regression model** to the data and plot the residuals to clearly identify outliers
  - *This allows you to quickly find outliers, while controlling for seasonality and trending*

DIMENSIONALITY REDUCTION

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DIMENSIONALITY REDUCTION



In this section we'll review the fundamentals of **dimensionality reduction**, discuss common Machine Learning and business use cases, and introduce core techniques like Principal Component Analysis (PCA)

TOPICS WE'LL COVER:

- Dimensionality Reduction Basics
- Principal Component Analysis
- Advanced Techniques
- Key Takeaways

GOALS FOR THIS SECTION:

- Understand how dimensionality reduction can be applied in business and Machine Learning contexts
- Explore the intuition behind core linear dimensionality reduction techniques, including Principal Component Analysis (PCA)
- Introduce more advanced models, including techniques for non-linear dimensionality reduction



DIMENSIONALITY REDUCTION

- Dimensionality Reduction Basics
- Principal Component Analysis
- Advanced Techniques
- Key Takeaways

Dimensionality reduction involves reducing the number of columns *(i.e. dimensions)* in a dataset while losing as little information as possible

- This can be used strictly as a ML optimization technique, or to help develop a better understanding of the data for business intelligence, market research, etc.

COMMON **BUSINESS** USE CASES:

- ✓ *Summarizing item-level sales data to better understand group or category performance*
- ✓ *Analyzing survey data to understand overarching trends and findings without parsing through individual question responses*
- ✓ *Understanding the main traits or characteristics of your customers (website activity, transaction patterns, survey response data, etc.)*

COMMON **ML** USE CASES:

- ✓ *Improving a model's predictive power by reducing redundant or unnecessary dimensions in the data*
- ✓ *Visualizing clusters in limited dimensions to spot-check cluster/segmentation outputs*
- ✓ *Reducing trivial correlations and removing multicollinearity to build more accurate and meaningful models*



PRINCIPAL COMPONENT ANALYSIS

- Dimensionality Reduction Basics
- Principal Component Analysis
- Advanced Techniques
- Key Takeaways

One of the most common dimensionality reduction techniques is **Principal Component Analysis** (PCA)

- In the simplest form, PCA finds lines that best fit through the observations in a data set, and uses those lines to create *new* dimensions to analyze

STEP 1 → **STEP 2** → **STEP 3** → **STEP 4** → **STEP 5**

Use high-dimensional data that could be more useful if it could be accurately represented with fewer dimensions

Find "components" or lines that fit through those dimensions, to potentially use as new dimensions to represent the data

Interpret the weighting of each component on the original dimensions to build an intuition for what they represent

Select the "principal" components which explain a material proportion of the variance in the data

Use those principal components in place of existing dimensions for further analysis

PRINCIPAL COMPONENT ANALYSIS

- Dimensionality Reduction Basics
- **Principal Component Analysis**
- Advanced Techniques
- Key Takeaways

EXAMPLE

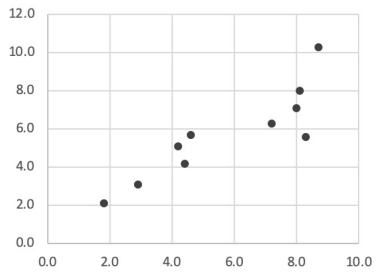
Student test scores by subject (n=10)

studentID	spelling	vocabulary	multiplication	geometry
1	1.8	2.1	5.4	5.6
2	2.9	3.1	6.6	5.7
3	4.6	5.7	2.8	1.6
4	8.0	7.1	3.5	3.4
5	8.1	8.0	1.3	2.2
6	8.3	5.6	7.0	9.1
7	4.4	4.2	9.1	7.3
8	7.2	6.3	8.1	7.8
9	4.2	5.1	9.9	9.3
10	8.7	10.3	8.2	8.7

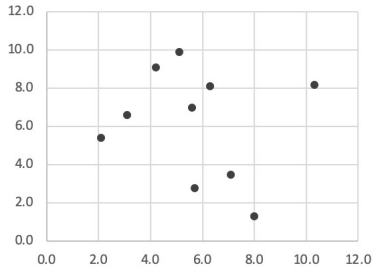
Plotting pairs of dimensions reveals some correlation between **Spelling/Vocabulary** and **Multiplication/Geometry** test scores

- Linear dimensionality reduction techniques will find lines that fit through *highly correlated* dimensions first

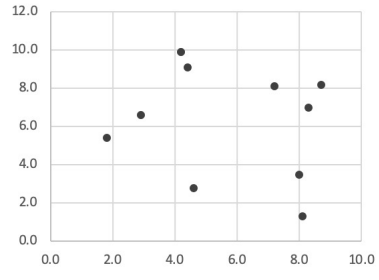
Spelling (X) / **Vocabulary** (Y)



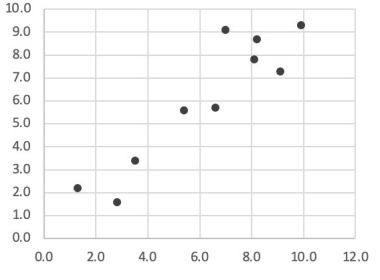
Vocabulary (X) / **Multiplication** (Y)



Spelling (X) / **Multiplication** (Y)



Multiplication (X) / **Geometry** (Y)



PRINCIPAL COMPONENT ANALYSIS

Dimensionality Reduction Basics

Principal Component Analysis

Advanced Techniques

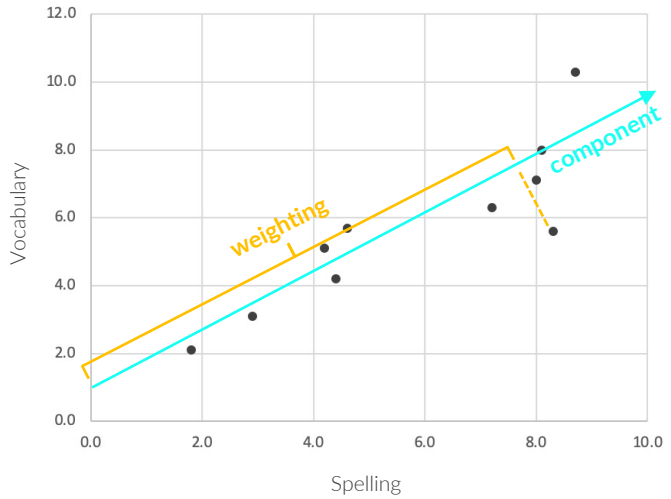
Key Takeaways

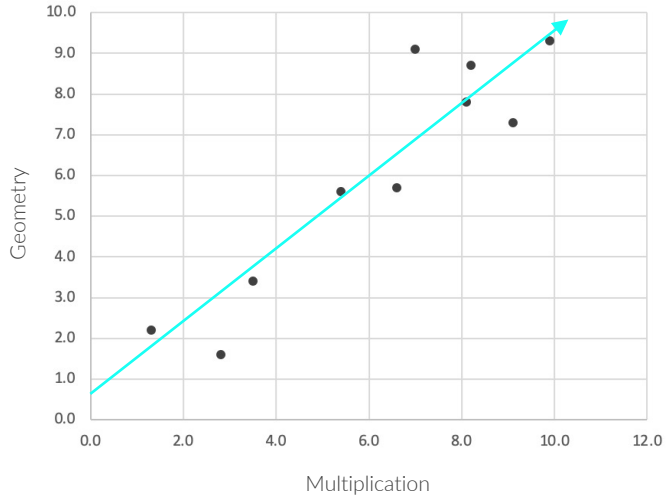
EXAMPLE *Student test scores by subject (n=10)*

studentID	spelling	vocabulary	multiplication	geometry
1	1.8	2.1	5.4	5.6
2	2.9	3.1	6.6	5.7
3	4.6	5.7	2.8	1.6
4	8.0	7.1	3.5	3.4
5	8.1	8.0	1.3	2.2
6	8.3	5.6	7.0	9.1
7	4.4	4.2	9.1	7.3
8	7.2	6.3	8.1	7.8
9	4.2	5.1	9.9	9.3
10	8.7	10.3	8.2	8.7

In the context of dimensionality reduction, these lines are called **components** or **factors**

Observations are given a **weighting** for each component, which essentially measures its position relative to the length of the line





PRINCIPAL COMPONENT ANALYSIS

Dimensionality Reduction Basics

Principal Component Analysis

Advanced Techniques

Key Takeaways

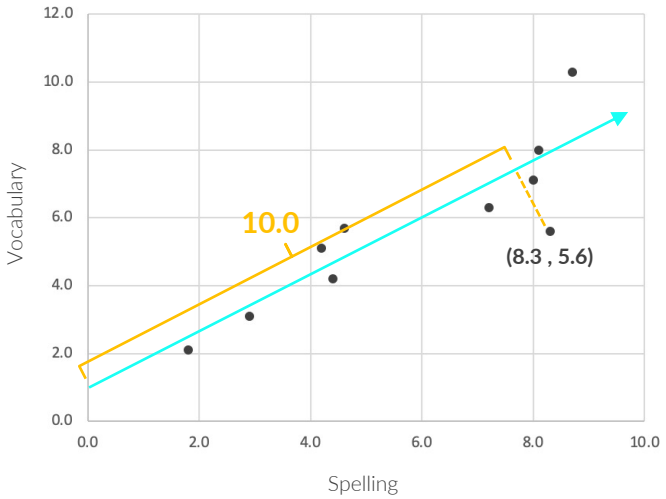
EXAMPLE

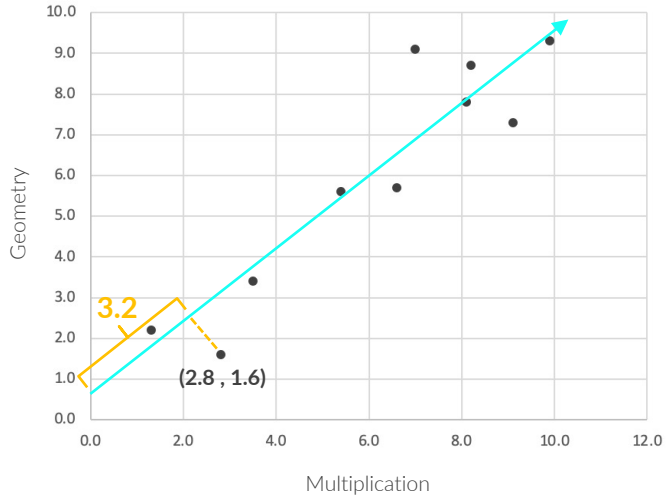
Student test scores by subject (n=10)

studentID	spelling	vocabulary	multiplication	geometry	component 1	component 2
1	1.8	2.1	5.4	5.6	2.8	7.8
2	2.9	3.1	6.6	5.7	4.3	8.7
3	4.6	5.7	2.8	1.6	7.3	3.2
4	8.0	7.1	3.5	3.4	10.7	4.9
5	8.1	8.0	1.3	2.2	11.4	2.6
6	8.3	5.6	7.0	9.1	10.0	11.5
7	4.4	4.2	9.1	7.3	6.1	11.6
8	7.2	6.3	8.1	7.8	9.6	11.3
9	4.2	5.1	9.9	9.3	6.6	13.6
10	8.7	10.3	8.2	8.7	13.5	12.0

Plot the component weightings for each point, and record them as **new dimensions** in the table

NOTE: We're looking at only 2 dimensions here, but PCA scales to ***any*** **number of dimensions**





PRINCIPAL COMPONENT ANALYSIS

Dimensionality Reduction Basics

Principal Component Analysis

Advanced Techniques

Key Takeaways

How do you determine the *meaning* of each component?

- PCA will generate many potential components, and it won't be immediately clear what they actually represent
- In practice, PCA models calculate weights (or "**loadings**") to help explain the relationship between each component and the original dimensions *(think of loadings like coefficients in a linear regression)*

	X_1	X_2	X_3	X_4	Y	
studentID	spelling	vocabulary	multiplication	geometry	language	component 2
1	1.8	2.1	5.4	5.6	2.8	7.8
2	2.9	3.1	6.6	5.7	4.3	8.7
3	4.6	5.7	2.8	1.6	7.3	3.2
4	8.0	7.1	3.5	3.4	10.7	4.9
5	8.1	8.0	1.3	2.2	11.4	2.6
6	8.3	5.6	7.0	9.1	10.0	11.5
7	4.4	4.2	9.1	7.3	6.1	11.6
8	7.2	6.3	8.1	7.8	9.6	11.3
9	4.2	5.1	9.9	9.3	6.6	13.6
10	8.7	10.3	8.2	8.7	13.5	12.0

$$Y = (0.72)^*X_1 + (0.69)^*X_2 + (-0.02)^*X_3 + (0.03)^*X_4$$

*High coefficients on X_1 (**spelling**) and X_2 (**vocabulary**), so we might interpret component 1 as **"language"***

PRINCIPAL COMPONENT ANALYSIS

Dimensionality Reduction Basics

Principal Component Analysis

Advanced Techniques

Key Takeaways

How do you determine the *meaning* of each component?

- PCA will generate many potential components, and it won't be immediately clear what they actually represent
- In practice, PCA models calculate weights (or "**loadings**") to help explain the relationship between each component and the original dimensions *(think of loadings like coefficients in a linear regression)*

	X_1	X_2	X_3	X_4		Y
studentID	spelling	vocabulary	multiplication	geometry	language	math
1	1.8	2.1	5.4	5.6	2.8	7.8
2	2.9	3.1	6.6	5.7	4.3	8.7
3	4.6	5.7	2.8	1.6	7.3	3.2
4	8.0	7.1	3.5	3.4	10.7	4.9
5	8.1	8.0	1.3	2.2	11.4	2.6
6	8.3	5.6	7.0	9.1	10.0	11.5
7	4.4	4.2	9.1	7.3	6.1	11.6
8	7.2	6.3	8.1	7.8	9.6	11.3
9	4.2	5.1	9.9	9.3	6.6	13.6
10	8.7	10.3	8.2	8.7	13.5	12.0

$$Y = (0.004)^*X_1 + (0.001)^*X_2 + (0.68)^*X_3 + (0.72)^*X_4$$

*High coefficients on X_3 (**multiplication**) and X_4 (**geometry**), so we might interpret component 2 as **"math"***

This requires human intuition; your model won't tell you what the components mean!

PRINCIPAL COMPONENT ANALYSIS

Dimensionality Reduction Basics

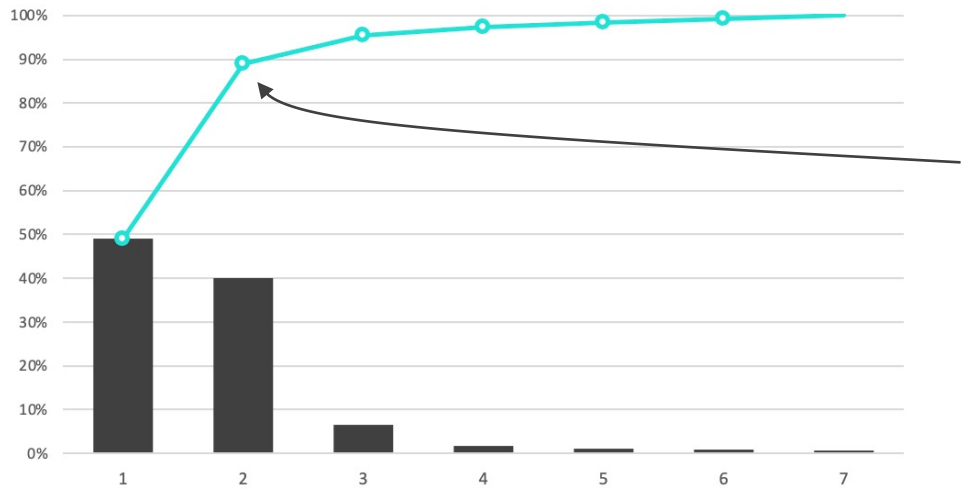
Principal Component Analysis

Advanced Techniques

Key Takeaways

How do you know how many components to keep?

- You can use a **Scree Plot** to visualize the cumulative variance explained by each component, and look for the "elbow" where additional components add relatively little value *(similar to a WSS chart for clustering)*
- The components up to and including the "elbow" become your **principal** components, and others can be disregarded



*In this plot, the first **2 components** explain most of the variance in the data, and additional components have minimal impact*

PRINCIPAL COMPONENT ANALYSIS



- Dimensionality Reduction Basics
- Principal Component Analysis
- Advanced Techniques
- Key Takeaways

studentID	spelling	vocabulary	multiplication	geometry	language	math
1	1.8	2.1	5.4	5.6	2.8	7.8
2	2.9	3.1	6.6	5.7	4.3	8.7
3	4.6	5.7	2.8	1.6	7.3	3.2
4	8.0	7.1	3.5	3.4	10.7	4.9
5	8.1	8.0	1.3	2.2	11.4	2.6
6	8.3	5.6	7.0	9.1	10.0	11.5
7	4.4	4.2	9.1	7.3	6.1	11.6
8	7.2	6.3	8.1	7.8	9.6	11.3
9	4.2	5.1	9.9	9.3	6.6	13.6
10	8.7	10.3	8.2	8.7	13.5	12.0

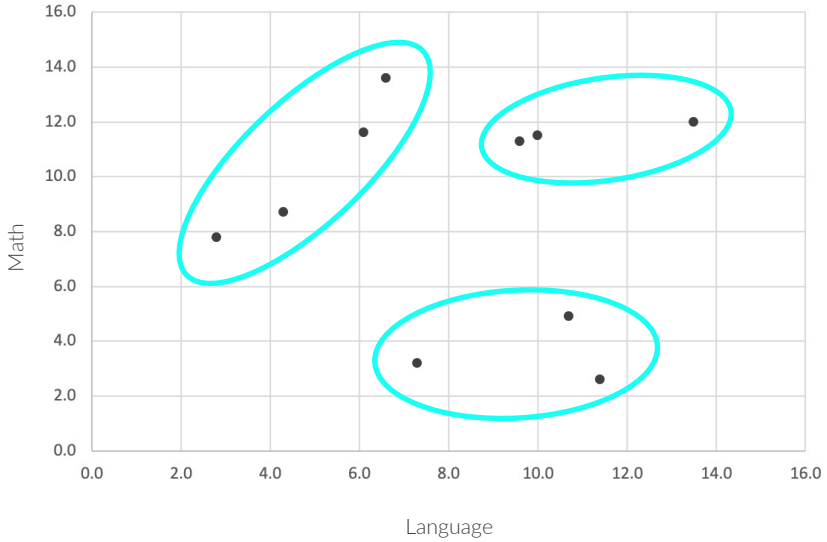
In this example, defining components for **language** and **math** helped us simplify and better understand the data set

- Using the new components we derived, we can conduct further analysis like predictive modeling, classification, clustering, etc.

PRINCIPAL COMPONENT ANALYSIS

- Dimensionality Reduction Basics
- **Principal Component Analysis**
- Advanced Techniques
- Key Takeaways

studentID	language	math
1	2.8	7.8
2	4.3	8.7
3	7.3	3.2
4	10.7	4.9
5	11.4	2.6
6	10.0	11.5
7	6.1	11.6
8	9.6	11.3
9	6.6	13.6
10	13.5	12.0



In this example, defining components for **language** and **math** helped us simplify and better understand the data set

- Using the new components we derived, we can conduct further analysis like predictive modeling, classification, clustering, etc.
- For example, clustering might help us understand student testing patterns *(most skew towards either language or math, while a few excel in both subjects)*



ADVANCED TECHNIQUES

Dimensionality Reduction Basics

Principal Component Analysis

Advanced Techniques

Key Takeaways



What about *non-linear* dimensionality reduction?

- Linear dimensionality reduction is the most well-researched and widely used method, and can be implemented across a broad range of BI and ML use cases
- Other, non-linear dimensionality reduction techniques do exist, but they are significantly more specialized and complex in terms of the underlying algorithms and math:
    - Kernel PCA
    - T-Distributed Stochastic Neighbors (T-SNE)
    - Uniform Manifold Approximation & Projection (UMAP)
    - Self-Organizing Maps

KEY TAKEAWAYS

- **Dimensionality reduction** is used to reduce the number of columns in a data set while losing as little information as possible
  - *This can be used for model optimization, or to identify overarching trends or patterns in a granular dataset*

- One of the most common dimensionality reduction techniques is known as **Principal Component Analysis** (PCA)
  - *PCA essentially fits lines through the observations in a data set to create new dimensions to analyze*

- Dimensionality reduction requires **human logic and intuition** to interpret what each component actually means
  - *This will help you simplify your data, understand it on a deeper level, and conduct further analysis*

WRAPPING UP

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WRAPPING UP

CONGRATULATIONS!

You now have a solid foundational understanding of unsupervised learning, including techniques like **clustering**, **association mining**, **outlier detection**, and **dimensionality reduction**.

We hope you've enjoyed the entire **Machine Learning for BI** series, and that you find an opportunity to put your skills to good use!



PART 1
QA & Data Profiling



PART 2
Classification



PART 3
Regression & Forecasting



PART 4
Unsupervised Learning